

as *angular impulse* or *moment of impulse*, and *angular momentum*, also called *moment of momentum*.

Angular momentum is a vector quantity and is usually given the symbol $\mathbf{H}$. For particle P of mass dm at location $\mathbf{R}_P$ with momentum $d\mathbf{L}$, the moment of momentum, $d\mathbf{H}$, of this single particle is defined, as implied by its name, as follows:

$$d\mathbf{H} = \mathbf{R}_P \times d\mathbf{L} = \mathbf{R}_P \times \mathbf{V}_P dm. \quad (a)$$

Analogous to the development in Sec. 12.12, we can express this equation as

$$\begin{aligned} d\mathbf{H} &= \mathbf{R}_P \times (\mathbf{V}_G + \boldsymbol{\omega} \times \mathbf{R}_{PG}) dm \\ &= \mathbf{R}_P \times \mathbf{V}_G dm + \mathbf{R}_P \times (\boldsymbol{\omega} \times \mathbf{R}_{PG}) dm \\ &= \mathbf{R}_P \times \mathbf{V}_G dm + \mathbf{R}_G \times (\boldsymbol{\omega} \times \mathbf{R}_{PG}) dm + \mathbf{R}_{PG} \times (\boldsymbol{\omega} \times \mathbf{R}_{PG}) dm. \end{aligned}$$

This equation can now be integrated over all particles of a rigid body:

$$\int d\mathbf{H} = \int \mathbf{R}_P dm \times \mathbf{V}_G + \mathbf{R}_G \times \left(\boldsymbol{\omega} \times \int \mathbf{R}_{PG} dm \right) + \int \mathbf{R}_{PG} \times (\boldsymbol{\omega} \times \mathbf{R}_{PG}) dm. \quad (b)$$

The left-hand side of this equation integrates to give the total angular momentum of the body about the origin of coordinates:

$$\int d\mathbf{H} = \mathbf{H}_O. \quad (c)$$

Recognizing the definition of the center of mass, we see that the first two terms on the right-hand side of Eq. (b) become

$$\int \mathbf{R}_P dm \times \mathbf{V}_G = m\mathbf{R}_G \times \mathbf{V}_G = \mathbf{R}_G \times \mathbf{L}, \quad (d)$$

$$\mathbf{R}_G \times \left(\boldsymbol{\omega} \times \int \mathbf{R}_{PG} dm \right) = \mathbf{R}_G \times (\boldsymbol{\omega} \times \mathbf{R}_{GG}) = \mathbf{0}. \quad (e)$$

As in Sec. 12.12, the final term of Eq. (b) must be expanded according to its components along the coordinate axes and integrated. Thus,

$$\begin{aligned} \int \mathbf{R}_{PG} \times (\boldsymbol{\omega} \times \mathbf{R}_{PG}) dm &= (I_G^{xx} \omega^x - I_G^{xy} \omega^y - I_G^{xz} \omega^z) \hat{\mathbf{i}} \\ &\quad + (-I_G^{yx} \omega^x - I_G^{yy} \omega^y - I_G^{yz} \omega^z) \hat{\mathbf{j}} \\ &\quad + (-I_G^{zx} \omega^x - I_G^{zy} \omega^y - I_G^{zz} \omega^z) \hat{\mathbf{k}}. \end{aligned} \quad (f)$$

Upon substituting Eqs. (c) through (f) into Eq. (b), we obtain

$$\begin{aligned}\mathbf{H}_O &= \mathbf{R}_G \times \mathbf{L} \\ &+ (+I_G^{xx}\omega^x - I_G^{xy}\omega^y - I_G^{xz}\omega^z)\hat{\mathbf{i}} \\ &+ (-I_G^{yx}\omega^x - I_G^{yy}\omega^y - I_G^{yz}\omega^z)\hat{\mathbf{j}} \\ &+ (-I_G^{zx}\omega^x - I_G^{zy}\omega^y - I_G^{zz}\omega^z)\hat{\mathbf{k}}.\end{aligned}\quad (g)$$

We can recognize that, if we had chosen a coordinate system with origin at the center of mass of the body, then $\mathbf{R}_G = \mathbf{0}$, and the above equation gives the angular momentum as

$$\mathbf{H}_G = H_G^x \hat{\mathbf{i}} + H_G^y \hat{\mathbf{j}} + H_G^z \hat{\mathbf{k}}, \quad (12.117)$$

where

$$\begin{aligned}H_G^x &= +I_G^{xx}\omega^x - I_G^{xy}\omega^y - I_G^{xz}\omega^z, \\ H_G^y &= -I_G^{yx}\omega^x + I_G^{yy}\omega^y - I_G^{yz}\omega^z, \\ H_G^z &= -I_G^{zx}\omega^x - I_G^{zy}\omega^y + I_G^{zz}\omega^z.\end{aligned}\quad (12.118)$$

However, when moments are taken about other than the center of mass, then the moment of momentum about an arbitrary point, O , is

$$\mathbf{H}_O = \mathbf{H}_G + \mathbf{R}_{GO} \times \mathbf{L}. \quad (12.119)$$

Now, as we did in Eq. (a) above, we can take the cross-product between the position vector of a particle and the terms of Newton's laws. Using an inertial coordinate system,

$$\sum \mathbf{R}_P \times d\mathbf{F} = \mathbf{R}_P \times \mathbf{A}_P dm. \quad (h)$$

If we consider that both force and acceleration may be functions of time, we can multiply both sides of this equation by dt and integrate between t_1 and t_2 . The result is

$$\sum \int_{t_1}^{t_2} \mathbf{R}_P \times d\mathbf{F}(t)dt = \mathbf{R}_P \times \mathbf{V}_P(t_2)dm - \mathbf{R}_P \times \mathbf{V}_P(t_1)dm.$$

After integrating over all particles of a rigid body, this yields

$$\sum \int_{t_1}^{t_2} \mathbf{M}_{ij}(t)dt = \mathbf{H}(t_2) - \mathbf{H}(t_1) = \Delta \mathbf{H}. \quad (12.120)$$

The integral on the left-hand side of this equation is the net result of all external moment impulses that occur on the body in the time interval t_1 to t_2 , and is called the *angular impulse*. On the right is the change in the moment of angular momentum that occurs over the same time interval as a result of the angular impulse.

If we now make the interval between t_1 and t_2 infinitesimal, divide Eq. (12.120) by this time interval, and take the limit, it becomes

$$\sum \mathbf{M}_{ij} = \frac{d\mathbf{H}}{dt}. \quad (12.121)$$

Therefore, *the resultant external moment acting upon a rigid body is equal to the time rate of change of its angular momentum*. If there is no net external moment acting on a body, then there can be no change in its angular momentum. Thus, with no net external moment,

$$\frac{d\mathbf{H}}{dt} = \mathbf{0} \quad \text{or} \quad \mathbf{H} = \text{constant} \quad (12.122)$$

is a statement of the *law of conservation of angular momentum*.

It would be wise now to review this section mentally, and to take careful note of how the coordinate axes may be selected for a particular application. This review will show that, whether stated or not, all results that were derived from Newton's law depend on the use of an inertial coordinate system. For example, the angular velocity used in Eq. (12.118) must be taken with respect to an absolute coordinate system so that the derivative of Eq. (12.121) contains the required absolute angular acceleration terms. Yet this seems contradictory, since the integration performed in finding the mass moments and products of inertia must often be done in a coordinate system attached to the body itself.

If the coordinate axes are chosen stationary, then the moments and products of inertia used in finding the angular momentum must be transformed to an inertial coordinate system, using the methods of Sec. 12.10, and therefore become functions of time. This complicates the use of Eqs. (12.120), (12.121), and (12.122). For this reason, it is usually preferable to choose the x , y , and z components to be directed along axes fixed in the moving body.

Equations (12.120), (12.121), and (12.122), being vector equations, can be expressed in any coordinate system, including body-fixed axes, and are still correct. This has the great advantage that the mass moments and products of inertia are constants for a rigid body. However, it must be kept in mind that the $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{k}}$ unit vectors are moving. They are functions of time in Eq. (12.117), and they have derivatives when using Eqs. (12.121) or (12.122). Therefore, the components of Eqs. (12.121), for example, cannot be found directly by differentiating Eqs. (12.118), but will be

$$\begin{aligned} \sum M_G^x &= I_G^{xx}\alpha^x - I_G^{xy}\alpha^y - I_G^{xz}\alpha^z + \omega^y H^z - \omega^z H^y, \\ \sum M_G^y &= -I_G^{yx}\alpha^x - I_G^{yy}\alpha^y - I_G^{yz}\alpha^z + \omega^z H^x - \omega^x H^z, \\ \sum M_G^z &= -I_G^{zx}\alpha^x - I_G^{zy}\alpha^y + I_G^{zz}\alpha^z + \omega^x H^y - \omega^y H^x. \end{aligned} \quad (12.123)$$

Here, we must be careful to find the *components* of $\boldsymbol{\omega}$ and $\boldsymbol{\alpha}$ along the axes of the body, even though they are absolute angular velocity and acceleration terms. In addition, we must remember when results are interpreted that these moment components are expressed along the body axes.

EXAMPLE 12.9

Figure 12.30 illustrates an offset flywheel on a turntable, which is a problem typical of situations occurring in the design or analysis of machines in which gyroscopic forces must be considered. A round plate designated body 2 rotates about its central axis with a constant angular velocity $\omega_2 = 5\hat{\mathbf{k}}$ rad/s. Mounted on this revolving plate are two bearings, *A* and *B*, which retain a shaft and a mass, 3, rotating at an angular velocity with respect to the rotating plate, $\omega_{3/2} = 350\hat{\mathbf{i}}$ rad/s. Rotating body 3 has a mass of 4.5 kg, a radius of gyration of 50 mm about its spin axis, and center of mass located at *G*. Calculate the bearing reactions at *A* and *B*.

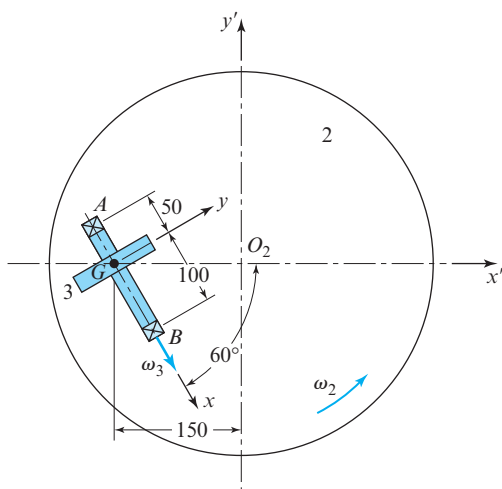


Figure 12.30 Offset flywheel on a turntable, with dimensions in millimeters.

ASSUMPTIONS

The weight of the shaft is negligible; body 3 rotates at a constant angular velocity with respect to the turntable; and bearing *B* can only support a radial load.

SOLUTION

We begin by choosing the local body coordinate system shown by axes *x* and *y* in Fig. 12.30. The origin is at the center of mass, and the axes are principal axes of inertia. Note that this coordinate system is fixed to body 2, not to body 3, and it is not stationary. Since body 3 is radially symmetric about this *x* axis, its mass properties remain constant.

From the given data and a kinematic analysis, we have

$$I_{G_3}^{xx} = m_3(k_3^x)^2 = (4.5 \text{ kg})(0.050 \text{ m})^2 = 0.01125 \text{ kg} \cdot \text{m}^2,$$

$$\omega_2 = 5\hat{\mathbf{k}}' \text{ rad/s} = 5\hat{\mathbf{k}} \text{ rad/s},$$

$$\omega_3 = \omega_2 + \omega_{3/2} = 350\hat{\mathbf{i}} + 5\hat{\mathbf{k}} \text{ rad/s},$$

$$\alpha_2 = \alpha_3 = \mathbf{0},$$

$$\begin{aligned}
\mathbf{R}_{GO_2} &= -(150 \cos 60^\circ) \hat{\mathbf{i}} - (150 \sin 60^\circ) \hat{\mathbf{j}} \text{ mm} \\
&= -0.075 \hat{\mathbf{i}} - 0.130 \hat{\mathbf{j}} \text{ m}, \\
\mathbf{V}_G &= \boldsymbol{\omega}_2 \times \mathbf{R}_{GO_2} = 0.650 \hat{\mathbf{i}} - 0.375 \hat{\mathbf{j}} \text{ m/s}, \\
\mathbf{R}_{AG} &= -0.050 \hat{\mathbf{i}} \text{ m}, \\
\mathbf{R}_{BG} &= 0.100 \hat{\mathbf{i}} \text{ m},
\end{aligned}$$

and the time rates of change of the unit vectors are

$$\begin{aligned}
\frac{d\hat{\mathbf{i}}}{dt} &= \boldsymbol{\omega}_2 \times \hat{\mathbf{i}} = 5 \hat{\mathbf{j}} \text{ rad/s}, \\
\frac{d\hat{\mathbf{j}}}{dt} &= \boldsymbol{\omega}_2 \times \hat{\mathbf{j}} = -5 \hat{\mathbf{i}} \text{ rad/s}, \\
\frac{d\hat{\mathbf{k}}}{dt} &= \boldsymbol{\omega}_2 \times \hat{\mathbf{k}} = \mathbf{0}.
\end{aligned}$$

From Eq. (12.113), we find the momentum, which is,

$$\begin{aligned}
\mathbf{L}_3 &= m_3 \mathbf{V}_G = (4.5 \text{ kg})(0.650 \hat{\mathbf{i}} - 0.375 \hat{\mathbf{j}} \text{ m/s}) \\
&= 2.93 \hat{\mathbf{i}} - 1.69 \hat{\mathbf{j}} \text{ kg} \cdot \text{m/s}.
\end{aligned}$$

From Eq. (12.115), we find the dynamic force effects attributable to the motion of the center of mass. Remembering that $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$ are moving and have derivatives, this equation gives

$$\begin{aligned}
\sum \mathbf{F}_{i3} &= \frac{d\mathbf{L}_3}{dt}, \\
\mathbf{F}_{A23} + \mathbf{F}_{B23} &= 2.93 \frac{d\hat{\mathbf{i}}}{dt} - 1.69 \frac{d\hat{\mathbf{j}}}{dt} = 8.45 \hat{\mathbf{i}} + 14.65 \hat{\mathbf{j}} \text{ N}, \\
F_{A23}^x + F_{B23}^x &= 8.45 \text{ N}, \\
F_{A23}^y + F_{B23}^y &= 14.65 \text{ N}, \\
F_{A23}^z + F_{B23}^z &= 0.
\end{aligned} \tag{1}$$

From Eqs. (12.118) and (12.117) we find the angular momentum, which is,

$$\begin{aligned}
H_{G_3}^x &= (0.01125 \text{ kg} \cdot \text{m}^2)(350 \text{ rad/s}) = 3.94 \text{ kg} \cdot \text{m}^2/\text{s}, \\
H_{G_3}^y &= 0, \\
H_{G_3}^z &= I_G^{zz}(5 \text{ rad/s}) = 5I_G^{zz} \text{ kg} \cdot \text{m}^2/\text{s}, \\
\mathbf{H}_{G_3} &= 3.94 \hat{\mathbf{i}} + 5I_G^{zz} \hat{\mathbf{k}} \text{ kg} \cdot \text{m}^2/\text{s}.
\end{aligned}$$

We can find the dynamic moment effects from Eq. (12.121); these are

$$\begin{aligned}\sum \mathbf{M}_{G_3} &= \frac{d\mathbf{H}_{G_3}}{dt}, \\ \mathbf{R}_{AG} \times \mathbf{F}_{A_{23}} + \mathbf{R}_{BG} \times \mathbf{F}_{B_{23}} &= 3.94 \frac{d\hat{\mathbf{i}}}{dt} + 5I_G^{zz} \frac{d\hat{\mathbf{k}}}{dt} \text{ N} \cdot \text{m}, \\ (-0.050\hat{\mathbf{i}} \text{ m}) \times \mathbf{F}_{A_{23}} + (0.100\hat{\mathbf{i}} \text{ m}) \times \mathbf{F}_{B_{23}} &= 3.94(5\hat{\mathbf{j}}) \text{ N} \cdot \text{m}.\end{aligned}$$

Equating the $\hat{\mathbf{j}}$ and $\hat{\mathbf{k}}$ components, respectively, gives

$$\begin{aligned}(0.050 \text{ m})F_{A_{23}}^z - (0.100 \text{ m})F_{B_{23}}^z &= 19.7 \text{ N} \cdot \text{m}, \\ -(0.050 \text{ m})F_{A_{23}}^y + (0.100 \text{ m})F_{B_{23}}^y &= 0.\end{aligned}\quad (2)$$

Since bearing B can only support a radial load, $F_{B_{23}}^x = 0$. Therefore, solving Eqs. (1) and (2) simultaneously, we find

$$\begin{aligned}F_{A_{23}}^x &= 8.45 \text{ N}, \\ F_{A_{23}}^y &= 9.77 \text{ N}, \\ F_{A_{23}}^z &= 131 \text{ N}, \\ F_{B_{23}}^y &= 4.88 \text{ N}, \\ F_{B_{23}}^z &= -131 \text{ N}.\end{aligned}$$

Therefore, the total bearing reactions at A and B are

$$\begin{aligned}\mathbf{F}_{A_{23}} &= 8.45\hat{\mathbf{i}} + 9.77\hat{\mathbf{j}} + 131\hat{\mathbf{k}} \text{ N}, & \text{Ans.} \\ \mathbf{F}_{B_{23}} &= 4.88\hat{\mathbf{j}} - 131\hat{\mathbf{k}} \text{ N}. & \text{Ans.}\end{aligned}$$

We note that the effect of the gyroscopic couple is to lift the front bearing upward and to push the rear bearing against the plate. We can also note that there is no x component in $\sum \mathbf{M}_{23}$; this justifies our assumption that $\alpha_3 = 0$ even though there is no motor controlling that shaft.

The general moment equation is given by Eq. (12.121) and can be written as

$$\sum \mathbf{M}_O = \dot{\mathbf{H}}_O, \quad (12.124)$$

where O denotes either a fixed point or the center of mass of the system. Recall that, in the derivation, the time derivative of the angular momentum is taken with respect to an absolute coordinate system. In many problems it is helpful to express the time derivative of the angular momentum in terms of components measured relative to a moving coordinate system xyz that has an angular velocity, $\boldsymbol{\Omega}$. The relation between the time derivative of the angular momentum in the moving xyz system and the absolute time derivative can be

written as

$$\dot{\mathbf{H}}_O = \left(\frac{dH_O}{dt} \hat{\mathbf{H}}_O \right)_{xyz} + \boldsymbol{\Omega} \times \mathbf{H}_O. \quad (12.125)$$

The first term on the right-hand side represents that part of $\dot{\mathbf{H}}_O$ caused by the change in the magnitude of $\mathbf{H}_O$, and the cross-product term represents that part caused by the change in the direction of $\mathbf{H}_O$. An application of this equation to a dynamic problem is demonstrated in the following example.

EXAMPLE 12.10

Bevel gear A is rolling around the fixed horizontal bevel gear B , as shown in Fig. 12.31. Gear A also rotates about the shaft OG with angular velocity ω_1 . The shaft OG , attached to a vertical shaft by clevis pin O , is initially at rest when it is given a constant angular acceleration, $\dot{\Omega}$. If the weight of gear A is 20 lb, and the weight of shaft OG is negligible, determine the tangential force and the normal force between the two gears as a function of the angle θ .

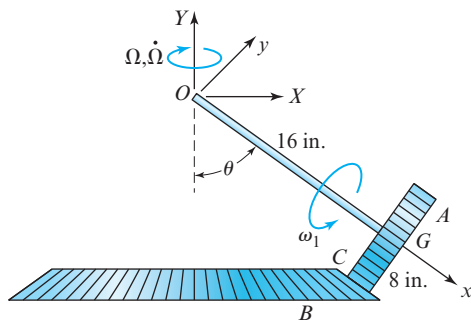


Figure 12.31 Meshing bevel gears.

SOLUTION

The coordinate system shown by the x and y axes in Fig. 12.31 is fixed to shaft OG and it is not stationary. The origin of this system is chosen at fixed point O , and the axes are principal axes of inertia. Since gear A is symmetric about the x axis, its mass properties remain constant.

The mass moments of inertia of gear A about point O are

$$I_O^{xx} = \frac{1}{2} m R^2 = \frac{1}{2} \left(\frac{20 \text{ lb}}{386 \text{ in/s}^2} \right) (8 \text{ in})^2 = 1.658 \text{ in} \cdot \text{lb} \cdot \text{s}^2, \quad (1)$$

$$\begin{aligned} I_O^{yy} &= I_O^{zz} = I_G^{yy} + m l^2 = 1/4 m R^2 + m l^2 \\ &= \left(\frac{20 \text{ lb}}{386 \text{ in/s}^2} \right) \left(\frac{(8 \text{ in})^2}{4} + (16 \text{ in})^2 \right) = 14.093 \text{ in} \cdot \text{lb} \cdot \text{s}^2, \end{aligned} \quad (2)$$

and, since these are principal axes, the products of inertia are

$$I_O^{xy} = I_O^{yz} = I_G^{xz} = 0. \quad (3)$$

The angular velocity and angular acceleration of gear A , respectively, are

$$\boldsymbol{\omega} = (\omega_1 + \Omega \cos \theta) \hat{\mathbf{i}} - \Omega \sin \theta \hat{\mathbf{j}} \text{ rad/s}, \quad (4)$$

$$\boldsymbol{\alpha} = (\dot{\omega}_1 + \dot{\Omega} \cos \theta) \hat{\mathbf{i}} - \dot{\Omega} \sin \theta \hat{\mathbf{j}} + \omega_1 \Omega \sin \theta \hat{\mathbf{k}} \text{ rad/s}^2. \quad (5)$$

The velocity of the point of contact, C , between the two gears can be written as

$$\mathbf{V}_C = \mathbf{V}_O + \boldsymbol{\omega} \times \mathbf{R}_{CO},$$

where $\mathbf{R}_{CO} = 16\hat{\mathbf{i}} - 8\hat{\mathbf{j}}$ in. Since the velocity of points O and C are zero then this equation can be written, with the aid of Eq. (4), as

$$\boldsymbol{\omega} \times \mathbf{R}_{CO} = [-(8 \text{ in})(\omega_1 + \Omega \cos \theta) + (16 \text{ in})\Omega \sin \theta] \hat{\mathbf{k}} = \mathbf{0}.$$

Rearranging this equation, the angular velocity of gear A relative to shaft OG is

$$\omega_1 = \Omega (2 \sin \theta - \cos \theta). \quad (6)$$

Differentiating this equation with respect to time gives

$$\dot{\omega}_1 = \dot{\Omega} (2 \sin \theta - \cos \theta). \quad (7)$$

Substituting Eqs. (6) and (7) into Eqs. (4) and (5), respectively, gives

$$\boldsymbol{\omega} = \Omega \sin \theta (2\hat{\mathbf{i}} - \hat{\mathbf{j}}), \quad (8)$$

$$\boldsymbol{\alpha} = \dot{\Omega} \sin \theta (2\hat{\mathbf{i}} - \hat{\mathbf{j}}) + \Omega^2 \sin \theta (2 \sin \theta - \cos \theta) \hat{\mathbf{k}}. \quad (9)$$

From Eq. (12.125), the time rate of change of the angular momentum of gear A can be written as

$$\dot{\mathbf{H}}_O = \left(\frac{dH_O}{dt} \right)_{xyz} \hat{\mathbf{k}} + \boldsymbol{\omega} \times \mathbf{H}_O. \quad (10)$$

From Eqs. (12.117) and (12.118), the angular momentum of gear A about point O can be written as

$$\mathbf{H}_O = I_O^{xx} \omega^x \hat{\mathbf{i}} + I_O^{yy} \omega^y \hat{\mathbf{j}} + I_O^{zz} \omega^z \hat{\mathbf{k}}. \quad (11)$$

Substituting Eqs. (1), (2), and (8) into this equation gives

$$\begin{aligned} \mathbf{H}_O &= (1.658 \text{ in} \cdot \text{lb} \cdot \text{s}^2) (2\Omega \sin \theta) \hat{\mathbf{i}} + (14.093 \text{ in} \cdot \text{lb} \cdot \text{s}^2) (-\Omega \sin \theta) \hat{\mathbf{j}} \\ &= \Omega \sin \theta (3.316\hat{\mathbf{i}} - 14.093\hat{\mathbf{j}} \text{ in} \cdot \text{lb} \cdot \text{s}^2). \end{aligned} \quad (12)$$

Differentiating Eq. (11) with respect to time gives

$$\left(\frac{dH_O}{dt} \hat{\mathbf{H}}_O \right)_{xyz} = I_O^{xx} \alpha^x \hat{\mathbf{i}} + I_O^{yy} \alpha^y \hat{\mathbf{j}} + I_O^{zz} \alpha^z \hat{\mathbf{k}}.$$

Substituting Eqs. (1), (2), and (9) into this equation gives

$$\begin{aligned} \left(\frac{dH_O}{dt} \hat{\mathbf{H}}_O \right)_{xyz} &= (1.658 \text{ in} \cdot \text{lb} \cdot \text{s}^2) (2\dot{\Omega} \sin \theta) \hat{\mathbf{i}} + (14.093 \text{ in} \cdot \text{lb} \cdot \text{s}^2) (-\dot{\Omega} \sin \theta) \hat{\mathbf{j}} \\ &\quad + (14.093 \text{ in} \cdot \text{lb} \cdot \text{s}^2) [\Omega^2 \sin \theta (2 \sin \theta - \cos \theta)] \hat{\mathbf{k}}, \end{aligned}$$

which can be written as

$$\left(\frac{dH_O}{dt} \hat{\mathbf{H}}_O \right)_{xyz} = \sin \theta [3.316 \dot{\Omega} \hat{\mathbf{i}} - 14.093 \dot{\Omega} \hat{\mathbf{j}} + 14.093 \Omega^2 (2 \sin \theta - \cos \theta) \hat{\mathbf{k}}] \text{ in} \cdot \text{lb} \cdot \text{s}^2. \quad (13)$$

Substituting Eqs. (8), (12), and (13) into Eq. (10) gives

$$\dot{\mathbf{H}}_O = \sin \theta [3.316 \dot{\Omega} \hat{\mathbf{i}} - 14.093 \dot{\Omega} \hat{\mathbf{j}} + \Omega^2 (3.316 \sin \theta - 14.093 \cos \theta) \hat{\mathbf{k}}] \text{ in} \cdot \text{lb} \cdot \text{s}^2. \quad (14)$$

From the free-body diagram of the shaft and the gear shown in Fig. 12.32, the sum of the moments about point O can be written as

$$\begin{aligned} \sum \mathbf{M}_O &= M_O^y \hat{\mathbf{j}} + \mathbf{R}_{GO} \times (20 \text{ lb}) (\cos \theta \hat{\mathbf{i}} - \sin \theta \hat{\mathbf{j}}) + \mathbf{R}_{GO} \times (N \hat{\mathbf{j}} + f \hat{\mathbf{k}}) \\ &= -(8 \text{ in}) f \hat{\mathbf{i}} + [M_O^y - (16 \text{ in}) f] \hat{\mathbf{j}} + [-(320 \text{ in} \cdot \text{lb}) \sin \theta + (16 \text{ in}) N] \hat{\mathbf{k}}, \end{aligned} \quad (15)$$

where f is the tangential force between the two gears at the point of contact C .

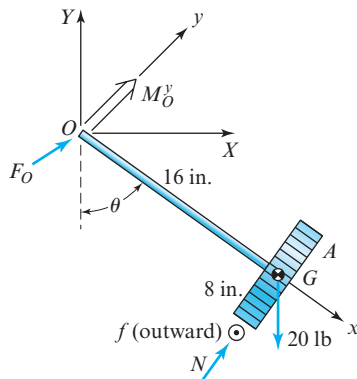


Figure 12.32 Free-body diagram of shaft OG and gear A .

Equating the $\hat{\mathbf{i}}$ components of Eqs. (14) and (15) gives

$$-(8 \text{ in})f = (3.316 \text{ in} \cdot \text{lb} \cdot \text{s}^2) \sin \theta \dot{\Omega}.$$

Therefore, the tangential force between the two gears is

$$f = -(0.415 \text{ lb} \cdot \text{s}^2) \sin \theta \dot{\Omega}. \quad \text{Ans.}$$

Equating the $\hat{\mathbf{j}}$ components of Eqs. (14) and (15) gives

$$M_O^y - (16 \text{ in})f = -(14.093 \text{ in} \cdot \text{lb} \cdot \text{s}^2) \sin \theta \dot{\Omega}.$$

Substituting the tangential force into this equation, and rearranging, the moment about point O can be written as

$$M_O^y = -(20.725 \text{ in} \cdot \text{lb} \cdot \text{s}^2) \sin \theta \dot{\Omega}.$$

Finally, equating the $\hat{\mathbf{k}}$ components of Eqs. (14) and (15) gives

$$-(320 \text{ in} \cdot \text{lb}) \sin \theta + (16 \text{ in})N = \Omega^2 \sin \theta (3.316 \sin \theta - 14.093 \cos \theta) \text{ in} \cdot \text{lb} \cdot \text{s}^2.$$

Rearranging this equation, the normal reaction force can be written as

$$N = 20 \sin \theta \text{ lb} + \Omega^2 \sin \theta (0.207 \sin \theta - 0.881 \cos \theta) \text{ lb} \cdot \text{s}^2. \quad \text{Ans.}$$

Note that the first term on the right-hand side of this equation represents the static gravitational effect, and the second term represents two opposing inertial effects. If gear A did not mesh with the fixed gear B , then the angular velocity ω_1 would be zero, and the centripetal acceleration of point G would decrease the normal reaction force N . Due to ω_1 , however, a gyroscopic moment is created that has the effect of forcing gear A downward. Note that if $\theta = \tan^{-1}(0.881/0.207) = 76.78^\circ$, then the two dynamic effects counterbalance each other and the normal reaction force reduces to its static value of $N = 20 \sin \theta \text{ lb}$.

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- [4] Meriam, J. L., and L. G. Kraige, 2007. *Engineering Mechanics: Dynamics*, 6th ed., New Jersey: Wiley.

PROBLEMS*

- 12.1** The steel bell crank is used as an oscillating cam follower. Using 0.282 lb/in^3 for the density of steel, find the mass moment of inertia of the lever about an axis through O .

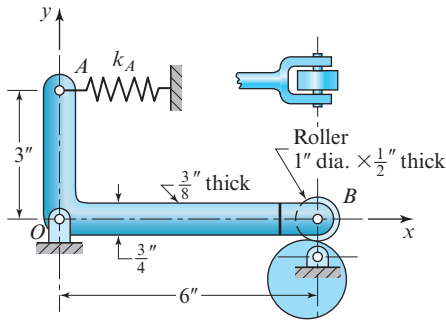


Figure P12.1

- 12.2** A 5-mm-by-50-mm-by-300-mm steel bar has two round steel disks, each 50 mm in diameter and 20 mm long, welded to one end. A small hole is drilled 25 mm from the other end. Using 7.80 Mg/m^3 for the density of steel, find the mass moment of inertia of the bar about an axis through the hole.

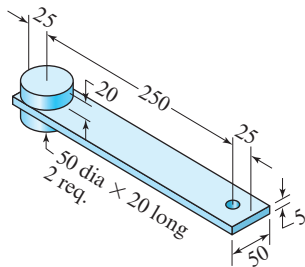


Figure P12.2 Dimensions in millimeters.

- 12.3** Determine the reaction forces at the joints and the external torque applied to input link 2 of the four-bar linkage in the posture shown. For the constant angular velocity $\omega_2 = 180\hat{k} \text{ rad/s}$, the known kinematics are $\alpha_3 = 4950\hat{k} \text{ rad/s}^2$, $\alpha_4 = -8900\hat{k} \text{ rad/s}^2$, $\mathbf{A}_{G_3} = 6320\hat{i} + 750\hat{j} \text{ ft/s}^2$, and $\mathbf{A}_{G_4} = 2280\hat{i} + 750\hat{j} \text{ ft/s}^2$. The weights and mass moments of inertia of the links are $w_3 = 0.708 \text{ lb}$, $w_4 = 0.780 \text{ lb}$, $I_{G_2} = 0.0258 \text{ in} \cdot \text{lb} \cdot \text{s}^2$, $I_{G_3} = 0.0154 \text{ in} \cdot \text{lb} \cdot \text{s}^2$, and $I_{G_4} = 0.0112 \text{ in} \cdot \text{lb} \cdot \text{s}^2$.

* Unless instructed otherwise, solve all problems without friction and without gravitational loads.

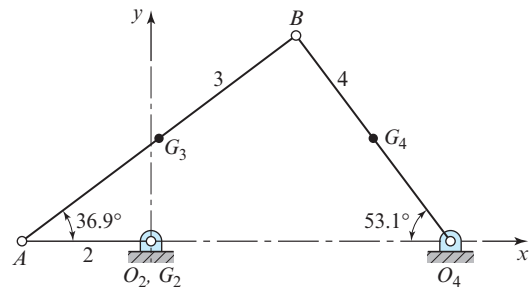


Figure P12.3 $R_{AO_2} = 3 \text{ in}$, $R_{O_4O_2} = 7 \text{ in}$, $R_{BA} = 8 \text{ in}$, $R_{BO_4} = 6 \text{ in}$, $R_{G_3A} = 4 \text{ in}$, and $R_{G_4O_4} = 3 \text{ in}$.

- 12.4** Determine the reaction forces at the joints and the external torque applied to input link 2 of the four-bar linkage in the posture shown. For the constant angular velocity $\omega_2 = 200\hat{k} \text{ rad/s}$, the known kinematics are $\alpha_3 = -6500\hat{k} \text{ rad/s}^2$, $\alpha_4 = -240\hat{k} \text{ rad/s}^2$, $\mathbf{A}_{G_3} = -3160\hat{i} + 262\hat{j} \text{ ft/s}^2$, and $\mathbf{A}_{G_4} = -800\hat{i} - 2110\hat{j} \text{ ft/s}^2$. The weights and mass moments of inertia of the links are $w_3 = 2.65 \text{ lb}$, $w_4 = 6.72 \text{ lb}$, $I_{G_2} = 0.0239 \text{ in} \cdot \text{lb} \cdot \text{s}^2$, $I_{G_3} = 0.0606 \text{ in} \cdot \text{lb} \cdot \text{s}^2$, and $I_{G_4} = 0.531 \text{ in} \cdot \text{lb} \cdot \text{s}^2$.

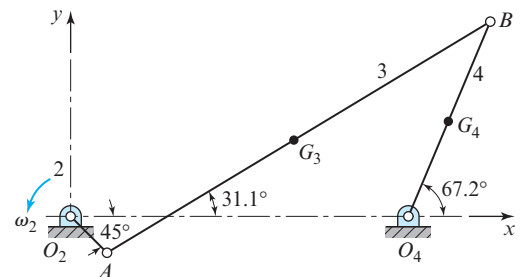


Figure P12.4 $R_{AO_2} = 2 \text{ in}$, $R_{O_4O_2} = 13 \text{ in}$, $R_{BA} = 17 \text{ in}$, $R_{BO_4} = 8 \text{ in}$, and $R_{G_4O_4} = 4 \text{ in}$.

- 12.5** Determine the reaction forces at the joints and the crank torque of the slider-crank linkage in the posture shown. For the constant angular velocity $\omega_2 = 210\hat{k} \text{ rad/s}$, the known kinematics are: $\alpha_3 = -7670\hat{k} \text{ rad/s}^2$, $\mathbf{A}_{G_3} = -7820\hat{i} + 4876\hat{j} \text{ ft/s}^2$, and $\mathbf{A}_{G_4} = -7850\hat{i} \text{ ft/s}^2$. The weights and mass moments of inertia of the links are $w_3 = 3.40 \text{ lb}$, $w_4 = 2.86 \text{ lb}$, $I_{G_2} = 0.352 \text{ in} \cdot \text{lb} \cdot \text{s}^2$, and $I_{G_3} = 0.108 \text{ in} \cdot \text{lb} \cdot \text{s}^2$.

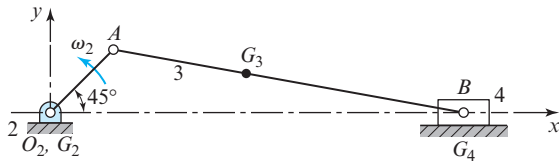


Figure P12.5 $R_{AO_2} = 3$ in, $R_{BA} = 12$ in, $R_{G_3A} = 4.5$ in

- 12.6** Determine the reaction forces at the joints and the crank torque of the slider-crank linkage in the posture shown if the external force acting through pin B of the piston is $\mathbf{F}_B = -800\hat{i}$ lb. For the constant angular velocity $\omega_2 = 160\hat{k}$ rad/s, the known kinematics are: $\alpha_3 = -3\,090\hat{k}$ rad/s², $\mathbf{A}_{G_2} = 2\,640$ ft/s² $\angle 150^\circ$, $\mathbf{A}_{G_3} = 6\,130$ ft/s² $\angle 158.3^\circ$, and $\mathbf{A}_{G_4} = 6\,280$ ft/s² $\angle 180^\circ$. The weights and mass moments of inertia of the links are $w_2 = 0.95$ lb, $w_3 = 3.50$ lb, $w_4 = 2.50$ lb, $I_{G_2} = 0.003\,69$ in $\cdot$ lb $\cdot$ s², and $I_{G_3} = 0.110$ in $\cdot$ lb $\cdot$ s².

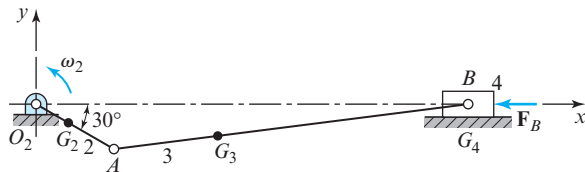


Figure P12.6 $R_{AO_2} = 3$ in, $R_{BA} = 12$ in, $R_{G_2O_2} = 1.25$ in, and $R_{G_3A} = 3.5$ in.

- 12.7** Determine the reaction forces at the joints and the torque applied to input link 2 of the four-bar linkage in the posture when $\theta_2 = 53^\circ$. For the constant angular velocity $\omega_2 = 12\hat{k}$ rad/s ccw, the known kinematics are $\theta_3 = 0.7^\circ$, $\theta_4 = 20.4^\circ$,

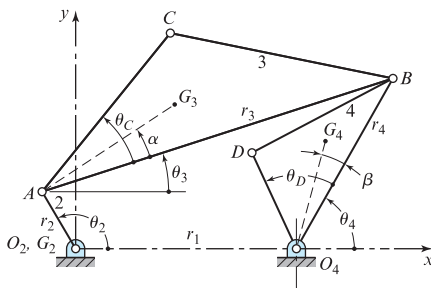


Figure P12.7 $R_{AO_2} = 0.3$ m, $R_{O_4O_2} = 0.9$ m, $R_{BA} = 1.5$ m, $R_{BO_4} = 0.8$ m, $\theta_C = 33^\circ$, $R_{CA} = 0.85$ m, $\theta_D = 53^\circ$, $R_{DO_4} = 0.4$ m, $\alpha = 16^\circ$, $R_{G_3A} = 0.65$ m, $\beta = 17^\circ$, and $R_{G_4O_4} = 0.45$ m.

$\alpha_3 = 85.6$ rad/s² cw, $\alpha_4 = 172$ rad/s² cw, $\mathbf{A}_{G_3} = 96.4$ m/s² $\angle 259^\circ$, and $\mathbf{A}_{G_4} = 97.8$ m/s² $\angle 270^\circ$. The masses and mass moments of inertia of the links are $m_2 = 5.2$ kg, $m_3 = 65.8$ kg, $m_4 = 21.8$ kg, $I_{G_2} = 2.3$ kg $\cdot$ m², $I_{G_3} = 4.2$ kg $\cdot$ m², and $I_{G_4} = 0.51$ kg $\cdot$ m².

- 12.8** Solve Problem 12.7 with external force $\mathbf{F}_D = 12\hat{i}$ kN acting at point D .

- 12.9** Make a complete kinematic and dynamic analysis of the four-bar linkage of Problem 12.7 using the data in the figure caption but in the posture when $\theta_2 = 170^\circ$. The constant angular velocity of input link 2 is $\omega_2 = 12$ rad/s ccw, and the external force at point D is $\mathbf{F}_D = 8.94$ kN $\angle 64.3^\circ$.

- 12.10** Repeat Problem 12.9 in the posture when $\theta_2 = 200^\circ$. The constant angular velocity of input link 2 is $\omega_2 = 12$ rad/s ccw, the external force at point C is $\mathbf{F}_C = 8.49$ kN $\angle 45^\circ$, and there is no external force at point D .

- 12.11** Make a complete dynamic analysis of the four-bar linkage of Problem 12.7, but in the posture when $\theta_2 = 270^\circ$. The constant angular velocity of input link 2 is $\omega_2 = 18$ rad/s ccw, and the external force at point D is $\mathbf{F}_D = 8.94$ kN $\angle 64.3^\circ$. The known kinematics are: $\theta_3 = 46.6^\circ$, $\theta_4 = 80.5^\circ$, $\alpha_3 = 178$ rad/s² cw, $\alpha_4 = 256$ rad/s² cw, $\mathbf{A}_{G_3} = 112$ m/s² $\angle 22.7^\circ$, and $\mathbf{A}_{G_4} = 119$ m/s² $\angle 352.5^\circ$.

- 12.12** Make a complete dynamic analysis of the four-bar linkage in the posture when $\theta_2 = 90^\circ$, $\theta_3 = 23.9^\circ$, and $\theta_4 = 91.7^\circ$. For the constant angular velocity $\omega_2 = 32$ rad/s ccw, the known kinematics are $\alpha_3 = 221$ rad/s² ccw, $\alpha_4 = 122$ rad/s² ccw, $\mathbf{A}_{G_3} = 88.6$ m/s² $\angle 255^\circ$,

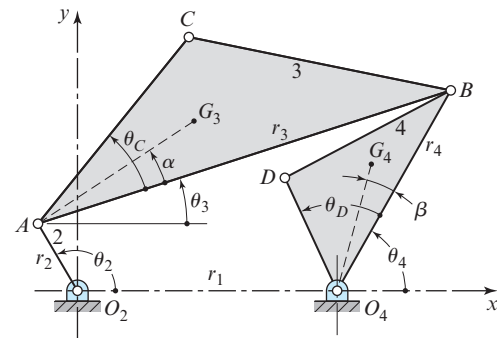


Figure P12.12 $R_{AO_2} = 120$ mm, $R_{O_4O_2} = 300$ mm, $R_{BA} = 320$ mm, $R_{BO_4} = 250$ mm, $\theta_C = 15^\circ$, $R_{CA} = 360$ mm, $\theta_D = 0^\circ$, $R_{DO_4} = 0$, $R_{G_2O_2} = 0$, $\alpha = 8^\circ$, $R_{G_3A} = 200$ mm, $\beta = 0^\circ$, and $R_{G_4O_4} = 125$ mm.

and $\mathbf{A}_{G_4} = 32.6 \text{ m/s}^2 \angle 244^\circ$. There is an external force $\mathbf{F}_C = 632 \text{ kN} \angle 342^\circ$ acting at point C . The masses and mass moments of inertia of the links are $m_2 = 0.5 \text{ kg}$, $m_3 = 4 \text{ kg}$, $m_4 = 1.5 \text{ kg}$, $I_{G_2} = 0.005 \text{ N} \cdot \text{m} \cdot \text{s}^2$, $I_{G_3} = 0.011 \text{ N} \cdot \text{m} \cdot \text{s}^2$, and $I_{G_4} = 0.0023 \text{ N} \cdot \text{m} \cdot \text{s}^2$.

12.13 Make a complete kinematic and dynamic analysis of the four-bar linkage in Problem 12.12 but in the posture when $\theta_2 = 260^\circ$.

12.14 Make a complete kinematic and dynamic analysis of the four-bar linkage in Problem 12.12 but in the posture when $\theta_2 = 300^\circ$.

12.15 Make a complete kinematic and dynamic analysis of the offset slider-crank linkage in the posture when $\theta_2 = 120^\circ$, and the constant angular velocity is $\omega_2 = 18 \text{ rad/s}$ cw. The masses and mass moments of inertia of the links are $m_2 = 2.5 \text{ kg}$, $m_3 = 7.4 \text{ kg}$, $m_4 = 2.5 \text{ kg}$, $I_{G_2} = 0.005 \text{ N} \cdot \text{m} \cdot \text{s}^2$, and $I_{G_3} = 0.013 \text{ N} \cdot \text{m} \cdot \text{s}^2$. The external forces acting at points B and C are $\mathbf{F}_B = -2000\hat{i} \text{ N}$ and $\mathbf{F}_C = -1000\hat{i} \text{ N}$, respectively.

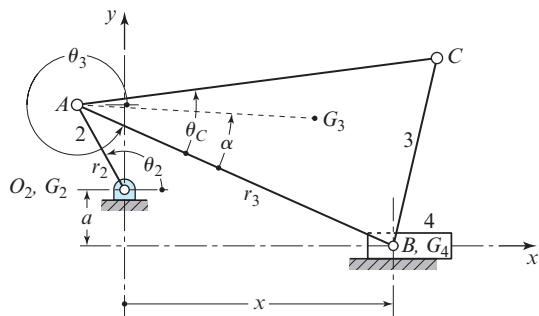


Figure P12.15 $a = 0.06 \text{ m}$, $R_{AO_2} = 0.1 \text{ m}$, $R_{BA} = 0.38 \text{ m}$, $\theta_C = 32^\circ$, $R_{CA} = 0.4 \text{ m}$, $\alpha = 22^\circ$, and $R_{G_3A} = 0.26 \text{ m}$.

12.16 Perform a kinematic and dynamic analysis of the offset slider-crank linkage of Problem 12.15 for a complete rotation of the crank. The forces are $\mathbf{F}_B = -1000\hat{i} \text{ N}$ and $\mathbf{F}_C = \mathbf{0}$ when the velocity of link 4 is to the right, and the forces are $\mathbf{F}_B = \mathbf{F}_C = \mathbf{0}$ when the velocity of link 4 is to the left. Plot the crank torque, T_{12} , versus the crank angle, θ_2 .

12.17 Make a complete dynamic analysis of the slider-crank linkage in the posture when $\theta_2 = 120^\circ$. For the constant angular velocity $\omega_2 = 24 \text{ rad/s}$ cw, the known kinematics are $\theta_3 = -9^\circ$,

$R_B = 0.374 \text{ m}$, $\alpha_3 = 89.3 \text{ rad/s}^2$ ccw, $\mathbf{A}_B = 40.6\hat{i} \text{ m/s}^2$, and $\mathbf{A}_{G_3} = 40.6\hat{i} - 22.6\hat{j} \text{ m/s}^2$. The constant crank torque is $T_{12} = 60 \text{ N} \cdot \text{m}$. The masses and mass moments of inertia of the links are $m_2 = 1.5 \text{ kg}$, $m_3 = 3.5 \text{ kg}$, $m_4 = 1.2 \text{ kg}$, $I_{G_2} = 0.010 \text{ N} \cdot \text{m} \cdot \text{s}^2$, and $I_{G_3} = 0.060 \text{ N} \cdot \text{m} \cdot \text{s}^2$.

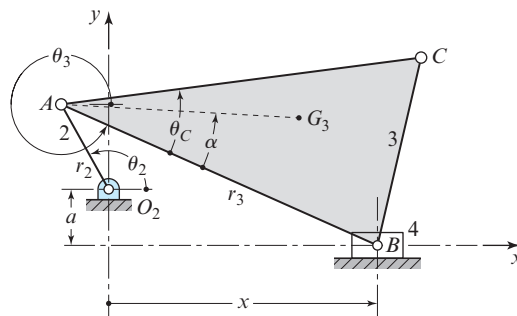


Figure P12.17 $a = 0$, $R_{AO_2} = 0.1 \text{ m}$, $R_{BA} = 0.45 \text{ m}$, $\theta_C = 0$, $R_{CB} = 0$, $\alpha = 0$, and $R_{G_3A} = 0.2 \text{ m}$.

12.18 Repeat Problem 12.17 in the posture where $\theta_2 = 240^\circ$. The known kinematics are $\theta_3 = 11.1^\circ$, $R_B = 0.392 \text{ m}$, $\alpha_3 = 112 \text{ rad/s}^2$ cw, $\mathbf{A}_B = 35.2\hat{i} \text{ m/s}^2$, and $\mathbf{A}_{G_3} = 31.6\hat{i} - 27.7\hat{j} \text{ m/s}^2$.

12.19 Make a complete kinematic and dynamic analysis of the offset slider-crank linkage in the posture when $\theta_2 = 120^\circ$ and the constant angular velocity $\omega_2 = 6 \text{ rad/s}$ ccw and $\alpha_2 = 0$. The external forces at points B and C are $\mathbf{F}_B = -50\hat{i} \text{ kN}$ and $\mathbf{F}_C = 80 \text{ kN} \angle -60^\circ$, respectively. The masses and mass moments of inertia of the links are $m_2 = 10 \text{ kg}$, $m_3 = 140 \text{ kg}$, $m_4 = 50 \text{ kg}$, $I_{G_2} = 2.0 \text{ N} \cdot \text{m} \cdot \text{s}^2$, and $I_{G_3} = 8.42 \text{ N} \cdot \text{m} \cdot \text{s}^2$.

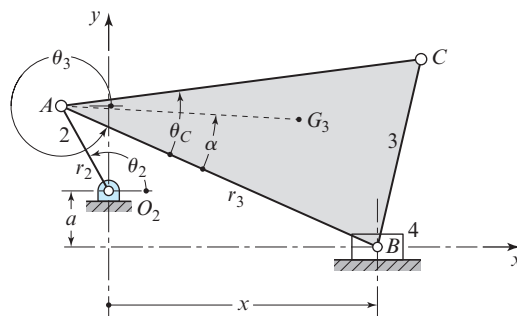


Figure P12.19 $a = 0.008 \text{ m}$, $R_{AO_2} = 0.25 \text{ m}$, $R_{CA} = 1.0 \text{ m}$, $\theta_C = -38^\circ$, $\alpha = -18^\circ$, and $R_{G_3A} = 0.75 \text{ m}$.

- 12.20** Find the driving torque and the reaction forces at the joints for the crossed linkage in the posture shown. For the constant angular velocity $\omega_2 = 10$ rad/s ccw, the known kinematics are $\omega_3 = 1.43$ rad/s cw, $\omega_4 = 11.43$ rad/s cw, $\alpha_3 = \alpha_4 = 84.8$ rad/s² ccw, and $\mathbf{A}_{G_3} = 25.92\hat{\mathbf{i}} + 24.58\hat{\mathbf{j}}$ ft/s². The external force at point C is $\mathbf{F}_C = -30\hat{\mathbf{j}}$ lb. The weights and mass moments of inertia of the links are $w_3 = 4$ lb, $I_{G_2} = I_{G_4} = 0.063$ in $\cdot$ lb $\cdot$ s², and $I_{G_3} = 0.497$ in $\cdot$ lb $\cdot$ s².

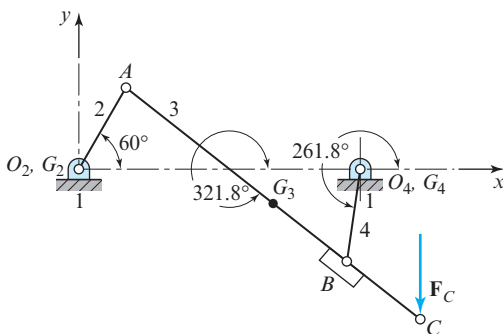


Figure P12.20 $R_{AO_2} = 6$ in, $R_{O_4O_2} = 18$ in, $R_{BA} = 18$ in, $R_{BO_4} = 6$ in, $R_{CA} = 24$ in, and $R_{G_3A} = 12$ in.

- 12.21** Find the driving torque and the reaction forces at the joints for Problem 12.20 under the same dynamic conditions but with crank 4 as the driver of the linkage.
- 12.22** Using the same force, $\mathbf{F}_C$, as in Problem 12.20, compute the crank torque and the reaction forces at the joints in the posture when $\theta_2 = 210^\circ$. For the constant angular velocity $\omega_2 = 10$ rad/s ccw, the known kinematics are $\theta_3 = 14.7^\circ$, $\theta_4 = 164.7^\circ$, $\omega_3 = 4.73$ rad/s ccw, $\omega_4 = 5.27$ rad/s cw, $\alpha_3 = \alpha_4 = 10.39$ rad/s cw, and $\mathbf{A}_{G_3} = 26$ ft/s² $\angle 20.85^\circ$.
- 12.23** Make a kinematic and dynamic analysis of the linkage for a complete rotation of the crank with the constant angular velocity $\omega_2 = 10$ rad/s ccw. The external force at point C is $\mathbf{F}_C = -500\hat{\mathbf{i}} + 886\hat{\mathbf{j}}$ lb for $90^\circ \leq \theta_2 \leq 300^\circ$, $\mathbf{F}_C = \mathbf{0}$ otherwise. The weights and the mass moments of inertias of the links are $w_3 = 222$ lb, $w_4 = 208$ lb, $I_{G_3} = 226$ in $\cdot$ lb $\cdot$ s², and $I_{G_4} = 264$ in $\cdot$ lb $\cdot$ s².
- 12.24** The motor is geared to a shaft on which a flywheel is mounted. The mass moments of inertia of the parts are flywheel, $I = 2.73$ in $\cdot$ lb $\cdot$ s²; flywheel

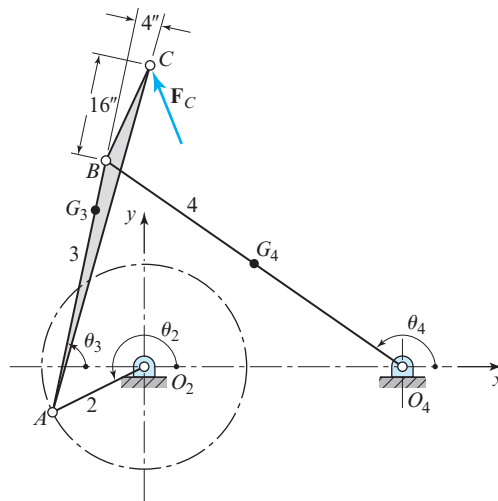


Figure P12.23 $R_{AO_2} = 16$ in, $R_{O_4O_2} = R_{BA} = 40$ in, $R_{BO_4} = 56$ in, $R_{G_3A} = 32$ in, and $R_{G_4O_2} = 20$ in.

shaft, $I = 0.0155$ in $\cdot$ lb $\cdot$ s²; gear, $I = 0.172$ in $\cdot$ lb $\cdot$ s²; pinion, $I = 0.00349$ in $\cdot$ lb $\cdot$ s², and motor, $I = 0.0864$ in $\cdot$ lb $\cdot$ s². If the motor has a starting torque of 75 in $\cdot$ lb, determine the angular acceleration of the flywheel shaft at the instant the motor is started.

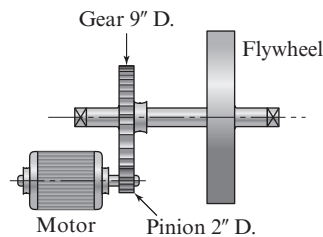


Figure P12.24

- 12.25** The disk cam of Problem 11.31 is driven at the constant input shaft speed $\omega_2 = 20$ rad/s ccw. Both the cam and the follower have been balanced so that the centers of mass of each are located at their respective fixed pivots. The mass and radius of gyration of the cam are 0.075 kg and 30 mm, respectively, and the mass and radius of gyration of the follower are 0.030 kg and 35 mm, respectively. At the instant shown, determine the torque, $\mathbf{T}_{12}$, required on the camshaft to produce this motion.

12.26 Repeat Problem 12.25 with the constant shaft speed $\omega_2 = 40$ rad/s ccw.

12.27 A rotating drum is pivoted at O_2 and is decelerated by the double-shoe brake mechanism. The weight and radius of gyration of the drum are 230 lb and 5.66 in, respectively. The brake is actuated by the force $\mathbf{P} = -100\hat{\mathbf{j}}$ lb. Assume that the contact points between the two shoes and the drum are C and D , where the coefficients of Coulomb friction are $\mu = 0.300$. Determine the angular deceleration of the drum and the reaction force, $\mathbf{F}_{12}$, at the fixed pivot.

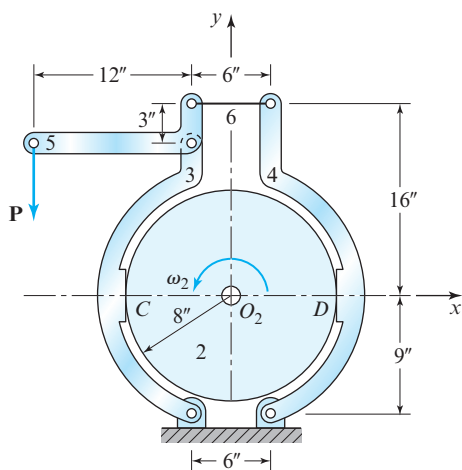


Figure P12.27

12.28 The length of link 4 is 0.20 m, symmetric about O_4 , and the ground bearing is midway between E and G_2 . Link 2 is in translation with velocity $\mathbf{V}_{G_2} = 0.114\hat{\mathbf{j}}$ m/s and acceleration $\mathbf{A}_{G_2} = -0.35\hat{\mathbf{j}}$ m/s², and the acceleration of the mass center of link 3 is $\mathbf{A}_{G_3} = +1.053\hat{\mathbf{i}} + 0.432\hat{\mathbf{j}}$ m/s². The kinematic coefficients of the linkage are $\theta'_3 = -11.5$ m⁻¹, $\theta''_3 = -380$ m⁻², $R'_{43} = +2$ m/m, and $R''_{43} = +40$ m/m² (where $\mathbf{R}_{43}$ is the vector from G_3 to G_4). A moment, M_{12} , acts on input link 2, and a torque, T_4 , acts on link 4. The masses and second moments of mass of the links are $m_2 = m_4 = 0.5$ kg, $m_3 = 1$ kg, $I_{G_2} = I_{G_4} = 2$ kg · m², and $I_{G_3} = 5$ kg · m². Gravity is in the negative z direction. Determine the internal reaction forces, the moment, M_{12} , and the torque, T_4 .

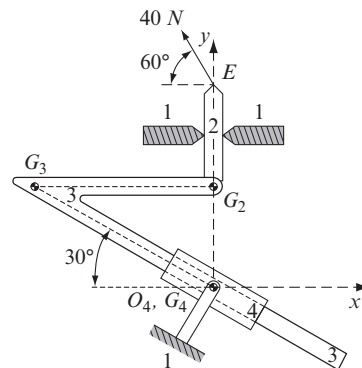


Figure P12.28 $R_{G_2O_4} = 0.15$ m and $R_{EG_2} = 0.20$ m.

12.29 The kinematic coefficients for the elliptic trammel linkage are $\theta'_3 = -2$ rad/m, $\theta''_3 = -6.928$ rad/m², $R'_4 = -1.732$ m/m, and $R''_4 = -8$ m/m². A linear spring is attached between O and A with a free length $L = 0.5$ m and spring constant $K = 2500$ N/m. A viscous damper with damping coefficient $C = 45$ N · s/m is connected between the ground and link 4. The masses and moments of inertia of the links are $m_2 = 0.75$ kg, $m_3 = 2.0$ kg, $m_4 = 1.5$ kg, $I_{G_2} = 0.25$ N · m · s², $I_{G_3} = 1.0$ N · m · s², and $I_{G_4} = 0.35$ N · m · s². The velocity, acceleration, and force acting on input link 2 are $\mathbf{V}_{A_2} = -5\hat{\mathbf{j}}$ m/s, $\mathbf{A}_{A_2} = -20\hat{\mathbf{j}}$ m/s², and $\mathbf{F} = -200\hat{\mathbf{j}}$ N, respectively. If gravity acts in the negative y direction, determine: (a) the first-order kinematic coefficients of the spring and the viscous damper; (b) the equivalent mass of the linkage; and (c) the horizontal force, P acting on link 4.

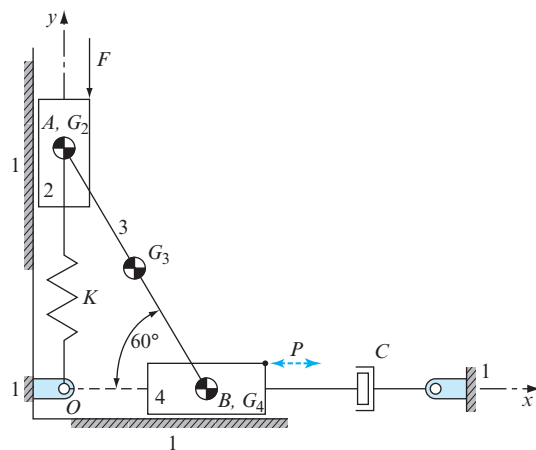


Figure P12.29 $R_{BA} = 1$ m and $R_{G_3G_2} = 0.5$ m.

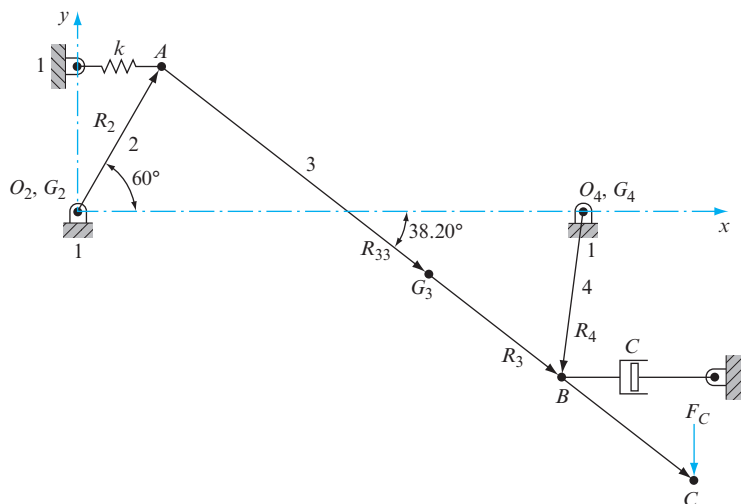


Figure P12.30 $R_{AO_2} = 6$ in, $R_{O_4O_2} = 18$ in, $R_{BA} = 18$ in, $R_{BO_4} = 6$ in, $R_{CA} = 24$ in, and $R_{G_3A} = 12$ in.

12.30 The input crank of the four-bar linkage is rotating with the constant angular velocity $\omega_2 = 10$ rad/s ccw. The angular acceleration of link 3 and the acceleration of the mass center of link 3 are $\alpha_3 = 84.8$ rad/s² ccw and $\mathbf{A}_{G_3} = 310\hat{\mathbf{i}} + 295\hat{\mathbf{j}}$ in/s², respectively. The masses and second moments of mass of the links are specified in Problem 12.20 with the exception that the weight of link 3 is $w_3 = 10$ lb. The spring has stiffness $k = 12$ lb/in and free length $R_0 = 4.5$ in. The viscous damper has a damping coefficient $C = 0.25$ lb · s/in. The external force acting at point C is $\mathbf{F}_C = -125\hat{\mathbf{j}}$ lb and gravity is in the negative y direction. Determine the equivalent mass moment of inertia of the linkage and the driving torque, T_2 .

12.31 For the Scotch-yoke linkage in the posture shown, the angle $\varphi = 30^\circ$, and the angular velocity and acceleration of input link 2 are $\omega_2 = 15\hat{\mathbf{k}}$ rad/s and $\alpha_2 = 2\hat{\mathbf{k}}$ rad/s², respectively. The accelerations of the centers of mass of the links are $\mathbf{A}_{G_2} = -5.4\hat{\mathbf{i}} + 11.3\hat{\mathbf{j}}$ m/s², $\mathbf{A}_{G_3} = -10.8\hat{\mathbf{i}} + 22.6\hat{\mathbf{j}}$ m/s², and $\mathbf{A}_{G_4} = 22.6\hat{\mathbf{j}}$ m/s². The masses and mass moments of inertia of the links are $m_2 = 5$ kg, $m_3 = 5$ kg, $m_4 = 15$ kg, $I_{G_2} = 0.02$ N · m · s², $I_{G_3} = 0.12$ N · m · s², and $I_{G_4} = 0.08$ N · m · s². Gravity is in the negative y direction, an external force $\mathbf{P} = 125\hat{\mathbf{j}}$ N is acting on link 4, and an unknown torque, T_2 , is acting on link 2. Determine the internal reaction forces and the torque, T_2 . Indicate the point(s) of contact between link 4 and the ground link.

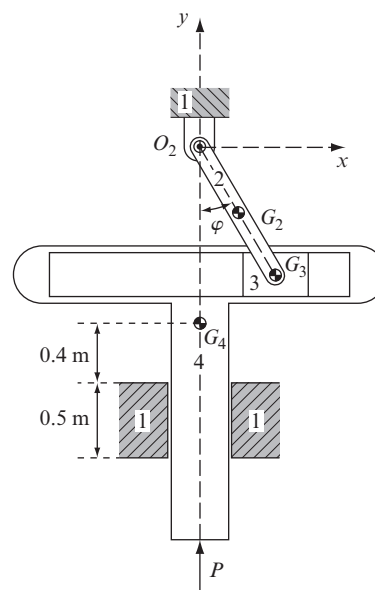


Figure P12.31 $R_2 = 1$ m

12.32 For the parallelogram four-bar linkage in the posture shown, the angular velocity and acceleration of input link 2 are $\omega_2 = 2$ rad/s ccw and $\alpha_2 = 1$ rad/s² ccw, respectively. The first- and second-order kinematic coefficients of the center of mass of link 3 are $x'_{G_3} = 0.141$ m/rad, $y'_{G_3} = -0.141$ m/rad, $x''_{G_3} = 0.141$ m/rad², and $y''_{G_3} = -0.141$ m/rad². The masses and second moments

of mass of the links are $m_2 = m_4 = 0.5$ kg, $I_{G_2} = I_{G_4} = 2$ kg $\cdot$ m², $m_3 = 1$ kg, and $I_{G_3} = 5$ kg $\cdot$ m². Gravity is in the negative z direction, the force $\mathbf{F}_C = 100\hat{i}$ N acts at point C , and torque $\mathbf{T}_4 = 10\hat{k}$ N $\cdot$ m acts on link 4. Determine: (a) the acceleration of the mass center of link 3; (b) the internal reaction forces, $\mathbf{F}_{23}$ and $\mathbf{F}_{43}$, and (c) the torque, $\mathbf{T}_2$.

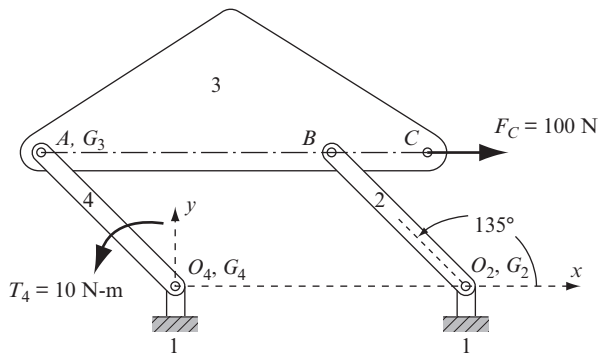


Figure P12.32 $R_{BO_2} = R_{AO_4} = 0.2$ m, $R_{BA} = R_{O_2O_4} = 0.3$ m, and $R_{CB} = 0.1$ m.

12.33 For the mechanism in the posture shown, the distance $R_{OG_3} = 2.5$ m and the velocity and acceleration of input link 2 are $\mathbf{V}_{G_2} = -10\hat{i}$ m/s and $\mathbf{A}_{G_2} = 10\hat{i}$ m/s², respectively. The first- and second-order kinematic coefficients of link 3 are $\theta'_3 = 0.20$ m⁻¹ and $\theta''_3 = -0.1386$ m⁻², respectively. The masses and second moments of mass of links 2 and 3 are $m_2 = 3$ kg, $m_3 = 5$ kg, $I_{G_2} = 1.5$ kg $\cdot$ m², and $I_{G_3} = 7.5$ kg $\cdot$ m². Gravity is in the negative y direction, force $\mathbf{F}_C = -10\hat{j}$ N

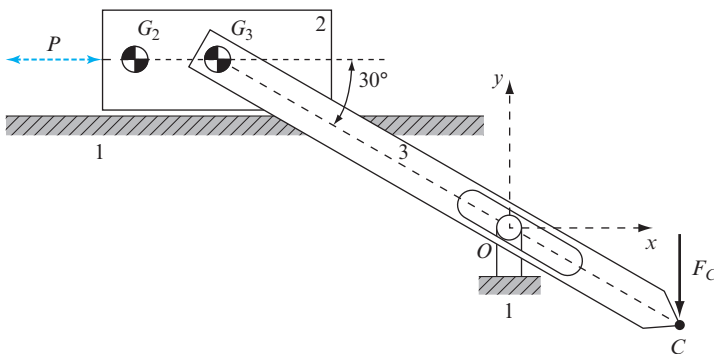


Figure P12.33 Link 2 is 1.5 m by 0.75 m, $R_{G_2G_3} = 0.5$ m, and $R_{CG_3} = 3.5$ m.

acts at point C , and the line of action of the unknown force, P , acting on link 2 is parallel to the x axis. Determine: (a) the internal reaction forces; (b) force P ; and (c) indicate the point(s) of contact of link 2 with the ground link.

12.34 For the slider-crank linkage in the posture when $\theta_2 = 45^\circ$, the angular velocity and acceleration of the input link 2 are $\omega_2 = 100\hat{k}$ rad/s and $\alpha_2 = 10\hat{k}$ rad/s², respectively. The postures, velocities, and accelerations of links 3 and 4 are provided in the table. Gravity is in the negative y direction, and the weights and mass moments of inertia of the links are: $w_3 = 3.40$ lb, $w_4 = 2.86$ lb, $I_{G_2} = 0.352$ in $\cdot$ lb $\cdot$ s², and $I_{G_3} = 0.108$ in $\cdot$ lb $\cdot$ s². The stiffness and unstretched length of the spring are $K_S = 20$ lb/in and $r_0 = 3$ in, respectively. The damping coefficient of the viscous damper is $C = 7$ lb $\cdot$ s/in. An external force, $\mathbf{F}_B$, acts horizontally at pin B on link 4, and the motor torque is $\mathbf{T}_2 = 60\hat{k}$ lb $\cdot$ in. Determine: (a) the first- and second-order kinematic coefficients of the linkage; (b) the equivalent mass moment of inertia; and (c) the external force, $\mathbf{F}_B$.

12.35 For the mechanism in the posture shown, massless link 4 is rolling on the ground link. The first- and second-order kinematic coefficients of links 3 and 4 are $\theta'_3 = 0$, $\theta'_4 = -1.0$ rad/m, $\theta''_3 = 1.0$ rad/m², and $\theta''_4 = 1.0$ rad/m². The velocity and acceleration of the mass center of input link 2 are $\mathbf{V}_{G_2} = 7\hat{i}$ m/s and $\mathbf{A}_{G_2} = -2\hat{i}$ m/s², respectively. The free length and spring rate of the spring, the damping constant of the viscous damper, the masses, and mass moments of inertia of links 2 and 3 are provided in the table. Gravity is in the negative y direction, and a horizontal

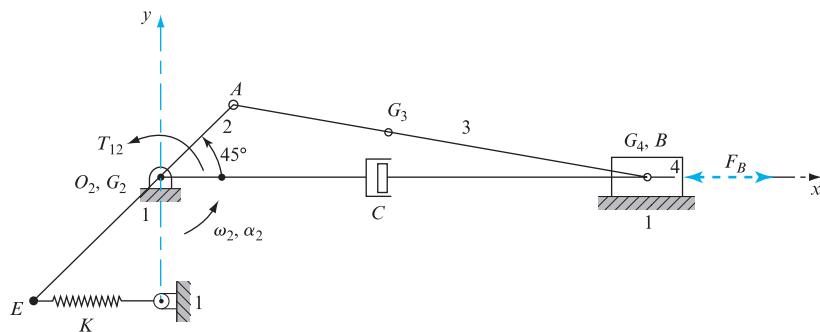


Figure P12.34 $R_{AO_2} = 3$ in, $R_{BA} = 12$ in, and $R_{G_3A} = 4.5$ in.

Table P12.34

θ_2 deg	θ_3 deg	R_{AO_2} in	R_{BA} in	R_{EO_2} in	ω_3 rad/s	V_{B_4} in/s	α_3 rad/s ²	A_{B_4} in/s ²
45	-10.18	3	12	5	-17.96	-250.23	1 736.2	-21 361

external force P acts on link 2. Determine: (a) the first-order kinematic coefficients of the spring and the damper; (b) the first- and second-order

kinematic coefficients of the mass center of link 3; (c) the equivalent mass; (d) the equation of motion; and (e) the force, P , acting on link 2.

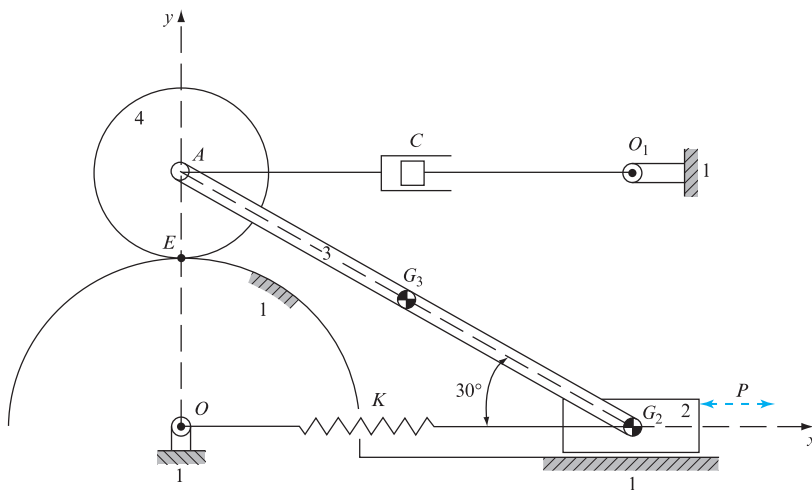


Figure P12.35 $R_4 = 1$ m, $R_{G_2A} = 6$ m, and $R_1 = 2$ m.

Table P12.35

R_o m	k N/m	C N·s/m	m_2 kg	m_3 kg	I_{G_2} kg·m ²	I_{G_3} kg·m ²
3	25	15	1.20	0.80	0.25	0.10

- 12.36** For the mechanism in the posture shown, the first- and second-order kinematic coefficients of links 3 and 4 are $\theta'_3 = -0.125$ rad/rad, $R'_4 = 1.299$ m/rad, $\theta''_3 = 0$, and $R''_4 = 0.094$ m/rad². The constant angular velocity of input link 2, which rolls without slipping on the inclined plane, is $\omega_2 = 20$ rad/s cw. The free length of the spring is 3 m, the spring rate is $k = 25$ N/m, and the damping constant of the viscous damper is $C = 15$ N · s/m. The masses and mass moments of inertia of links 2 and 4 are $m_2 = 7$ kg, $m_4 = 4$ kg, $I_{G_2} = 1.8$ kg · m², and $I_{G_4} = 22$ kg · m². The mass of link 3 is negligible compared to the masses of links 2 and 4, and gravity is in the negative y direction. Determine: (a) the first- and second-order kinematic coefficients of the mass centers of links 2 and 4; (b) the equivalent mass moment of inertia; and (c) the magnitude and direction of the torque acting on link 2.
- 12.37** The two-throw opposed-crank crankshaft is mounted in bearings at A and G. Each crank

has an eccentric weight of 6 lb, which may be considered located at a radius of 2 in from the axis of rotation and at the center of each throw (points C and E). It is proposed to locate weights at B and F to reduce the bearing reactions, caused by the rotating eccentric cranks, to zero. Determine the magnitudes of these weights if they are to be mounted 3 in from the axis of rotation.

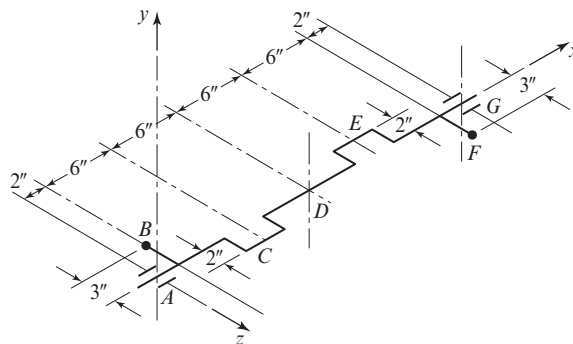
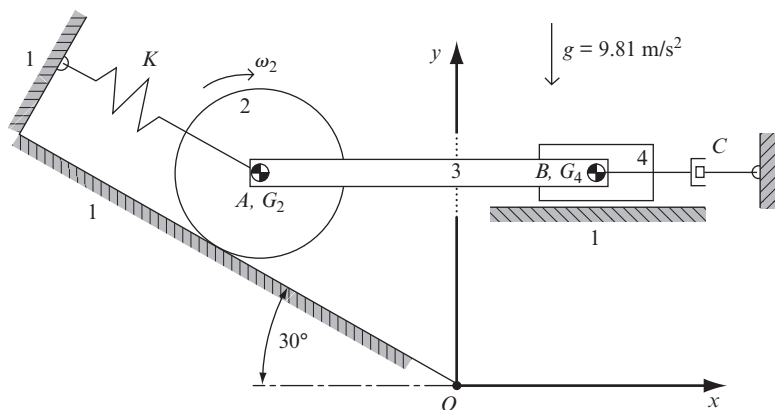


Figure P12.37

Figure P12.36 $\rho_2 = 1.5$ m, $R_{BA} = R_{G_4G_2} = 6$ m, and $R_{AO_S} = 2.5$ m.

- 12.38** The two-throw crankshaft is mounted in bearings at *A* and *F*, with the cranks spaced 90° apart. Each crank may be considered to have an eccentric weight of 6 lb at the center of the throw and 2 in from the axis of rotation. It is proposed to eliminate the rotating bearing reactions, which the crank would cause, by mounting additional correction weights on 3-in arms at points *B* and *E*. Calculate the magnitudes and angular locations of these weights.

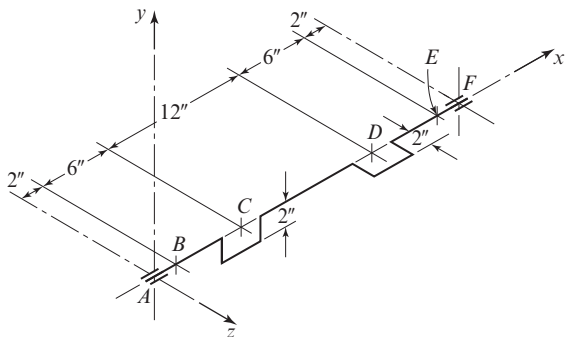


Figure P12.38

- 12.39** Solve Problem 12.38 when the angle between the two throws is reduced from 90° to 0° .
- 12.40** The connecting rod weighs 7.90 lb and is pivoted on a knife edge and caused to oscillate as a pendulum. The rod is observed to complete 64.5 oscillations in 1 min. Determine the mass moment of inertia of the rod about the center of mass.

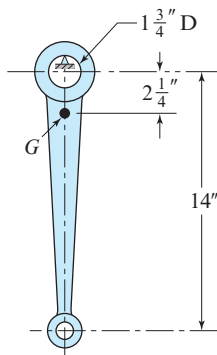


Figure P12.40

- 12.41** The gear is suspended on a knife edge at the rim and caused to oscillate as a pendulum. The period of oscillation is observed to be 1.08 s. Assume that the center of mass and the axis of rotation are coincident. If the weight of the gear is 41 lb,

find the mass moment of inertia and the radius of gyration of the gear.

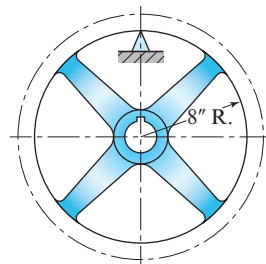


Figure P12.41

- 12.42** The wheel is mounted on a shaft in bearings with very low frictional resistance to rotation. At one end of the shaft and on the outboard side of the bearings is connected a rod with a weight, W_b , secured to its end. The weight, W_b , is displaced from equilibrium, and the assembly is permitted to oscillate. If the weight of the pendulum arm is neglected, show that the mass moment of inertia of the wheel can be written as

$$I = W_b l \left(\frac{\tau^2}{4\pi^2} - \frac{l}{g} \right).$$

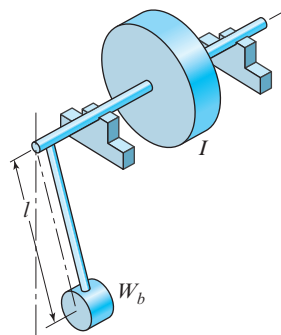


Figure P12.42

- 12.43** If the weight of the pendulum arm, W_a , is not neglected in Problem 12.42, but is assumed to be uniformly distributed over the length, l , show that the mass moment of inertia of the wheel can be written as

$$I = l \left[\frac{\tau^2}{4\pi^2} \left(W_b + \frac{W_a}{2} \right) - \frac{l}{g} \left(W_b + \frac{W_a}{3} \right) \right],$$

where W_a is the weight of the arm.

12.47 It frequently happens in motor-driven machinery that the greatest torque is exerted when the motor is first turned on, because of the fact that some motors are capable of delivering more starting torque than running torque. Analyze the bearing reactions of Problem 12.46 again, but this time use a starting torque equal to 250% of the full-load torque. Assume a normal-load torque and a speed of zero. How does this starting condition affect the forces on the mounting bolts?

12.48 The gear-reduction unit of Problem 12.46 is running at 600 rev/min when the motor is suddenly turned off, without changing the resisting-load torque. Solve Problem 12.46 for this condition.

12.49 The differential gear train has gear 1 fixed and is driven by rotating shaft 5 at 500 rev/min in the direction shown. Gear 2 has fixed bearings constraining it to rotate about the positive y axis, which remains vertical; this is the output shaft. Gears 3 and 4 have bearings connecting them to the ends of the carrier arm that is integral with shaft 5. The pitch diameters of gears 1 and 2 are both 8.0 in, while the pitch diameters of gears 3 and 4 are both 6.0 in. All gears have 20° pressure angles and are each 0.75 in thick, and all are made of steel with density 0.286 lb/in^3 . The mass of shaft 5 and all gravitational loads are negligible. The output shaft torque loading is $\mathbf{T} = -100\hat{\mathbf{j}} \text{ ft}\cdot\text{lb}$, as shown. Note that the coordinate axes shown rotate with input shaft 5. Determine the driving torque required and the

forces and moments in each of the bearings. (*Hint:* It is reasonable to assume through symmetry that $F_{13}^t = F_{14}^t$. It is also necessary to recognize that only compressive loads, not tension, can be transmitted between gear teeth.)

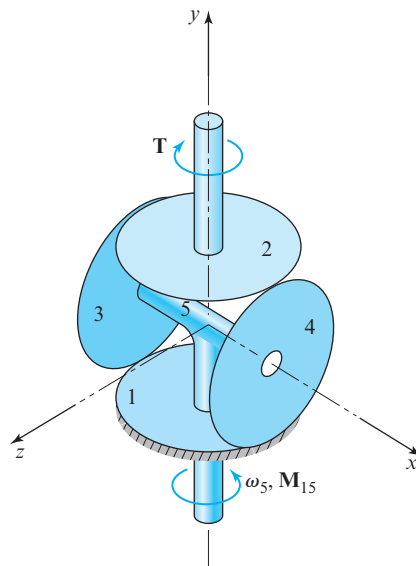


Figure P12.49 Differential gear train.

12.50 Arms 2 and 3 of the flyball governor are pivoted to block 6, which remains at the height shown but is free to rotate around the y axis. Block 7

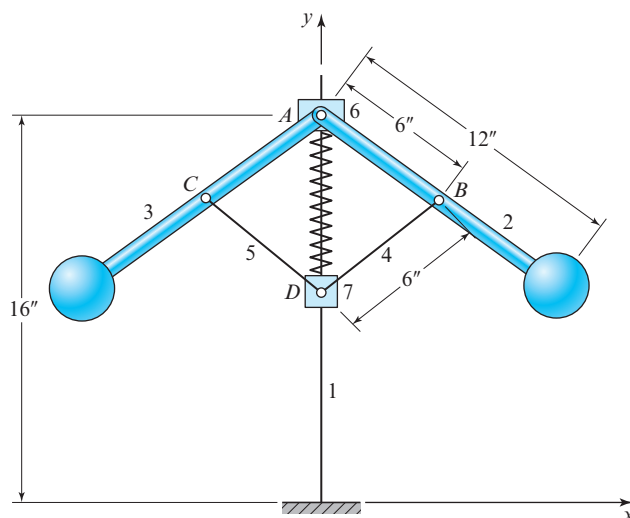


Figure P12.50 Flyball governor.

also rotates about, and is free to slide along, the y axis. Links 4 and 5 are pivoted at both ends between the two arms and block 7. The two balls at the ends of links 2 and 3 weigh 3.5 lb each, and all other masses are negligible in comparison; gravity acts in the $-\hat{j}$ direction. The spring between links 6 and 7 has a stiffness of 1.0 lb/in and would be unloaded if block 7 were at a height of $\mathbf{R}_D = 11\hat{j}$ in. All moving links rotate about the y axis with angular velocities of $\omega\hat{j}$. Make a graph of height R_D versus rotational speed ω in rev/min, assuming that the changes in speed are slow.

- 12.51** For the mechanism in the posture shown, the pin at the center of the wheel is sliding in the slot in link 2, and wheel 3 is rolling without slipping on the ground link. The constant angular velocity of input link 2 is $\omega_2 = 1.50\hat{k}$ rad/s, the angular acceleration of the wheel is $\alpha_3 = -62.354\hat{k}$ rad/s², and the acceleration of the center of mass of the wheel is $\mathbf{A}_{G_3} = 3.1177\hat{i}$ m/s². The masses and mass moments of inertia of link 2 and the wheel are $m_2 = 15$ kg, $m_3 = 25$ kg, $I_{G_2} = 0.105$ N · m · s², and $I_{G_3} = 0.016$ N · m · s², respectively. Gravity acts in the negative y direction. The force acting at point C is $\mathbf{F}_C = 75\hat{i}$ N, and there is a torque, $\mathbf{T}_{12}$, acting on link 2 at the crankshaft, O_2 . Determine the torque, $\mathbf{T}_{12}$, and the minimum coefficient of friction between the wheel and the ground.

- 12.52** For the mechanism in the posture shown, $R_{AO_2} = 68$ in, the position of the mass center of block 3 is 4 in below pin A, and the kinematic coefficients

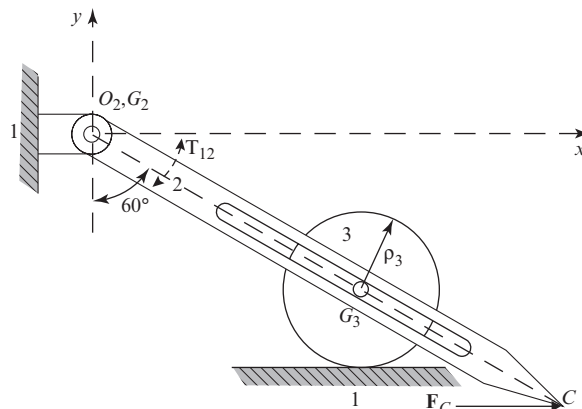


Figure P12.51 $R_{G_3O_2} = 200$ mm, $R_{CO_2} = 350$ mm, and $\rho_3 = 50$ mm.

of link 3 are $R'_3 = 57.72$ in/rad and $R''_3 = 66.68$ in/rad² (where $\mathbf{R}_3$ is the vertical vector from point C fixed in link 3 to the x axis). The angular velocity and acceleration of the input link 2 are $\omega_2 = 12\hat{k}$ rad/s and $\alpha_2 = 15\hat{k}$ rad/s², respectively. The masses and mass moments of inertia of links 2 and 3 are $m_2 = 19.84$ lb, $m_3 = 15.43$ lb, $I_{G_2} = 7.434$ lb · in², and $I_{G_3} = 6.726$ lb · in², respectively. The torque acting on link 2 at the crankshaft O_2 is $\mathbf{T}_{12} = -450\hat{k}$ in · lb. Gravity is in the negative y direction. Determine the internal reaction forces and the force $\mathbf{P}$ acting at point B on link 3. Is link 3 sliding or tipping on the ground link in this posture?

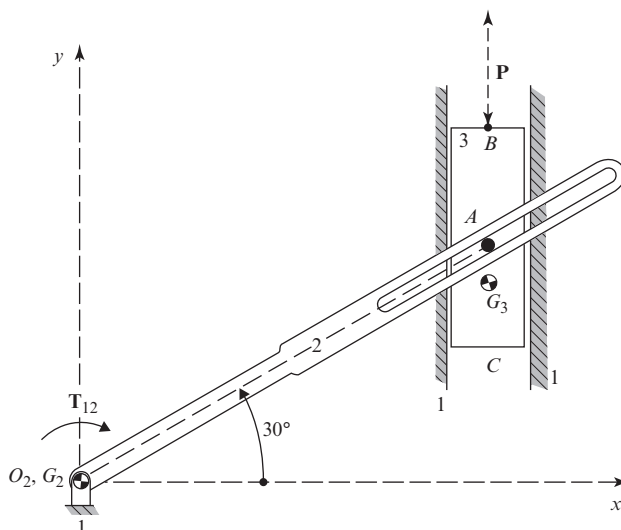


Figure P12.52 Block 3 is 24 in by 8 in and pin A is centrally located.

Table P12.53

O_2A mm	O_2C mm	CD mm	AD mm	AB mm	O_4B mm	O_2O_4 mm	θ_3 deg	θ_4 deg	θ_5 deg
25	40	45	30	50	60	75	41.2	114.5	4.0
		θ'_3 rad/rad	θ'_4 rad/rad	θ'_5 rad/rad	θ''_3 rad/rad	θ''_4 rad/rad ²	θ''_5 rad/rad ²		
		-0.425	0.140	-0.188	0.333	0.563	1.810		

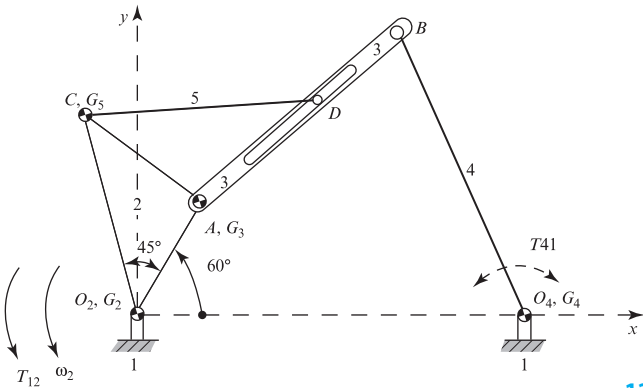


Figure P12.53 Planar mechanism.

12.53 For the mechanism in the posture shown, the constant angular velocity of input link 2 is $\omega_2 = 26\hat{k}$ rad/s, and the accelerations of the centers of mass of links 3 and 5 are $\mathbf{A}_{G_3} = -8.45\hat{i} - 14.64\hat{j}$ m/s² and $\mathbf{A}_{G_5} = 7.00\hat{i} - 26.12\hat{j}$ m/s², respectively. The masses and mass moments of inertia of the links are $m_2 = m_3 = m_4 = m_5 = 7$ kg and $I_{G_2} = I_{G_3} = I_{G_4} = I_{G_5} = 0.013$ N · m · s², respectively. The input torque is $\mathbf{T}_{12} = 0.50\hat{k}$ N · m, and gravity is in the negative z direction. The link lengths, position variables, and first- and second-order kinematic coefficients are shown in Table P12.53. Determine the internal reaction forces between links 3 and 5 and between links 2 and 5.

12.54 For the linkage in the posture shown, the angular velocity and acceleration of input link 2 are $\omega_2 = -5\hat{k}$ rad/s and $\alpha_2 = 3\hat{k}$ rad/s², respectively. The kinematic coefficients are $R'_{AO_2} = -10.40$ in/rad, $R'_{CA} = 14.80$ in/rad, $R''_{AO_2} = -18.00$ in/rad², $R''_{CA} = -14.80$ in/rad², $x'_C = -14$ in/rad, $y'_C = 14$ in/rad, $x''_C = -14$ in/rad², and $y''_C = -14$ in/rad². Gravity is in the negative y direction. The distances, the free length and spring rate of the rectilinear spring, the damping constant of the viscous damper, and the weights and second moments of mass about the mass centers of links 2, 3, and 4 are shown in Table P12.54.

Determine: (a) the first-order kinematic coefficients of the spring and the damper; (b) the kinetic energy of the linkage; and (c) the torque, $\mathbf{T}_{12}$.

Table P12.54

R_{CO_2} in	R_{AO_2} in	R_{CD} in	R_{S_0} in	K lb/in	C lb · s/in	m_2 lb	m_3 lb	m_4 lb	I_{G_2} lb · in · s ²	I_{G_3} lb · in · s ²	I_{G_4} lb · in · s ²
20	17.93	20	24	0.28	0.084	1.1	11	8.8	27.4	64	55

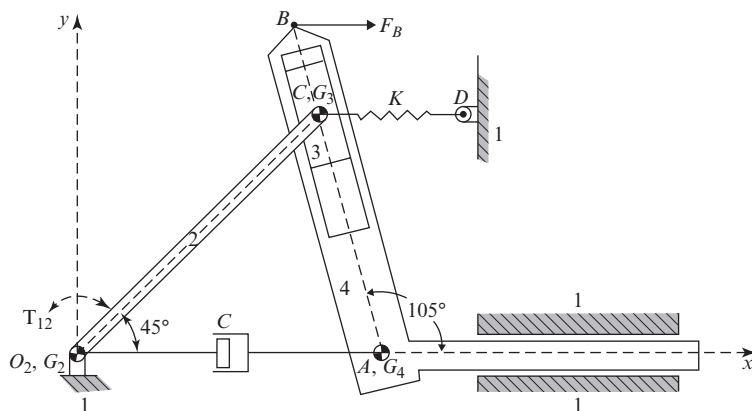


Figure P12.54 $F_C = 56\hat{i}$ lb.

- 12.55** For the elliptic trammel linkage in the posture shown, the velocity and acceleration of input link 2 are $\mathbf{V}_{A_2} = -3\hat{j}$ m/s and $\mathbf{A}_{A_2} = 7\hat{j}$ m/s², respectively. The kinematic coefficients, masses, and second moments of mass about the mass centers of the links are shown in Table P12.55.

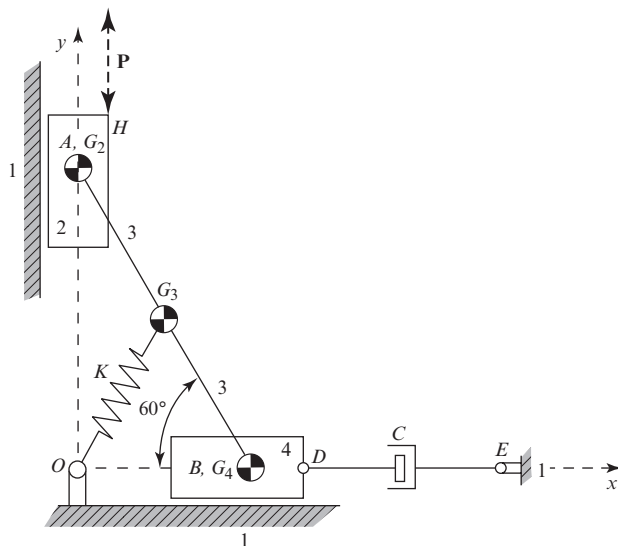


Figure P12.55 $AG_3 = G_3B = 1.25$ m.

The stiffness and free length of the spring are $K = 200$ N/m and $R_0 = 0.5$ m, respectively, and the coefficient of the viscous damper is $C = 15$ N·s/m. The only friction in the linkage is between links 1 and 4, where the coefficient of friction is $\mu = 0.35$. The normal force at the point of contact between links 1 and 4 is $\mathbf{F}_{41}^n = 500\hat{j}$ N, and gravity is vertically downward. Determine: (a) the first-order kinematic coefficients of the spring and the damper, (b) the kinetic energy of the linkage, and (c) the external force $\mathbf{P}$ acting at point H .

Table P12.55

θ'_3 rad/m	θ''_3 rad/m ²	R'_4 m/m	R''_4 m/m ²	m_2 kg	m_3 kg	m_4 kg	I_{G_2} kg·m ²	I_{G_3} kg·m ²	I_{G_4} kg·m ²
-0.800	-1.109	-1.732	-3.200	3.5	6.0	8.0	2.50	7.25	1.75

12.56 For the linkage in the posture shown, the angular velocity and acceleration of the input link 2 are $\omega_2 = -5\hat{\mathbf{k}} \text{ rad/s}$ and $\alpha_2 = 3\hat{\mathbf{k}} \text{ rad/s}^2$, respectively, and the acceleration of the mass center of link 3 are $\alpha_3 = 15.75\hat{\mathbf{k}} \text{ rad/s}^2$ and $\mathbf{A}_{G_3} = 9.60\hat{\mathbf{i}} - 15.20\hat{\mathbf{j}} \text{ in/s}^2$, respectively. The weights and second moments of mass of the links are $w_2 = w_3 = 13.2 \text{ lb}$, $w_4 = 2.2 \text{ lb}$, $I_{G_2} = I_{G_3} = 45.69 \text{ lb} \cdot \text{in} \cdot \text{s}^2$, and $I_{G_4} = 4.56 \text{ lb} \cdot \text{in} \cdot \text{s}^2$. The torque acting on link 4 is $\mathbf{T}_{14} = -17.7\hat{\mathbf{k}} \text{ in} \cdot \text{lb}$, and there is an unknown torque, $\mathbf{T}_{12}$, acting on link 2. Gravity is in the negative y direction. Determine: (a) the magnitude, direction, and location of the reaction force between links 3 and 4, and (b) the magnitude and direction of the torque, $\mathbf{T}_{12}$.

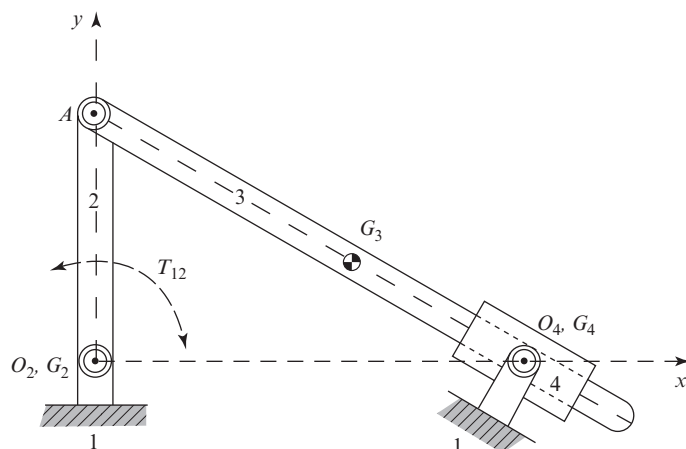


Figure P12.56 $O_2A = 2.0 \text{ in}$, $AO_4 = 4.0 \text{ in}$, and $AG_3 = 2.40 \text{ in}$.

12.57 For the mechanism in the posture shown, a force $F_2 = 15 \text{ N}$ is applied at point A on input link 2, which causes the link to move up the inclined plane with a velocity $V_{G_2} = 0.5 \text{ m/s}$. The first and second-order kinematic coefficients (where $R_3 = R_{G_2O_3}$) are shown in Table P12.57.

A linear spring with a free length $R_{S0} = 10 \text{ mm}$ and a spring constant $K = 2500 \text{ N/m}$ is attached to link 2 at point B and has a current length of 30 mm. A viscous damper with a damping coefficient $C = 80 \text{ N} \cdot \text{s/m}$ is attached to link 2 at point D. The masses and mass moments of inertia of links 2 and 3 are $m_2 = 3 \text{ kg}$, $m_3 = 5 \text{ kg}$, $I_{G_2} = 0.000035 \text{ kg} \cdot \text{m}^2$, and $I_{G_3} = 0.000070 \text{ kg} \cdot \text{m}^2$. Gravity is in the negative y direction. Determine the first- and second-order kinematic coefficients of the mass center of link 2, and the first-order kinematic coefficients of the spring and damper. Write the equation of motion for the mechanism in symbolic form and then determine the acceleration of link 2.

Table P12.57

$\theta'_3 \text{ rad/m}$	$R'_3 \text{ m/m}$	$\theta''_3 \text{ rad/m}^2$	$R''_3 \text{ m/m}^2$
-12.4	-0.500	-180	10.7

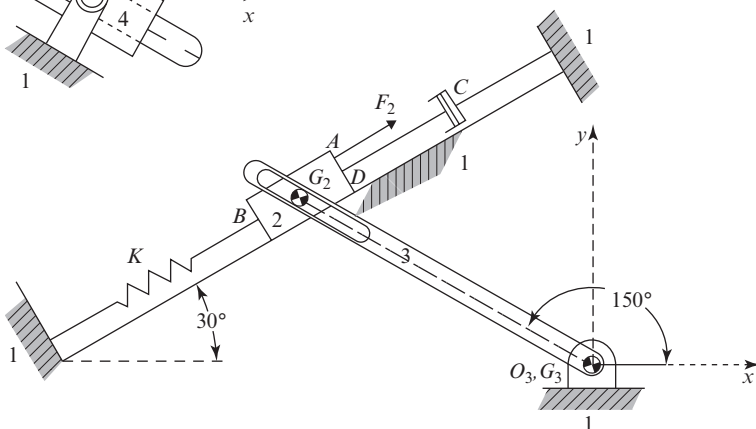


Figure P12.57 $G_3G_2 = 70 \text{ mm}$ and $BG_2 = 10 \text{ mm}$.

12.58 For the mechanism in the posture shown, the constant angular velocity of input link 2 is $\omega_2 = 3\hat{\mathbf{k}}$ rad/s. The free length of the linear spring attached between ground pin O_1 and pin A is 20 in, and a viscous damper is attached between pin O and link 3. The torque acting on link 2 is $\mathbf{T}_{12} = -13.3\hat{\mathbf{k}}$ in · lb, there is a horizontal force, $\mathbf{P}$, acting at pin A , and gravity is in the negative y direction. The known data (where $\mathbf{R}_2$ is the vector from bearing O_2 to mass center G_3 , and $\mathbf{R}_3$ is the vector from ground pin O to mass center, G_3) are shown in Table P12.58.

Determine: (a) the first-order kinematic coefficients of the spring and the damper, (b) the potential energy of the spring, (c) the equivalent mass moment of inertia, and (d) the force $\mathbf{P}$.

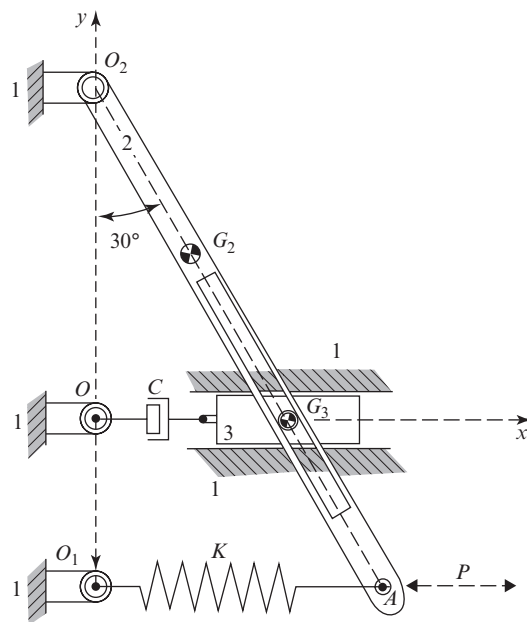


Figure P12.58 $R_{AO_2} = 4.8$ in, $R_{G_3O_2} = 3.6$ in, and $R_{G_2O_2} = 1.8$ in.

Table P12.58

R'_2 in/rad	R'_3 in/rad	R''_2 in/rad ²	R''_3 in/rad ²	m_2 lb	m_3 lb	I_{G_2} lb · in ²	I_{G_3} lb · in ²	K lb/in	C lb · s/in
2.080	4.160	6.000	4.800	15.44	26.46	2 646	7 938	0.082 7	0.413

12.59 For the linkage in the posture shown, the kinematic coefficients of link 3 are $\theta'_3 = 0.25 \text{ rad/rad}$ and $\theta''_3 = 0.60 \text{ rad/rad}^2$. The angular velocity and acceleration of the input link 2 are $\omega_2 = -5\hat{\mathbf{k}} \text{ rad/s}$ and $\alpha_2 = 3\hat{\mathbf{k}} \text{ rad/s}^2$, respectively. The free length and stiffness of the horizontal spring are $R_{S0} = 30 \text{ mm}$ and $K = 2\,000 \text{ N/m}$, respectively. The damping coefficient of the viscous damper is $C = 150 \text{ N} \cdot \text{s/m}$. The masses and mass moments of inertia of the links are $m_2 = m_3 = 6 \text{ kg}$, $m_4 = 1 \text{ kg}$, $I_{G_2} = I_{G_3} = 5 \text{ kg} \cdot \text{m}^2$, and $I_{G_4} = 4 \text{ kg} \cdot \text{m}^2$. Input torque $\mathbf{T}_{12} = -10\hat{\mathbf{k}} \text{ N} \cdot \text{m}$, and there is a horizontal force, $\mathbf{F}$, acting at the mass center of link 3. Gravity is in the negative y direction. Determine: (a) the first-order kinematic coefficients of the horizontal spring and the viscous damper, (b) the kinetic energy of this linkage, and (c) the force, $\mathbf{F}$.

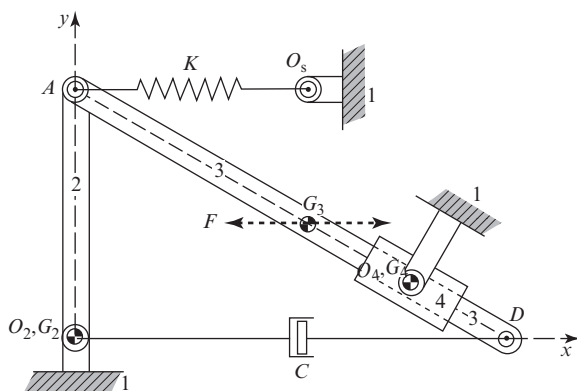


Figure P12.59 $O_2A = 50 \text{ mm}$, $AO_S = 52 \text{ mm}$, $AO_4 = 80 \text{ mm}$, $AD = 100 \text{ mm}$, and $AG_3 = 60 \text{ mm}$.

12.60 For the mechanism in the posture shown, the velocity and acceleration of input link 2 down slope EA are $V_A = 8 \text{ in/s}$ and $A_A = 24 \text{ in/s}^2$, respectively. The wheel, link 4, is rolling without slipping on the ground link. The free length of the linear spring is $R_{S0} = 8 \text{ in}$, and the damping coefficient of the viscous damper is $C = 1.4 \text{ lb} \cdot \text{s/in}$. Gravity acts vertically downward. The dimensions, first- and second-order kinematic coefficients ($\mathbf{R}_{44}$ is the vector from point O to pin B), weights, and second moments of mass about the mass centers of links 2, 3, and 4 are shown in Table P12.60.

Determine: (a) the first-order kinematic coefficients of the spring and the damper, (b) the kinetic energy of the mechanism, and (c) the stiffness of the spring.

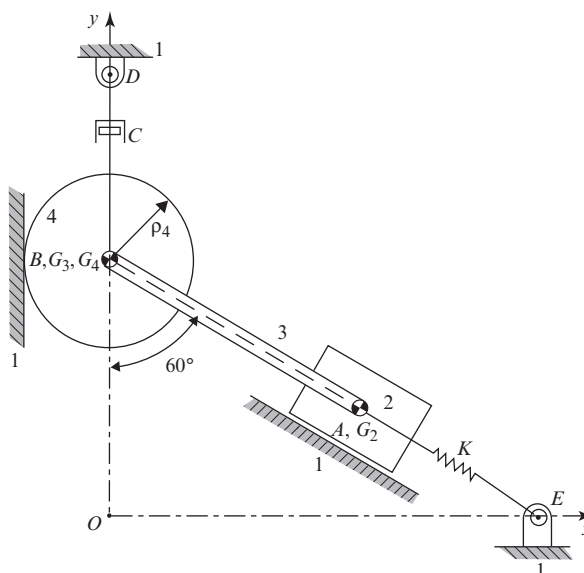
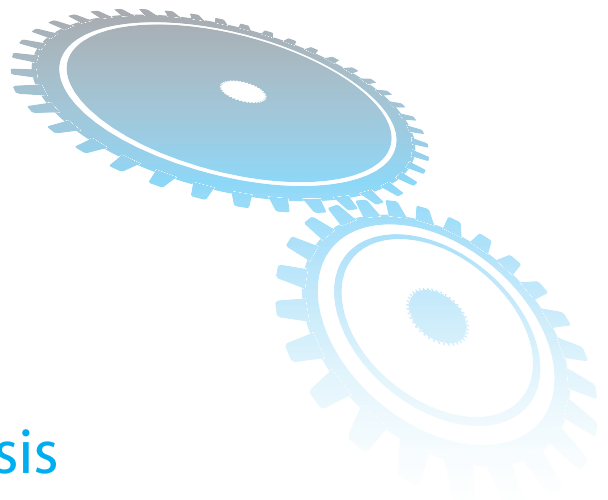


Figure P12.60 Planar mechanism.

Table P12.60

EA in	AB in	BD in	ρ_4 in	θ'_3 rad/in	θ''_3 rad/in ²	R'_{44} in/in	R''_{44} in/in ²	w_2 lb	w_3 lb	w_4 lb	I_{G_2} lb · in ²	I_{G_3} lb · in ²	I_{G_4} lb · in ²
10	24	14	6	-0.0723	0.0090	2.0	-0.25	1.1	4.4	8.8	12	211	158



13 Vibration Analysis

The existence of vibrating elements in a mechanical system produces unwanted noise, high stresses, wear, poor reliability, and, frequently, premature failure of one or more of the parts. The moving parts of all machines are inherently vibration producers, and for this reason, engineers must expect vibrations to exist in the devices they design. But there is a great deal they can do during the design of the system to anticipate a vibration problem and to minimize its undesirable effects.

Sometimes it is necessary to build a vibratory system into a machine—a vibratory conveyor, for example. Under these conditions the engineer must understand the mechanics of vibration to obtain an optimal design.

13.1 DIFFERENTIAL EQUATIONS OF MOTION

Any motion that exactly repeats itself after a certain interval of time is a periodic motion and is called a *vibration*. Vibrations may be either free or forced. A mechanical element is said to have a *free vibration* if the periodic motion continues after the cause of the original disturbance is removed, but if a vibratory motion persists because of the continuing existence of a disturbing force, then it is called a *forced vibration*. Any free vibration of a mechanical system eventually ceases because of loss of energy. In vibration analysis, we often take account of these energy losses using a single factor called the *damping factor*. Thus, a heavily damped system is one in which the vibration decays rapidly. The *period* of a vibration is the time for a single cycle of the motion after which repetition begins; the *frequency* is the number of cycles or periods occurring in unit time. The *natural frequency* is the frequency of a free vibration. If the forcing frequency becomes equal to the natural frequency of a system, then *resonance* is said to occur.

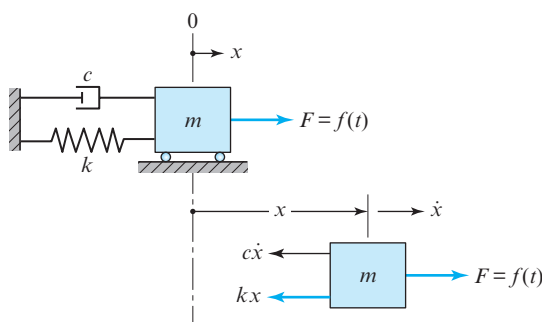
We shall also use the terms *steady-state vibration* to indicate that a motion is repeating itself exactly in each successive cycle and *transient vibration* to indicate a vibration-like motion that is changing in character. If a periodic force operates on a mechanical system, the resulting motion is transient in character when the force first begins to act, but after an interval of time, the transient decays, owing to damping, and the resulting motion is termed a *steady-state vibration*.

The word *response* is frequently used in discussing vibratory systems. The words *response*, *behavior*, and *performance* have roughly the same meaning when used in dynamic analysis. Thus, we can apply an external force having a sine-wave relationship with time to a vibrating system to determine how the system “responds,” or “behaves,” when the frequency of the force is varied. A plot showing the vibration amplitude along the ordinate and the forcing frequency along the abscissa is described as a *performance* or *response* curve for the system. Sometimes it is useful to apply arbitrary input disturbances or forces to a system. These may not resemble the force characteristics that a real system receives in use at all, yet the response of the system to these chosen disturbances can provide much useful information about the system.

Vibration analysis is sometimes called *elastic-body analysis* or *deformable-body analysis*, since, as we shall see, it is elasticity in a mechanical system that allows vibration. When a rotating shaft has a torsional vibration, this means that a mark on the circumference at one end of the shaft is successively ahead of and then behind a corresponding mark on the other end of the shaft. In other words, torsional vibration of a shaft is the alternate twisting and untwisting of the rotating material and requires elasticity for its existence. We shall begin our study of vibration by assuming that elastic parts have no mass and that heavy parts are absolutely rigid, that is, they have no elasticity. Of course these assumptions are never true, and so, in the course of our studies, we must also learn to correct for the effects of making these assumptions.

Figure 13.1 shows an idealized vibrating system having a mass m guided to move only in the x direction. The mass is connected to a fixed frame through the spring k , and the dashpot, c . The assumptions used are as follows:

1. The spring and the dashpot are massless.
2. The mass is absolutely rigid.
3. All damping is concentrated in the dashpot.



 **Figure 13.1**

It turns out that a great many mechanical systems can be analyzed quite accurately using these assumptions.

The elasticity of the system of Fig. 13.1 is completely represented by the spring. The *stiffness* is designated as k and is defined by the equation

$$k = \frac{F}{x}, \quad (13.1)$$

where F is the force required to deflect the spring a distance x .

Similarly, the friction, or damping, is assumed to be entirely viscous damping (later we shall examine other kinds of friction) and is designated using a *coefficient of viscous damping*, c . Thus,

$$c = \frac{F}{\dot{x}}, \quad (13.2)$$

where F is the force required to move the mass at a velocity of $\dot{x}$.

Note that we are using the notation for time derivatives that is customary in vibration theory. Thus,

$$\dot{x} = \frac{dx}{dt} \quad \text{and} \quad \ddot{x} = \frac{d^2x}{dt^2}.$$

This is called *Newton's notation*, since Newton was the first to use it. It is used only when derivatives are with respect to time.

The vibrating system of Fig. 13.1 has one degree of freedom, since the position of the mass is completely defined by a single coordinate. An external force, $F = f(t)$, is shown acting upon the mass. Thus, this system is classified as a forced, single-degree-of-freedom system with damping.

The equation of motion of the system is written by displacing the mass in the positive direction, giving it a positive velocity, and then summing all forces, including the inertia force. As shown in Fig. 13.1, there are three forces acting: a spring force, kx , acting in the negative direction, a damping force, $c\dot{x}$, also acting in the negative direction, and an external force, $f(t)$, acting in the positive direction. Summing these forces together with the inertia force gives

$$\sum F = -kx - c\dot{x} + f(t) + (-m\ddot{x}) = 0$$

or

$$m\ddot{x} + c\dot{x} + kx = f(t). \quad (13.3)$$

Equation (13.3) is an important equation in dynamic analysis, and we shall eventually solve it for many specialized conditions. It is a linear differential equation of the second order.

Consider next the idealized torsional vibrating system of Fig. 13.2. Here, a disk having a mass moment of inertia, I , is mounted upon the end of a weightless shaft having a torsional spring constant, k , defined by

$$k = \frac{T}{\theta}, \quad (13.4)$$

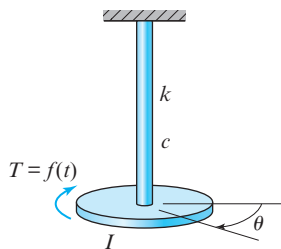


Figure 13.2

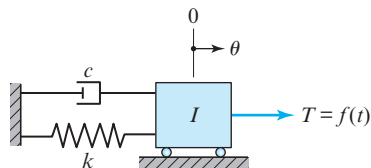


Figure 13.3

where T is the torque necessary to produce an angular deflection, θ , of the shaft. In a similar manner, the torsional viscous damping coefficient is defined by

$$c = \frac{T}{\dot{\theta}}. \quad (13.5)$$

Note that we are using the same symbols for denoting these torsional parameters as for rectilinear ones. The nature of the problem or of the differential equation reveals whether the system is rectilinear or torsional, and so there is no confusion in this usage.

Next, designating an external torque forcing function by $T = f(t)$, we find that the differential equation for the torsional system is

$$\sum M = -k\theta - c\dot{\theta} + f(t) + (-I\ddot{\theta}) = 0$$

or

$$I\ddot{\theta} + c\dot{\theta} + k\theta = f(t), \quad (13.6)$$

which is of the same form as Eq. (13.3). Thus, with appropriate substitutions, the solution of Eq. (13.6) is the same as that of Eq. (13.3). Note, too, that this means we can *simulate* a torsional system, as in Fig. 13.3, merely by substituting torsional notation for the rectilinear notation.

It has been said that mathematics is a human invention and, therefore, that it can never perfectly describe nature [1]. However, the linear differential equation frequently does such an excellent job of simulating the action of a mechanical system that one wonders whether differential equations are not an exception to this statement. Perhaps people just happened to discover them, as they have other curiosities of nature.

EXAMPLE 13.1

Figure 13.4a shows a vibrating system in which a time-dependent displacement, $y = y(t)$, excites a spring-mass system through a viscous dashpot. Write the differential equation of this system.

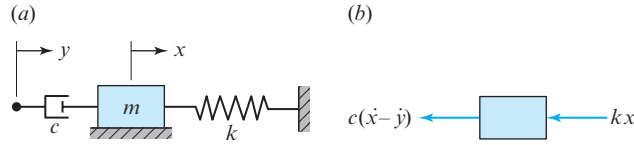


Figure 13.4 (a) System; (b) forces acting on the mass.

SOLUTION

We begin by assuming an arbitrary relationship between the coordinates and their first derivatives, say, $x > y$ and $\dot{x} > \dot{y}$. We also assume senses for x and $\dot{x}$, say, $x > 0$ and $\dot{x} > 0$. Regardless of the assumptions made, the resulting differential equation is the same. With these assumptions, the forces acting on the mass are as shown in Fig. 13.4b. Note that these external forces act in the negative direction because of the arbitrary assumptions made. Summing the forces in the x direction and including the inertia force gives

$$\sum F = -kx - c(\dot{x} - \dot{y}) + (-m\ddot{x}) = 0. \quad (1)$$

Such equations are usually rearranged with the response derivative terms on the left-hand side and the forcing term on the right, that is,

$$m\ddot{x} + c\dot{x} + kx = c\dot{y}. \quad \text{Ans.}$$

13.2 A VERTICAL MODEL

The system of Fig. 13.1 moves in the horizontal direction, and, as a result, gravity has no effect on its motion. Let us now turn the system to the vertical direction and, also incidentally, make the damping and the external forces zero. This yields the model of Fig. 13.5. We choose the origin of the coordinate system to correspond with the position of the mass where the spring has no force, and we choose the symbol x to denote the vertical displacement. Thus, when the mass is given a positive displacement, the spring force is made up of two components. One of these is $k\delta_{\text{st}}$, where $\delta_{\text{st}} = W/k$ is the distance the spring is deflected when the weight is suspended from it. The other component is kx . Summing the forces acting on the mass gives

$$\sum F = -k(x + \delta_{\text{st}}) + W + (-m\ddot{x}) = 0.$$

Rearranging this equation gives

$$m\ddot{x} + kx = W - k\delta_{\text{st}},$$

and, because of the definition $\delta_{\text{st}} = W/k$, this equation reduces to the homogeneous form

$$m\ddot{x} + kx = 0. \quad (13.7)$$

Note that Eq. (13.3) is identical to Eq. (13.7) if the damping force, $c\dot{x}$, and the external force, $f(t)$, are made zero.

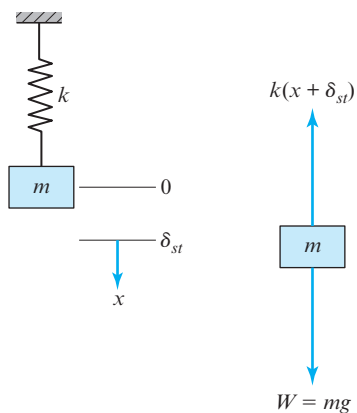


Figure 13.5

13.3 SOLUTION OF THE DIFFERENTIAL EQUATION

If Eq. (13.7) is arranged into the form

$$\ddot{x} = -\frac{k}{m}x, \quad (a)$$

then we can see that the function used for x has to be of the same form as the negative of its second derivative. One such function is

$$x = A \sin bt, \quad (b)$$

for which

$$\dot{x} = Ab \cos bt \quad (c)$$

and

$$\ddot{x} = -Ab^2 \sin bt. \quad (d)$$

Substituting Eqs. (b) and (d) into Eq. (a) produces

$$-Ab^2 \sin bt = -\frac{k}{m}A \sin bt. \quad (e)$$

Then, dividing both sides by $(-A \sin bt)$ leaves

$$b^2 = \frac{k}{m}.$$

Thus, Eq. (b) is a solution, provided that $b = \sqrt{k/m}$. By an exactly parallel development, it is clear that

$$x = B \cos bt \quad (f)$$

is also a solution to Eq. (a). A general solution is obtained by adding Eqs. (b) and (f), that is,

$$x = A \sin \sqrt{\frac{k}{m}}t + B \cos \sqrt{\frac{k}{m}}t. \quad (g)$$

Mathematically, the constants A and B are the constants of integration, since, theoretically, we could integrate Eq. (a) twice to get the solution. Unfortunately, most differential equations cannot be solved in this manner, and we must resort to cleverness to obtain the solution. Physically, constants A and B represent the manner in which the vibration is started, or, to put this another way, the state of the motion at the instant $t = 0$.

We define the quantity

$$\omega_n = \sqrt{\frac{k}{m}}, \quad (13.8)$$

which is called the *natural frequency* of the system; it has units of radians per second.

The solution, Eq. (g), can now be written in the form

$$x = A \sin \omega_n t + B \cos \omega_n t. \quad (13.9)$$

As an illustration of the meaning of constants A and B , suppose we move the mass by a distance x_0 and release it with zero velocity at the instant $t = 0$. Then the starting conditions are

$$\text{at } t = 0, \quad x = x_0, \quad \dot{x} = 0.$$

Substituting the first of these into Eq. (13.9) gives

$$x_0 = A(0) + B(1)$$

or

$$B = x_0.$$

Next, taking the first time derivative of Eq. (13.9) gives

$$\dot{x} = A\omega_n \cos \omega_n t - B\omega_n \sin \omega_n t, \quad (h)$$

and substituting the other condition into this equation yields

$$0 = A\omega_n(1) - B\omega_n(0)$$

or

$$A = 0.$$

Then, substituting A and B into Eq. (13.9) gives the result

$$x = x_0 \cos \omega_n t. \quad (13.10)$$

If we start the motion with the initial conditions,

$$\text{at } t = 0, \quad x = 0, \quad \dot{x} = v_0,$$

the result is

$$x = \frac{v_0}{\omega_n} \sin \omega_n t. \quad (13.11)$$

But, if we start the motion with

$$\text{at } t = 0, \quad x = x_0, \quad \dot{x} = v_0,$$

the result is

$$x = \frac{v_0}{\omega_n} \sin \omega_n t + x_0 \cos \omega_n t, \quad (13.12)$$

which is the most general form of the solution. We can substitute Eq. (13.12) together with its second derivative into the differential equation and so demonstrate its validity.

The three solutions [Eqs. (13.10), (13.11), and (13.12)] are represented graphically in Fig. 13.6 using *phasors* to generate the trigonometric functions. Phasors are not vectors in the classic sense, since phasors can be manipulated in ways that are not defined for vectors. They are complex numbers, however, and they may be added and subtracted just as vectors are.

The ordinate of the graph of Fig. 13.6 is displacement, x , and the abscissa can be considered as the time axis or as the angular displacement, $\omega_n t$, of the phasors for a given time after the motion has commenced. The phasors x_0 and v_0/ω_n are shown in their initial positions, and as time passes, these rotate counterclockwise with an angular velocity of ω_n and generate the displacement curves shown. Figure 13.6 shows that the phasor x_0 starts from a maximum positive displacement and the phasor v_0/ω_n starts from a zero displacement. These, therefore, are very special, and the most general form is that given by Eq. (13.12), in which motion begins at some intermediate point.

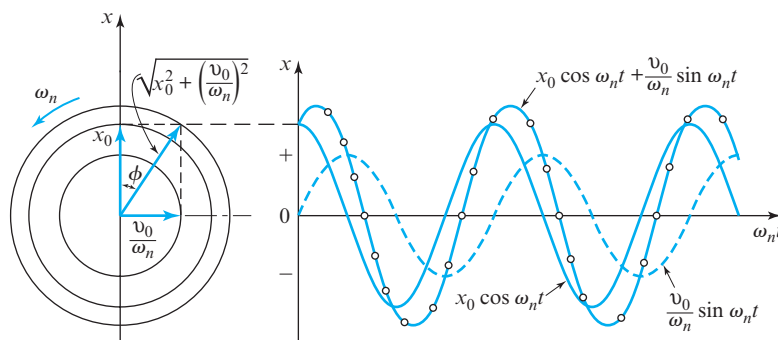


Figure 13.6 Use of phasors to generate time-displacement diagrams.

In Eq. (13.8), the quantity $\omega_n = \sqrt{k/m}$ was defined as the natural frequency of the system. For most systems, the natural frequency is a constant since the mass and the spring constant do not vary. Since one cycle is equal to 2π radians, the period of a free vibration is

$$\tau = \frac{2\pi}{\omega_n} = 2\pi \sqrt{\frac{m}{k}}, \quad (13.13)$$

where τ is usually expressed in seconds per cycle. Although the natural units for ω_n are radians per second, frequency can also be defined as the reciprocal of the period and this gives

$$f = \frac{\omega_n}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{k}{m}}, \quad (13.14)$$

where f is expressed in hertz (cycles/s or s^{-1}) and is abbreviated as Hz.*

A study of Fig. 13.6 indicates that one should also be able to express the motion by the equation†

$$x = X_0 \cos(\omega_n t - \phi), \quad (13.15)$$

where X_0 and ϕ are the constants of integration whose values depend upon the initial conditions. These constants can be obtained directly from the trigonometry of Fig. 13.6 and are

$$X_0 = \sqrt{x_0^2 + \left(\frac{v_0}{\omega_n}\right)^2} \quad \text{and} \quad \phi = \tan^{-1} \frac{v_0/\omega_n}{x_0}. \quad (13.16)$$

Equation (13.15) can now be written in the form

$$x = \sqrt{x_0^2 + \left(\frac{v_0}{\omega_n}\right)^2} \cos(\omega_n t - \phi). \quad (13.17)$$

This is a particularly convenient form of the equation, since the coefficient is the *amplitude* of the vibration. The amplitude is the maximum displacement of the mass. The angle ϕ is called the *phase angle*, and it denotes the angular lag of the motion with respect to the cosine function.

* Named after German scientist Heinrich Rudolf Hertz (1857–1894).

† The motion can also be expressed in the form

$$x = X_0 \sin(\omega_n t + \psi),$$

where X_0 and ψ are the constants of integration. This is probably not a good way of expressing it, however, because, in the study of forced vibration, it might imply that the output can lead the input, which, of course, is not possible.

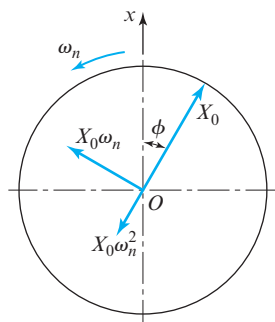


Figure 13.7 Phase relationship of displacement, velocity, and acceleration phasors.

The velocity and acceleration are obtained by successively differentiating Eq. (13.15), that is,

$$\dot{x} = -X_0\omega_n \sin(\omega_n t - \phi) \text{ and } \ddot{x} = -X_0\omega_n^2 \cos(\omega_n t - \phi),$$

where the velocity amplitude is $X_0\omega_n$, and the amplitude of the acceleration is $X_0\omega_n^2$. The displacement, velocity, and acceleration phasors are shown in Fig. 13.7 to clarify their phase relationships. All three phasors maintain fixed relative phase angles as they rotate together at the constant angular velocity ω_n . The velocity leads the displacement by 90° and the acceleration is 180° out of phase with the displacement.

13.4 STEP INPUT FORCING

Let us again consider the vibrating system of Fig. 13.1, this time adding to the system a force F applied to the mass that, in general, might be a function of time. As a start, however, let us assume that this force is constant and acting in the positive x direction. As before, we consider the damping to be zero. For this condition Eq. (13.3) is written

$$\ddot{x} + \frac{k}{m}x = \frac{F}{m}. \quad (13.18)$$

The general solution to this equation is

$$x = A \cos \omega_n t + B \sin \omega_n t + \frac{F}{k}, \quad (13.19)$$

where A and B are the constants of integration and where the natural frequency is

$$\omega_n = \sqrt{\frac{k}{m}}, \quad (a)$$

as before. We note that the quantity F/k has units of length. The physical interpretation is that a force F applied to a spring of stiffness k would produce a deformation of the spring of magnitude F/k .

It is interesting to consider starting the motion when the system is at rest. Thus, with the system in equilibrium at $x = 0$, we apply a constant force F at the instant $t = 0$ and

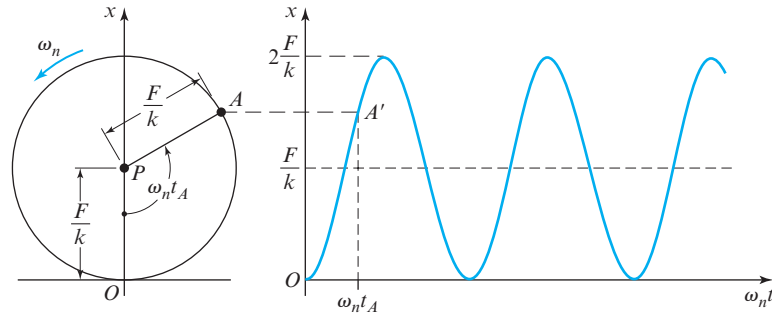


Figure 13.8 Response of an undamped vibrating system to a constant force.

observe the behavior of the system. Since the system is motionless at the instant the force is applied, the starting conditions are $x = 0$ and $\dot{x} = 0$ when $t = 0$. Substituting these conditions into Eq. (13.19) produces the following values for the two constants of integration:

$$A = -\frac{F}{k} \quad \text{and} \quad B = 0.$$

The equation for the motion is obtained by substituting these back into Eq. (13.19). This gives

$$x = \frac{F}{k}(1 - \cos \omega_n t). \quad (13.20)$$

This equation is plotted in Fig. 13.8 to show how the system behaves. Figure 13.8 shows that the application of a constant force F produces a vibration of amplitude F/k about a position of equilibrium displaced a distance F/k from the origin. This is evident from Eq. (13.20) since it contains a positive constant term,

$$x_1 = \frac{F}{k},$$

and a negative trigonometric term,

$$x_2 = -\frac{F}{k} \cos \omega_n t.$$

For $t = 0$ these two terms become

$$x_1 = \frac{F}{k} \quad \text{and} \quad x_2 = -\frac{F}{k}.$$

On the phase diagram (Fig. 13.8) the motion is represented by the phasor PA of length F/k rotating counterclockwise at ω_n rad/s. This phasor starts from the posture PO when $t = 0$, rotates about P as a center, and generates the circle of radius F/k .

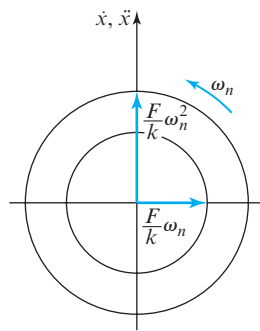


Figure 13.9 Starting postures of the velocity and acceleration phasors.

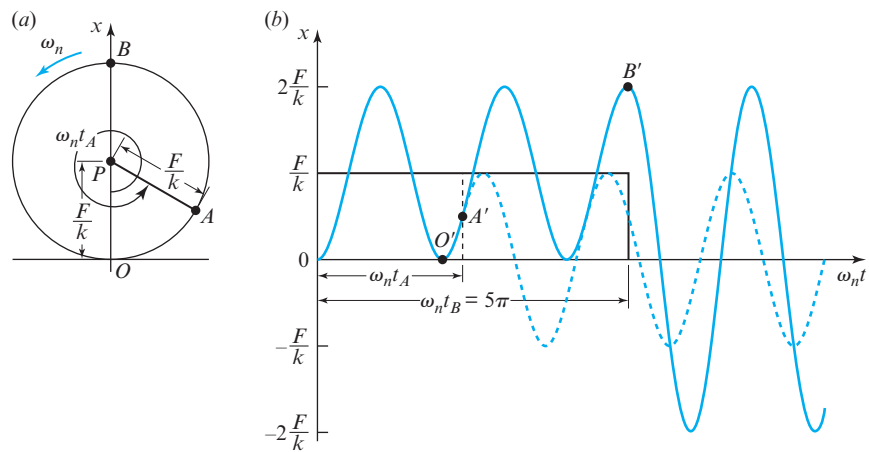


Figure 13.10 (a) Phase diagram; (b) displacement diagram.

The velocity and acceleration are obtained by successive differentiation of Eq. (13.20) and are

$$\dot{x} = \frac{F}{k} \omega_n \sin \omega_n t \quad (13.21)$$

$$\ddot{x} = \frac{F}{k} \omega_n^2 \cos \omega_n t. \quad (13.22)$$

These can be represented as phasors too, as shown in Fig. 13.9. Their starting postures are determined by calculating their values for $\omega_{nt} = 0$. Velocity-time and acceleration-time graphs can also be plotted for these phasors employing the same methods used to obtain the displacement-time plot of Fig. 13.8.

Figure 13.10 shows the results when the force is just as suddenly removed. If, for example, the force is removed at time $t_B = 5\pi/\omega_n$ when the phasor occupies the posture PB and the displacement diagram has progressed to B' , the resulting motion has an amplitude

about the zero axis of $2F/k$. The diagram shows an amplitude of F/k about the zero axis if the force is removed at time $t_A = 9\pi/4\omega_n$ at position A' on the displacement diagram. Finally, note that if the force is removed at $t = 2\pi/\omega_n$ at point O' on the displacement diagram, the resulting motion is zero.

13.5 PHASE-PLANE REPRESENTATION

The phase-plane method is a graphic method of solving transient vibration problems that is quite easy to understand and use. The method eliminates the necessity for solving differential equations, some of which can be very difficult, and even enables solutions to be obtained when the functions involved are not expressed in algebraic form. Engineers must concern themselves as much with transient disturbances and motions of machine parts as with steady-state motions. The phase-plane method presents the physics of the problem with so much clarity that it serves as an excellent vehicle for the study of mechanical transients.

Before introducing the details of the phase-plane method, it is of value to demonstrate how the displacement-time and the velocity-time relations are generated by a single rotating phasor. We have already observed that a free undamped vibrating system has a displacement equation that can be expressed in the form

$$x = X_0 \cos(\omega_n t - \phi) \quad (13.23)$$

and that its velocity is

$$\dot{x} = -X_0 \omega_n \sin(\omega_n t - \phi). \quad (13.24)$$

The displacement given by Eq. (13.23) can be represented by the projection on a vertical axis of a phasor of length X_0 rotating at ω_n rad/s in the counterclockwise direction (Fig. 13.11a). The angle $(\omega_n t - \phi)$ in this example is measured from the vertical axis. Similarly, the velocity can be represented on the same vertical axis by the projection of another phasor of length $X_0 \omega_n$ rotating at the same angular velocity, but leading X_0 by a phase angle of 90° as shown in Fig. 13.11a. Therefore, the angular location of the velocity phasor is measured from the horizontal axis. If we take the coordinate system containing the velocity phasor and rotate it backward (clockwise) through an angle of 90° ,

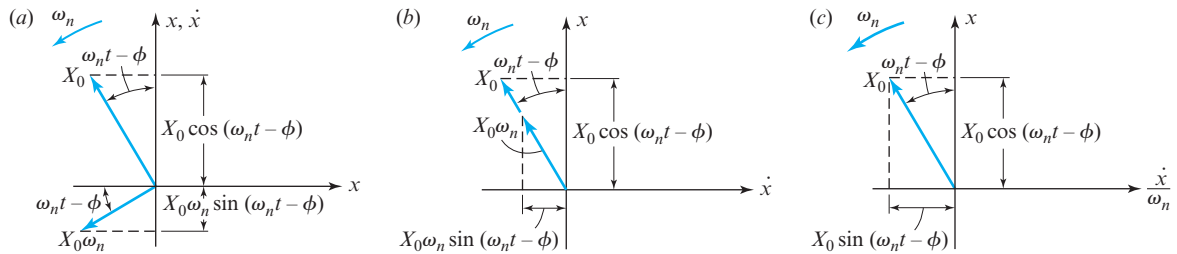


Figure 13.11

then the velocity and displacement phasors become colinear and their angular locations can be measured from the same vertical axis. This step has been taken in Fig. 13.11*b*, where the displacement is still measured along the same vertical axis. But, having rotated the coordinate system in which the velocity is measured, we see that the velocity is obtained by projecting the $X_0\omega_n$ phasor to the horizontal axis. Thus, we can now measure velocities on a separate axis from displacements. Note, too, that the direction of positive velocities is to the right.

Our final step is taken by noting that the velocity phasor differs in length from the displacement phasor by the constant factor, ω_n . Thus, instead of plotting velocities, if we plot the quantity $\dot{x}/\omega_n$, then we will have a quantity that is proportional to velocity. This step has been taken in Fig. 13.11*c*, where the horizontal axis is designated the $\dot{x}/\omega_n$ axis. With this change the projection of the phasor X_0 on the x axis gives the displacement and its projection on the $\dot{x}/\omega_n$ axis gives a quantity that is directly proportional to the velocity.

The displacement-time and the velocity-time graphs of a free undamped vibration have been plotted in Fig. 13.12 to demonstrate how the vector diagram is related to them. A point A on the phase diagram corresponds with A' on the displacement diagram and with A'' on the velocity diagram. Note that quantities obtained from the velocity plot must be multiplied by ω_n to obtain the actual velocity.

It is possible to arrive at these same conclusions in a different manner. If Eq. (13.23) is squared, we obtain

$$x^2 = X_0^2 \cos^2(\omega_n t - \phi). \quad (a)$$

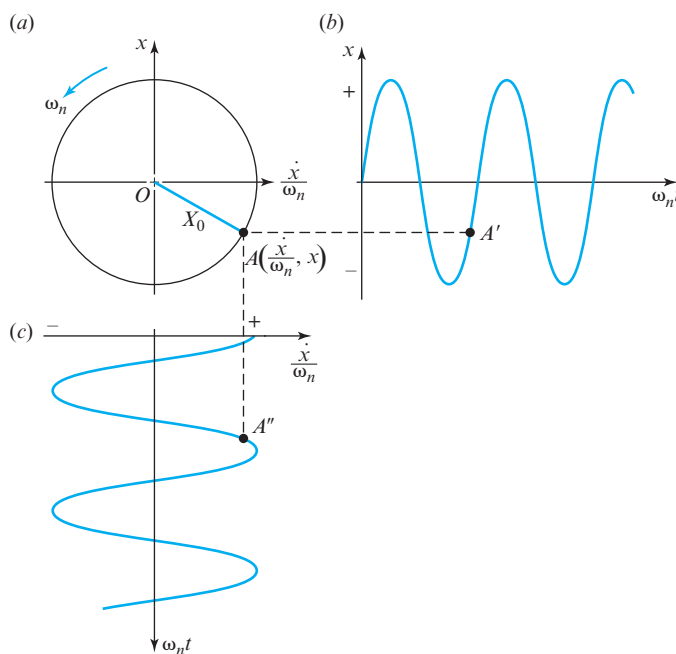


Figure 13.12 (a) Phase diagram; (b) displacement diagram; (c) velocity diagram.

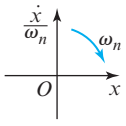


Figure 13.13

Next, dividing Eq. (13.24) by ω_n and squaring the result gives

$$\left(\frac{\dot{x}}{\omega_n}\right)^2 = X_0^2 \sin^2(\omega_n t - \phi). \quad (b)$$

Then, adding Eqs. (a) and (b) gives

$$x^2 + \left(\frac{\dot{x}}{\omega_n}\right)^2 = X_0^2. \quad (13.25)$$

This is the equation of a circle having the amplitude X_0 as its radius and with its center at the origin of a coordinate system having the axes x and $\dot{x}/\omega_n$. Thus, Eq. (13.25) describes the circle of Fig. 13.12a, where OA is the amplitude, X_0 , of the motion. The coordinate system $x, \dot{x}/\omega_n$ defines the position of points in a region called the *phase plane*.

The phase plane is widely used in the solution of nonlinear differential equations. Such equations occur frequently in the study of vibrations and feedback-control systems. When the phase plane is used to solve nonlinear differential equations, it is customary to arrange the axes as shown in Fig. 13.13, with ω_n considered positive in the clockwise direction instead of the counterclockwise direction as we use it here. This arrangement of the axes seems more logical to some, since $\dot{x}/\omega_n$ is a function of x . However, for the analysis of transient disturbances to mechanical systems and for analyzing cam mechanisms, the arrangement of the axes as in Fig. 13.12 is more appropriate, and this is the one we shall employ throughout Chap. 13. The reason for this is that we employ the phase-plane method, not as an end in itself, as is done when it is used for the solution of nonlinear problems, but as a tool to obtain the transient response.

13.6 PHASE-PLANE ANALYSIS

As a first example of the use of the phase-plane method, we shall consider the vibrating system of Fig. 13.14. This is a spring-mass system having a spring attached to a frame that can be repositioned at will. To begin the motion, we might consider the system as initially at rest and then, at $t = 0$, we suddenly move the frame a distance x_1 to the right. The effect of this sudden motion is to compress the spring an amount x_1 and to shift the equilibrium position of the mass a distance x_1 to the right. For $t < 0$, the mass is in equilibrium at the position shown in Fig. 13.14. For $t > 0$, the equilibrium position of the mass is a distance x_1 to the right of the position shown. If the motion of the frame occurs instantaneously, the initial conditions are

$$\text{at } t = 0, \quad x = -x_1, \quad \text{and} \quad \dot{x} = 0, \quad (a)$$

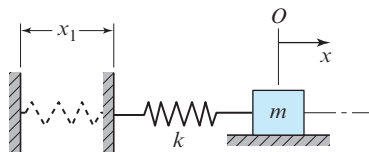


Figure 13.14

where the motion is now measured from the shifted equilibrium position. If these initial conditions are used to evaluate the constants of integration of Eq. (13.15), we obtain

$$X_0 = -x_1 \quad \text{and} \quad \phi = 0$$

so that

$$x = -x_1 \cos \omega_n t. \quad (b)$$

Next, recall from Eq. (13.20) that the application of a constant force to an undamped spring-mass system gives as the equation of motion

$$x = \frac{F}{k} - \frac{F}{k} \cos \omega_n t. \quad (c)$$

By rearranging this, we obtain

$$x - \frac{F}{k} = -\frac{F}{k} \cos \omega_n t. \quad (d)$$

If we let $x_1 = F/k$, then Eq. (d) becomes

$$x - x_1 = -x_1 \cos \omega_n t. \quad (e)$$

But, if the origin of x is shifted a distance x_1 , then Eq. (e) is the same as Eq. (b). Thus, shifting of the frame of a vibrating system through a distance x_1 is equivalent to adding a constant force $F = kx_1$ acting upon the mass. This problem is shown in Fig. 13.15, where the earlier origin is O and the new origin is O_1 . These are separated, then, by the distance x_1 . At the instant $t = 0$, the origin O is shifted to O_1 , initiating the motion. A phasor, O_1A , equal in magnitude to x_1 , begins its rotation from the posture O_1O . Rotating at ω_n rad/s, the projection of this phasor on the x axis describes the displacement of the mass. This phasor then continues to rotate until something else happens to the system. Arriving at point A , it has traversed an angle $\omega_n t_A$, and the corresponding point on the displacement-time diagram is A' .

The phase-plane method permits us to move the origin in any manner we may choose and at any instant in time that we choose. In the example above, the origin was shifted a

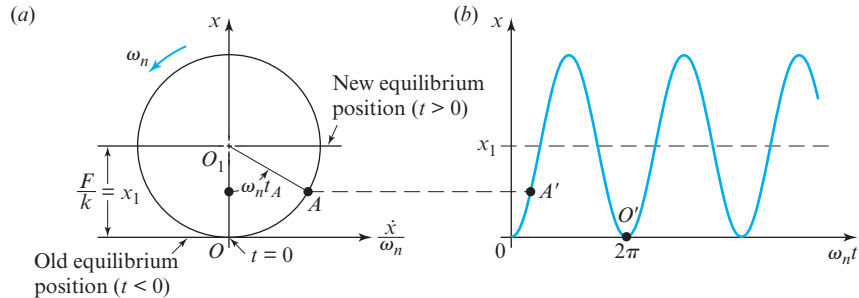


Figure 13.15 (a) Phase-plane diagram; (b) displacement-time diagram.

distance x_1 , and we have seen that this is equivalent to the sudden application of a force $F = kx_1$ applied to the mass in the positive direction. Suppose that we permit the phasor to make one complete revolution of 360° . It then generates one complete cycle of displacement and returns to point O . If, at this instant, we shift the frame back to its original position, we might reasonably ask: What are the displacement and velocity at the instant of making this shift? The phase-plane diagram shows that both the displacement and velocity are zero at the end of a cycle. The displacement and velocity are exactly the quantities we require to determine the motion of the system in the next era. Since these are both zero, there is no motion after returning the frame to its original position. It will be recalled that these are exactly the results that we obtained in analyzing square-wave forcing functions when the force endured for an integral number of cycles. Thus, the phasor generates the displacement diagram through the angle $\omega_n t = 2\pi$ in this case, and at this point, we return the frame to its original position. The initial conditions are now $x = 0$, $\dot{x} = 0$; consequently, the mass stops its motion completely.

Let us now create a vibration by shifting the frame from $x = 0$ to $x = x_1$, waiting a period of time Δt , then shifting the frame back to $x = 0$. This constitutes a square-wave forcing function, or a step disturbance to the system, and the phase-plane and displacement diagrams for such a motion are shown in Fig. 13.16. The motion can be described in three eras. During the first, from t_0 to t_1 , everything is at rest. At $t = t_1$, the frame suddenly moves to the right a distance $x = x_1$, compressing the spring. The duration of the second era is from t_1 to t_2 (Δt), and at time t_2 , the frame suddenly returns to its original position. The third era represents time when $t \geq t_2$. The position of the frame during these three eras is shown in the displacement diagram. The displacement diagram also describes the motion of the mass, and the phase-plane diagram explains why it moves as it does. Between t_0 and t_1 , the mass is at rest and nothing happens. At time t_1 , the frame shifts from O to O_1 on the phase-plane diagram, which is the distance x_1 . If we designate the original frame position as the origin of the x , $\dot{x}/\omega_n$ system, then the conditions at $t = t_1$ are $x = +x_1$ and $\dot{x} = 0$. However, the motion of the mass about O is equal to its motion about O_1 plus the distance from O to O_1 . Therefore

$$x = -x_1 \cos \omega_n t' + x_1 = x_1(1 - \cos \omega_n t'). \quad (f)$$

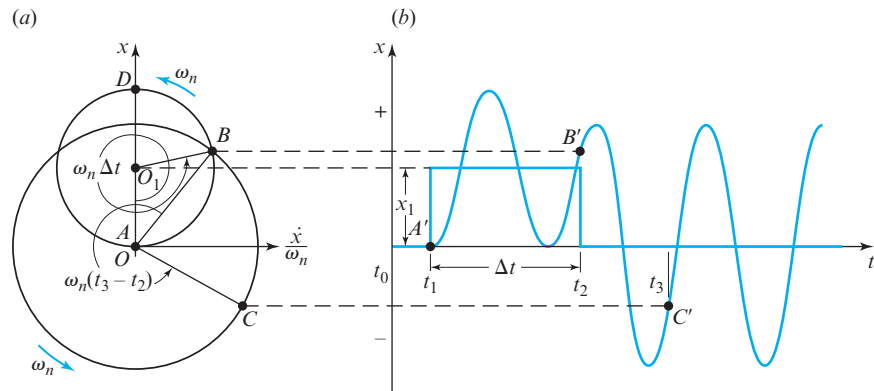


Figure 13.16 A step disturbance.

Also,

$$\dot{x} = x_1 \omega_n \sin \omega_n t', \quad (g)$$

where t' is understood to begin at time t_1 . As shown in Fig. 13.16b, the mass vibrates about position $x = x_1$ during this era. The vibration begins at point A on the phase-plane diagram and continues as a free vibration for time Δt . During this period, the phasor O_1A rotates counterclockwise with angular velocity ω_n rad/s until at time t_2 it occupies the posture O_1B . At this instant, the third era begins when the frame suddenly returns through the distance x_1 to its original position. Thus the starting conditions for the third era are the same as the ending conditions for the second and are

$$x_2 = x_1(1 - \cos \omega_n \Delta t) \quad \text{and} \quad \dot{x}_2 = x_1 \omega_n \sin \omega_n \Delta t. \quad (h)$$

Substituting these conditions into Eq. (13.12) and rearranging gives the equation of motion for the third era as

$$x = x_1(1 - \cos \omega_n \Delta t) \cos \omega_n t'' + x_1 \sin \omega_n \Delta t \sin \omega_n t'', \quad (i)$$

where t'' is the time measured from the start of the third era. Equation (i) can be transformed into an equivalent expression containing only a single trigonometric term and a phase angle as in Eq. (13.15), but we shall not do so here. The third era, as we have seen, begins at point B on the phase-plane diagram (B' on the displacement diagram); the motion of point B as it moves about a circle with center at O in the counterclockwise direction describes the motion of the mass. Thus, at instant t_3 , the phasor OB has rotated through the angle $\omega_n(t_3 - t_2)$ and is located at C . The corresponding point on the displacement diagram is C' .

The extension of the phase-plane method to any number of steps, taken in either or both the positive or negative x direction, should now be apparent. In each case the starting conditions for the next era are taken equal to the ending conditions for the previous era. The equations of motion should be written for each era with time counted from the start of that era. We can now understand that if the third era of Fig. 13.16 had begun at point D on the phase-plane diagram, instead of at point B , then the resulting motion would have an amplitude twice as large as that in the second era.

13.7 TRANSIENT DISTURBANCES

Any action that destroys the equilibrium of a vibratory system may be called a *disturbance* to that system. A *transient disturbance* is any action that endures for only a relatively short period of time. The analyses in the several preceding sections have dealt with transient disturbances having a stepwise relationship to time. Since all machine parts have elasticity and inertia, forces do not come into existence instantaneously in real life.* Consequently,

* If a force could truly be applied instantaneously, then, according to Newton, an acceleration would occur instantaneously. It would still take at least one instant to integrate this to become a change in velocity, and another instant to integrate into a change in position. Finally, according to Hooke's law, our elastic body would display a force.

we usually expect to encounter forcing functions that vary smoothly with time. Although the step forcing function is not true to nature, it is our purpose in this section to demonstrate how the step function is used with the phase-plane method to obtain a good approximation of the vibration of a system excited by a “natural” disturbance.

The procedure is to plot the disturbance as a function of time, to divide this into steps, and then to use the steps successively to make a phase-plane plot. The resulting displacement and velocity diagrams can then be obtained by graphically projecting points from the phase-plane diagram, as previously explained. It turns out that very accurate results can frequently be obtained using only a small number of steps. Of course, as in any graphic solution, better results are obtained when a larger number of steps are employed and when the work is plotted to a larger scale. It is difficult to set up general rules for selecting the size of the steps to be used. For slowly vibrating systems and for relatively smooth forcing functions, the step width can be quite large, but even a slowly varying system requires shorter steps if the forcing function has numerous sharp peaks and valleys, that is, if it has a great deal of frequency content. For smooth forcing functions and slowly vibrating systems, a step size such that the phasor sweeps out an angle of 180° is probably about the largest that one should use. It is a good idea to check the step size during the construction of the phase-plane diagram. Too great a step size causes a discontinuity in the slope of two curves at the point of adjacency of the two. If this occurs, then the step can immediately be decreased and the procedure resumed.

Figure 13.17 shows how to find the heights of the steps. The first step for the forcing function in Fig. 13.17 has been given a duration of Δt_1 and a height of h_1 . This height is obtained by constructing a horizontal line across the force curve such that the areas of the two shaded triangles are equal. For a step size of Δt_1 , the phasor sweeps out an angle of $\omega_n \Delta t_1$. Thus, the angular rotation of the phasor is obtained simply by multiplying the step size in seconds by the natural frequency of the system.

Figure 13.18 shows a four-step forcing function that we may assume has been deduced from a natural function. The widths and heights of the steps have been plotted to scale and are Δt_1 and h_1 for the first step, Δt_2 and h_2 for the second step, and so on. Note that the fourth step has a negative h_4 . The motion begins at $t = 0$ when a phasor of length h_1 starts rotating about A as a center from the initial posture AO . This phasor rotates through an angle $\omega_n \Delta t_1$ and generates the portion of the response curve contained in the interval Δt_1 .

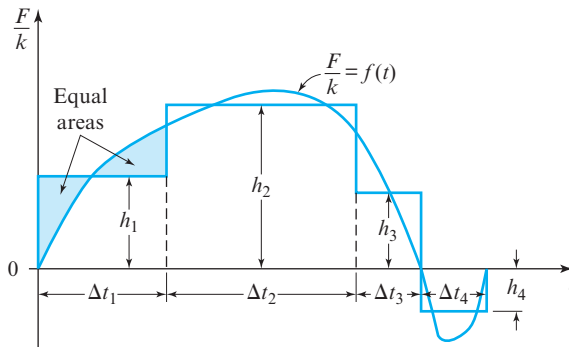


Figure 13.17 Finding the heights of the steps.

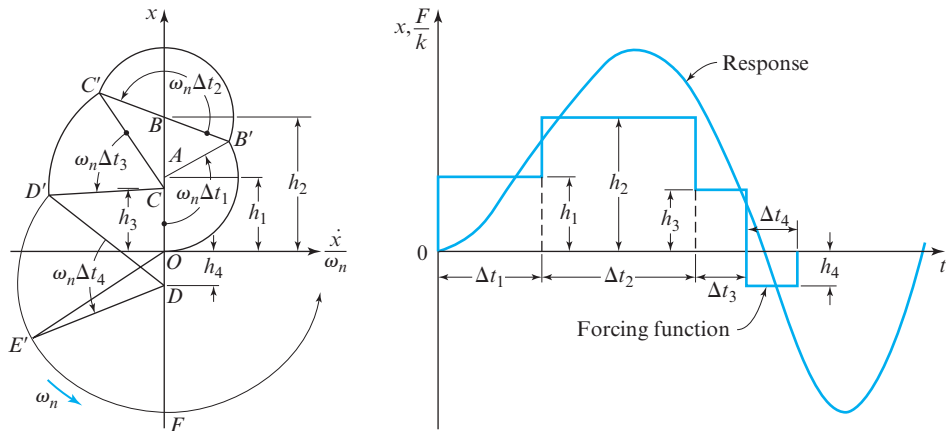


Figure 13.18 Construction of the phase-plane and displacement diagrams.

At the end of this period of time, the second step begins with the center of rotation of the phasor shifting from A to B . The length of the phasor for the duration of the second step is the distance BB' . This phasor then rotates through the angle $\omega_n \Delta t_2$ about a center at B until it arrives at the posture BC' . At this instant, the third step begins. The center of rotation shifts to C , and the phasor CC' rotates through the angle $\omega_n \Delta t_3$. At the end of the fourth step the phasor has arrived at E' . The center of rotation now shifts to the origin O , and the motion continues as a free vibration of amplitude OE' until something else (not shown) happens to the system.

EXAMPLE 13.2

Measurements on a mechanical vibrating system show that the mass has a weight of 16.90 lb and that the springs combine to give an equivalent spring rate of 30 lb/in. A transient force resembling the first half-cycle of a sine wave operates on the system. The maximum value of the force is 10 lb, and it is applied for 0.120 s. Determine the response.

ASSUMPTION

Since the system is observed to vibrate quite freely, damping is neglected.

SOLUTION

The natural frequency of the system is

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{(30 \text{ lb/in})}{(16.90 \text{ lb})/(386 \text{ in/s}^2)}} = 26.2 \text{ rad/s.}$$

The period and frequency are

$$\tau = \frac{2\pi}{\omega_n} = \frac{2\pi \text{ rad/cycle}}{26.2 \text{ rad/s}} = 0.240 \text{ s/cycle} \quad \text{and} \quad f = \frac{1}{\tau} = \frac{1}{0.240 \text{ s/cycle}} = 4.17 \text{ Hz.}$$

We first replace the entire forcing function by a single step disturbance. The height of the step should be the time average of the forcing function for the duration of the step. The average ordinate of a half-cycle of the sine wave is

$$h_1 = \frac{1}{\pi} \int_0^{\pi} \left[\frac{F_{\max}}{k} \sin bt \right] d(bt) = \frac{2F_{\max}}{\pi k} = 0.637 \frac{10 \text{ lb}}{30 \text{ lb/in}} = 0.212 \text{ in.} \quad (1)$$

This is the only value required to solve the problem. The graphic solution is shown in Fig. 13.19, together with the force and the step, all plotted to the same time scale. As shown, the amplitude of a vibration resulting from a single step is $X_0 = 2h_1 = 0.424 \text{ in.}$

To check the accuracy of a single step, we shall solve the example again using three force steps. In Fig. 13.20, the first step has a height of h_1 and is taken from O to B' with a radius AO , the second has a height h_2 and is from B' to C' with a radius BB' , and the third has a height h_3 and is from C' to D' with a radius CC' . The vibration is a free motion after point D' . These steps need not be of equal size, but it is convenient in this example to make them so. For the three steps of equal size, the ordinate of each part of the half-cycle of the sine wave of Eq. (1) can be written as

$$h_1 = h_3 = \frac{1}{\pi/3} \int_0^{\pi/3} \left[\frac{F_{\max}}{k} \sin bt \right] d(bt) = \frac{3}{2\pi} \left(\frac{F_{\max}}{k} \right) = \frac{3}{2\pi} \left(\frac{10 \text{ lb}}{30 \text{ lb/in}} \right) = 0.159 \text{ in,}$$

$$h_2 = \frac{1}{\pi/3} \int_{\pi/3}^{2\pi/3} \left[\frac{F_{\max}}{k} \sin bt \right] d(bt) = \frac{3}{\pi} \left(\frac{F_{\max}}{k} \right) = \frac{3}{\pi} \left(\frac{10 \text{ lb}}{30 \text{ lb/in}} \right) = 0.318 \text{ in.}$$

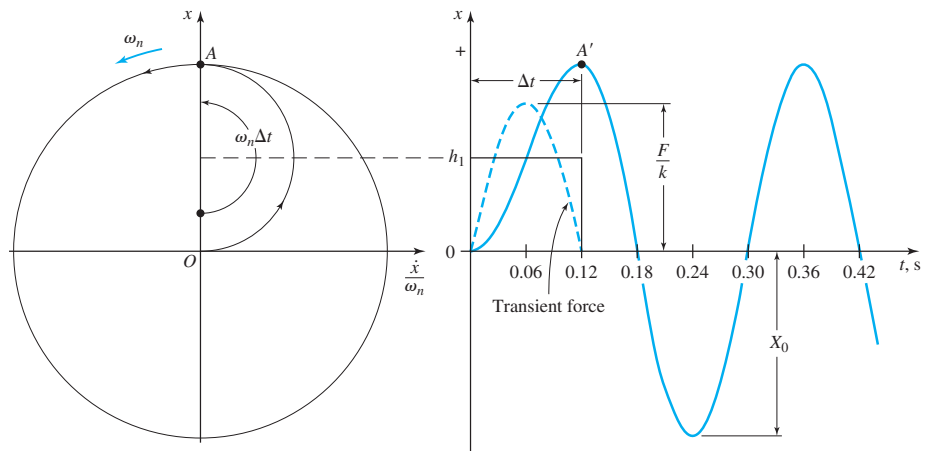


Figure 13.19 Response determined with only a single step of force.

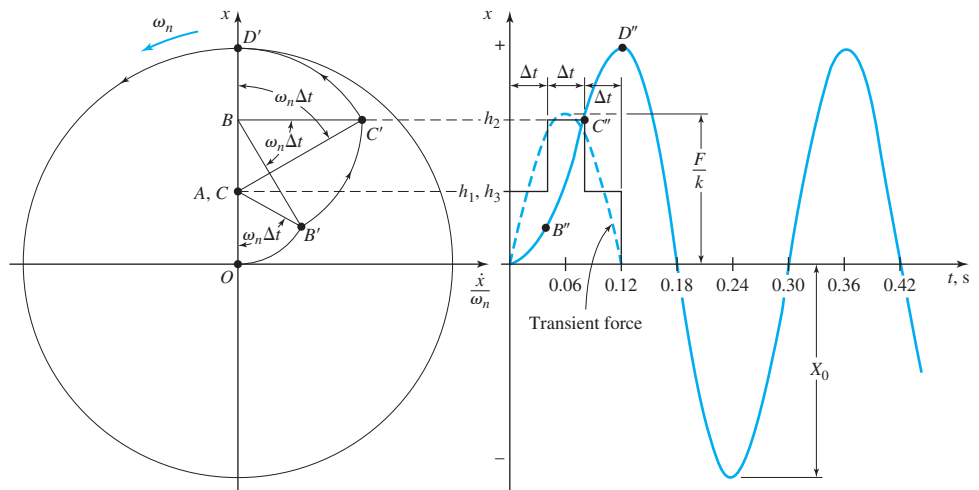


Figure 13.20 Response determined in three force steps.

The construction and results are shown in Fig. 13.20. Note that the amplitude ($X_0 = h_1 + h_2 + h_3 = 0.636$ in) is larger than that for a single step. However, it is doubtful whether the extra labor of a five-step solution is justified.

13.8 FREE VIBRATION WITH VISCOUS DAMPING

Combining all the friction in a system and representing it totally as viscous damping is the method most widely used in the simulation of vibration in mechanical systems. This is primarily because it results in a linear differential equation that is convenient to solve, not necessarily because this truly represents the response of the real system. The differential equation for an unforced, viscously damped system was derived in Eq. (13.3), and, with no externally applied force, this is

$$m\ddot{x} + c\dot{x} + kx = 0.$$

Dividing by m gives

$$\ddot{x} + \frac{c}{m}\dot{x} + \frac{k}{m}x = 0. \quad (13.26)$$

Inspection of this equation indicates that the time variation of x and its derivatives must be alike if the relation is to be satisfied. The exponential function satisfies this requirement, so it is not unreasonable to speculate that the solution might be of the form

$$x = Ae^{st}, \quad (a)$$

where A and s are values still to be determined. The first and second time derivatives of Eq. (a) are

$$\dot{x} = Ase^{st} \text{ and } \ddot{x} = As^2e^{st}.$$

Substituting these into Eq. (13.26) and dividing out the common factors gives

$$s^2 + \frac{c}{m}s + \frac{k}{m} = 0. \quad (b)$$

Thus, Eq. (a) is a solution of Eq. (13.26), provided that the quantity s is selected to satisfy Eq. (b). The roots of Eq. (b) are

$$s = -\frac{c}{2m} \pm \sqrt{\left(\frac{c}{2m}\right)^2 - \frac{k}{m}}, \quad (c)$$

and so Eq. (a) can be written

$$x = Ae^{s_1 t} + Be^{s_2 t}, \quad (d)$$

where s_1 and s_2 are the two roots of Eq. (c), and A and B are the constants of integration.

The value of damping that makes the radical of Eq. (c) zero has special significance; we shall call it the *critical-damping coefficient* and designate it by the symbol c_c . With critical damping the radical is zero, and so

$$\frac{c_c}{2m} = \sqrt{\frac{k}{m}} = \omega_n \quad \text{or} \quad c_c = 2m\omega_n. \quad (13.27)$$

It is also convenient to define a *damping ratio* ζ , which is the ratio of the actual to the critical damping. Thus,

$$\zeta = \frac{c}{c_c} = \frac{c}{2m\omega_n}. \quad (13.28)$$

After some algebraic manipulation, Eq. (c) can be written

$$s = \left(-\zeta \pm \sqrt{\zeta^2 - 1}\right) \omega_n. \quad (e)$$

For the case in which the actual damping is larger than the critical, $\zeta > 1$ and the radical is real. This is analogous to movement of the mass in a very thick viscous fluid, such as molasses, and there is no vibration. Although this case does have application in certain mechanical systems, because of space limitations, we shall not pursue it further here. It is not difficult to find that, if the mass is displaced and released in a system with more than critical damping, the mass returns slowly to its position of equilibrium without overshooting.

When the damping is less than critical, $\zeta < 1$ and the radical of Eq. (e) is imaginary. It is then written as

$$s = \left(-\zeta \pm j\sqrt{1 - \zeta^2}\right) \omega_n, \quad (f)$$

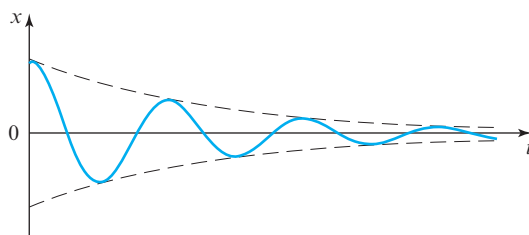


Figure 13.21

where we have used the notation $j = \sqrt{-1}$. Substituting these roots into Eq. (d) gives

$$x = e^{-\zeta \omega_n t} \left(A e^{j\sqrt{1-\zeta^2} \omega_n t} + B e^{-j\sqrt{1-\zeta^2} \omega_n t} \right). \quad (g)$$

Since x must be real, A and B are complex conjugates, and this equation can be written as

$$x = e^{-\zeta \omega_n t} \left[(A + B) \cos \omega_d t + j(A - B) \sin \omega_d t \right], \quad (h)$$

where ω_d is the natural frequency of the damped vibration, and

$$\omega_d = \sqrt{1 - \zeta^2} \omega_n. \quad (13.29)$$

It is expedient to transform Eq. (h) into a form having only a single trigonometric term. Making this transformation gives

$$x = X_0 e^{-\zeta \omega_n t} \cos(\omega_d t - \phi), \quad (13.30)$$

where X_0 and ϕ are the constants of integration. It is clear that Eq. (13.30) reduces to Eq. (13.15) if the damping is zero. The constants of integration can also be found in exactly the same manner.

Equation (13.30) is the product of a trigonometric function and a decreasing exponential function. The resulting motion is therefore oscillatory with an exponentially decreasing amplitude, as shown in Fig. 13.21. The frequency does *not* depend upon amplitude and is less than that of an undamped system, as is indicated by the factor $\sqrt{1 - \zeta^2}$ in Eq. (13.29).

13.9 DAMPING OBTAINED BY EXPERIMENT

The classic method of obtaining the damping coefficient is by an experiment in which the system is disturbed in some manner—say, by striking it with a hammer—and then recording the decaying response by means of a strain gauge, rotary potentiometer, solar cell, or other transducer. If the friction is primarily viscous, the result resembles Fig. 13.21. The rate of decay can then be measured, and, by means of the analysis to follow, the viscous-damping coefficient can be calculated.

The recording also provides a qualitative indicator of the predominating friction. In many mechanical systems, recordings reveal that the first portion of the decay is curved as in Fig. 13.21, but the later part decays at a linear rate instead. A linear rate of decay signifies Coulomb (sliding) friction. Still another type of decay sometimes found is curved at the beginning but then flattens out for small amplitudes and requires a much greater time to decay to zero. This kind of damping is proportional to the square of the velocity. Most mechanical systems have several kinds of friction present, and the investigator is usually interested only in the predominant kind. For this reason he or she should analyze that portion of the decay record in which the amplitudes are closest to those actually experienced in normal operation of the system.

Perhaps the best method of obtaining an average damping coefficient is to use a number of cycles of decay, if they can be obtained, rather than a single cycle. If we take any response curve, such as that of Fig. 13.21, and measure the amplitude of the n th and also of the $(n + N)$ th cycle, then these measurements are taken when the cosine term of Eq. (13.30) is approximately unity; so

$$x_n = X_0 e^{-\zeta \omega_n t_n} \quad \text{and} \quad x_{n+N} = X_0 e^{-\zeta \omega_n (t_n + N\tau)},$$

where τ is the period of vibration and N is the number of cycles of motion between the amplitude measurements. The *logarithmic decrement*, δ_N , is defined as the natural logarithm of the ratio of these two amplitudes and is

$$\delta_N = \ln \frac{x_n}{x_{n+N}} = \ln \frac{X_0 e^{-\zeta \omega_n t_n}}{X_0 e^{-\zeta \omega_n (t_n + N\tau)}} = \ln e^{\zeta \omega_n N\tau} = \zeta \omega_n N\tau. \quad (13.31)$$

The period of the vibration is

$$\tau = \frac{2\pi}{\omega_d} = \frac{2\pi}{\omega_n \sqrt{1 - \zeta^2}}. \quad (13.32)$$

Therefore, the logarithmic decrement can be written in the form

$$\delta = \frac{\delta_N}{N} = \frac{2\pi\zeta}{\sqrt{1 - \zeta^2}}. \quad (13.33)$$

Measurements of many damping ratios indicate that a value of $\zeta < 0.20$ can be expected for most machine systems, with a value of $\zeta = 0.10$ or less being the most probable. For this range of values, the radical in Eq. (13.33) can be taken as approximately unity, giving

$$\delta \approx 2\pi\zeta \quad (13.34)$$

as an approximate formula.

EXAMPLE 13.3

Let the vibrating system of Example 13.2 have a dashpot attached that exerts a force of 0.25 lb on the mass when the mass has a velocity of 1 in/s. Find the critical damping constant, the logarithmic decrement, and the ratio of two consecutive maxima.

SOLUTION

The data from Example 13.2 are repeated here for convenience:

$$W = 16.90 \text{ lb}, k = 30 \text{ lb/in}, \omega_n = 26.2 \text{ rad/s}.$$

The critical damping constant is

$$c_c = 2m\omega_n = 2 \left(\frac{16.90 \text{ lb}}{386 \text{ in/s}^2} \right) (26.2 \text{ rad/s}) = 2.29 \text{ lb} \cdot \text{s/in.} \quad \text{Ans.}$$

The viscous damping constant is $c = F/\dot{x} = 0.25 \text{ lb} \cdot \text{s/in}$. Therefore the damping ratio is

$$\zeta = \frac{c}{c_c} = \frac{0.25 \text{ lb} \cdot \text{s/in}}{2.29 \text{ lb} \cdot \text{s/in}} = 0.109.$$

The logarithmic decrement is obtained from Eq. (13.33):

$$\delta = \frac{2\pi\zeta}{\sqrt{1-\zeta^2}} = \frac{2\pi(0.109)}{\sqrt{1-(0.109)^2}} = 0.689. \quad \text{Ans.}$$

The ratio of two consecutive maxima is

$$\frac{x_{n+1}}{x_n} = e^{-\delta} = e^{-0.689} = 0.502. \quad \text{Ans.}$$

Therefore, for this amount of damping, each peak amplitude is approximately 50% of that of the previous cycle.

13.10 PHASE-PLANE REPRESENTATION OF DAMPED VIBRATION

We have seen that when undamped mechanical systems are subjected to transient forces, the resulting motion (mathematically, at least) endures forever without decreasing in amplitude. We know, however, that energy losses always exist, although sometimes only in minute amounts, and that these losses eventually cause vibration to cease. In the case of vibrating systems known to be acted upon by transient forces, it is often desirable to introduce additional damping as one means of decreasing the number of cycles of vibration. For this reason, the phase-plane solution of damped vibration is particularly important.

We have seen that the phase-plane diagram of an undamped free vibration is generated by a phasor of constant length rotating at a constant angular velocity. In the case of a damped free vibration, the phase-plane diagram is generated by a phasor whose length is decreasing exponentially with time. Thus, the trajectory for such a motion is a spiral instead of a circle.

It turns out that a simple spiral will give incorrect values for the velocity if plotted on the same $x, \dot{x}/\omega_n$ axes as used for undamped free vibration. The reason for this is that

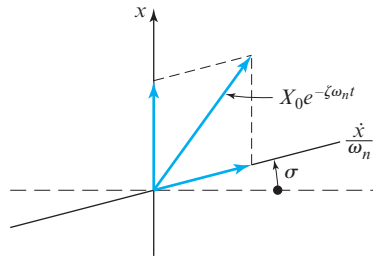


Figure 13.22 Damping phase angle σ is applied only to the velocity.

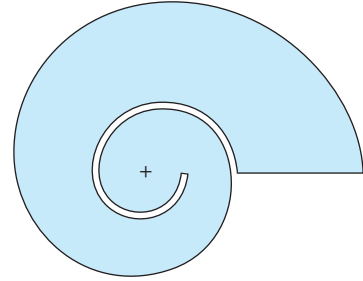


Figure 13.23 Spiral template for $\zeta = 0.15$.

the phase relationship between the velocity and displacement is not 90° as it is for an undamped system. Instead, for a damped system, the phase angle depends upon the amount of damping present.

An easy way to obtain a phase-plane diagram for damped motion is to tilt or rotate the $\dot{x}/\omega_n$ axis through angle σ , as shown in Fig. 13.22. The angle σ is called the *damping phase angle*, and the following analysis demonstrates how this angle comes about.

Taking the time derivative of Eq. (13.30) gives

$$\dot{x} = -X_0 \zeta \omega_n e^{-\zeta \omega_n t} \cos\left(\sqrt{1 - \zeta^2} \omega_n t - \phi\right) - X_0 \sqrt{1 - \zeta^2} \omega_n e^{-\zeta \omega_n t} \sin\left(\sqrt{1 - \zeta^2} \omega_n t - \phi\right)$$

or

$$\frac{\dot{x}}{\omega_n} = -X_0 e^{-\zeta \omega_n t} \left[\zeta \cos\left(\sqrt{1 - \zeta^2} \omega_n t - \phi\right) + \sqrt{1 - \zeta^2} \sin\left(\sqrt{1 - \zeta^2} \omega_n t - \phi\right) \right]. \quad (a)$$

Designating $\sin \sigma = \zeta$ and $\cos \sigma = \sqrt{1 - \zeta^2}$ and noting the trigonometric identity

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta,$$

we see that Eq. (a) can be written as

$$\frac{\dot{x}}{\omega_n} = -X_0 e^{-\zeta \omega_n t} \sin\left(\sqrt{1 - \zeta^2} \omega_n t - \phi + \sigma\right). \quad (13.35)$$

As shown in Fig. 13.22, the $\dot{x}/\omega_n$ axis is rotated counterclockwise through the angle σ when plotting the phase-plane diagram. The velocity diagram is then obtained by projecting perpendicular to the $\dot{x}/\omega_n$ axis. If the damping is zero, note that $\sqrt{1 - \zeta^2} = 1$, $e^{-\zeta \omega_n t} = 1$, $\sigma = 0$, and Eqs. (13.30) and (13.35) reduce to the same set of equations that we employed to develop the phase-plane analysis of undamped motion.

In analyzing a damped system that is acted upon by transient forces by the phase-plane method, it is very helpful to construct transparent spiral templates, as shown in Fig. 13.23. These are made from a sheet of clear plastic that is placed over a drawing of the spiral so that the spiral and its center can be scribed on the plastic using the point of a pair of

dividers. The scribed curve can then be trimmed with a pair of manicurist's scissors or a sharp razor blade. Since only one template is needed for each damping ratio, these can be stored for use in future problems. Do not forget to label each template with the value of the damping ratio for which it is constructed.

The spirals can be plotted directly from a table calculated using Eq. (13.30), but this is tedious. A more rapid method, which is accurate enough for graphic purposes, is to approximate each quadrant of the spiral by a circular arc. To do this it is necessary only to calculate the change in the length of the phasor in a quarter of a rotation. From Eq. (13.30), the length of the phasor is $x = X_0 e^{-\zeta \omega_n t}$ and its angular velocity is $\omega_n \sqrt{1 - \zeta^2}$. Therefore, for a 90° rotation,

$$\omega_n \sqrt{1 - \zeta^2} t = \frac{\pi}{2} \quad \text{or} \quad \omega_n t = \frac{\pi}{2\sqrt{1 - \zeta^2}}. \quad (b)$$

Thus, the length, after rotation through 90° , is

$$x_{90^\circ} = X_0 e^{-\zeta \pi / 2 \sqrt{1 - \zeta^2}}. \quad (c)$$

To demonstrate an example of the construction of the spirals, let us employ a damping ratio $\zeta = 0.15$ and begin with a phasor 2.50 in long. Then, after 90° of rotation, the length, from Eq. (c), is

$$x_{90^\circ} = (2.50 \text{ in}) e^{-0.15\pi/2\sqrt{1-(0.15)^2}} = 1.97 \text{ in}.$$

The construction is shown in Fig. 13.24 and is explained as follows: Construct axes 1 and 2 at right angles to each other and lay off 2.50 in to A and 1.97 in to B on axes 1 and 2, respectively. Draw two lines through the origin at 45° to the 1, 2 axes. The perpendicular bisector of AB intersects one of these 45° lines at P_1 ; using P_1 as a center, strike an arc from A to B . Line P_1B crosses the other 45° line defining the center P_2 of arc BC . This process is continued in the same fashion with point P_3 defining the center of the arc CD , and so on.

The template is used by placing its center at the same point in the phase-plane diagram from which a circle arc would be constructed for undamped motion. The template is then turned until the spiral coincides with the point from which the spiral diagram is to be started.

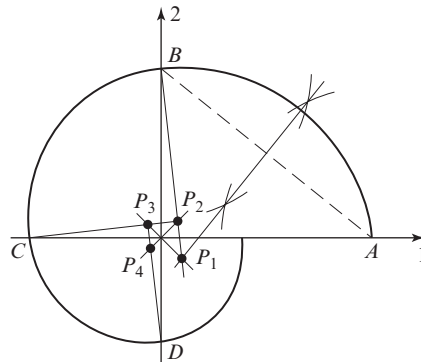


Figure 13.24

EXAMPLE 13.4

A vibrating system has an 80-lb mass mounted upon springs with an equivalent spring rate of 360 lb/in. A dashpot is included in the system and is estimated to produce 15% of critical damping. A transient force on the system is assumed to act in three steps as follows: 720 lb for 0.050 s, -270 lb for 0.063 s, and 180 lb for 0.038 s. The negative sign on the second force step simply means that its direction is opposite to the direction of the first and third steps. Determine the response of the system assuming no motion when the force function begins to act.

SOLUTION

The undamped natural frequency is

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{(360 \text{ lb/in})}{(80 \text{ lb})/(386 \text{ in/s}^2)}} = 41.7 \text{ rad/s},$$

and so the damped natural frequency is

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} = (41.7 \text{ rad/s}) \sqrt{1 - (0.15)^2} = 41.2 \text{ rad/s}.$$

The period of the motion is $\tau = 2\pi/\omega_d = (2\pi \text{ rad/cycle}) / (41.2 \text{ rad/s}) = 0.152 \text{ s/cycle}$. The F/k values are

$$\frac{F_1}{k} = \frac{720 \text{ lb}}{360 \text{ lb/in}} = 2.00 \text{ in}, \quad \frac{F_2}{k} = \frac{-270 \text{ lb}}{360 \text{ lb/in}} = -0.75 \text{ in}, \quad \frac{F_3}{k} = \frac{180 \text{ lb}}{360 \text{ lb/in}} = 0.50 \text{ in}.$$

These three steps are plotted to scale in Fig. 13.25. The angular duration of each step for the phase-plane diagram is obtained by multiplying the damped natural frequency by the time of each step and converting to degrees. This gives

$$\begin{aligned} \text{Step 1} \quad \omega_d \Delta t_1 \frac{180^\circ}{\pi \text{ rad}} &= (41.2 \text{ rad/s})(0.050 \text{ s}) \frac{180^\circ}{\pi \text{ rad}} = 120^\circ \\ \text{Step 2} \quad \omega_d \Delta t_2 \frac{180^\circ}{\pi \text{ rad}} &= (41.2 \text{ rad/s})(0.063 \text{ s}) \frac{180^\circ}{\pi \text{ rad}} = 150^\circ \\ \text{Step 3} \quad \omega_d \Delta t_3 \frac{180^\circ}{\pi \text{ rad}} &= (41.2 \text{ rad/s})(0.038 \text{ s}) \frac{180^\circ}{\pi \text{ rad}} = 90^\circ. \end{aligned}$$

The $\dot{x}/\omega_n$ axis must be rotated $\sigma = \sin^{-1}(0.15) = 8.63^\circ$ in the counterclockwise direction, as indicated in the phase-plane diagram.

The construction of Fig. 13.25 is explained as follows: We project F_1/k to A on the x axis. A phasor AO, with center at A, begins rotating at the instant $t = 0$. This phasor rotates through an angle of 120° to AA' while exponentially decreasing in length. At A' , the step function changes the origin of the phasor to B, and it rotates 150° about this point to B' , again exponentially decreasing in length. At B' the origin again shifts, this time to C, and the phasor rotates 90° to C' , again exponentially decreasing. Now the origin shifts to O, and the phasor continues its rotation and its decreasing magnitude with O as the origin until the motion becomes too small to follow. Note that points on the spiral diagram are projected

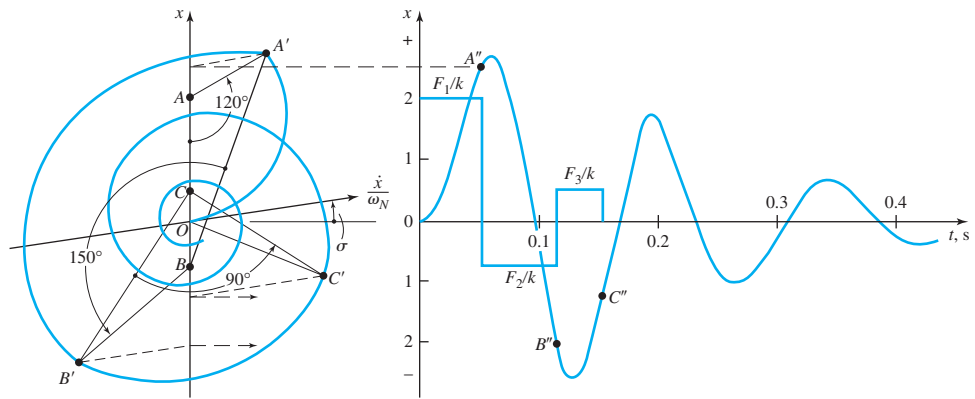


Figure 13.25 Phase-plane diagram.

parallel to the $\dot{x}/\omega_n$ axis to the x axis and then horizontally to the displacement diagram. The displacement diagram shows the resulting response. The maximum amplitude occurs on the first positive half-cycle and is 2.72 in. In a similar manner, the velocity diagram, although not shown, would be projected perpendicular to the $\dot{x}/\omega_n$ axis. Points on the spiral would project parallel to the x axis until they intersect the $\dot{x}/\omega_n$ axis.

13.11 RESPONSE TO PERIODIC FORCING

Differential equations of the form

$$m\ddot{x} + kx = ky(\omega t), \quad (13.36)$$

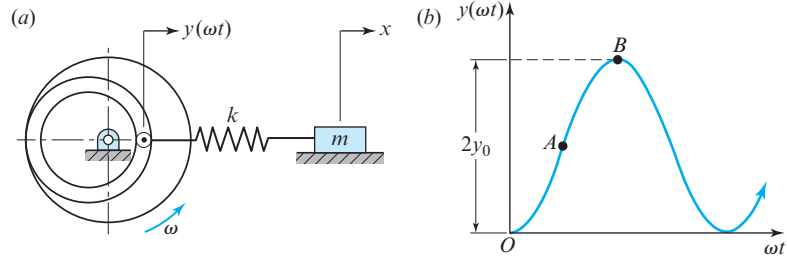
where $y(\omega t)$ is a periodic forcing function, occur frequently in mechanical systems. Let us define $y(\omega t + n\pi) = y(\omega t)$, where n is an integer; we shall then be concerned in this section only with small values of n , say, n less than 8. An example of Eq. (13.36) is shown in Fig. 13.26 in which a roller riding in the groove of a face cam generates the motion from point O (Fig. 13.26b) of

$$y(\omega t) = y_0(1 - \cos \omega t). \quad (13.37)$$

This, then, is the motion applied to the left end of a spring, k , connected to a sliding mass, m , with friction assumed to be zero. Note, in Fig. 13.26b, that we choose Eq. (13.37) to represent the driving function and that we have assumed that the roller is in its extreme left position when time is zero and cam rotation begins.

Alternatively, we could have chosen point A in Fig. 13.26b to correspond to the start of the motion. For this choice, the driving function would have been

$$y_A(\omega t) = y_0 \sin \omega t.$$



 **Figure 13.26**

One of the reasons for introducing this problem is to show an interesting relationship: Fig. 13.26a represents a single-degree-of-freedom vibrating system, but, when Eq. (13.36) or a similar equation is solved, the solution often looks more like that of a two-degree-of-freedom system. Novices are sometimes quite surprised when they first view this solution, and they are unable to explain what is happening.

Substituting Eq. (13.37) into Eq. (13.36) gives the equation to be solved as

$$m\ddot{x} + kx = ky_0(1 - \cos \omega t). \quad (13.38)$$

The complementary solution is

$$x' = A \cos \omega_n t + B \sin \omega_n t,$$

and the particular solution is

$$x'' = C + D \cos \omega t. \quad (a)$$

The second derivative of this equation is

$$\ddot{x}'' = -D\omega^2 \cos \omega t. \quad (b)$$

By substituting Eqs. (a) and (b) into Eq. (13.38), we find that $C = y_0$ and

$$D = -\frac{ky_0}{k - m\omega^2} = -\frac{y_0}{1 - (\omega/\omega_n)^2}.$$

Therefore, the complete solution is

$$x = A \cos \omega_n t + B \sin \omega_n t + y_0 \left[1 - \frac{\cos \omega t}{1 - (\omega/\omega_n)^2} \right]. \quad (13.39)$$

The velocity is

$$\dot{x} = -A\omega_n \sin \omega_n t + B\omega_n \cos \omega_n t + \frac{y_0\omega \sin \omega t}{1 - (\omega/\omega_n)^2}. \quad (13.40)$$

Since we are starting from rest, where $x(0) = \dot{x}(0) = 0$, Eqs. (13.39) and (13.40) give

$$A = \frac{(\omega/\omega_n)^2 y_0}{1 - (\omega/\omega_n)^2} \text{ and } B = 0.$$

Equation (13.39) then becomes

$$x = \frac{(\omega/\omega_n)^2 y_0 \cos \omega_n t}{1 - (\omega/\omega_n)^2} - \frac{y_0 \cos \omega t}{1 - (\omega/\omega_n)^2} + y_0. \quad (13.41)$$

The first term on the right-hand side of Eq. (13.41) is called the *starting transient*. Note that this is a vibration at the natural frequency, ω_n , not at the forcing frequency, ω . The usual physical system contains a certain amount of friction, which, as we shall see in the sections to follow, causes this term to die out after a period of time. The second and third terms on the right represent the *steady-state* solution and these denote another component of the vibration at the forcing frequency, ω . Thus, the total motion is composed of a constant plus two vibrations of differing amplitudes and frequencies. If Fig. 13.26a were a cam-and-follower system, spring k would probably be quite stiff; consequently, ω_n would be quite large compared with ω . This would cause the amplitude of the starting transient to be comparatively small, and a phase-plane plot would resemble Fig. 13.27. Here we can see that the motion is formed by the sum of a fixed phasor of length y_0 and two phasors of different lengths and different rotational velocities.

A computer solution for the special case $\omega/\omega_n = 1/3$ is shown in Fig. 13.28. Part (a) is a plot of the forcing function, and Part (b) is the response given by Eq. (13.41). This case is of interest since there are short dwells at the extremes of the motion. One can verify the existence of these dwells by constructing a plot similar to Fig. 13.27 or by substituting $\omega/\omega_n = 1/3$ into Eq. (13.41) and investigating the solution in the region for which ωt is near zero or π .

Examination of Eq. (13.41) for $\omega_n \rightarrow \infty$ indicates that $\omega/\omega_n = 0$, and the solution becomes

$$x = y_0 (1 - \cos \omega t),$$

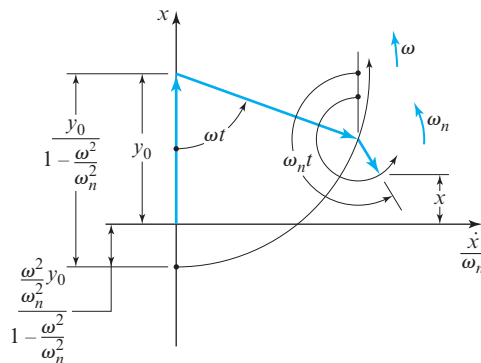


Figure 13.27

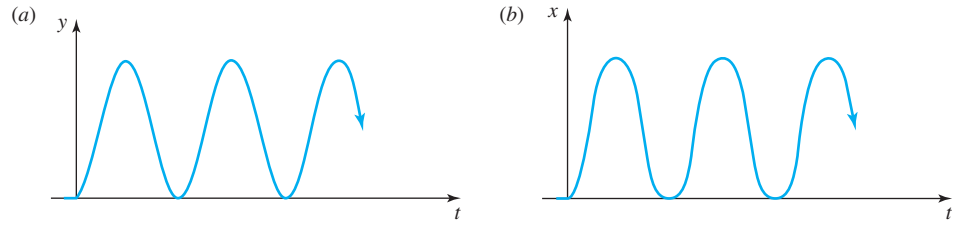


Figure 13.28 (a) Forcing function at frequency ω ; (b) response. The amplitude scales for both figures are the same.

which is the rigid-body solution. If we replace the spring with a rigid member, then k and, therefore, ω_n , become very large, and so the mass exactly follows the cam motion.

Resonance is defined by the case in which $\omega = \omega_n$. Rearranging Eq. (13.41) to

$$x = y_0 \frac{(\omega/\omega_n)^2 \cos \omega_n t - \cos \omega t + 1 - (\omega/\omega_n)^2}{1 - (\omega/\omega_n)^2},$$

we see that $x = 0/0$ for $\omega/\omega_n = 1$. However, l'Hôpital's rule enables us to find the solution; thus

$$\lim_{\omega \rightarrow \omega_n} (x) = y_0 \left(1 - \cos \omega_n t + \frac{\omega_n t}{2} \sin \omega_n t \right).$$

A plot of this equation, obtained by numerically solving the differential equation, is shown in Fig. 13.29. Note that the sine term grows and very quickly dominates the response.

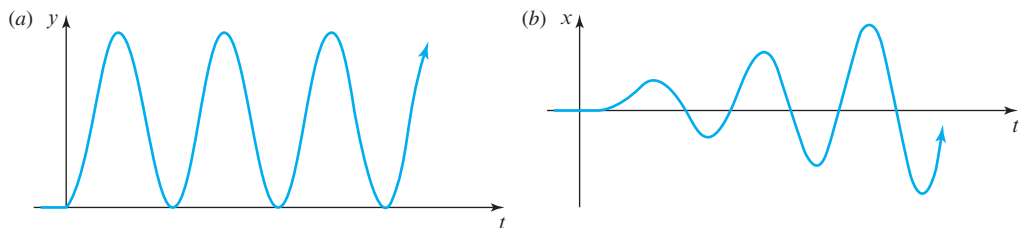


Figure 13.29 (a) Forcing function; (b) response at resonance. The amplitude scale was increased so the plot stays within reasonable bounds.

13.12 HARMONIC FORCING

We have seen that the action of any temporary forcing function on a damped system is to create a vibration, but the vibration decays when the applied force is removed. A machine element that is connected to or is a part of any moving machinery is often subject to forces that vary periodically with time. Since all metal machine parts have both mass and elasticity, the opportunity for vibration always exists. Many machines do operate at fairly constant speeds and uniform output, and it is not difficult to see that vibratory forces may exist that have a fairly constant amplitude over a period of time. Of course, these varying forces do change in magnitude when the machine speed or output changes, but there is a rather broad class of vibration problems that can be analyzed and corrected using the assumption of a periodically varying force of constant amplitude. Sometimes these forces exhibit a time characteristic that is very similar to that of a sine wave. At other times, they are quite complex and have to be analyzed as a Fourier series. Such a series is the sum of a number of sine and cosine terms, and the resultant motion is the sum of the responses to the individual terms. In this text, we shall study only the motion resulting from the application of a single sinusoidal force.

In Sec. 13.11, we examined an undamped system subjected to a periodic force and discovered that the solution contains components at two frequencies. One of these components exhibits the forcing frequency, whereas the other occurs at the natural frequency. Real systems always have damping present, and this causes the component at the (damped) natural frequency to become insignificant after a period of time; the motion that remains contains only the forcing frequency and is termed the *steady-state motion*.

To illustrate steady-state motion, we shall solve the equation

$$m\ddot{x} + c\dot{x} + kx = F_0 \cos \omega t. \quad (13.42)$$

The solution contains two parts, as before. The first, the transient or complementary, part is the solution to the homogeneous equation

$$m\ddot{x} + c\dot{x} + kx = 0. \quad (13.43)$$

This we solved in Sec. 13.8, and we obtained [Eq. (h)]

$$x' = e^{-\zeta \omega_n t} (C_1 \cos \omega_d t + C_2 \sin \omega_d t), \quad (13.44)$$

where C_1 and C_2 are the constants of integration. Since the forcing is harmonic, the particular solution is obtained by assuming a solution of the form

$$x'' = A \cos \omega t + B \sin \omega t, \quad (13.45)$$

where ω is the frequency of the forcing function, and where A and B are constants yet to be determined but are not the constants of integration. Taking successive time derivatives of Eq. (13.45) produces

$$\dot{x}'' = -A\omega \sin \omega t + B\omega \cos \omega t, \quad (a)$$

$$\ddot{x}'' = -A\omega^2 \cos \omega t - B\omega^2 \sin \omega t, \quad (b)$$

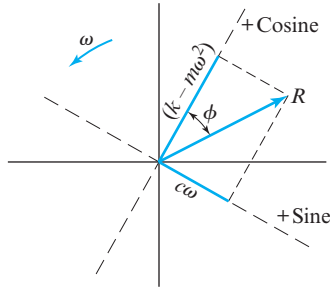


Figure 13.30

and substituting Eqs. (13.45), (a), and (b) into the differential equation, Eq. (13.42), yields

$$(kA + c\omega B - m\omega^2 A) \cos \omega t + (kB + c\omega A - m\omega^2 B) \sin \omega t = F_0 \cos \omega t.$$

Next, separating the terms, we obtain

$$\begin{aligned} (k - m\omega^2)A + c\omega B &= F_0, \\ -c\omega A + (k - m\omega^2)B &= 0. \end{aligned} \quad (c)$$

Solving Eqs. (c) for A and B and substituting the results into Eq. (13.45), we obtain

$$x'' = \frac{F_0}{(k - m\omega^2)^2 + c^2\omega^2} [(k - m\omega^2) \cos \omega t + c\omega \sin \omega t]. \quad (d)$$

Now, when we create the phase diagram (Fig. 13.30) and plot the amplitudes of the two trigonometric terms, we find the resultant is

$$R = \sqrt{(k - m\omega^2)^2 + c^2\omega^2}, \quad (e)$$

lagging the direction of the positive cosine by a phase angle of

$$\phi = \tan^{-1} \left(\frac{c\omega}{k - m\omega^2} \right). \quad (13.46)$$

Substituting Eq. (e) into Eq. (d) and dividing out common factors gives

$$x'' = \frac{F_0 \cos(\omega t - \phi)}{\sqrt{(k - m\omega^2)^2 + c^2\omega^2}}, \quad (13.47)$$

and so the complete solution is

$$x = x' + x'' = e^{-\zeta\omega_n t} (C_1 \cos \omega_d t + C_2 \sin \omega_d t) + \frac{F_0 \cos(\omega t - \phi)}{\sqrt{(k - m\omega^2)^2 + c^2\omega^2}}. \quad (13.48)$$

The first term of Eq. (13.48) is called the *transient* term, because the exponential factor causes that term to decay in a short period of time. When this happens, only the second term remains, and, since this has no end, it is called the *steady-state solution*.

When we are interested only in the steady-state term, we write the solution in the form of Eq. (13.47), omitting the prime marks on x .

Now we write Eq. (13.47) in the form

$$x = X \cos(\omega t - \phi), \quad (13.49)$$

and find the successive derivatives to be

$$\dot{x} = -\omega X \sin(\omega t - \phi), \quad (f)$$

$$\ddot{x} = -\omega^2 X \cos(\omega t - \phi). \quad (g)$$

These are then substituted into Eq. (13.42), and the forcing term is brought to the left-hand side of the equation. This gives

$$-m\omega^2 X \cos(\omega t - \phi) - c\omega X \sin(\omega t - \phi) + kX \cos(\omega t - \phi) - F_0 \cos \omega t = 0. \quad (h)$$

The amplitudes of the trigonometric terms are identified as

$$m\omega^2 X = \text{inertia force};$$

$$c\omega X = \text{damping force};$$

$$kX = \text{spring force}; \text{ and}$$

$$F_0 = \text{exciting force}.$$

Our next step is to plot these phasors on the phase diagram. We are interested only in the steady-state solution; it has no end, and so, although the phasors have a definite phase relation with each other, they can exist anywhere on the diagram at the instant we examine them.

We begin Fig. 13.31a by selecting one of the directions, say, the direction of $\cos \omega t$. The direction of $\cos(\omega t - \phi)$ lags that of $\cos \omega t$ by the angle ϕ . These automatically

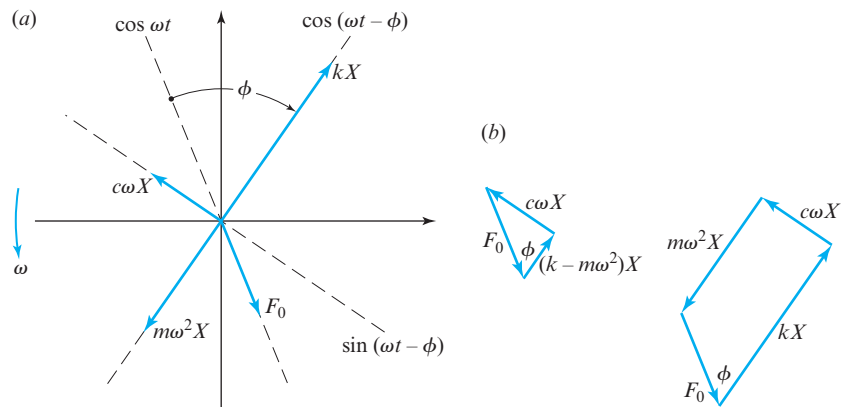


Figure 13.31

define the direction of $\sin(\omega t - \phi)$, lagging $\cos(\omega t - \phi)$ by 90° . Having defined this set of directions, we lay off each phasor in the proper direction, being sure not to forget the signs.

Since Eq. (h) states that the sum of these four phasors is zero, we can construct a phasor or vector polygon of them. This has been done in Fig. 13.31b; from the figure, we see

$$\left[(k - m\omega^2)^2 + c^2\omega^2 \right] X^2 = F_0^2$$

or

$$X = \frac{F_0}{\sqrt{(k - m\omega^2)^2 + c^2\omega^2}} \quad (i)$$

and

$$\phi = \tan^{-1} \left(\frac{c\omega}{k - m\omega^2} \right). \quad (13.50)$$

Consequently, Eq. (13.49) becomes

$$x = \frac{F_0 \cos(\omega t - \phi)}{\sqrt{(k - m\omega^2)^2 + c^2\omega^2}}. \quad (13.51)$$

These equations can be simplified by introducing the symbols

$$\omega_n = \sqrt{\frac{k}{m}}, \quad \zeta = \frac{c}{c_c}, \quad c_c = 2m\omega_n.$$

They can then be written in the form

$$\frac{X}{F_0/k} = \frac{1}{\sqrt{(1 - \omega^2/\omega_n^2)^2 + (2\zeta\omega/\omega_n)^2}} \quad (13.52)$$

and

$$\phi = \tan^{-1} \left(\frac{2\zeta\omega/\omega_n}{1 - \omega^2/\omega_n^2} \right), \quad (13.53)$$

which is a dimensionless form that is convenient for analysis. It is interesting to note that the quantity F_0/k is the deflection that a spring of rate k would experience if acted upon by force F_0 .

If various values are selected for the frequency and damping ratio, curves can be plotted showing the effects upon the motion of varying these quantities. This has been done in Figs. 13.32 and 13.33. Note that the amplitude is very large for small damping ratios near resonance (that is, where ω/ω_n is near unity). Note, too, that the peak amplitudes occur slightly before resonance is reached. Figure 13.33 also indicates that the phase angle increases very rapidly near resonance.

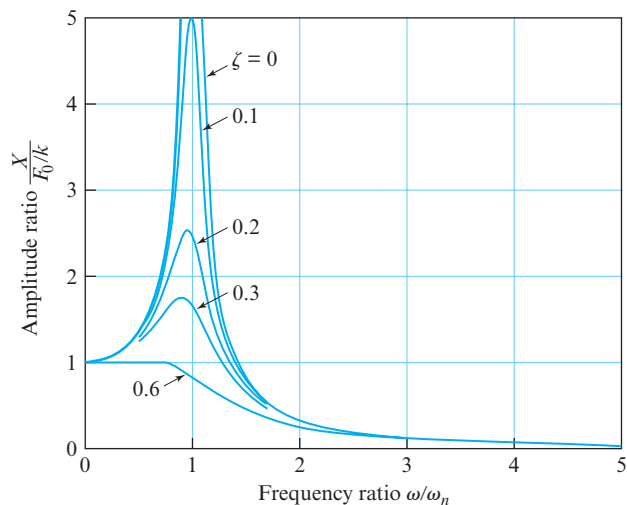


Figure 13.32 Relative displacement of a damped forced system as a function of frequency and damping ratios.

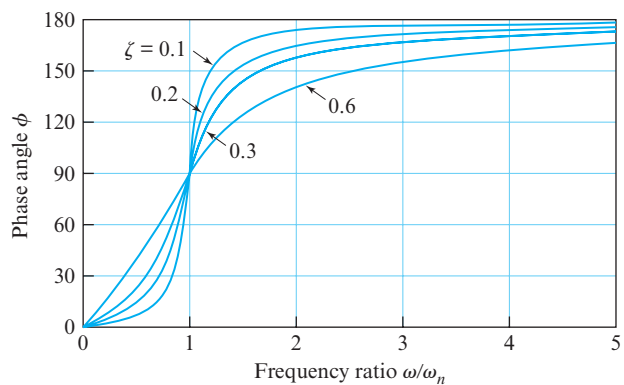


Figure 13.33 Relationship of the phase angle to frequency and damping ratios.

13.13 FORCING CAUSED BY UNBALANCE

Quite frequently the exciting force in a machine (Chap. 15) is due to the rapid rotation of an unbalanced mass. In this case, the differential equation is written

$$m\ddot{x} + c\dot{x} + kx = m_u e \omega^2 \cos \omega t, \quad (13.54)$$

where m = vibrating mass, m_u = unbalanced mass, and e = eccentricity. The solution to this equation is the same as that for Eq. (13.51) with F_0 replaced by $m_u e \omega^2$. Thus, the

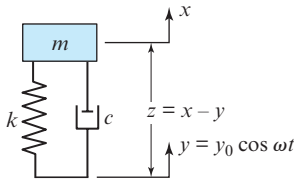


Figure 13.35

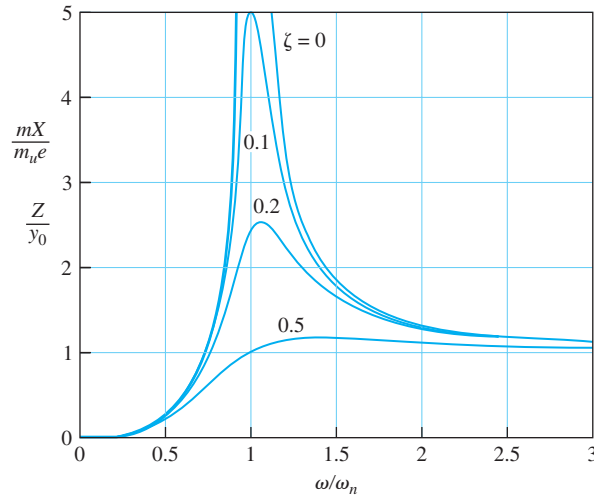


Figure 13.34 Plot of magnification factor versus frequency ratio.

motion of the vibrating mass is

$$x = \frac{m_u e \omega^2 \cos(\omega t - \phi)}{\sqrt{(k - m\omega^2)^2 + c^2 \omega^2}}. \quad (13.55)$$

If we make the same transformation we used to obtain Eq. (13.52), we get

$$\frac{mX}{m_u e} = \frac{\omega^2 / \omega_n^2}{\sqrt{[1 - (\omega^2 / \omega_n^2)]^2 + (2\zeta \omega / \omega_n)^2}}. \quad (13.56)$$

This is plotted in Fig. 13.34. In one case, the main mass is excited by a rotating unbalanced mass m_u . The plot gives the nondimensional absolute response of the mass, m . In another case, as shown in Fig. 13.35, the mass is excited by moving the far end of a spring and dashpot, which is connected to the mass, with harmonic motion. The plot then gives the nondimensional response Z/y_0 relative to the end of the spring-dashpot combination. This plot shows that for high speeds where the frequency ratio is greater than unity, we can reduce the amplitude only by reducing the mass and eccentricity of the rotating unbalance.

13.14 RELATIVE MOTION

In many cases—in vibration-measuring instruments, for example—it is useful to know the response of a system relative to another moving system. Thus, in Fig. 13.35, we might wish to learn the relative response, z , instead of the absolute response, x .

The differential equation for this system is

$$m\ddot{x} + c\dot{x} + kx = ky + c\dot{y}. \quad (13.57)$$

Note that since $\dot{z} = \dot{x} - \dot{y}$ and $\ddot{z} = \ddot{x} - \ddot{y}$, then, written in terms of z and y , Eq. (13.57) becomes

$$m\ddot{z} + c\dot{z} + kz = -m\ddot{y}$$

or

$$m\ddot{z} + c\dot{z} + kz = m\omega^2 y_0 \cos \omega t. \quad (13.58)$$

So the solution is the same as that of Eq. (13.54), and we can write

$$\frac{Z}{y_0} = \frac{\omega^2 / \omega_n^2}{\sqrt{[1 - (\omega^2 / \omega_n^2)]^2 + (2\zeta \omega / \omega_n)^2}}. \quad (13.59)$$

The response chart of Fig. 13.34 also applies to the relative response of such a system.

13.15 ISOLATION

The investigations so far enable the engineer to design mechanical equipment so as to minimize vibration and other dynamic problems, but since machines are inherently vibration generators, in many cases, it is impractical to eliminate all such motion. In such cases, a more practical and economic vibration approach is that of reducing as much as possible the difficulties that the vibratory motion causes. This may take any of several directions, depending upon the nature of the problem. For example, everything economically feasible may have been undertaken to eliminate vibrations in a machine, but the residuals may still be strong and may cause objectionable noise by transmitting these vibrations to the base structure. In another case, a piece of equipment, such as a computer monitor, which is not of itself a vibration generator, may receive objectionable vibrations from another source. Both of these problems can be solved by isolating the equipment from the support. In analyzing these problems, we may be interested, therefore, in the *isolation of forces* or in the *isolation of motions*.

In Fig. 13.36a, the exciting-force amplitude is constant; however, in Fig. 13.36c, it depends upon ω since it is due to a rotating unbalance. In both parts (a) and (c), the support is to be isolated from the vibrating mass, but in part (b), the support is vibrating. In parts (a) and (c) the forces acting against the support must be transmitted through the spring and dashpot, since these are the only connections. The spring and damping forces are phasors that are always at right angles to each other, and so the force transmitted through these connections is

$$F_{tr} = \sqrt{(kX)^2 + (c\omega X)^2} = kX \sqrt{1 + \left(\frac{2\zeta\omega}{\omega_n}\right)^2}. \quad (13.60)$$

The *transmissibility*, T , is a nondimensional ratio that defines the percentage of the exciting force transmitted to the frame. For example, for the system of Fig. 13.36a, X is given by Eq. (13.52), and so the transmissibility is

$$T = \frac{F_{tr}}{F_0} = \frac{\sqrt{1 + (2\zeta\omega/\omega_n)^2}}{\sqrt{[1 - (\omega/\omega_n)^2]^2 + (2\zeta\omega/\omega_n)^2}}. \quad (13.61)$$

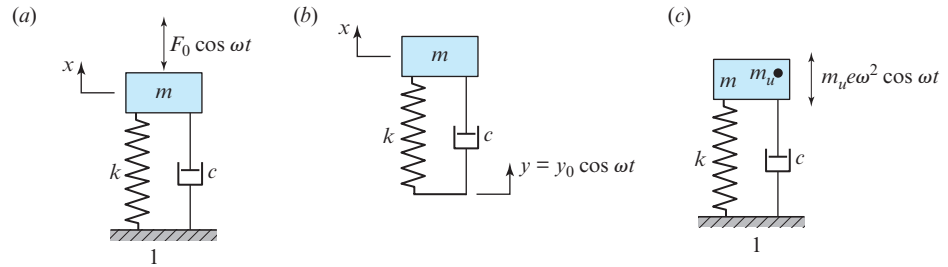


Figure 13.36 (a) Exciting-force amplitude is constant; (b) support is vibrating; (c) exciting force depends upon ω .

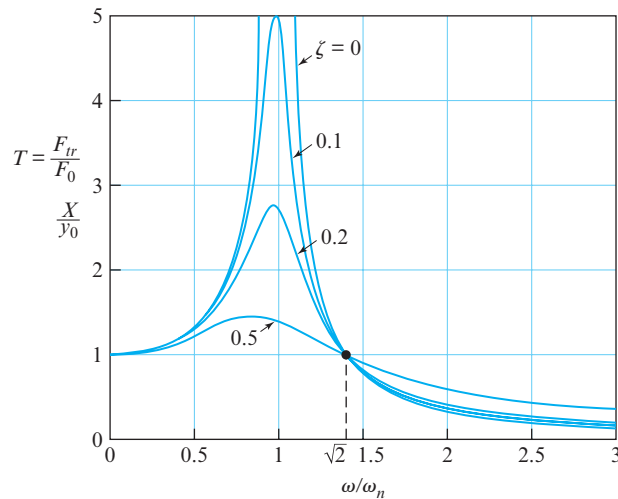


Figure 13.37 Plot of transmissibility versus frequency ratio.

Figure 13.37 is a plot of: (1) force transmissibility versus frequency ratio for a system in which a steady-state periodic forcing function is applied directly to the mass, and (2) amplitude-magnification factor for a system in which the far end of a spring and dashpot are driven by a sinusoidal displacement of amplitude y_0 (shown in Fig. 13.36b). The plots give the non dimensional absolute response of the mass and show that the isolator can be designed so that ω/ω_n is large and the damping is small.

Finally, the transmissibility for the system of Fig. 13.36c, using Eqs. (13.56) and (13.60), is

$$T = \frac{F_{tr}/k}{(m_u/m)e} = \frac{(\omega/\omega_n)^2 \sqrt{1 + (2\zeta\omega/\omega_n)^2}}{\sqrt{[1 - (\omega/\omega_n)^2]^2 + (2\zeta\omega/\omega_n)^2}}. \quad (13.62)$$

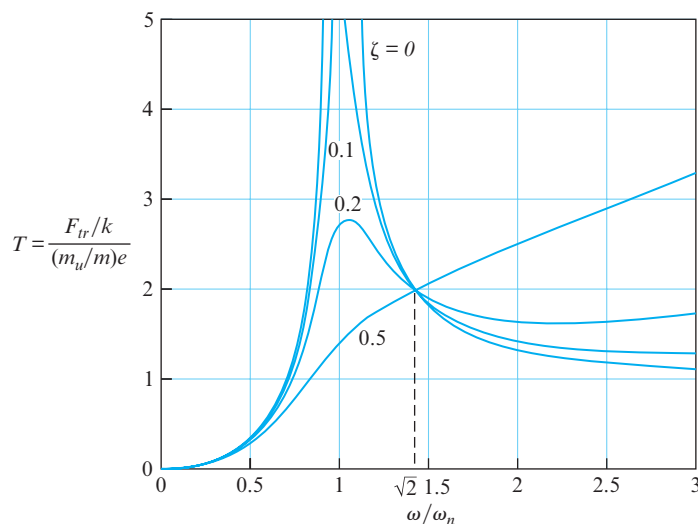


Figure 13.38 Plot of transmissibility versus frequency ratio.

In the plot of Fig. 13.38, for large values of ω/ω_n , the damping actually serves to increase the transmissibility. Therefore, the only practical solution, in general, is to balance the equipment as perfectly as possible to minimize the product $m_u e$.

We choose the complex-operator method for the solution of the system of Fig. 13.36*b*. Note that this system is the same as that of Sec. 13.14, except that now we are interested in the absolute motion of the mass. We begin by defining the forcing function as

$$y = y_0 e^{j\omega t}.$$

Then, the time derivative is

$$\dot{y} = j\omega y_0 e^{j\omega t}.$$

Also, assuming a solution in the form

$$x = X e^{j\omega t}$$

gives velocity and acceleration of the form

$$\dot{x} = j\omega X e^{j\omega t} \quad \text{and} \quad \ddot{x} = -\omega^2 X e^{j\omega t}.$$

Substituting these into Eq. (13.57) and dividing through by the exponential produces

$$-m\omega^2 X + j c \omega X + k X = k y_0 + j c \omega y_0,$$

and so

$$\frac{X}{y_0} = \frac{k + jc\omega}{(k - m\omega^2) + jc\omega}. \quad (a)$$

The product of two complex values is a new quantity whose magnitude is the product of the magnitudes of the original complex values and whose angle is the sum of the angles of the original complex values. Complex values can also be divided; in this case, we divide their magnitudes and subtract their angles.

In this case we are not interested in the phase and so Eq. (a) becomes

$$\frac{X}{y_0} = \frac{\sqrt{k^2 + c^2\omega^2}}{\sqrt{(k - m\omega^2)^2 + c^2\omega^2}}.$$

Finally, dividing both numerator and denominator by m , we obtain

$$\frac{X}{y_0} = \frac{\sqrt{1 + (2\zeta\omega/\omega_n)^2}}{\sqrt{[1 - (\omega/\omega_n)^2]^2 + (2\zeta\omega/\omega_n)^2}}, \quad (13.63)$$

which matches Eq. (13.61) and is plotted in Fig. 13.37.

13.16 RAYLEIGH'S METHOD

For any single-degree-of-freedom system, quantity W/k represents the static deflection y_0 of a spring of stiffness k due to a weight W acting upon it. Rearranging the natural frequency equation by substituting $m = W/g$ gives

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{g}{W/k}} = \sqrt{\frac{g}{y_0}}. \quad (13.64)$$

This equation is, incidentally, very useful for determining natural frequencies of mechanical systems, since static deflection can usually be measured quite easily.

The static deflection of a cantilever beam caused by a concentrated weight W on the end is [4]

$$y_0 = \frac{Wl^3}{3EI},$$

where l is the length of the cantilever, E is its modulus of elasticity, and I is the second moment of area of its cross section. Substituting this value of y_0 into Eq. (13.64) produces

$$\omega_n = \sqrt{\frac{g}{y_0}} = \sqrt{\frac{3EIg}{Wl^3}},$$

so that the natural frequency in hertz is

$$f = \frac{1}{2\pi} \sqrt{\frac{3EIg}{Wl^3}}.$$

In our investigation of the mechanics of vibration we have been concerned with systems whose motions can be described using a single coordinate. In the cases of vibrating beams and rotating shafts, there may be many masses involved or the mass may be distributed. A coupled set of differential equations must be written, with one equation for each mass or each element of mass of the system, and these equations must be solved simultaneously if we are to obtain the equations of motion of a multimass system. Although a number of optional approaches are available, here we shall present an energy method, which is due to Lord Rayleigh [5], because of its importance in the study of vibration.

It is probable that Eq. (13.64) first suggested to Rayleigh the idea of employing the static deflection to find the natural frequency of a system. If we consider a freely vibrating system without damping, then, during its motion, no energy is added to the system, nor is any lost. Yet when the mass has velocity, kinetic energy exists, and when the spring is compressed or extended, potential energy exists. Since no energy is created or destroyed, the maximum kinetic energy of a system must be the same as the maximum potential energy. This is the basis of Rayleigh's method; a mathematical statement of it is

$$E_{\max} = V_{\max}, \quad (13.65)$$

where E denotes the kinetic energy and V denotes the potential energy in the system.

In order to see how it works, let us apply the method to a simple cantilever beam. We begin by assuming that the motion, y , of any point along the beam is harmonic and of the form

$$y = y_0 \sin \omega_n t. \quad (a)$$

The potential energy is maximum when a spring is fully extended or compressed and occurs when $\sin \omega_n t = 1$. That is,

$$V_{\max} = \frac{1}{2} W y_0. \quad (b)$$

The velocity of the weight is given by

$$\dot{y} = y_0 \omega_n \cos \omega_n t. \quad (c)$$

The kinetic energy reaches a maximum when the velocity is maximum, that is, when $\cos \omega_n t = 1$. The maximum kinetic energy is

$$E_{\max} = \frac{1}{2} m \dot{y}_{\max}^2 = \frac{1}{2} \frac{W}{g} (y_0 \omega_n)^2. \quad (d)$$

Applying Eq. (13.65),

$$\frac{1}{2} \frac{W}{g} (y_0 \omega_n)^2 = \frac{1}{2} W y_0 \quad \text{or} \quad \omega_n = \sqrt{\frac{g}{y_0}}, \quad (e)$$

which is identical with Eq. (13.64).

It is true in multimass systems that the dynamic deflection curves are not the same as the static deflection curves. The importance of the method, however, is that any *reasonable* deflection curve can be used in the process. The static deflection curve is a reasonable one, and hence it gives a good approximation. Rayleigh also shows that the correct curve always gives the lowest value for the natural frequency (a lower bound), although we shall not demonstrate this fact here. It is sufficient to know that, if many different deflection curves are assumed, the one giving the lowest natural frequency is the best.

Rayleigh's method is now applied to a multimass system composed of weights W_1, W_2, W_3 , and so on, by assuming, as before, a deflection of each mass according to the equation

$$y_i = y_{0i} \sin \omega_n t. \quad (f)$$

The maximum deflections are therefore y_{01}, y_{02}, y_{03} , and so on, and the maximum velocities are $y_{01} \omega_n, y_{02} \omega_n, y_{03} \omega_n$, and so on. The maximum potential energy for the system is

$$V_{\max} = \frac{1}{2} W_1 y_{01} + \frac{1}{2} W_2 y_{02} + \frac{1}{2} W_3 y_{03} + \cdots = \frac{1}{2} \sum W_i y_{0i}. \quad (g)$$

The maximum kinetic energy for the system is

$$E_{\max} = \frac{W_1}{2g} y_{01}^2 \omega_n^2 + \frac{W_2}{2g} y_{02}^2 \omega_n^2 + \frac{W_3}{2g} y_{03}^2 \omega_n^2 + \cdots = \frac{\omega_n^2}{2g} \sum W_i y_{0i}^2. \quad (h)$$

Applying Rayleigh's principle and solving for the frequency yields

$$\omega_n = \sqrt{\frac{g \sum W_i y_{0i}}{\sum W_i y_{0i}^2}} \quad \text{or} \quad f = \frac{1}{2\pi} \sqrt{\frac{g \sum W_i y_{0i}}{\sum W_i y_{0i}^2}}. \quad (13.66)$$

Equation (13.66), commonly called the Rayleigh-Ritz equation, can be applied to a beam consisting of several masses or to a rotating shaft having masses in the form of gears, pulleys, flywheels, and the like, mounted upon it. If the speed of the rotating shaft should become equal to the natural frequency given by Eq. (13.66), then violent vibrations occur. For this reason, it is common practice to designate this frequency as the *critical speed*.

13.17 FIRST AND SECOND CRITICAL SPEEDS OF A SHAFT

This section presents an approach to obtain exact solutions to the first and second critical speeds of a rotating shaft using the method of *influence coefficients*.

The deflection at location i of a shaft due to a load of unit magnitude at location j is denoted a_{ij} and is referred to as an *influence coefficient*. Note the reciprocal relationship

between the definition of an influence coefficient and the definition of spring stiffness, that is,

$$a_{ij} = \frac{1}{k_{ij}}. \quad (13.67)$$

So, for a shaft of negligible mass supporting two disks with mass (for example, gears, pulleys, or flywheels), the influence coefficients are as follows:

a_{11} = the deflection at location 1 caused by a load of unit magnitude at location 1;

a_{12} = the deflection at location 1 caused by a load of unit magnitude at location 2;

a_{21} = the deflection at location 2 caused by a load of unit magnitude at location 1;

a_{22} = the deflection at location 2 caused by a load of unit magnitude at location 2.

Maxwell's Reciprocity Theorem This theorem states that the deflection at location i caused by a unit load at location j is equal to the deflection at location j caused by a unit load at location i :

$$a_{ij} = a_{ji}. \quad (13.68)$$

The total deflection at location 1 caused by both disks can be written, by superposition, as

$$y_1 = a_{11}W_1 + a_{12}W_2. \quad (13.69)$$

Similarly, the total deflection at location 2 caused by both disks can be written as

$$y_2 = a_{21}W_1 + a_{22}W_2. \quad (13.70)$$

Therefore, for a rotating shaft with two masses, the total deflections at locations 1 and 2 can be written from Eqs. (13.69) and (13.70) as

$$y_1 = a_{11}(m_1y_1\omega^2) + a_{12}(m_2y_2\omega^2)$$

and

$$y_2 = a_{21}(m_1y_1\omega^2) + a_{22}(m_2y_2\omega^2).$$

Rearranging these two equations and writing them in matrix form gives

$$\begin{bmatrix} a_{11}m_1\omega^2 - 1 & a_{12}m_2\omega^2 \\ a_{21}m_1\omega^2 & a_{22}m_2\omega^2 - 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \quad (13.71)$$

A nontrivial solution requires that the determinant of the (2×2) coefficient matrix must be zero, that is

$$(a_{11}m_1\omega^2 - 1)(a_{22}m_2\omega^2 - 1) - (a_{12}m_2\omega^2)(a_{21}m_1\omega^2) = 0,$$

which can be written as

$$m_1 m_2 (a_{11} a_{22}) \omega^4 - (a_{11} m_1 + a_{22} m_2) \omega^2 + 1 - m_1 m_2 (a_{12} a_{21}) \omega^4 = 0.$$

Rearranging this equation and dividing by ω^4 , we have

$$1/\omega^4 - (a_{11} m_1 + a_{22} m_2) 1/\omega^2 + m_1 m_2 (a_{11} a_{22} - a_{12} a_{21}) = 0. \quad (13.72)$$

Equation (13.72) is commonly referred to as the *frequency equation*. If we substitute $Z = 1/\omega^2$, then we have a quadratic equation in Z , that is,

$$Z^2 - (a_{11} m_1 + a_{22} m_2) Z + m_1 m_2 (a_{11} a_{22} - a_{12} a_{21}) = 0. \quad (13.73)$$

Then, the two solutions for Z can be written as

$$Z_1, Z_2 = \left(\frac{a_{11} m_1 + a_{22} m_2}{2} \right) \pm \sqrt{\left(\frac{a_{11} m_1 + a_{22} m_2}{2} \right)^2 - m_1 m_2 (a_{11} a_{22} - a_{12} a_{21})}. \quad (13.74)$$

This equation is commonly referred to as the *exact equation* for the first two critical speeds of the system.

This procedure can easily be extended to a rotating shaft supporting n disks. In this case, Eq. (13.71) contains an $(n \times n)$ matrix, and Eq. (13.73) is an n^{th} -order polynomial in $Z = 1/\omega^2$. The n critical speeds of the system can then be determined from this polynomial.

Dunkerley's Method The Dunkerley equation is an approximation for the first critical speed of a rotating shaft supporting several mass disks. Note that the coefficient of Z in Eq. (13.73) is the sum of the roots of the polynomial; therefore, for two mass disks, this is,

$$\frac{1}{\omega_1^2} + \frac{1}{\omega_2^2} = a_{11} m_1 + a_{22} m_2. \quad (13.75)$$

This is a single equation in two unknowns, ω_1 and ω_2 . To obtain an approximation to the first critical speed, assume that

$$\frac{1}{\omega_1^2} \gg \frac{1}{\omega_2^2}. \quad (13.76)$$

Therefore, Eq. (13.75) can be written as

$$\frac{1}{\omega_1^2} = a_{11} m_1 + a_{22} m_2. \quad (13.77)$$

From Eq. (13.77), the Dunkerley approximation for the first critical speed can be written as

$$\omega_1 = \frac{1}{\sqrt{a_{11} m_1 + a_{22} m_2}}. \quad (13.78)$$

Note that the Dunkerley approximation always underestimates the first critical speed (a lower bound). However, in general, the Rayleigh-Ritz equation [Eq. (13.66)], overestimates the first critical speed (an upper bound).

EXAMPLE 13.5

A rotating steel shaft that is 4 ft long and has a 0.5-in diameter is simply supported in two bearings as shown in Fig. 13.39. Two 80-lb gears are attached to the shaft. One gear is 12 in to the right of the left bearing, and the other is 12 in to the left of the right bearing. The mass of the shaft is negligible compared to the masses of the gears. The influence coefficients are known to be $a_{11} = a_{22} = 5.504 \times 10^{-5}$ in/lb and $a_{12} = a_{21} = 4.278 \times 10^{-5}$ in/lb. Determine the first critical speed of the system using (1) the Rayleigh equation, (2) the exact equation, and (3) the Dunkerley equation.

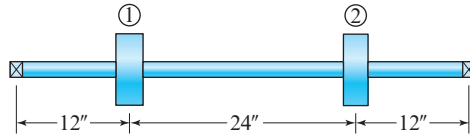


Figure 13.39 Two gears on a rotating shaft.

SOLUTION

- (1) Rayleigh's equation [Eq. (13.66)] can be written as

$$\omega_1^2 = \frac{g(W_1 y_1 + W_2 y_2)}{W_1 y_1^2 + W_2 y_2^2}. \quad (1)$$

Since the two gears have the same weight, $W_1 = W_2$, Eq. (1) can be written

$$\omega_1^2 = \frac{g(y_1 + y_2)}{y_1^2 + y_2^2}. \quad (2)$$

Note from symmetry that the deflection is $y_1 = y_2 = y$; therefore, Eq. (2) can be written

$$\omega_1^2 = \frac{g}{y}. \quad (3)$$

The total deflections at locations 1 and 2 from Eqs. (13.69) and (13.70) are

$$y_1 = a_{11} W_1 + a_{12} W_2, \quad (4)$$

$$y_2 = a_{21} W_1 + a_{22} W_2. \quad (5)$$

Substituting the given data into Eq. (4) or Eq. (5) gives

$$y = y_1 = y_2 = 80 \text{ lb}(a_{11} + a_{12}).$$

Therefore, the deflection is

$$y = 80 \text{ lb}(5.504 + 4.278) \times 10^{-5} \text{ in/lb} = 782.56 \times 10^{-5} \text{ in.} \quad (6)$$

Substituting Eq. (6) into Eq. (3) gives

$$\omega_1^2 = \frac{386 \text{ in/s}^2 \times 10^5}{782.56 \text{ in}} = 49\,325 \text{ rad}^2/\text{s}^2.$$

Therefore, the first critical speed by Rayleigh's method is

$$\omega_1 = 222 \text{ rad/s.} \quad \text{Ans.}$$

- (2) From Eq. (13.74), the exact equation for the first two critical speeds of the system is

$$Z_1, Z_2 = \left(\frac{a_{11}m_1 + a_{22}m_2}{2} \right) \pm \sqrt{\left(\frac{a_{11}m_1 + a_{22}m_2}{2} \right)^2 - m_1m_2(a_{11}a_{22} - a_{12}a_{21})}. \quad (7)$$

Substituting the given data into this equation gives

$$Z_1, Z_2 = 1.141 \times 10^{-5} \text{ s}^2 \pm \sqrt{(1.141^2 - 0.516) \cdot 10^{-10} \text{ s}^2},$$

which can be written as

$$\frac{1}{\omega_1^2}, \frac{1}{\omega_2^2} = (1.141 \pm 0.886) \cdot 10^{-5} \text{ s}^2$$

or as

$$\omega_1^2 = 49\,334 \text{ rad}^2/\text{s}^2 \text{ and } \omega_2^2 = 392\,157 \text{ rad}^2/\text{s}^2. \quad (8)$$

Therefore, the first and second critical speeds from the exact equation are

$$\omega_1 = 222 \text{ rad/s} \quad \text{and} \quad \omega_2 = 626 \text{ rad/s.} \quad \text{Ans.}$$

Note that for this example the answer for the first critical speed from the Rayleigh equation is the same as the answer from the exact equation.

- (3) The Dunkerley equation for the rotating shaft supporting the two mass discs is

$$\frac{1}{\omega_1^2} = a_{11}m_1 + a_{22}m_2. \quad (9)$$

Substituting the given data into this equation gives

$$\frac{1}{\omega_1^2} = 2 \left(\frac{80 \text{ lb}}{386 \text{ in/s}^2} \right) 5.504 \cdot 10^{-5} \text{ in/lb} = 2.281 \cdot 10^{-5} \text{ s}^2. \quad (10)$$

Therefore, the first critical speed for the system is

$$\omega_1 = 209 \text{ rad/s.}$$

Ans.

Note that this answer is less than the first critical speed obtained from the exact equation.

EXAMPLE 13.6

The rotating shaft shown in Fig. 13.40 supports two flywheels, each of mass m . The first critical speed of the system has been determined experimentally to be 300 rad/s and the first critical speed of the shaft alone is 600 rad/s. If one of the flywheels is removed from the shaft, then determine the first critical speed of the system.

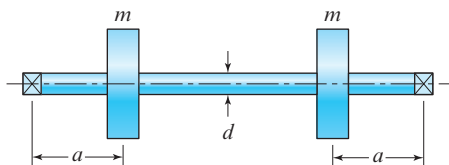


Figure 13.40 Two flywheels on a rotating shaft.

SOLUTION

The first natural frequency of the shaft carrying the two flywheels can be written as

$$\frac{1}{\omega_1^2} = \frac{1}{\omega_{11}^2} + \frac{1}{\omega_{22}^2} + \frac{1}{\omega_{ss}^2} \quad (1a)$$

or as

$$\frac{1}{\omega_1^2} = ma_{11} + ma_{22} + \frac{1}{\omega_{ss}^2}, \quad (1b)$$

where ω_{ss} is the critical speed of the shaft alone.

From symmetry, we note that the influence coefficient $a_{11} = a_{22}$; therefore, Eq. (1b) can be written as

$$2ma_{11} = \frac{1}{\omega_1^2} - \frac{1}{\omega_{ss}^2}. \quad (2)$$

Substituting the given critical speeds into Eq. (2) gives

$$2ma_{11} = \frac{1}{(300 \text{ rad/s})^2} - \frac{1}{(600 \text{ rad/s})^2} = \frac{1}{120\,000} \text{ s}^2. \quad (3)$$

For the massless shaft with only one flywheel, the first natural frequency can be written from Eq. (1a) as

$$\frac{1}{\omega_1^2} = \frac{1}{\omega_{11}^2} + \frac{1}{\omega_{ss}^2} \quad (4)$$

or as

$$\frac{1}{\omega_1^2} = ma_{11} + \frac{1}{\omega_{ss}^2}. \quad (5)$$

Substituting Eq. (3) into Eq. (5) gives

$$\frac{1}{\omega_1^2} = \frac{1}{2} \left[\frac{1}{120\,000} \text{ s}^2 \right] + \frac{1}{\omega_{ss}^2}. \quad (6)$$

Finally, substituting the critical speed of the shaft $\omega_{ss} = 600 \text{ rad/s}$ into Eq. (6) gives

$$\frac{1}{\omega_1^2} = \frac{1}{2} \left[\frac{1}{120\,000} \text{ s}^2 \right] + \frac{1}{(600 \text{ rad/s})^2} = 6.944 \cdot 10^{-6} \text{ s}^2. \quad (7)$$

Therefore, the first critical speed of the system is

$$\omega_1 = 379.5 \text{ rad/s.} \quad \text{Ans.}$$

In the investigation of rectilinear vibrations to follow, we will see that there are as many natural frequencies in a multimass vibrating system as there are degrees of freedom. Therefore, a three-mass torsional system has three degrees of freedom and, consequently, three natural frequencies. Equation (13.66) gives only the first or lowest of these frequencies.

It is probable that the critical speed can be determined by Eq. (13.66), for most cases, within about 5%. This is because of the assumption that the static and dynamic deflection curves are identical. Bearings, couplings, belts, and so on, all have an effect upon the spring constant and the damping. Shaft vibration usually occurs over an appreciable range of shaft speed, and for this reason, Eq. (13.66) is accurate enough for many engineering purposes.

As a general rule, shafts have many discontinuities—such as shoulders, keyways, holes, and grooves—to locate and secure various gears, pulleys, and other shaft-mounted masses. These diametral changes usually require deflection analysis using numeric integration. For such problems, Simpson's rule is easy to apply using a computer, a calculator, or by hand calculations [3].

13.18 TORSIONAL SYSTEMS

Figure 13.41 shows a shaft supported on bearings at *A* and *B* with two masses connected at the ends. The masses represent rotating machine parts—an engine and its flywheel, for example. We wish to study the possibility of free vibration of the system when it rotates

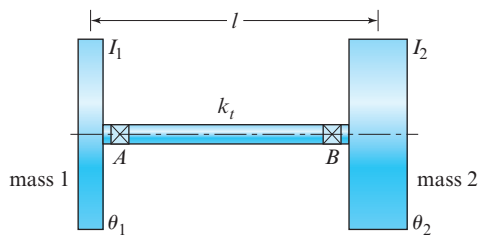


Figure 13.41

at constant angular velocity. To investigate the motion of each mass, it is necessary to picture a reference system fixed to the shaft and rotating with the shaft at the same angular velocity. Then, we can measure the angular displacement of either mass by finding the instantaneous angular location of a mark on the mass relative to one of the rotating axes. Thus, we define θ_1 and θ_2 as the angular displacements of mass 1 and mass 2, respectively, with respect to the rotating axes.

Now, assuming no damping, Eq. (13.6) is written for each mass,

$$\begin{aligned} I_1 \ddot{\theta}_1 + k_t (\theta_1 - \theta_2) &= 0, \\ I_2 \ddot{\theta}_2 + k_t (\theta_2 - \theta_1) &= 0, \end{aligned} \quad (a)$$

where the angle $(\theta_1 - \theta_2)$ represents the total twist of the shaft. A solution is assumed in the form

$$\theta_i = \gamma_i \sin \omega_n t.$$

Then, the acceleration is

$$\ddot{\theta}_i = -\gamma_i \omega_n^2 \sin \omega_n t.$$

If these equations are substituted into Eqs. (a), we obtain

$$\begin{aligned} (k_t - I_1 \omega_n^2) \gamma_1 - k_t \gamma_2 &= 0, \\ -k_t \gamma_1 + (k_t - I_2 \omega_n^2) \gamma_2 &= 0, \end{aligned} \quad (b)$$

which can be written in matrix form as

$$\begin{bmatrix} k_t - I_1 \omega_n^2 & -k_t \\ -k_t & k_t - I_2 \omega_n^2 \end{bmatrix} \begin{bmatrix} \gamma_1 \\ \gamma_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

For a nontrivial solution, the determinant of the coefficient matrix must be zero; that is

$$(k_t - I_1 \omega_n^2)(k_t - I_2 \omega_n^2) - k_t^2 = 0,$$

so that

$$\omega_n^2 \left(\omega_n^2 - k_t \frac{I_1 + I_2}{I_1 I_2} \right) = 0$$

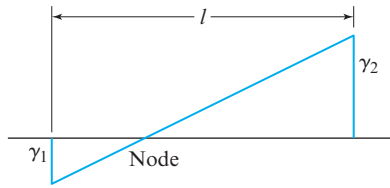


Figure 13.42

and

$$\omega_{n1} = 0 \text{ and } \omega_{n2} = \sqrt{k_t \frac{I_1 + I_2}{I_1 I_2}}. \quad (13.79)$$

Substituting $\omega_{n1} = 0$ into Eqs. (b), we find that $\gamma_1 = \gamma_2$. Therefore, for this case, the masses rotate together without any relative displacement and there is no vibration.

Substituting the value of ω_{n2} into Eqs. (b), it is determined that

$$\gamma_2 = -\frac{I_1}{I_2} \gamma_1,$$

indicating that the motion of the second mass is opposite to that of the first and proportional to I_1/I_2 . Figure 13.42 is a graphic representation of the vibration. Here, l is the distance between the two masses, and γ_1 and γ_2 are the instantaneous angular displacements plotted to scale. If a line is drawn connecting the ends of the angular displacements, this line crosses the axis at a *node*, which is a location on the shaft having zero angular displacement. It is convenient to designate the configuration of the vibrating system as a *mode* of vibration. This system has two modes of vibration, corresponding to the two frequencies, although we have seen that one of them is degenerate. A system of n masses would have n modes corresponding to n different frequencies.

For multimass torsional systems, the *Holzer tabulation method* [2] can be used to find all the natural frequencies. It is easy to use and avoids solving the simultaneous differential equations.

13.19 REFERENCES

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PROBLEMS

- †13.1** Derive the differential equation of motion for each system, and write the formula for the natural frequency ω_n for each system.

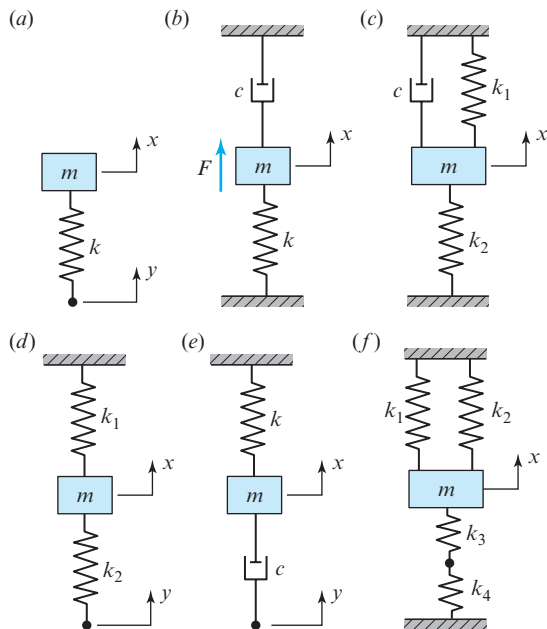


Figure P13.1

- †13.2** Evaluate the constants of integration of the solution to the differential equation for an undamped free system, using the following sets of starting conditions: (a) $x = x_0$, $\dot{x} = 0$; (b) $x = 0$, $\dot{x} = v_0$; (c) $x = x_0$, $\ddot{x} = a_0$; and (d) $x = x_0$, $x = b_0$. For each case, transform the solution to a form containing a single trigonometric term.

- †13.3** A system like Fig. 13.5 has $m = 1$ kg and an equation of motion $x = 20\cos(8\pi t - \pi/4)$ mm. Determine: (a) spring constant k ; (b) static deflection δ_{st} ; (c) period; (d) frequency in hertz; and (e) velocity, acceleration, and spring force at $t = 0.20$ s. Plot a phase diagram to scale showing the displacement, velocity, acceleration, and spring-force phasors at this instant.

- †13.4** Weight W_1 drops through distance h and collides with W_2 plastic impact (a coefficient of restitution of zero). Derive the differential equation

of motion of the system, and determine the amplitude of the resulting motion of W_2 .

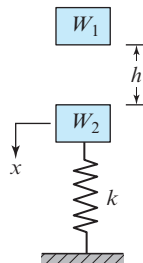


Figure P13.4

- 13.5** The vibrating system has $k_1 = k_3 = 875$ N/m, $k_2 = 1\,750$ N/m, and $W = 40$ N. What is the natural frequency in hertz?

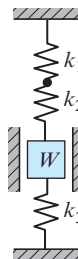
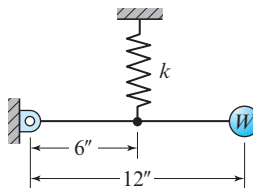


Figure P13.5

- 13.6** Weight $W = 15$ lb is connected to a pivoted rod that is assumed to be weightless but rigid. A spring having a rate of $k = 60$ lb/in is connected to the center of the rod and holds the system in static equilibrium at the posture shown. Assuming that the rod can vibrate with a small amplitude, determine the period of the motion.

Figure P13.6 $\ell = 12$ in.

- 13.7** The upside-down pendulum of length l is retained by two springs connected a distance a from the pivot. The springs have been positioned such that the pendulum is in static equilibrium when it is in the vertical posture. (a) For small amplitudes, find the natural frequency of this system. (b) Find the ratio l/a at which the system becomes unstable.

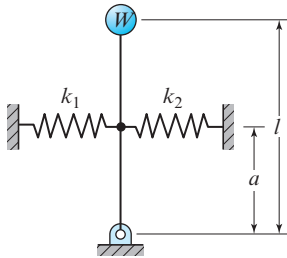


Figure P13.7

- 13.8** Write the differential equation for the system, and find the natural frequency. Find the response x if y is a step input of height y_0 . Find the relative response $z = x - y$ to this step input.

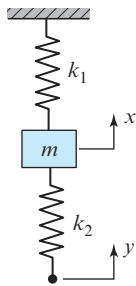


Figure P13.8

- 13.9** An undamped vibrating system consists of a spring whose scale is 35 kN/m and a mass of 1.2 kg. A step force $F = 50$ N is exerted on the mass for 0.040 s. (a) Write the equations of motion of the system for the era in which the force acts and for the era that follows. (b) What are the amplitudes in each era? (c) Sketch a time plot of the displacement.
- 13.10** A round shaft whose torsional spring constant is k_t in $\cdot \text{lb}/\text{rad}$ connects two wheels having mass moments of inertia I_1 and I_2 . Show that the system is likely to vibrate torsionally with a frequency of

$$\omega_n = \sqrt{\frac{k_t(I_1 + I_2)}{I_1 I_2}}.$$

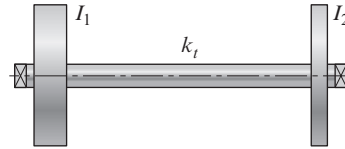


Figure P13.10

- 13.11** A motor is connected to a flywheel by a 5/8-in diameter steel shaft 36 in long. Using the methods of this chapter, it can be demonstrated that the torsional spring rate of the shaft is 4 700 in $\cdot \text{lb}/\text{rad}$. The mass moments of inertia of the motor and flywheel are 24.0 and 56.0 in $\cdot \text{lb} \cdot \text{s}^2$, respectively. The motor is turned on for 2 s, and during this period, it exerts a constant torque of 200 in $\cdot \text{lb}$ on the shaft. (a) What speed in revolutions per minute does the shaft attain? (b) What is the natural frequency of vibration of the system? (c) Assuming no damping, what is the amplitude of the vibration of the system in degrees during the first era? During the second era?

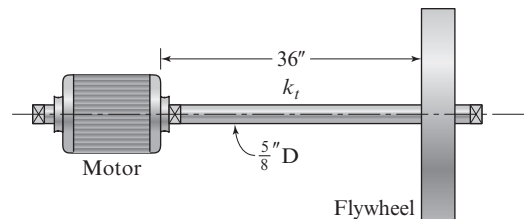


Figure P13.11

- 13.12** The weight of the mass of a vibrating system is 10 lb, and it has a natural frequency of 1 Hz. Using the phase-plane method, plot the response of the system to the given force function. What is the final amplitude of the motion?

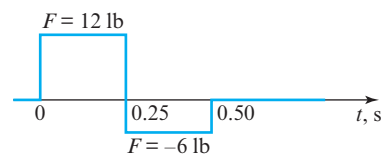


Figure P13.12

- 13.13** An undamped vibrating system has a spring scale of 200 lb/in and a weight of 50 lb. Find the response and the final amplitude of vibration of the system if it is acted upon by the given forcing function. Use the phase-plane method.

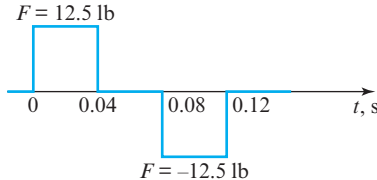


Figure P13.13

- 13.14** A vibrating system has a spring rate of $k = 400$ lb/in and a weight of $W = 80$ lb. Plot the response of this system to the given forcing function using: (a) three steps, and (b) six steps.

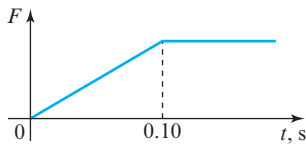


Figure P13.14 $F_{\max} = 200$ lb.

- 13.15** What is the value of the coefficient of critical damping for a spring-mass-damper system in which $k = 56$ kN/m and $m = 40$ kg? If the actual damping is 20% of critical, what is the natural frequency of the system? What is the period of the damped system? What is the value of the logarithmic decrement?
- 13.16** A vibrating system has a spring scale of $k = 3.5$ kN/m and a mass $m = 15$ kg. When disturbed, it is observed that the amplitude decays to one-fourth of its original value in 4.80 s. Find the damping coefficient and the damping factor.
- 13.17** A vibrating system has $k = 300$ lb/in, $W = 90$ lb, and damping equal to 20% of critical. (a) What is the damped natural frequency ω_d of the system? (b) What are the period and the logarithmic decrement?
- 13.18** Solve Problem 13.14 using damping equal to 15% of critical.
- 13.19** A damped vibrating system has an undamped natural frequency of 10 Hz and a weight of 800 lb. The damping ratio is 0.15. Using the phase-plane

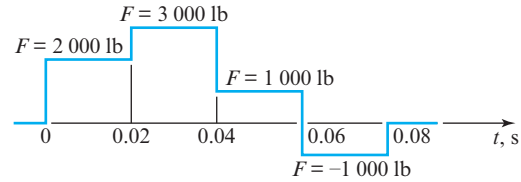


Figure P13.19

method, determine the response of the system to the given forcing function.

- 13.20** A vibrating system has a spring rate of 3 000 lb/in, a damping factor of 55 lb · s/in, and a weight of 800 lb. It is excited by a harmonically varying force $F_0 = 100$ lb at a frequency of 435 cycles per minute. (a) Calculate the amplitude of the forced vibration and the phase angle between the vibration and the force. (b) Plot several cycles of the displacement-time and force-time diagrams.
- 13.21** A spring-mounted mass has $k = 525$ kN/m, $c = 9\,640$ N · s/m, and $m = 360$ kg. This system is excited by a force having an amplitude of 450 N at a frequency of 4.80 Hz. Find the amplitude and phase angle of the resulting vibration and plot several cycles of the force-time and displacement-time diagrams.
- 13.22** When a 6 000-lb press is mounted upon structural-steel floor beams, it causes them to deflect 0.75 in. If the press has a reciprocating unbalance of 420 lb and it operates at a speed of 80 rev/min, how much of the force will be transmitted from the floor beams to other parts of the building? Assume no damping. Can this mounting be improved?
- 13.23** Four vibration mounts are used to support a 450-kg machine that has a rotating unbalance of 0.35 kg · m and runs at 300 rev/min. The vibration mounts have damping equal to 30% of critical. What must the spring constant of the mounting be if 20% of the exciting force is transmitted to the foundation? What is the resulting amplitude of motion of the machine?
- 13.24** A 600-mm-long steel shaft is simply supported by two bearings at A and C. Flywheels 1 and 2 are attached to the shaft at locations B and D, respectively. Flywheel 1 at location B weighs 50 N, flywheel 2 at location D weighs 20 N, and the weight of the shaft can be neglected. The known

stiffness coefficients are $k_{11} = 25\,000\text{ N/m}$, $k_{12} = 50\,000\text{ N/m}$, and $k_{22} = 40\,000\text{ N/m}$. Determine (a) the first and second critical speeds of the shaft using the exact solution, and (b) the first critical speed using the Dunkerley, and (c) Rayleigh-Ritz approximations. (d) If flywheel 2 is then placed at location B and flywheel 1 is placed at location D , determine the first critical speed of the new system using the Dunkerley approximation.

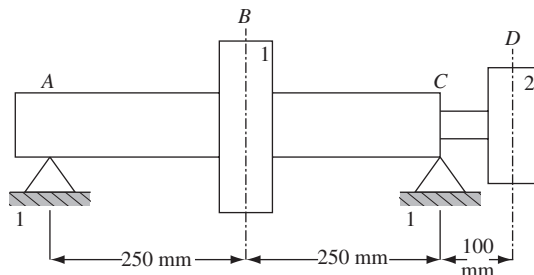


Figure P13.24 Steel shaft simply supported by two rolling element bearings.

- 13.25** The first critical speeds of a rotating shaft with two mass disks, obtained from three different mathematical techniques, are 110 rad/s, 112 rad/s, and 100 rad/s, respectively. (a) Which values correspond to the first critical speed of the shaft from the exact solution, the Dunkerley approximation, and the Rayleigh-Ritz approximation? (b) If the influence coefficients are $a_{11} = a_{22} = 10^{-4}\text{ m/N}$ and the masses of the two disks are the same, that is, $m_1 = m_2 = m$, then use the Dunkerley approximation to calculate the mass m . (c) If the influence coefficients are $a_{11} = a_{22} = 10^{-4}\text{ m/N}$ and the masses of the two disks are specified as $m_1 = m_2 = m = 0.5\text{ kg}$, use the Rayleigh-Ritz approximation to calculate the influence coefficient a_{12} .

- 13.26** A steel shaft is simply supported by two rolling element bearings at A and B . The length of the shaft is 1.45 m, and two flywheels with weight 300 N are attached to the shaft at locations shown. One flywheel is 0.35 m to the right of the left bearing at A , and the other flywheel is 0.35 m to the left of the right bearing at B . The weight of the shaft can be neglected. The influence coefficients are specified as $a_{11} = 126 \times 10^{-5}\text{ mm/N}$ and $a_{21} = 92.5 \times 10^{-5}\text{ mm/N}$. (a) Determine the first and second critical speeds of the shaft using the exact solution. Determine the first critical speed

of the shaft using: (b) the Dunkerley approximation, and (c) the Rayleigh-Ritz equation.

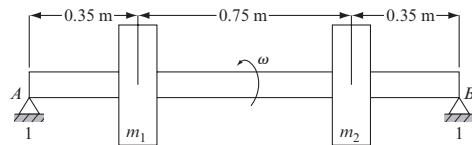


Figure P13.26 Steel shaft simply supported by two rolling element bearings.

- 13.27** A steel shaft is simply supported by two rolling element bearings at A and C . The length of the shaft is 0.6 m, and two flywheels are attached to the shaft at the locations B and D as shown. The flywheel at location B weighs 200 N, and the flywheel at location D weighs 90 N. The weight of the shaft can be neglected. It was determined that with flywheel 1 alone, the first critical speed of the shaft is 800 rad/s, and with flywheel 2 alone, the first critical speed of the shaft is 1 200 rad/s. (a) Determine the first critical speed for the two-mass system. (b) If the two flywheels are interchanged (that is, flywheel 2 is placed at location B and flywheel 1 is placed at location D), determine the first critical speed of the new system using the Dunkerley approximation.

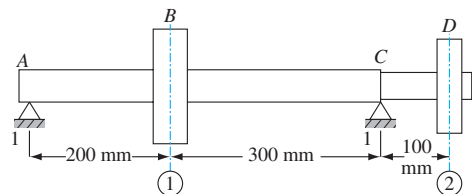


Figure P13.27 Steel shaft simply supported by two rolling element bearings.

- 13.28** A steel shaft, which is 50 inches in length, is simply supported by two bearings at B and D . Flywheels 1 and 2 are attached to the shaft at A and C , respectively. The flywheel at location A weighs 15 lbs, the flywheel at location C weighs 30 lbs, and the weight of the shaft can be neglected. The stiffness coefficients are specified as $k_{11} = 2.5 \times 10^4\text{ lb/in}$ and $k_{22} = 4.0 \times 10^4\text{ lb/in}$. (a) Determine the first critical speed for the two-mass system using the Dunkerley approximation. (b) If the two flywheels are interchanged (that is, flywheel 2 is placed at location A and flywheel 1 is placed at location C), determine the first critical speed of the new system.

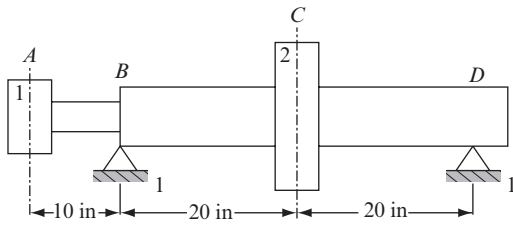


Figure P13.28 Steel shaft simply supported by two rolling element bearings.

- 13.29** The weights of two gears rigidly attached to a shaft at two different locations, denoted as 1 and 2, are $W_1 = 200$ N and $W_2 = 350$ N, respectively. The shaft is rotating counterclockwise with a constant operating speed $\omega = 100$ rad/s. From a deflection analysis of the shaft, the known influence coefficients are $a_{11} = 35 \times 10^{-6}$ mm/N, $a_{12} = 50 \times 10^{-6}$ mm/N, and $a_{22} = 90 \times 10^{-6}$ mm/N. Neglecting gravity and the mass of the shaft, determine the first and second critical speeds of the shaft using the exact equation. Is the operating speed of the shaft acceptable? Determine the first critical speed of the shaft using: (a) the Rayleigh-Ritz method; and (b) the Dunkerley approximation.
- 13.30** The weights of two masses m_1 and m_2 that are rigidly attached to the rotating shaft are 31.5 lb and 13.5 lb, respectively. The shaft is rotating counterclockwise with a constant angular velocity $\omega = 100$ rad/s. From a deflection analysis, the influence coefficients for the shaft are $a_{11} =$

3.56×10^{-6} in/lb, $a_{22} = 21.36 \times 10^{-6}$ in/lb, and $a_{12} = a_{21} = 7.12 \times 10^{-6}$ in/lb. Neglecting the mass of the shaft, determine the first and second critical speeds of the shaft using the exact equation. Is the operating speed of the shaft acceptable? Determine the first critical speed of the shaft using the Rayleigh-Ritz equation and the Dunkerley approximation. Determine the first critical speed of the shaft if mass m_1 is moved to location 2 and mass m_2 is moved to location 1.

- 13.31** The first and second critical speeds of a rotating shaft with two flywheels rigidly attached are $\omega_1 = 375$ rad/s and $\omega_2 = 615$ rad/s. The weights of the flywheels are $W_1 = 65$ N and $W_2 = 80$ N and the known influence coefficients of the shaft are $a_{11} = 5.90 \times 10^{-4}$ mm/N and $a_{21} = 2.74 \times 10^{-4}$ mm/N. Determine the influence coefficient, a_{22} .
- 13.32** A shaft rotating with a constant angular velocity is simply supported at A and B. Gears C and D are rigidly attached to the shaft at locations 1 and 2, respectively. The weight of gear C at location 1 is 13 N, and the weight of gear D at location 2 is 23 N. The known influence coefficients of the shaft are $a_{11} = 6.8 \times 10^{-6}$ m/N, $a_{12} = 5.3 \times 10^{-6}$ m/N, and $a_{22} = 7.9 \times 10^{-6}$ m/N. Determine the first critical speed of the shaft using: (a) the Dunkerley approximation, and (b) the Rayleigh-Ritz method. If gear C is moved to location 2 and gear D is moved to location 1, then calculate the new first critical speed using: (c) the Dunkerley approximation, and (d) the Rayleigh-Ritz method.

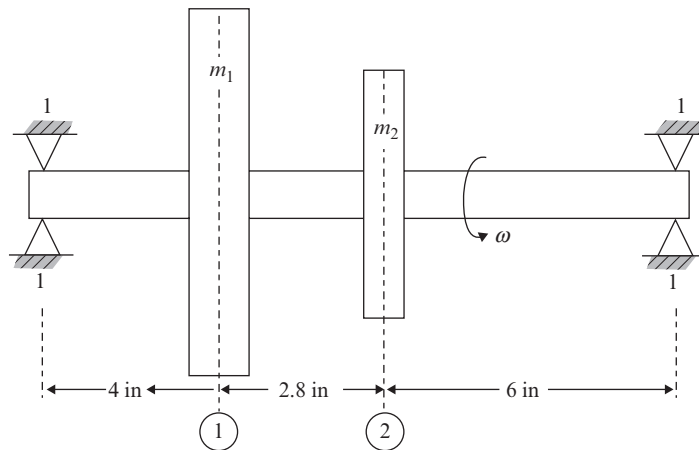


Figure P13.30 Rotating shaft with two mass disks.

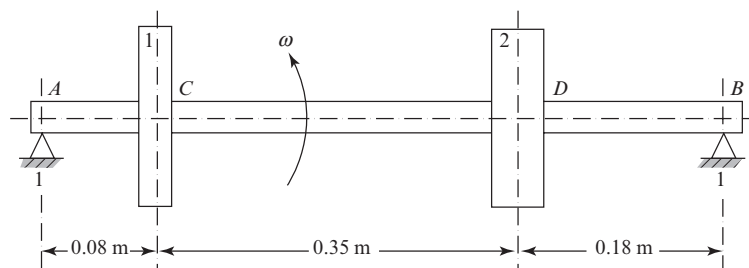


Figure P13.32 Rotating shaft with two gears.

13.33 A shaft is simply supported at A and B and is rotating with constant angular velocity ω . Two identical flywheels C and D of unknown mass are rigidly attached to the shaft (at locations 1 and 2). The known influence coefficients are $a_{11} = 41.1 \times 10^{-6}$ in/lb, $a_{12} = 29.4 \times 10^{-6}$ in/lb, and $a_{22} = 53.3 \times 10^{-6}$ in/lb. The first critical speed of the shaft, obtained from the exact equation, is 135 rad/s. Determine: (a) the masses of the two flywheels, and (b) the second critical speed of the shaft.

13.34 The 600-mm-long shaft is simply supported by the bearings at A and D. Flywheel 1 at location B weighs 30 N, and flywheel 2 at location C weighs 15 N. The weight of the shaft can be neglected. The stiffness coefficients of the shaft are $k_{11} = 1.0 \times 10^6$ N/m and $k_{22} = 5.0 \times 10^6$ N/m. (a) Using the Dunkerley approximation, determine the first critical speed of the shaft. (b) If the flywheels are interchanged (that is, if flywheel 2 is at location B, and flywheel 1 is at location C), then use the Dunkerley approximation to determine the first critical speed of the new system.

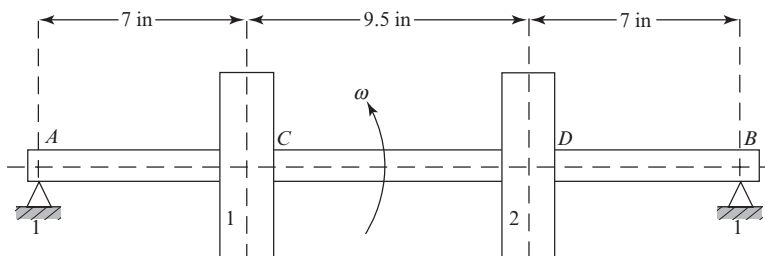


Figure P13.33 Rotating shaft with two identical flywheels.

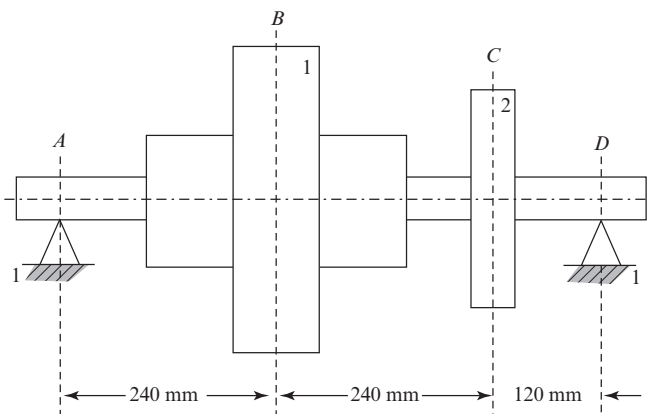


Figure P13.34 Shaft simply supported at A and D.

13.35 Three identical flywheels, each weighing 19 lb, are rigidly attached to a rotating shaft. The known influence coefficients of the shaft are: $a_{11} = 2.25 \times 10^{-4}$ in/lb, $a_{22} = 10.75 \times 10^{-4}$ in/lb, $a_{33} = 6.25 \times 10^{-4}$ in/lb, $a_{12} = 0.80 \times 10^{-4}$ in/lb, $a_{13} = 0.15 \times 10^{-4}$ in/lb, and $a_{23} = 1.0 \times 10^{-4}$ in/lb. Use the Rayleigh-Ritz method to determine the first critical speed of the shaft.

13.36 The first and second critical speeds of a rotating shaft supporting two identical gears, each with a mass of 6 kg, are $\omega_1 = 500$ rad/s and $\omega_2 = 1\,120$ rad/s, respectively. If the influence coefficient $a_{11} = 2a_{22}$, then determine the numeric values of the influence coefficients of the shaft.

13.37 The shaft is simply supported by the bearings at A and D. Flywheel 1 at location B weighs 5.6 lb, flywheel 2 at location C weighs 17.9 lb, and the weight of the shaft can be neglected. The influence coefficients of the shaft are $a_{11} = 1.8 \times 10^{-8}$ in/lb, $a_{12} = 0.7 \times 10^{-8}$ in/lb, and $a_{22} = 9.8 \times 10^{-8}$ in/lb. Determine the first critical speed of the shaft using: (a) the Dunkerley approximation, and (b) the Rayleigh-Ritz approximation.

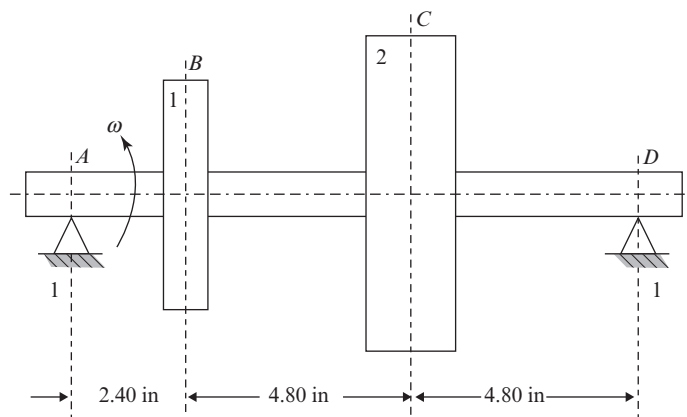


Figure P13.37 Shaft simply supported at A and D.

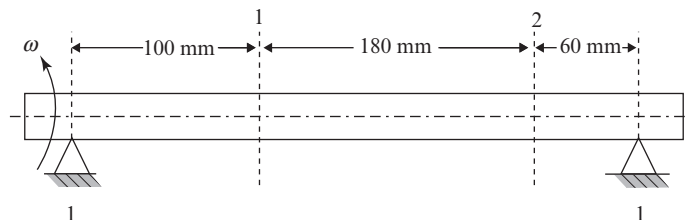


Figure P13.38 Simply supported shaft.

13.38 A shaft with negligible mass is simply supported by two bearings. When a gear with a mass of 7 kg is attached to the shaft at location 1, the first critical speed is measured as 1 200 rad/s. After a second identical gear is attached to the shaft at location 2, the first and second critical speeds of the shaft are measured as 500 rad/s and 1 800 rad/s, respectively. Using the exact solution to the first and second critical speeds of a rotating shaft, determine the four influence coefficients of this shaft.

13.39 The 72-in-long shaft is simply supported by the bearings at B and D. Flywheel 1 at location A weighs 9 lb, and flywheel 2 at location C weighs 16 lb. The weight of the shaft can be neglected. The stiffness coefficients of the shaft are $k_{11} = 5.6 \times 10^3$ lb/in, $k_{12} = 33.6 \times 10^3$ lb/in, and $k_{22} = 22.4 \times 10^3$ lb/in. Determine the first and second critical speeds of the shaft using the exact equation. Then, determine the first critical speed of the shaft using: (a) the Rayleigh-Ritz method, and (b) the Dunkerley approximation.

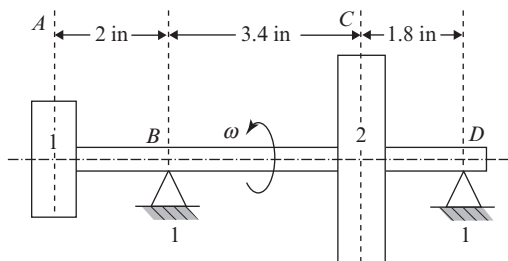


Figure P13.39 Shaft simply supported at B and D.

- 13.40** The steel shaft is simply supported by two bearings at A and D. Two gears are rigidly attached to the shaft at locations B and C. Gear 1 at location B weighs 150 N, gear 2 at location C weighs 90 N, and the weight of the shaft can be neglected. The stiffness coefficients are $k_{11} = 2.5 \times 10^4$ N/m, $k_{12} = 5.0 \times 10^4$ N/m, and $k_{22} = 4.0 \times 10^4$ N/m. Determine the first critical speed of the shaft using: (a) the Dunkerley approximation, and (b) the Rayleigh-Ritz method.

- 13.41** The first critical speeds of a rotating shaft with two mass disks, obtained from three different mathematical techniques, are 150 rad/s, 152 rad/s, and 153 rad/s, respectively. The influence coefficients $a_{11} = a_{22} = 3 \times 10^{-4}$ m/N. (a) Specify which value corresponds to the first critical speed from the exact solution, the Rayleigh-Ritz method, and the Dunkerley approximation. (b) If the masses of the two disks are identical, then use the Dunkerley approximation to calculate the mass of each disk. (c) If the masses of the two disks are specified as $m_1 = m_2 = 0.10$ kg, then use the Rayleigh-Ritz method to calculate the influence coefficient a_{12} .

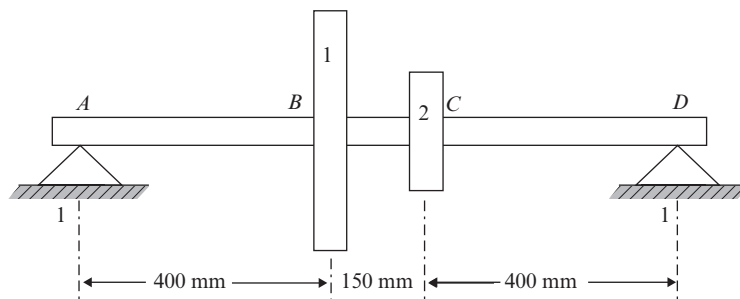
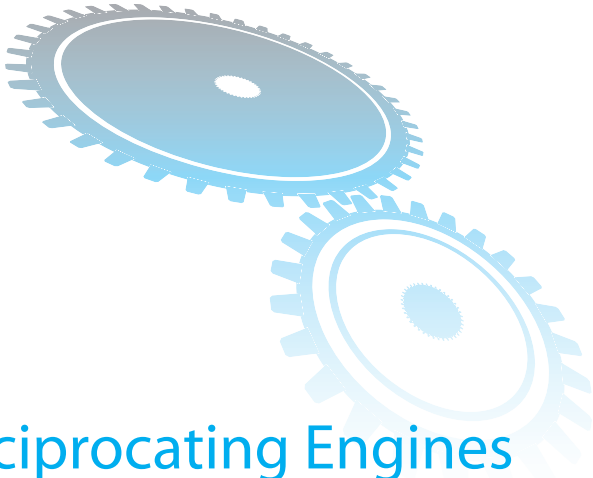


Figure P13.40 Shaft simply supported at A and D.



14 Dynamics of Reciprocating Engines

The purpose of this chapter is to apply fundamentals—kinematic and dynamic analysis—in a complete investigation of a particular category of machines. The reciprocating engine has been selected for this purpose, since it has reached a high state of development and is of more general interest than most other machines. For our purposes, however, another type of machine involving interesting dynamic situations would serve just as well. The primary objective of the chapter is to demonstrate methods of applying fundamentals to the dynamic analysis of machines.

14.1 ENGINE TYPES

The descriptions and characteristics of all the engine types that have been conceived and constructed over the years would fill many books. Here, our purpose is to describe very briefly a few of the engine types that are currently in general use. This exposition is not intended to be complete. Furthermore, since you are anticipated to be mechanically inclined and generally familiar with internal combustion engines, the primary purpose of this section is merely to record things that you already know and to furnish nomenclature for the remainder of the chapter.

In this chapter, we classify engines according to their intended application, the combustion cycle used, and the number and arrangement of cylinders. Thus, we refer to aircraft engines, automotive engines, marine engines, and stationary engines, for example, all so named because of the applications for which they are designed. Similarly, one might have in mind an engine designed on the basis of the *Otto cycle*, in which the fuel and air are mixed before compression and in which combustion takes place with no excess air, or on the basis of the *Diesel cycle*, in which the fuel is injected near the end of compression and combustion takes place with a good deal of excess air. The Otto-cycle engine uses

quite volatile fuels and ignition is by spark, but the Diesel-cycle engine operates on fuels of lower volatility and ignition occurs because of compression.

Both Diesel- and Otto-cycle engines may use either a *two-stroke cycle* or a *four-stroke cycle*, depending upon the number of piston strokes comprising the complete combustion cycle. In the four-stroke cycle, the exhaust ports open near the end of each power stroke to permit exhaust gases to flow out during the exhaust stroke. Next, the exhaust ports close and inlet ports open and permit entry of a fuel-air mixture during the suction stroke. Then, all ports are closed and the piston compresses the fuel-air mixture. Finally, the fuel-air mixture is ignited, causing an expansion or power stroke, and the cycle begins again. Note that each four-stroke cycle requires two revolutions of the crank. In contrast, many outboard or lawn mower engines use a two-stroke cycle or are simply two-stroke engines. Note that a two-stroke engine has only an expansion stroke and a compression stroke and that both occur during each revolution of the crank.

The four-stroke engine uses four piston strokes in a single combustion cycle corresponding to two revolutions of the crank. The events corresponding to the four strokes are: (1) suction, or the input stroke; (2) the compression stroke; (3) expansion, or the power stroke; and (4) the exhaust stroke.

Multi-cylinder engines are also broadly classified according to how the cylinders are arranged with respect to each other and the crankshaft. Thus, an *in-line* engine is one in which all cylinder axes form a single plane containing the crankshaft, and, in that plane, the pistons are all on the same side of the crankshaft. Figure 14.1 is a schematic drawing of a three-cylinder in-line engine with the three cranks spaced at 120° of crank rotation. A firing-order diagram for four-stroke operation is included for interest.

Figure 14.2 shows a cutaway view of a five-cylinder in-line engine of a recent passenger car. We can see here that the plane formed by the centerlines of the cylinders was not mounted vertically to keep the height of the engine smaller to fit low-profile styling requirements. Figure 14.2 also includes a view of the relative location of the camshaft and the overhead valves.

A V-type engine uses two banks of one or more in-line cylinders, all connected to a single crankshaft. Figure 14.3 shows some typical crank arrangements. The pistons in

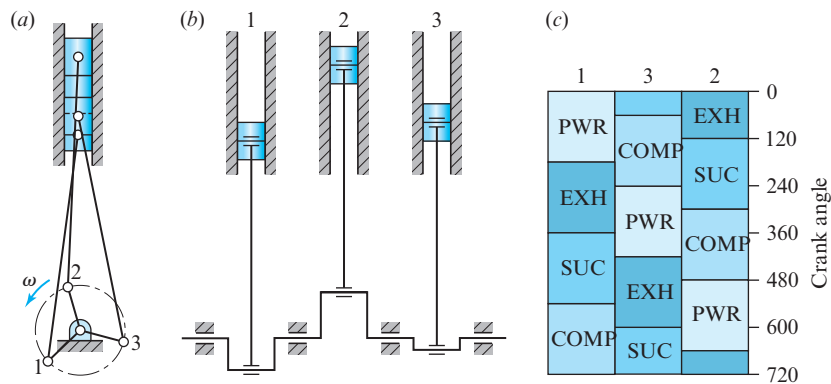


Figure 14.1 Three-cylinder in-line engine: (a) front view; (b) side view; (c) firing order.

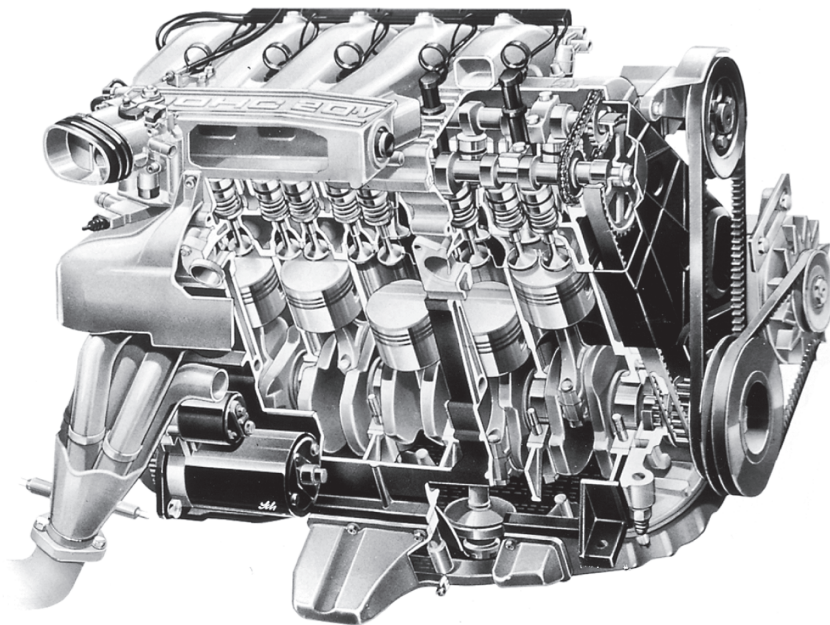


Figure 14.2 Five-cylinder in-line engine from an Audi Coupe Quattro. (Courtesy of Audi of America, Inc., Troy, MI.)

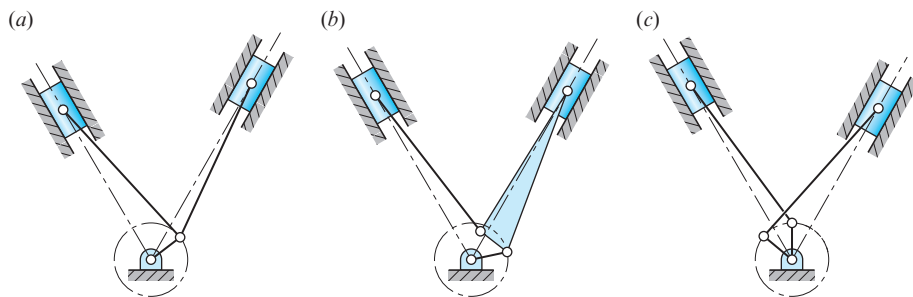


Figure 14.3 Crank arrangements of different V engines: (a) single crank throw per pair of cylinders—connecting rods interlock with each other and are of *fork-and-blade* design; (b) single crank throw per pair of cylinders—the *master connecting rod* carries a bearing for the *articulated rod*; (c) separate crank throws connect to staggered rods and pistons.

the right and left banks of Figs. 14.3a and 14.3b may be in the same plane, but those in Fig. 14.3c are in different planes.

If the V angle is increased to 180° , the result is called an *opposed-piston engine*. An opposed engine may have the two piston axes coincident or offset, and the rods may connect to the same crank throw or to separate crank throws 180° apart.

A *radial engine* is one having multiple cylinder axes arranged radially about the crank center. Radial engines use a master connecting rod for one cylinder, and the remaining pistons are connected to the master rod by articulated rods, somewhat the same as for the V engine of Fig. 14.3*b*.

Figures 14.4 to 14.6 show, respectively, the piston-connecting-rod assembly, the crankshaft, and the block of a V6 truck engine. These are included as typical of engine design to demonstrate the forms of important parts of an engine. The following specifications also give a general idea of the performance and design characteristics of typical engines, together with the sizes of parts used in them.

The reader will note that all quantities throughout this chapter are quoted in SI units rather than US customary units. Because of the international nature of the automotive and related industries, engines today are all designed in SI units, even in the United States. Although units of horsepower and gallons are still seen in advertisements, the power of an engine is measured in kilowatts (kW), and engine displacement is quoted in liters (L)* by knowledgeable engineers throughout the industry.

GMC Truck and Coach Division, General Motors Corporation, Pontiac, Michigan One of the GMC V6 truck engines is shown in Fig. 14.7. These engines are manufactured in four displacements, and they include one model, a V12 (11.5 L), which is described as a “twin six,” since many of the V6 parts are interchangeable with it. Data included here are for the 6.57-L engine. Typical performance curves are exhibited in Figs. 14.8, 14.9, and 14.10.



Figure 14.4

Piston-connecting-rod assembly for a 5.75-L V6 truck engine. (Courtesy of GMC Truck and Coach Division, General Motors Corporation, Pontiac, MI.)

* The unit for volume of an engine or cylinder is the liter. The international symbol for liter is lower case “l,” which can easily be confused with the numeral “1.” Therefore, the symbol “L” is used throughout this chapter.

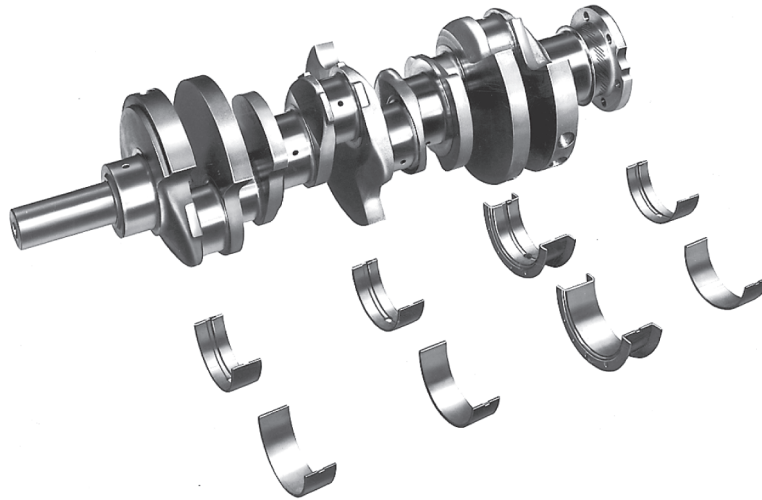


Figure 14.5 Cast crankshaft for a 5-L V6 truck engine. (Courtesy of GMC Truck and Coach Division, General Motors Corporation, Pontiac, MI.)

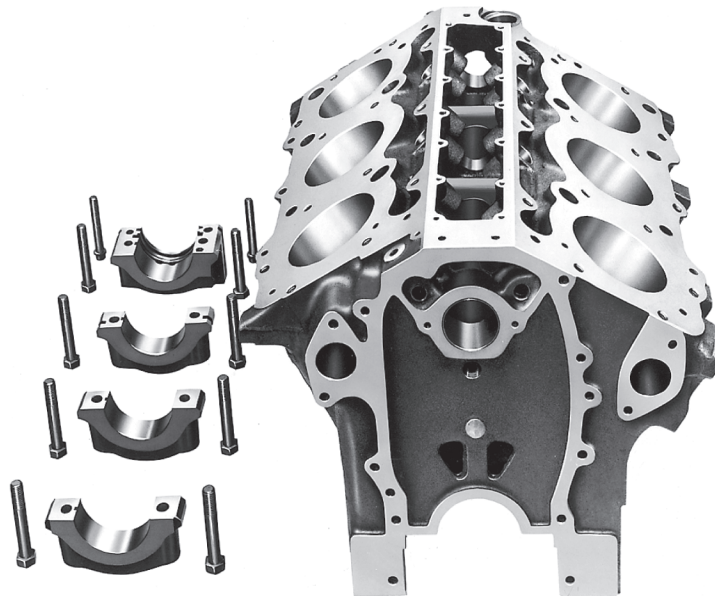


Figure 14.6 Block for a 5-L V6 truck engine. The same casting is used for a 5.75-L engine by boring for larger pistons. (Courtesy of GMC Truck and Coach Division, General Motors Corporation, Pontiac, MI.)

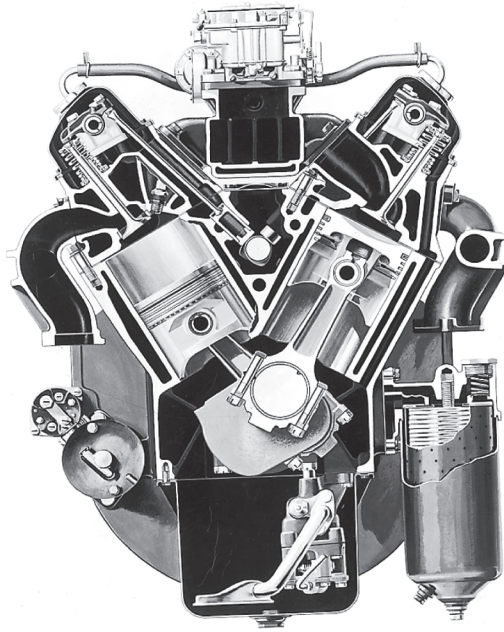


Figure 14.7 Cross-sectional view of a 6.57-L V6 truck engine. (Courtesy of GMC Truck and Coach Division, General Motors Corporation, Pontiac, MI.)

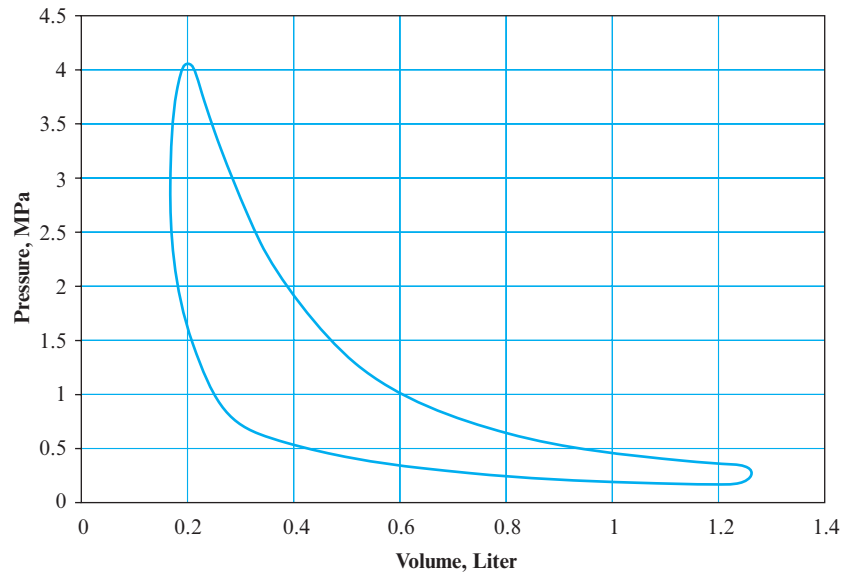


Figure 14.8 Typical indicator diagram for a 6.57-L V6 truck engine; conditions unknown. (Courtesy of GMC Truck and Coach Division, General Motors Corporation, Pontiac, MI.)

The specifications are as follows: 60° V design; bore = 123.8 mm; stroke = 90.4 mm; connecting rod length = 182.6 mm; compression ratio = 7.50:1; cylinders are numbered 1, 3, and 5 from front to rear on the left bank, and 2, 4, and 6 on the right bank; and the firing order is 1, 6, 5, 4, 3, 2. The crank arrangement is shown in Fig. 14.11.

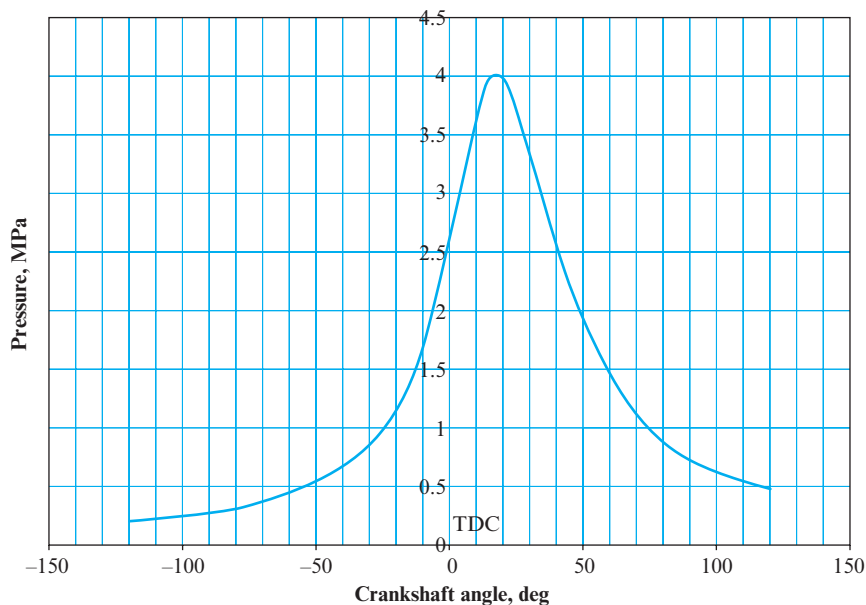


Figure 14.9 Pressure-crank-angle curve for the 6.57-L V6 truck engine; these data were taken from a running engine. (Courtesy of GMC Truck and Coach Division, General Motors Corporation, Pontiac, MI.)

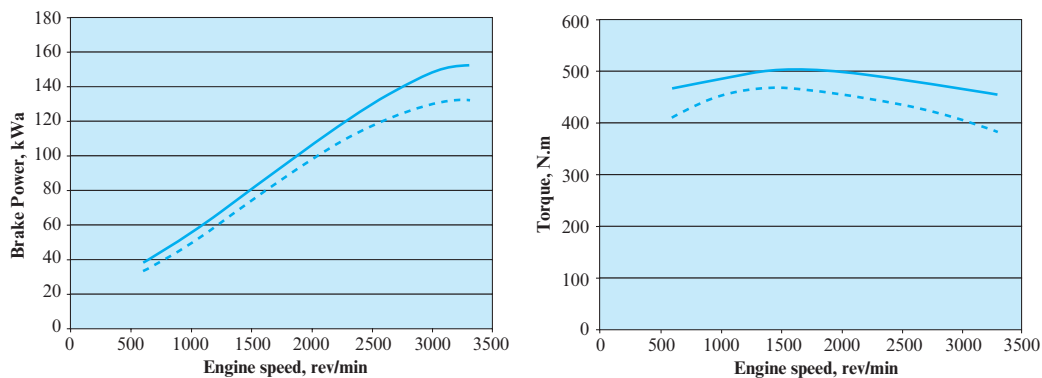


Figure 14.10 Power and torque characteristics of the 6.57-L V6 truck engine. Solid curve is the net output as installed; dashed curve is the maximum output without accessories. Note that maximum torque occurs at very low engine speed. (Courtesy of GMC Truck and Coach Division, General Motors Corporation, Pontiac, MI.)

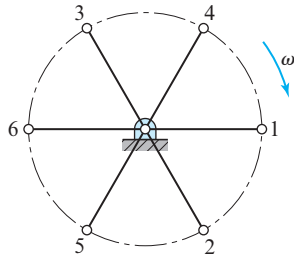


Figure 14.11 Front view of a V6 engine showing crank arrangement and direction of rotation.

14.2 INDICATOR DIAGRAMS

Experimentally, an instrument called an *engine indicator* is used to measure the variation in pressure within a cylinder. This instrument constructs a graph, during operation of the engine, which is known as an *indicator diagram*. Known constraints of the indicator make it possible to study the diagram and determine the relationship between the gas pressure and the crank angle for the particular running conditions in existence at the time the diagram is recorded.

When an engine is in the design stage, it is necessary to estimate a diagram from theoretic considerations. From such an approximation, a pilot model of the proposed engine is designed and built, and the actual indicator diagram is taken and compared with the theoretically estimated one. This provides much useful information for the design of the production model.

An indicator diagram for the ideal air-standard cycle is shown in Fig. 14.12 for a four-stroke engine. During compression, the cylinder volume changes from v_1 to v_2 , and the cylinder pressure changes from p_1 to p_2 . The relationship, at any point of this stroke, is given by the polytropic gas law as

$$pv^k = p_1v_1^k = \text{constant}. \quad (14.1)$$

In an actual indicator card, the corners at points 2 and 3 are rounded, and the line joining these points is curved. This is explained by the fact that combustion is not instantaneous, and ignition occurs before the end of the compression stroke. An actual card is also rounded at points 4 and 1, since the valves do not operate simultaneously.

The polytropic exponent, k , in Eq. (14.1) is often taken to be about 1.30 for both compression and expansion, although differences probably exist.

The relationship between the power developed and the dimensions of the engine is given by

$$P_b = \frac{p_b l a n (1\,000 \text{ N/m}^2/\text{kPa})}{(1\,000 \text{ N} \cdot \text{m/s/kW})(1\,000 \text{ mm/m})(60 \text{ s/min})} = \frac{p_b l a n}{60\,000}, \quad (14.2)$$

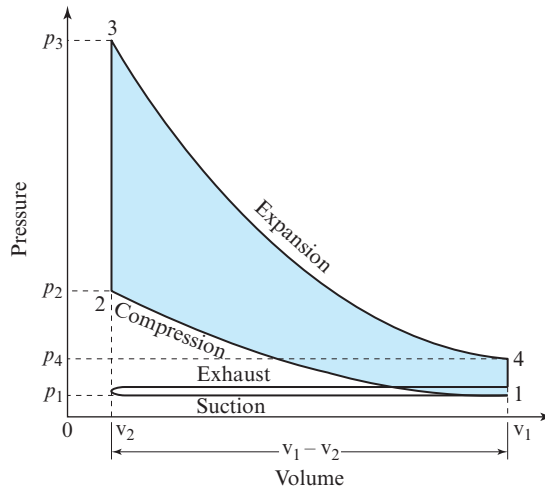


Figure 14.12 An ideal indicator diagram for a four-stroke engine.

where

- P_b = brake power per cylinder (kW);
- p_b = brake mean effective pressure (kPa);
- l = length of the stroke (mm);
- a = piston area (m^2);
- n = number of working strokes per minute.

The amount of power that can be obtained from one liter of piston displacement varies considerably, depending upon the engine type. For typical automotive engines, it ranges from about 25 to 45 kW/L, with a current average of perhaps 32. On the other hand, many marine diesel engines have ratios varying from 4.5 to 9 kW/L. About the best that can be done in designing a new engine is to use standard references to discover what others have done with the same types of engines, and then choose a value that seems to be reasonably attainable.

For many engines, the bore-to-stroke ratio varies from about 0.75 to 1.00. The tendency in automotive-engine design seems to be toward shorter-stroke engines to reduce engine height. Decisions on bore-stroke ratio and power per unit displacement volume are helpful in solving Eq. (14.2) to obtain suitable dimensions once the power, speed, and number of cylinders have been decided upon.

The ratio of the brake mean effective pressure, p_b , to the indicated mean effective pressure, p_i , obtained experimentally from an indicator card, is the mechanical efficiency, e_m :

$$e_m = \frac{p_b}{p_i}. \quad (14.3)$$

Differences between a theoretic and an experimentally determined indicator diagram can be accounted for by applying a correction called a *card factor*. The card factor is

defined by

$$f_c = \frac{p_i}{p'_i}, \quad (14.4)$$

where p'_i is the theoretic indicated mean effective pressure, and the card factor is usually about 0.90 to 0.95.

Since the *compression ratio* (Fig. 14.12) is defined as

$$R = \frac{v_1}{v_2}, \quad (14.5)$$

the work done during compression is

$$U_c = \int_{v_2}^{v_1} p dv = p_1 v_1^k \int_{v_2}^{v_1} \frac{dv}{v^k} = \frac{p_1 v_1}{k-1} (R^{k-1} - 1). \quad (a)$$

The displacement volume can be written as

$$v_1 - v_2 = v_1 - \frac{v_1}{R} = \frac{v_1(R-1)}{R}. \quad (b)$$

Substituting v_1 from Eq. (b) into Eq. (a), the work done during compression can be written as

$$U_c = \frac{p_1(v_1 - v_2)}{k-1} \frac{R^k - R}{R-1}. \quad (c)$$

The work done during expansion is the area under the curve between points 3 and 4 of Fig. 14.12. This is determined in the same manner; the result is

$$U_e = \frac{p_4(v_1 - v_2)}{k-1} \frac{R^k - R}{R-1}. \quad (d)$$

The net amount of work accomplished in a cycle is the difference between Eqs. (c) and (d), and it must be equal to the product of the theoretic indicated mean effective pressure and the displacement volume. That is,

$$U = U_e - U_c = \frac{p_4(v_1 - v_2)}{k-1} \frac{R^k - R}{R-1} - \frac{p_1(v_1 - v_2)}{k-1} \frac{R^k - R}{R-1} = p'_i(v_1 - v_2). \quad (14.6)$$

If the exponent is the same for expansion as for compression, Eq. (14.6) can be solved to give

$$p_4 = p'_i(k-1) \frac{R-1}{R^k - R} + p_1. \quad (e)$$

Substituting the theoretic indicated mean effective pressure, p'_i [Eq. (14.4)], into Eq. (e) gives

$$p_4 = (k-1) \frac{R-1}{R^k - R} \frac{p_i}{f_c} + p_1. \quad (14.7)$$

Equations (14.1) and (14.7) can be used to create the theoretical indicator diagram. The corners are then rounded off so that the pressure at point 3 is made about 75% of that given by Eq. (14.1). As a check, the area of the diagram can be measured and divided by the displacement volume; the result should equal the indicated mean effective pressure.

14.3 DYNAMIC ANALYSIS—GENERAL

Recall that Chaps. 11 and 12 present both graphic and analytic methods for finding the static and dynamic forces in a mechanism. The balance of this chapter is devoted to an analysis of the dynamics of a single cylinder of an operating engine. To simplify this work, it is assumed that the engine is running at a constant crankshaft speed and that gravitation forces and friction forces can be neglected in comparison with dynamic-force effects. The gas forces and inertia forces are determined separately in the upcoming sections; then, these forces are combined (Sec. 14.7), using the principle of superposition (Sec. 12.5) to obtain the total bearing forces and crankshaft torque.

The subject of engine balancing is treated in Chap. 15.

14.4 GAS FORCES

Analytic or graphic force analysis methods can be used to solve this gas-force problem. The advantage of the analytic methods is that they can be programmed for automatic computation throughout a cycle. In contrast, the graphic technique must be repeated for each crank posture until a complete cycle of operation (720° for a four-stroke engine) is completed.

From Chap. 12, we recall that dynamic force analysis requires knowing the acceleration of the piston. Recall from Chaps. 3 and 4 that the equations for the approximate position [Eq. (3.16a)] and acceleration [Eq. (4.22a)] of the piston can be written as

$$x = l - \frac{r^2}{4l} + r \left(\cos \omega t + \frac{r}{4l} \cos 2\omega t \right), \quad (14.8a)$$

$$\ddot{x} = -r\omega^2 \left(\cos \omega t + \frac{r}{l} \cos 2\omega t \right) - r\alpha \left(\sin \omega t + \frac{r}{2l} \sin 2\omega t \right). \quad (14.8b)$$

For constant crank angular velocity, Eq. (14.8b) reduces to

$$\ddot{x} = -r\omega^2 \left(\cos \omega t + \frac{r}{l} \cos 2\omega t \right). \quad (14.8c)$$

We assume that the moving parts of the engine are massless so that gravity, inertia forces, and torques are zero. We also assume that the effects of friction can be neglected. These assumptions allow us to trace the force effects of the gas pressure from the piston to the crankshaft without the complicating effects of other forces. These other effects, except for friction, will be added later, using superposition.

Referring to Fig. 14.13, we designate a gas-force vector $\mathbf{P}$ as defined or obtained using the methods of Sec. 14.2. Reactions caused by this force are designated using a single prime. Thus, $\mathbf{F}'_{14}$ is the force from the cylinder wall acting against the piston, and $\mathbf{F}'_{34}$ is

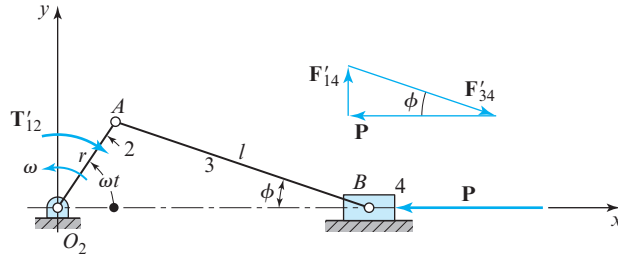


Figure 14.13 Graphic analysis of forces resulting from gas pressure.

the force of the connecting rod acting onto the piston at the piston pin. The force polygon in Fig. 14.13 shows the relation among $\mathbf{P}$, $\mathbf{F}'_{14}$, and $\mathbf{F}'_{34}$. Thus, we have

$$\mathbf{F}'_{14} = P \tan \phi \hat{\mathbf{j}}. \quad (14.9)$$

The quantity $\tan \phi$ appears frequently in expressions throughout the chapter; therefore, it is convenient to develop an expression for $\tan \phi$ in terms of the crank angle, $\theta = \omega t$. Recall from Sec. 3.9, that this quantity can be written as

$$\tan \phi = \frac{(r/l) \sin \omega t}{\cos \phi} = \frac{(r/l) \sin \omega t}{\sqrt{1 - [(r/l) \sin \omega t]^2}}. \quad (a)$$

Recall that the binomial expansion for a variable x can be written as

$$(1 - x)^n = 1 - nx + \frac{n(n-1)x^2}{2!} - \frac{n(n-1)(n-2)x^3}{3!} + \frac{n(n-1)(n-2)(n-3)x^4}{4!} - \dots. \quad (b)$$

Therefore, we can replace the denominator on the right-hand side of Eq. (a) by

$$\frac{1}{\sqrt{1 - [(r/l) \sin \omega t]^2}} = \left\{ 1 - [(r/l) \sin \omega t]^2 \right\}^{-\frac{1}{2}} \approx 1 + \frac{r^2}{2l^2} \sin^2 \omega t, \quad (c)$$

where only the first two terms of the expansion have been retained. Substituting Eq. (c) into Eq. (a) we have

$$\tan \phi = \frac{r}{l} \sin \omega t \left(1 + \frac{r^2}{2l^2} \sin^2 \omega t \right). \quad (14.10)$$

Then, substituting this equation into Eq (14.9), the force from the cylinder wall acting against the piston is

$$\mathbf{F}'_{14} = P \frac{r}{l} \sin \omega t \left(1 + \frac{r^2}{2l^2} \sin^2 \omega t \right) \hat{\mathbf{j}}. \quad (d)$$

The trigonometry of Fig. 14.13 indicates that the *wrist pin* (piston-pin) bearing force has a magnitude of

$$F'_{34} = \frac{P}{\cos \phi} = \frac{P}{\sqrt{1 - [(r/l) \sin \omega t]^2}} = P \left(1 + \frac{r^2}{2l^2} \sin^2 \omega t \right), \quad (e)$$

or, in vector notation,

$$\mathbf{F}'_{34} = P\hat{\mathbf{i}} - F'_{14}\hat{\mathbf{j}} = P\hat{\mathbf{i}} - P \tan \phi \hat{\mathbf{j}}. \quad (14.11)$$

By taking moments about the crank center, we find that the torque, $\mathbf{T}'_{21}$, delivered by the crank to the shaft is the product of the force, $\mathbf{F}'_{14}$, and the piston coordinate, x , that is

$$\mathbf{T}'_{21} = F'_{14}x\hat{\mathbf{k}}.$$

Substituting Eqs. (d) and (14.8a) into this equation, the torque is

$$\mathbf{T}'_{21} = P \frac{r}{l} \sin \omega t \left(1 + \frac{r^2}{2l^2} \sin^2 \omega t \right) \left[l - \frac{r^2}{4l} + r \left(\cos \omega t + \frac{r}{4l} \cos 2\omega t \right) \right] \hat{\mathbf{k}}. \quad (f)$$

Multiplying these terms, and (with only a very small error) neglecting second or higher powers of (r/l) gives

$$\mathbf{T}'_{21} = Pr \sin \omega t \left(1 + \frac{r}{l} \cos \omega t \right) \hat{\mathbf{k}}. \quad (14.12)$$

This is the torque delivered to the crankshaft by the gas force; the counterclockwise direction is positive.

14.5 EQUIVALENT MASSES

Problems 12.5 and 12.6, at the end of Chap. 12, focused on the slider-crank linkage, which is the basis for most engine mechanisms. The dynamics of this linkage were analyzed using the methods presented in Chap. 12. In Chap. 14, we are concerned with the same problem. However, here we will show certain simplifications that are customarily used to reduce the complexity of the algebraic solution process. These simplifications are approximations and do introduce certain errors. In Chap. 14, we demonstrate the simplifications and comment on the errors that they introduce.

In analyzing the inertia forces caused by the connecting rod of an engine, it is often convenient to picture a portion of the mass of the rod as concentrated at the crankpin A with the remaining portion lumped at the wrist pin, B (Fig. 14.14). The reason for this is that the crankpin moves on a circle and the wrist pin on a straight line, and both of these motions are quite easy to analyze. However, the center of mass, G_3 , of the connecting rod is somewhere between the crankpin and the wrist pin, and its motion is more complex and consequently more difficult to determine accurately in algebraic form.

The mass of the connecting rod, m_3 , is correctly located at the centroid, G_3 . However, we can mentally divide this mass into two mass particles; one particle, m_{3B} , is concentrated at wrist pin B , whereas the other particle, m_{3P} , is concentrated at the center of percussion,

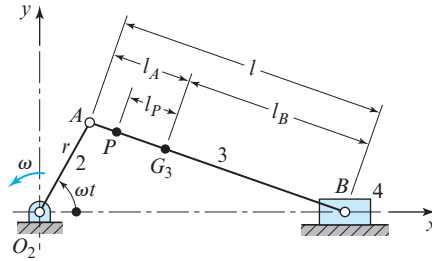


Figure 14.14

P (Sec. 12.6), for oscillation of the rod about point B . The division of these masses is dynamically equivalent to the original rod if (1) the total mass is identical, (2) the position of the centroid G_3 is unchanged, and (3) the mass moment of inertia is the same. Writing these three conditions in equation form gives

$$m_3 = m_{3B} + m_{3P}, \quad (a)$$

$$m_{3B}l_B = m_{3P}l_P, \quad (b)$$

and

$$I_G = m_{3B}l_B^2 + m_{3P}l_P^2. \quad (c)$$

These three equations can be solved to give the mass particles, m_{3B} and m_{3P} , and the location of the center of percussion, l_P . The procedure is as follows. Solving Eqs. (a) and (b) simultaneously gives the mass particles at crankpin A and wrist pin B :

$$m_{3B} = m_3 \frac{l_P}{l_B + l_P} \quad \text{and} \quad m_{3P} = m_3 \frac{l_B}{l_B + l_P}. \quad (14.13)$$

Then substituting these equations into Eq. (c) gives

$$I_G = m_3 \frac{l_P}{l_B + l_P} l_B^2 + m_3 \frac{l_B}{l_B + l_P} l_P^2 = m_3 l_P l_B \quad (d)$$

or

$$l_P l_B = \frac{I_G}{m_3}. \quad (e)$$

This equation demonstrates that the two distances, l_P and l_B , are dependent on each other. Therefore, if l_B is known, then l_P becomes set by Eq. (e). The result can be shown to be purely geometric by substituting $I_G = m_3 k_G^2$ into the equation, and rearranging, to give

$$l_P = \frac{k_G^2}{l_B}. \quad (14.14)$$

Note that this answer is in complete agreement with Eq. (12.20).

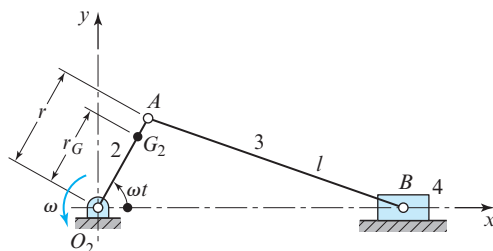


Figure 14.15

In the usual connecting rod, the center of percussion, P , is close to crankpin A , and it is usual to assume that they are coincident. Thus, if we let $l_P = l_A$, Eqs. (14.13) reduce to

$$m_{3B} = \frac{m_3 l_A}{l} \text{ and } m_{3A} = \frac{m_3 l_B}{l}. \quad (14.15)$$

We note again that the equivalent masses, obtained by Eqs. (14.15), are not exact because of the assumption made, but they are close enough for ordinary connecting rods. This approximation, however, would not be valid for the master connecting rod of a radial engine, since the crankpin end has bearings for all of the other connecting rods as well as its own bearing.

For estimating and checking purposes, about two-thirds of the mass of the usual connecting rod is concentrated at crankpin A , and the remaining one-third is concentrated at wrist pin B .

Figure 14.15 shows an engine linkage in which the mass of crank m_2 is not balanced, as evidenced by the fact that the center of gravity, G_2 , is located outward along the crank distance r_G from the axis of rotation. In the inertia force analysis, another simplification is made by replacing this by an equivalent mass, m_{2A} , at the crankpin. For equivalence, this requires

$$m_2 r_G = m_{2A} r \quad \text{or} \quad m_{2A} = m_2 \frac{r_G}{r}. \quad (14.16)$$

14.6 INERTIA FORCES

Using the methods of the previous section, we begin by locating equivalent masses at the crankpin and at the wrist pin. Thus,

$$m_A = m_{2A} + m_{3A}, \quad (14.17)$$

$$m_B = m_{3B} + m_4. \quad (14.18)$$

Equation (14.17) states that the rotating mass, m_A , located at the crankpin is composed of the equivalent mass, m_{2A} , of the crank, and the mass, m_{3A} , of a portion of the connecting rod. Of course, if the crank is balanced (Chap. 15), including balancing the m_{3A} portion of the connecting rod, all of its mass is located at the axis of rotation, and m_A is then zero. Equation (14.18) indicates that the reciprocating mass, m_B , located at the wrist pin is

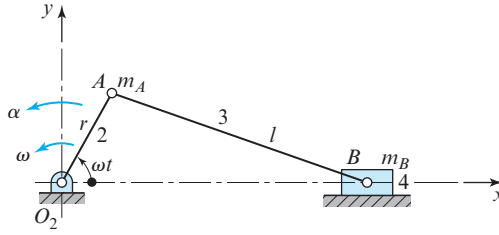


Figure 14.16

composed of the equivalent mass, m_{3B} , of the other portion of the connecting rod and the mass, m_4 , of the piston assembly.

Figure 14.16 shows the (unbalanced) slider-crank linkage with masses m_A and m_B located at crankpin A and wrist pin B, respectively.

If we designate the angular velocity and angular acceleration of the crank as ω and α , respectively, the position vector of crankpin A relative to the origin, O_2 , is

$$\mathbf{R}_A = r \cos \omega t \hat{\mathbf{i}} + r \sin \omega t \hat{\mathbf{j}}. \quad (a)$$

Differentiating this equation twice with respect to time, the acceleration of crankpin A is

$$\mathbf{A}_A = (-r\alpha \sin \omega t - r\omega^2 \cos \omega t) \hat{\mathbf{i}} + (r\alpha \cos \omega t - r\omega^2 \sin \omega t) \hat{\mathbf{j}}. \quad (14.19)$$

The inertia force of the rotating mass is then

$$-m_A \mathbf{A}_A = m_A r (\alpha \sin \omega t + \omega^2 \cos \omega t) \hat{\mathbf{i}} + m_A r (-\alpha \cos \omega t + \omega^2 \sin \omega t) \hat{\mathbf{j}}. \quad (14.20)$$

Since the analysis is usually performed for a constant angular velocity (that is, $\alpha = 0$), then Eq. (14.20) reduces to

$$-m_A \mathbf{A}_A = m_A r \omega^2 (\cos \omega t \hat{\mathbf{i}} + \sin \omega t \hat{\mathbf{j}}). \quad (14.21)$$

The acceleration of the piston is known from Eq. (14.8b); therefore, the inertia force of the reciprocating mass is

$$-m_B \mathbf{A}_B = \left[m_B r \omega^2 \left(\cos \omega t + \frac{r}{l} \cos 2\omega t \right) + m_B r \alpha \left(\sin \omega t + \frac{r}{2l} \sin 2\omega t \right) \right] \hat{\mathbf{i}}. \quad (14.22)$$

For constant angular velocity of the crank, Eq. (14.8c), then Eq. (14.22) can be written as

$$-m_B \mathbf{A}_B = m_B r \omega^2 \left(\cos \omega t + \frac{r}{l} \cos 2\omega t \right) \hat{\mathbf{i}}. \quad (14.23)$$

Adding Eqs. (14.21) and (14.23) gives the total inertia force for all of the moving parts (for constant angular velocity). The components in the x and y directions are

$$F^x = (m_A + m_B) r \omega^2 \cos \omega t + \left(m_B \frac{r}{l} \right) r \omega^2 \cos 2\omega t \quad (14.24)$$

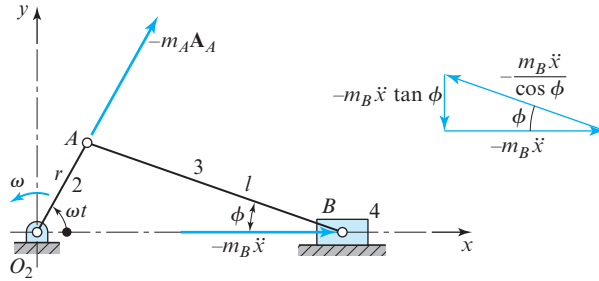


Figure 14.17

and

$$F^y = m_A r \omega^2 \sin \omega t. \quad (14.25)$$

It is customary to refer to the portion of the force occurring at the frequency ω rad/s as the *primary inertia force* and the portion occurring at 2ω rad/s as the *secondary inertia force*. We note that the x component, which is in the direction of the cylinder axis, has a primary part varying directly with the crankshaft speed and a secondary part varying at twice the crankshaft speed. On the other hand, the y component, perpendicular to the cylinder axis, has only a primary part, and it therefore varies directly with the crankshaft speed.

We proceed now to a determination of the inertia torque. As shown in Fig. 14.17, the inertia force caused by the mass at the crankpin A has no moment about O_2 and, therefore, produces no inertia torque. Consequently, we need consider only the inertia force given by Eq. (14.23) caused by the reciprocating mass, m_B .

From the force polygon of Fig. 14.17, the inertia torque exerted by the engine on the crankshaft is

$$\mathbf{T}_{21}'' = -(-m_B A_B \tan \phi) x \hat{\mathbf{k}}. \quad (b)$$

Substituting the expressions for x , A_B , and $\tan \phi$, as given by Eqs. (14.8a), (14.8c), and (14.10), into Eq. (b), the inertia torque can be written as

$$\begin{aligned} \mathbf{T}_{21}'' = & -m_B r \omega^2 \left(\cos \omega t + \frac{r}{l} \cos 2\omega t \right) \left[\frac{r}{l} \sin \omega t \left(1 + \frac{r^2}{2l^2} \sin^2 \omega t \right) \right] \\ & \left[l - \frac{r^2}{4l} + r \left(\cos \omega t + \frac{r}{4l} \cos 2\omega t \right) \right] \hat{\mathbf{k}}. \end{aligned} \quad (c)$$

Terms that are proportional to the second and higher powers of r/l are now neglected in performing the indicated products. Therefore, Eq. (c) can be written as

$$\mathbf{T}_{21}'' = -m_B r^2 \omega^2 \sin \omega t \left(\frac{r}{2l} + \cos \omega t + \frac{3r}{2l} \cos 2\omega t \right) \hat{\mathbf{k}}. \quad (d)$$

Substituting the identities

$$2 \sin \omega t \cos \omega t = \sin 2\omega t \quad (e)$$

and

$$2 \sin \omega t \cos 2\omega t = \sin 3\omega t - \sin \omega t \quad (f)$$

into Eq. (d), and rearranging, gives

$$\mathbf{T}_{21}'' = \frac{m_B}{2} r^2 \omega^2 \left(\frac{r}{2l} \sin \omega t - \sin 2\omega t - \frac{3r}{2l} \sin 3\omega t \right) \hat{\mathbf{k}}. \quad (14.26)$$

Equation (14.26) is the inertia torque exerted by the engine on the crankshaft in the positive direction. A clockwise, or negative, inertia torque of the same magnitude is, of course, exerted by the frame onto the engine.

The assumed distribution of the connecting-rod mass results in a moment of inertia that is greater than the true value. Consequently, the torque given by Eq. (14.26) is not the exact value. In addition, terms proportional to the second- and higher-order powers of r/l were neglected in simplifying Eq. (c). These two approximations are of about the same magnitude and are quite small for ordinary connecting rods having $r/l \leq 1/4$.

14.7 BEARING LOADS IN A SINGLE-CYLINDER ENGINE

The designer of a reciprocating engine must know the values of the forces acting upon the bearings and how these forces vary during a cycle of operation. These are necessary to proportion and select the bearings properly and are also needed for the design of other engine parts. This section is an investigation of the force exerted by the piston against the cylinder wall and the forces acting against the piston pin and against the crankpin. The main bearing forces are investigated in a later section, since they depend upon the action within the cylinders of the engine.

The resultant total bearing loads are made up of the following components:

1. Gas-force components, designated by a single prime,
2. Inertia force due to mass m_4 of the piston assembly, designated by a double prime,
3. Inertia force of part m_{3B} of the connecting rod assigned to the wrist-pin end, designated by a triple prime,
4. Connecting-rod inertia force of part m_{3A} at the crankpin end, designated by a quadruple prime.

Equations for the gas-force components have been determined in Sec. 14.4, and references are made to them in finding the total bearing loads.

Figure 14.18 is a graphic analysis of the forces in the engine linkage with zero gas force and subjected only to inertia force resulting from the mass, m_4 , of the piston assembly. Figure 14.18a shows the posture of the linkage selected for analysis, and the inertia force, $-m_4 \mathbf{A}_B$, is shown acting upon the piston. In Fig. 14.18b the free-body

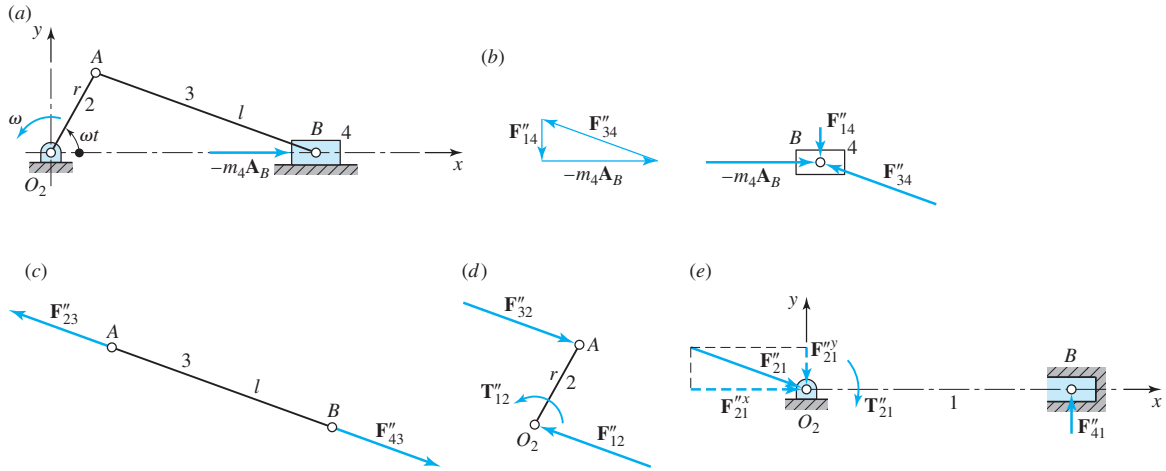


Figure 14.18 Graphic analysis of forces resulting from mass m_4 .

diagram of the piston is shown together with the polygon from which the forces are obtained. Figures 14.18c through 14.18e, respectively, show the free-body diagrams showing forces acting upon the connecting rod, crank, and frame.

In Fig. 14.18e, we note that the torque $\mathbf{T}_{21}''$ balances the couple formed by forces $\mathbf{F}_{41}''$ and $\mathbf{F}_{21}''$. However, force $\mathbf{F}_{21}''$ at the crank center remains unopposed by any other force. This is a very important observation that we shall reserve for discussion in a separate section.

The following forces are of interest to us:

1. The force $\mathbf{F}_{41}''$ of the piston against the cylinder wall,
2. The force $\mathbf{F}_{34}''$ of the connecting rod against the wrist-pin,
3. The force $\mathbf{F}_{32}''$ of the connecting rod against the crankpin,
4. The force $\mathbf{F}_{12}''$ of the engine block against the crank.

By methods similar to those presented earlier in the chapter, the analytic expressions are determined to be:

$$\mathbf{F}_{41}'' = -m_4 \ddot{x} \tan \phi \hat{\mathbf{j}}, \quad (14.27)$$

$$\mathbf{F}_{34}'' = m_4 \ddot{x} \hat{\mathbf{i}} - m_4 \ddot{x} \tan \phi \hat{\mathbf{j}}, \quad (14.28)$$

$$\mathbf{F}_{32}'' = -\mathbf{F}_{34}'', \quad (14.29)$$

$$\mathbf{F}_{12}'' = -\mathbf{F}_{32}'' = \mathbf{F}_{34}'', \quad (14.30)$$

where m_4 is the mass of the piston assembly, $\ddot{x}$ is the acceleration of the piston [Eq. (14.8b)], and the quantity $\tan \phi$ can be written in terms of the crank angle [Eq. (14.10)].

Figure 14.19 shows only those forces that result because of part m_{3B} of the mass of the connecting rod that is assumed to be concentrated at the wrist pin center.

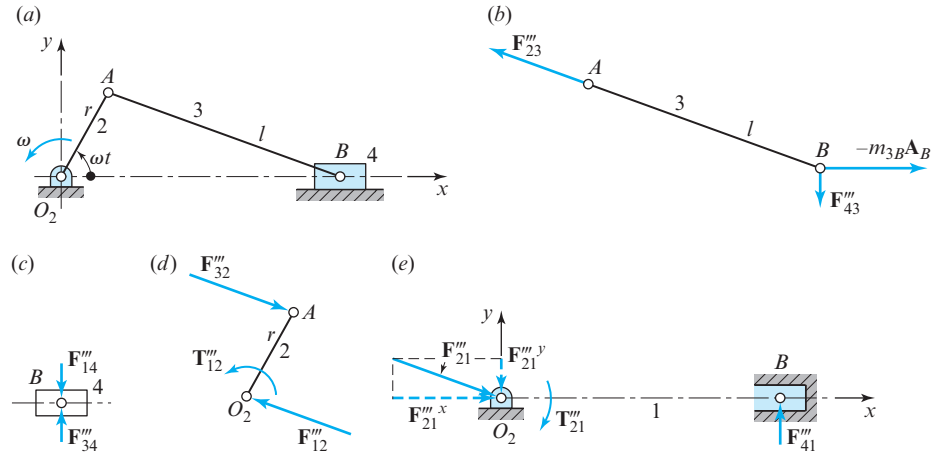


Figure 14.19 Graphic analysis of the forces resulting from mass m_{3B} .

Thus, Fig. 14.19b is a free-body diagram of the connecting rod showing the inertia force, $-m_{3B}\mathbf{A}_B$, acting at wrist pin B .

We note that it is incorrect when finding the bearing loads to add m_{3B} and m_4 together and then to compute a resultant inertia force, although such a procedure would seem to be simpler. The reason for this is that m_4 is the mass of the piston assembly, and the corresponding inertia force acts on the piston side of the wrist pin. But m_{3B} is part of the connecting-rod mass, and hence its inertia force acts on the connecting-rod side of the wrist pin. Thus, adding the two would yield correct results for the crankpin load and the force of the piston against the cylinder wall but would give *incorrect* results for the wrist-pin load.

The forces on the piston pin, the crank, and the frame resulting from the inertia of mass m_{3B} are shown in Figs. 14.19c, 14.19d, and 14.19e, respectively. The equations for these forces for a crank having uniform angular velocity are determined to be

$$\mathbf{F}_{41}''' = -m_{3B}\ddot{x}\tan\phi\hat{\mathbf{j}}, \quad (14.31)$$

$$\mathbf{F}_{34}''' = \mathbf{F}_{41}''', \quad (14.32)$$

$$\mathbf{F}_{32}''' = -m_{3B}\ddot{x}\hat{\mathbf{i}} + m_{3B}\ddot{x}\tan\phi\hat{\mathbf{j}}, \quad (14.33)$$

$$\mathbf{F}_{12}''' = -\mathbf{F}_{32}'''. \quad (14.34)$$

Figure 14.20 shows the forces that result from part m_{3A} of the connecting-rod mass that is assumed concentrated at the crankpin end. Whereas a counterweight attached to the crank balances the reaction at O_2 , it cannot make $\mathbf{F}_{32}'''$ zero. Thus, the crankpin force exists no matter whether or not the rotating mass of the connecting rod is balanced. This force is

$$\mathbf{F}_{32}''' = m_{3A}r\omega^2 (\cos\omega t\hat{\mathbf{i}} + \sin\omega t\hat{\mathbf{j}}). \quad (14.35)$$

The last step is to sum the above individual component equations to obtain the resultant bearing loads. For example, the total force of the piston against the cylinder wall is found

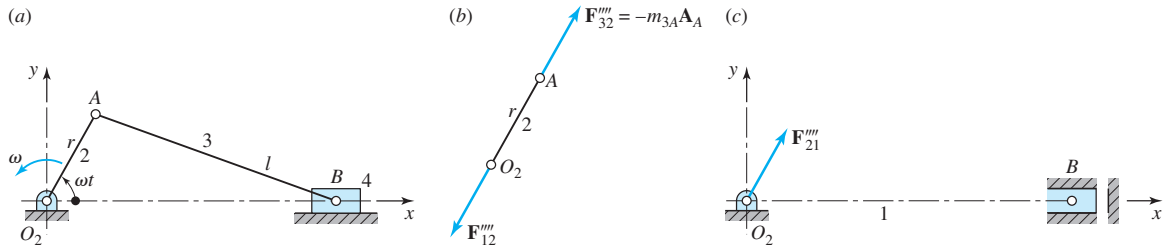


Figure 14.20 Graphic analysis of the forces resulting from mass m_{3A} .

by summing Eqs. (14.9), (14.27), and (14.31), with due regard for subscripts and signs. When simplified, the result is

$$\mathbf{F}_{41} = \mathbf{F}'_{41} + \mathbf{F}''_{41} + \mathbf{F}'''_{41} = -[P + (m_{3B} + m_4)\ddot{x}] \tan \phi \hat{\mathbf{j}}. \quad (14.36)$$

The forces on the wrist pin, the crankpin, and the crankshaft are determined in a similar manner and are

$$\mathbf{F}_{34} = (m_4\ddot{x} + P)\hat{\mathbf{i}} - [(m_{3B} + m_4)\ddot{x} + P] \tan \phi \hat{\mathbf{j}}, \quad (14.37)$$

$$\begin{aligned} \mathbf{F}_{32} = & [m_{3A}r\omega^2 \cos \omega t - (m_{3B} + m_4)\ddot{x} - P]\hat{\mathbf{i}} \\ & + \left\{ m_{3A}r\omega^2 \sin \omega t + [(m_{3B} + m_4)\ddot{x} + P] \tan \phi \right\} \hat{\mathbf{j}} \end{aligned} \quad (14.38)$$

$$\mathbf{F}_{21} = \mathbf{F}_{32}. \quad (14.39)$$

The torque delivered by the crankshaft to the load is called the *crankshaft torque*, and it is the negative of the moment of the couple formed by forces $\mathbf{F}_{41}$ and $\mathbf{F}_{21}^y$. Therefore, it is obtained from the equation

$$\mathbf{T}_{21} = -F_{41}x\hat{\mathbf{k}} = [(m_{3B} + m_4)\ddot{x} + P]x \tan \phi \hat{\mathbf{k}}. \quad (14.40)$$

14.8 SHAKING FORCES OF ENGINES

The inertia force caused by the reciprocating masses is shown acting in the positive direction in Fig. 14.21a. In Fig. 14.21b, the forces acting upon the engine block caused by these inertia forces are shown. The resultant forces are $\mathbf{F}_{21}$, the force exerted by the crankshaft on the main bearings, and a positive couple formed by forces $\mathbf{F}_{41}$ and $\mathbf{F}_{21}^y$. The force $\mathbf{F}_{21}^x = -m_B\mathbf{A}_B$ is termed the *shaking force*, and the couple $T = xF_{41}$ is called the *shaking moment*. As indicated by Eqs. (14.23) and (14.26), the magnitude and the direction of this force and couple change with ωt ; consequently, the shaking force induces rectilinear vibration of the block in the x direction, and the shaking couple induces a torsional vibration of the block about the crank center.

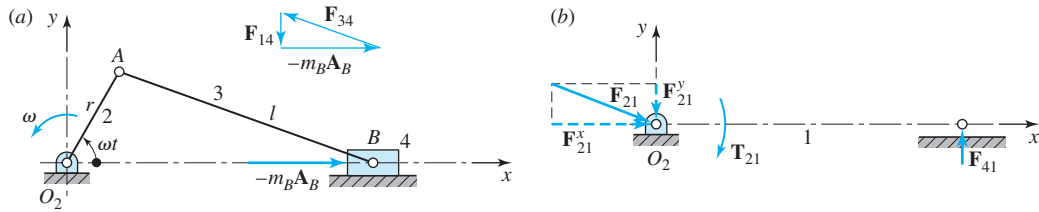


Figure 14.21 Dynamic forces due to the reciprocating masses; the primes have been omitted for clarity.

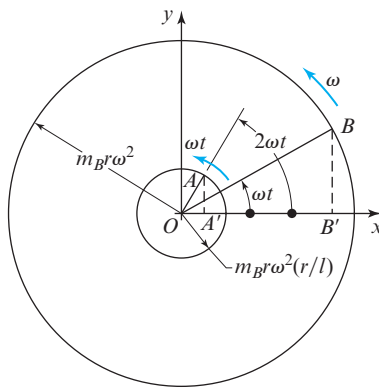


Figure 14.22 Circle diagram illustrating inertia forces. The total shaking force is $OA' + OB'$.

A graphic representation of the inertia force is possible if Eq. (14.23) is rearranged as

$$F = m_B r \omega^2 \cos \omega t + m_B r \omega^2 \frac{r}{l} \cos 2\omega t, \quad (14.41)$$

where we write $F = F_{21}^x$ for simplicity of notation. The first term of Eq. (14.41) is represented by the x projection of a vector of length $m_B r \omega^2$ rotating at ω rad/s. This is the primary part of the inertia force. The second term is similarly represented by the x projection of a vector of length $m_B r \omega^2 (r/l)$ rotating at 2ω rad/s; this is the secondary part. Such a diagram is shown in Fig. 14.22 for $r/l = 1/4$. The total inertia or shaking force is the algebraic sum of the horizontal projections of these two vectors.

14.9 COMPUTATION HINTS

This section contains suggestions for using a computer or programmable calculator in solving the dynamics of an engine mechanism. Many of the ideas, however, are also useful for readers using nonprogrammable calculators, as well as for checking purposes.

Indicator Diagrams It would be very convenient if a subprogram for computing the gas forces could be devised and the results used directly in a main program to compute all the resultant bearing forces and crankshaft torques. Unfortunately, the theoretical indicator diagram must be manipulated by hand to obtain a reasonable approximation to the

experimental data. This manipulation can be done graphically or with a computer having a graphic display. The procedure is illustrated by the following example.

EXAMPLE 14.1

Determine the pressure-versus-piston displacement relation for a six-cylinder engine having a displacement of 2.30 L, a compression ratio of 8, and a brake power of 42.5 kW at 2 400 rev/min. Use a mechanical efficiency of 75%, a card factor of 0.85, a suction pressure of 100 kPa, and a polytropic exponent of 1.30.

SOLUTION

Rearranging Eq. (14.2), we find the brake mean effective pressure as follows:

$$p_b = \frac{(42.5 \text{ kW}) (1\,000 \text{ (N} \cdot \text{m/s)/kW}) (60 \text{ s/min})}{(0.002\,30 \text{ m}^3) (2\,400 \text{ rev/min}) (0.5 \text{ cycle/rev}) (1\,000 \text{ (N/m}^2\text{)/kPa})} = 923.9 \text{ kPa.}$$

Then, from Eq. (14.3), the indicated mean effective pressure is

$$p_i = \frac{p_b}{e_m} = \frac{923.9 \text{ kPa}}{0.75} = 1\,232 \text{ kPa.}$$

Next, we determine the pressure, p_4 , on the theoretical diagram of Fig. 14.12. Substituting the information into Eq. (14.7) gives

$$p_4 = (1.3 - 1) \left(\frac{8 - 1}{8^{1.3} - 8} \right) \frac{1\,232 \text{ kPa}}{0.85} + 100 \text{ kPa} = 355 \text{ kPa.}$$

The volume difference, $v_1 - v_2$, in Fig. 14.12 is the volume swept out by the piston. Therefore,

$$v_1 - v_2 = l a = \frac{2.30 \text{ L}}{6 \text{ cyl}} = 0.383 \text{ L/cyl.}$$

Substituting this displacement volume and the compression ratio $R = 8$ into Eq. (b) of Sec. 14.2 gives

$$v_1 = \frac{R(v_1 - v_2)}{R - 1} = \frac{8(0.383 \text{ L})}{8 - 1} = 0.438 \text{ L.}$$

Therefore,

$$v_2 = 0.438 \text{ L} - 0.383 \text{ L} = 0.055 \text{ L.}$$

Then the percentage clearance C is

$$C = \frac{v_2}{v_1 - v_2} (100)\% = \frac{0.055 \text{ L}}{0.383 \text{ L}} (100)\% = 14.4\%.$$

Expressing volumes as percentages of the displacement volume enables us to write Eq. (14.1) in the form

$$p_x(X + C)^k = p_1(100 + C)^k,$$

where X is the percentage of piston travel measured from the head end of the stroke. Thus, the formula

$$p_x = p_1 \left(\frac{100 + C}{X + C} \right)^k = 1\,232 \text{ kPa} \left(\frac{114.4}{X + 14.4} \right)^{1.3} \quad (1)$$

is used to compute the pressure during the compression stroke for any piston position between $X = 0$ and $X = 100\%$. For the expansion stroke, Eq. (14.1) becomes

$$p_x = p_4 \left(\frac{100 + C}{X + C} \right)^k = 355 \text{ kPa} \left(\frac{114.4}{X + 14.4} \right)^{1.3}. \quad (2)$$

Equations (1) and (2) are easy to program for machine computation. The results can be displayed and recorded, or printed, for graphic use. Alternatively, the results can be displayed for hand calculation.

Figure 14.23 shows the plotted results of the computation using $\Delta X = 5\%$ increments. Note particularly how the results are rounded to obtain a smooth indicator diagram. This rounding, of course, produces results that are not exactly duplicated in subsequent trials. The greatest differences occur in the vicinity of point B .

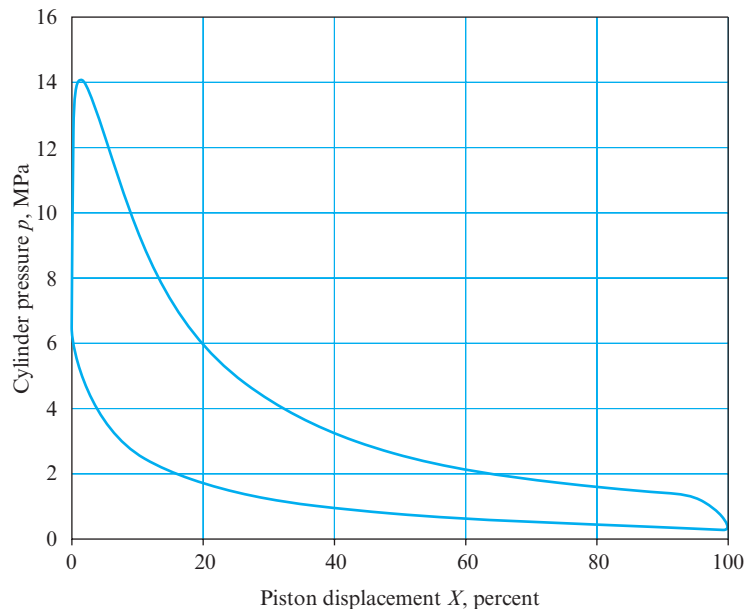


Figure 14.23 The diagram was rounded by hand. The peak cylinder pressure is about 75% of the maximum computed pressure at the beginning of the expansion stroke.

Force Analysis In a computer analysis the values of pressure are read from a diagram like Fig. 14.23. Most analysts prefer to tabulate these data; a table is usually constructed with the first column containing values of the crank angle, ωt . For a four-stroke engine, values of this angle are entered from 0° to 720° .

Values of the piston position, x , corresponding to each crank angle, ωt , are obtained from Eq. (3.15). Then, the corresponding piston displacement, X , in percentage is obtained from the equation

$$X = \frac{r + l - x}{2r}(100\%). \quad (14.42)$$

Some care must be taken in tabulating X and the corresponding pressures. Then, the gas forces corresponding to each value of ωt are computed using the piston area.

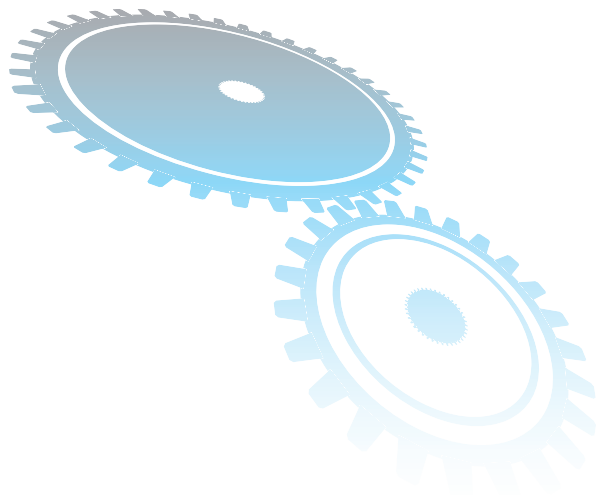
The remainder of the analysis is perfectly straightforward; we use Eqs. (14.8c), (14.10), and (14.36) through (14.40), in that order.

PROBLEMS

- 14.1** A one-cylinder, four-stroke engine has a compression ratio of 7.6 and develops brake power of 2.25 kW at 3 000 rev/min. The crank length is 22 mm with a 60-mm bore. Develop and plot a rounded indicator diagram using a card factor of 0.90, a mechanical efficiency of 72%, a suction pressure of 100 kPa, and a polytropic exponent of 1.30.
- 14.2** Construct a rounded indicator diagram for a four-cylinder, four-stroke gasoline engine having an 85-mm bore, a 90-mm stroke, and a compression ratio of 6.25. The operating conditions to be used are 22.4 kW at 1 900 rev/min. Use a mechanical efficiency of 72%, a card factor of 0.90, a suction pressure of 100 kPa, and a polytropic exponent of 1.30.
- 14.3** Construct an indicator diagram for a V6 four-stroke gasoline engine having a 100-mm bore, a 90-mm stroke, and a compression ratio of 8.40. The engine develops 150 kW at 4 400 rev/min. Use a mechanical efficiency of 72%, a card factor of 0.88, a suction pressure of 100 kPa, and a polytropic exponent of 1.30.
- 14.4** A single-cylinder, two-stroke gasoline engine develops 30 kW at 4 500 rev/min. The engine has an 80-mm bore, a stroke of 70 mm, and a compression ratio of 7.0. Develop a rounded indicator diagram for this engine using a card factor of 0.990, a mechanical efficiency of 65%, a suction pressure of 100 kPa, and a polytropic exponent of 1.30.
- 14.5** The engine of Prob. 14.1 has a connecting rod 80 mm long and a mass of 0.100 kg, with the mass center 10 mm from the crankpin end. Piston mass is 0.180 kg. Find the bearing reactions and the crankshaft torque during the expansion stroke corresponding to a piston displacement of $X = 30\%$ ($\omega t = 60^\circ$). To find p_e , see the answer to Prob. 14.1 in Appendix B.
- 14.6** Repeat Prob. 14.5, but do the computations for the compression cycle ($\omega t = 660^\circ$).
- 14.7** Make a complete force analysis of the engine of Prob. 14.5. Plot a graph of the crankshaft torque versus crank angle for 720° of crank rotation.
- 14.8** The engine of Prob. 14.3 uses a connecting rod 300 mm long. The masses are $m_{3A} = 0.80$ kg, $m_{3B} = 0.38$ kg, and $m_4 = 1.64$ kg. Find all the bearing reactions and the crankshaft torque for one cylinder of the engine during the expansion stroke at a piston displacement of $X = 30\%$ ($\omega t = 63.2^\circ$). The pressure should be obtained from the indicator diagram, Fig. AP14.3 in Appendix B.

- 14.9** Repeat Prob. 14.8, but do the computations for the same position in the compression cycle ($\omega t = 656.8^\circ$).
- 14.10** Additional data for the engine of Prob. 14.4 are $l_3 = 110$ mm, $R_{G_3A} = 15$ mm, $m_4 = 0.24$ kg, and $m_3 = 0.13$ kg. Make a complete force analysis of the engine, and plot a graph of the crankshaft torque versus crank angle for 360° of crank rotation.
- 14.11** The four-stroke engine of Prob. 14.1 has a stroke of 66 mm and a connecting rod length of 183 mm. The mass of the rod is 0.386 kg, and the center of mass is 42 mm from the crankpin. The piston assembly has mass of 0.576 kg. Make a complete force analysis for one cylinder of this engine for 720° of crank rotation. Use 110 kPa for the exhaust pressure and 70 kPa for the suction pressure. Plot a graph to show the variation of the crankshaft torque with the crank angle. Use Fig. 14.23 for the pressures.

15 Balancing



Balancing is the process of correcting or eliminating unwanted inertia forces and moments in rotating machinery. In previous chapters, we have seen that shaking forces on the frame can vary significantly during a cycle of operation. Such forces can cause vibrations that at times may reach dangerous amplitudes. Even if they are not dangerous, vibrations increase component stresses and subject bearings to repeated loads that may cause parts to fail prematurely by fatigue. Thus, in the design of machinery, it is not sufficient merely to avoid operation near the critical speeds; we must eliminate, or at least reduce, the dynamic forces that produce these vibrations in the first place.

Tolerances used in the manufacture of machinery are typically set as tight as possible without prohibitively increasing the cost of manufacture. In general, however, it is more economic to produce parts that are not exactly true and then to subject them to a balancing procedure than it is to produce such perfect parts that no correction is needed. Therefore, each part produced is an individual case, in that no two parts can normally be expected to require the same corrective measures. Thus, determining the unbalance and the required correction is the principal topic in the study of balancing.

15.1 STATIC UNBALANCE

Figure 15.1 shows a disk-shaft combination resting on rigid horizontal rails so that the shaft, which is assumed to be perfectly straight, can roll without friction. A reference system xyz is attached to the disk and moves with it (with the z axis as the axis of the shaft). Simple experiments to determine whether the disk is statically unbalanced can be conducted as follows. We can roll the disk gently by hand and permit it to coast until it comes to rest. Then we can mark with chalk the lowest point of the periphery of the disk. After we repeat this four or five times, if the chalk marks are scattered at different locations

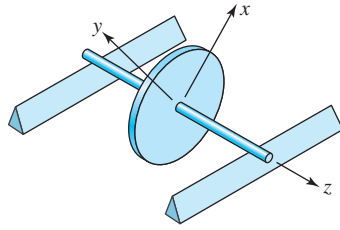


Figure 15.1 Static balancing experiment.

around the periphery, the disk is in static balance. If all of the chalk marks are clustered, the disk is statically unbalanced, which means that the axis of the shaft and the center of mass of the disk are not coincident. The position of the chalk marks with respect to the xy axes indicates the angular location of the unbalance but *not* the amount.

It is unlikely that any of the marks will be located 180° from the remaining cluster, although it is theoretically possible to obtain static equilibrium with the unbalance above the shaft axis. If static unbalance is found to exist, it can be corrected by drilling out material at the chalk mark cluster or by adding mass to the periphery 180° from the cluster. Since the amount of unbalance is unknown, the correction magnitude must be found by trial and error.

15.2 EQUATIONS OF MOTION

If an unbalanced disk and shaft of mass m is mounted in bearings and caused to rotate at an angular speed of ω then an inertia force of magnitude $mr_G\omega^2$ exists (where r_G , that is, the eccentricity, is the distance from the shaft axis, O , to the mass center, G , of the disk), as shown in Fig. 15.2. The figure also shows the force acting on the shaft that produces the rotating bearing reactions F_A and F_B .

To determine the equation of motion of the system, we specify m as the total mass and m_u as the unbalanced mass. We also let k signify the shaft stiffness, a value that describes the amount of force necessary to bend the shaft a unit distance when applied at O . Thus, k has units of pounds per inch or Newtons per meter. Let c be the coefficient of viscous damping as defined in Eq. (13.2). Selecting any x coordinate normal to the shaft axis, we

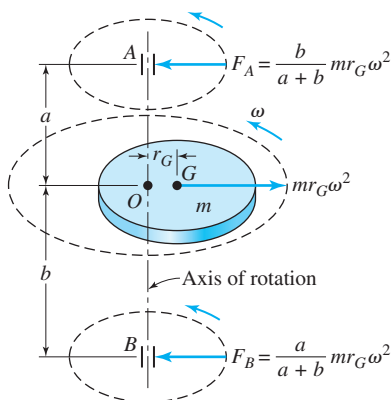


Figure 15.2 Unbalanced disk and shaft.

can now write

$$\sum F_O = -kx - c\dot{x} - m\ddot{x} + m_u r_G \omega^2 \cos \omega t = 0,$$

or, upon rearranging,

$$m\ddot{x} + c\dot{x} + kx = m_u r_G \omega^2 \cos \omega t. \quad (a)$$

Recall that the solution to this differential equation was studied in Secs. 13.12 and 13.13. From Eq. (13.55), the deflection of the shaft can be written as

$$x = \frac{m_u r_G \omega^2 \cos(\omega t - \phi)}{\sqrt{(k - m\omega^2)^2 + c^2 \omega^2}}, \quad (b)$$

where ϕ is the phase angle between force $m_u r_G \omega^2$ and amplitude X of the shaft vibration. The *phase angle* is given by Eq. (13.50) and is repeated here for convenience

$$\phi = \tan^{-1} \left(\frac{c\omega}{k - m\omega^2} \right). \quad (c)$$

Certain simplifications can be made in Eq. (b) to clarify its meaning. First, consider the term $k - m\omega^2$ in the denominator. If this term were zero, the amplitude of x would be very large, since it would be limited only by the damping constant, c , which is usually very small. The value of ω that makes the term $k - m\omega^2$ equal to zero is denoted as ω_n and is called the *natural angular velocity*, *critical speed*, or *natural frequency*. Therefore,

$$\omega_n = \sqrt{\frac{k}{m}}, \quad (15.1)$$

which agrees with the definition in Sec. 13.2, Eq. (13.8).

In the study of free or unforced vibrations (Sec. 13.8), it is found that a certain value of the viscous damping factor, c , results in no vibration at all. This special value is called the *critical coefficient of viscous damping* [Eq. (13.27)], that is,

$$c_c = 2m\omega_n. \quad (15.2)$$

The *damping ratio* ζ is the ratio of the actual to the critical damping, Eq. (13.28), that is,

$$\zeta = \frac{c}{c_c} = \frac{c}{2m\omega_n}. \quad (15.3)$$

For most machine systems in which damping is not deliberately introduced, ζ is generally in the range $0.015 \leq \zeta \leq 0.120$.

Recall that the deflection can be written from Eq. (13.15) or Eq. (13.23) as

$$x = X \cos(\omega t - \phi). \quad (d)$$

Substituting this equation into Eq. (b), dividing the numerator and denominator of the result by k , and introducing Eqs. (15.1) and (15.3), we obtain the ratio given by Eq. (13.56),

that is,

$$\frac{mX}{m_u r_G} = \frac{(\omega/\omega_n)^2}{\sqrt{(1 - \omega^2/\omega_n^2)^2 + (2\zeta\omega/\omega_n)^2}}, \quad (15.4)$$

where $r_G = e$. This is the equation for the amplitude ratio of the vibration of a rotating disk-shaft combination. If we neglect damping, that is, set $\zeta = 0$, and let $m = m_u$, then the amplitude of the vibration corresponding to any frequency ratio ω/ω_n can be written as

$$X = r_G \frac{(\omega/\omega_n)^2}{1 - (\omega/\omega_n)^2}. \quad (15.5)$$

Figure 15.3 is a plot of Eq. (15.5) where the amplitude is the ordinate and the frequency ratio is the abscissa. When rotation is just beginning, ω is much less than ω_n , and the graph indicates that the amplitude of the vibration is very small. As the shaft speed increases, the amplitude also increases and theoretically becomes infinite at the critical speed. As the shaft goes through the critical speed, the amplitude changes to a negative value and decreases as the shaft speed increases further. In the high-speed range, the disk is rotating about its own center of mass, which has then become coincident with the centerline of the bearings.

In summary, we note the following interesting conclusions:

- The amplitude approaches infinity when the denominator becomes zero, that is, when the speed of rotation is equal to the natural frequency.
- Regardless of how small r_G is (even if it is zero), trouble can always be expected when $\omega = \omega_n$.

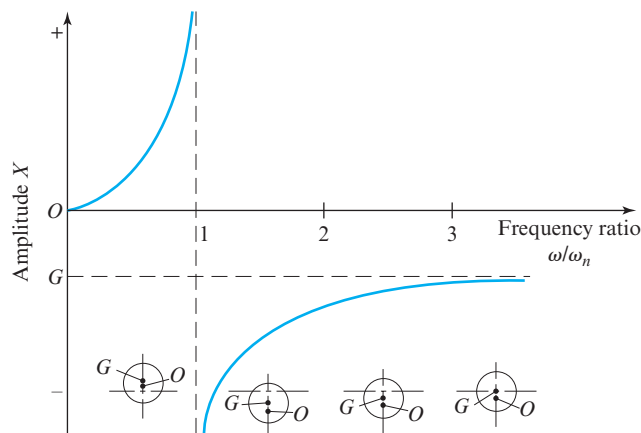


Figure 15.3 The small figures below the graph indicate the relative locations of points O and G for four typical frequency ratios.

- (c) The amplitude switches from positive to negative as the speed passes through the natural frequency.
- (d) The amplitude never returns to zero, no matter how much the shaft speed is increased, but reaches a limiting value of $-r_G$.

The discussion in Secs. 15.1 and 15.2 demonstrates that statically unbalanced rotating systems produce undesirable vibrations and bearing reactions. Using static balancing equipment, the eccentricity can be reduced, but it is impossible in practice to make it exactly zero; [see (b)]. When the operating speed must be higher than the natural frequency, the machine should be designed so that, on startup, it passes through the natural frequency quickly to avoid the buildup of dangerous vibrations.

15.3 STATIC BALANCING MACHINES

The first purpose of a balancing machine is to indicate whether a mechanical part is in balance. If it is out of balance, the machine should measure the unbalance by indicating its *magnitude* and *location*.

Static balancing machines are used only for parts whose axial dimensions are small, such as gears, fans, and impellers, and the machines are often called *single-plane balancers*, since the mass must lie practically in a single plane. In the sections to follow, we discuss balancing in several planes, but it is important to note here that if several disks are to be mounted upon a shaft that is to rotate, the parts should be individually statically balanced before mounting. Although it is possible, instead, to balance the assembly in two planes after the parts are mounted, additional external moments inevitably come into existence when this is done.

Static balancing is essentially a weighing process in which the part is acted upon either by gravity or by inertia force. We have seen that the disk and shaft of Sec. 15.1 could be balanced by placing it on parallel rails, rocking it, and permitting it to seek equilibrium. In this case, the location of the unbalance is determined through the force of gravity. Another method of balancing the disk is to rotate it at a predetermined speed. Then, the bearing reactions can be measured and their magnitudes used to indicate the amount of unbalance. Since the part is rotating while the measurements are taken, a stroboscope is used to indicate the location of the required correction.

When machine parts are manufactured in quantity, a balancer is required that will measure both the amount and location of the unbalance and give the correction directly and quickly. Also, time can be saved if it is not necessary to make the part rotate. Such a balancing machine is shown in Fig. 15.4. This machine is essentially a pendulum that can tilt in any direction, as shown by the schematic drawing of Fig. 15.5a.

When an unbalanced specimen is mounted on the balancing machine, the pendulum tilts. The direction of the tilt gives the location of the unbalance, whereas the tilt angle, θ (Fig. 15.5b), indicates the magnitude of the unbalance. Some damping is employed to reduce oscillations of the pendulum.



Figure 15.4 Aircraft propeller assembly balancer. (Courtesy of Schenck Ro Tec Corporation, Auburn Hills, MI.)

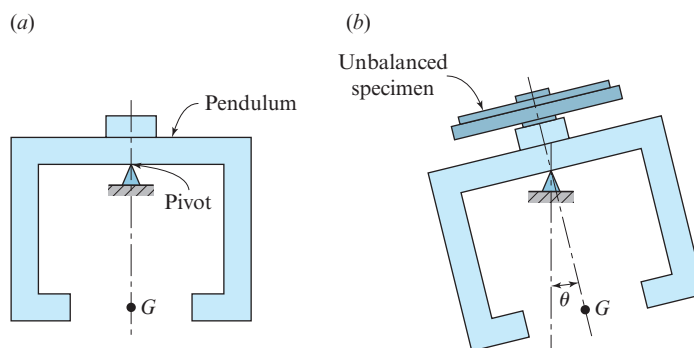


Figure 15.5 Operation of a static balancing machine.

15.4 DYNAMIC UNBALANCE

Figure 15.6 shows a long rotor with distributed mass that is mounted in bearings at A and B . Suppose that two equal masses m_1 and m_2 are placed at opposite ends of the rotor and at equal distances r_1 and r_2 from the axis of rotation. Since the masses are equal and on opposite sides of the rotational axis, placing the rotor on rails (as described in Sec. 15.1) demonstrates that it is statically balanced in all angular orientations. If the rotor is placed in bearings and caused to rotate at an angular velocity ω , then m_1 and m_2 create inertia forces of magnitudes $m_1 r_1 \omega^2$ and $m_2 r_2 \omega^2$, respectively. These inertia forces produce bearing reaction forces $\mathbf{F}_A$ and $\mathbf{F}_B$ that rotate with the rotor. Therefore, a rotating part may be statically balanced and, at the same time, dynamically unbalanced.

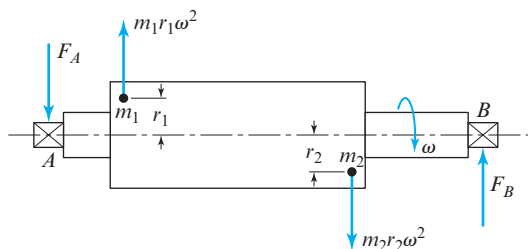


Figure 15.6 If $m_1 = m_2$ and $r_1 = r_2$, the rotor is statically balanced but dynamically unbalanced.

For the single-mass system in Fig. 15.7a, both bearing reaction forces are in the same plane and in the same direction. Therefore, the system is statically unbalanced. For the two-mass system (with equal masses m) in Fig. 15.7b, the static unbalance creates a couple tending to turn the shaft end over end. The shaft is in static equilibrium because of the opposite couple formed by the bearing reactions. Note that the bearing reaction forces F_A and F_B are still in the same plane, however, they now act in opposite directions. Therefore, the system is dynamically unbalanced. Note also that a rotating-mass system may be modeled as either a discrete or a distributed system. Procedures for static and dynamic balancing of such systems are presented in the following section.

In the general case, accurate distribution of the mass along the axis of a part depends upon the configuration of the part, but inaccuracies occur in machining and also in casting and in forging. Other errors or unbalances may be caused by improper boring, keys, and assembly. It is the designer's responsibility to design a rotating part so that a line joining all mass centers is a straight line coinciding with the axis of rotation. However, perfect parts and perfect assembly are seldom attained; consequently, a line from one end of the part to the other, joining all mass centers, is usually a spatial curve that may only occasionally cross or coincide with the axis of rotation. Before balancing, therefore, a part is usually out of balance, both statically and dynamically. This is the most general kind of unbalance, and when the part is supported by two bearings, we can expect the magnitudes as well as the directions of the rotating bearing reactions to be different.

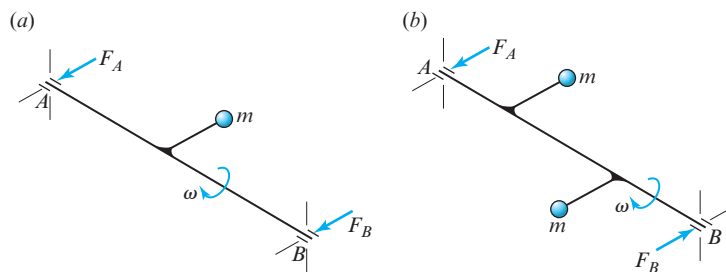


Figure 15.7 (a) Static unbalance; (b) dynamic unbalance.

15.5 ANALYSIS OF UNBALANCE

In this section we see how to analyze either a discrete mass or a distributed mass unbalanced rotating system and determine proper corrections using graphic methods, vector methods, and numeric methods. This section shows that a system can be balanced either by adding or removing correcting masses.

Graphic Analysis The two equations

$$\sum \mathbf{F} = \mathbf{0} \quad \text{and} \quad \sum \mathbf{M} = \mathbf{0} \quad (a)$$

are used to determine the amounts and orientations of the corrections. We begin by noting that the magnitude of the inertia force, $m\mathbf{R}\omega^2$, of an eccentric rotating mass is proportional to its $m\mathbf{R}$ product. Thus, for Fig. 15.8a, vector quantities (proportional to the inertia force of each of the three masses) $m_1\mathbf{R}_1$, $m_2\mathbf{R}_2$, and $m_3\mathbf{R}_3$, act in radial directions, as indicated. The first of Eqs. (a) is applied by constructing a force polygon as shown in Fig. 15.8b. Since this polygon requires another vector, $m_c\mathbf{R}_c$, for closure, the magnitude of the correction is m_cR_c , and its orientation is parallel to vector $m_c\mathbf{R}_c$. The three masses of Fig. 15.8 are assumed to rotate in a single plane, and so this is the correction for a case of static unbalance.

When the rotating masses are in different planes, both of Eqs. (a) must be used. Note that taking the sum of moments about a point in one of the correction planes eliminates that unknown correction mass. This is illustrated in the following example.

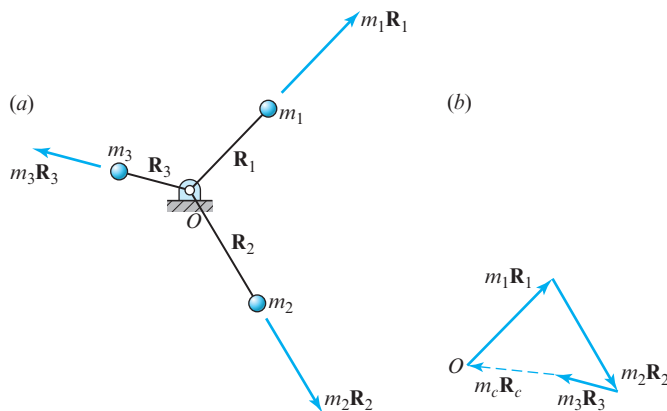


Figure 15.8 (a) Three-mass system rotating in a single plane; (b) inertia force polygon.

EXAMPLE 15.1

The end view of a shaft with masses m_1, m_2 , and m_3 at radial distances R_1, R_2 , and R_3 , respectively, is shown in Fig. 15.9a. A side view of the shaft showing left and right correction planes and axial distances l_1, l_2 , and l_3 to the three masses is shown in Fig. 15.9b. Determine the magnitudes and angular orientations of the corrections to be added in each correction plane.

SOLUTION

Applying the second of Eqs. (a), the sum of the moments of the inertia forces about point A (in order to eliminate the moment of the left correction mass) gives

$$\sum \mathbf{M}_A = m_1 l_1 \mathbf{R}_1 + m_2 l_2 \mathbf{R}_2 + m_3 l_3 \mathbf{R}_3 + m_R l_R \mathbf{R}_R = \mathbf{0}. \quad (1)$$

This is a vector equation in which the directions of vectors can be drawn parallel, respectively, to vectors $\mathbf{R}_j$ in Fig. 15.9a. Consequently, the moment polygon in Fig. 15.9c can be constructed. The closing vector, $m_R l_R \mathbf{R}_R$, gives the magnitude and orientation of the correction required for the right correction plane.

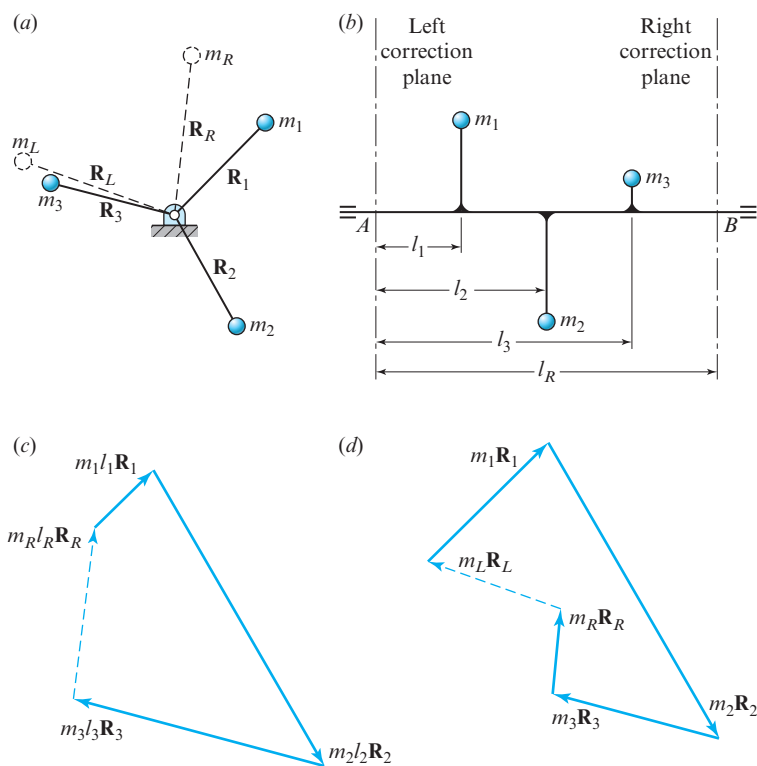


Figure 15.9 Graphic analysis of unbalance.

Although Fig. 15.9c has been called a *moment polygon*, it is worth noting that the vectors making up this polygon consist of quantities proportional to the moment magnitudes and are drawn in the position vector directions. A true moment polygon would require rotating this polygon 90° cw, since each moment vector is properly expressed as $I\hat{\mathbf{k}} \times m\mathbf{R}\omega^2$.

Applying the first of Eqs. (a),

$$\sum \mathbf{F} = m_1 \mathbf{R}_1 + m_2 \mathbf{R}_2 + m_3 \mathbf{R}_3 + m_R \mathbf{R}_R + m_L \mathbf{R}_L = \mathbf{0}. \quad (2)$$

By constructing the force polygon of Fig. 15.9d, this equation is solved for the left correction, $m_L \mathbf{R}_L$.

After solving for the $m_R \mathbf{R}_R$ and $m_L \mathbf{R}_L$ products, the separate quantities m_R , R_R , m_L , and R_L can be determined, since, in general, two of these magnitudes are specified in the problem statement or are chosen by the designer.

Vector Analysis This section shows how to analyze an unbalanced rotating system and determine proper corrections using a vector-algebraic approach. The following two examples illustrate this method.

EXAMPLE 15.2

Figure 15.10 represents a rotating system that has been idealized for illustrative purposes. A weightless shaft that is supported in bearings at A and B rotates with an angular velocity $\omega = 100\hat{\mathbf{i}}$ rad/s. Three mass particles, with weights $w_1 = 2$ oz, $w_2 = 1$ oz, and $w_3 = 1.5$ oz, are connected to the shaft and rotate with it, causing unbalance. Determine the magnitudes and orientations of the bearing reactions at A and B .

SOLUTION

The inertia force caused by each rotating mass particle is

$$m_1 r_1 \omega^2 = \frac{(2 \text{ oz})(3 \text{ in})(100 \text{ rad/s})^2}{(386 \text{ in/s}^2)(16 \text{ oz/lb})} = 9.72 \text{ lb},$$

$$m_2 r_2 \omega^2 = \frac{(1 \text{ oz})(2 \text{ in})(100 \text{ rad/s})^2}{(386 \text{ in/s}^2)(16 \text{ oz/lb})} = 3.24 \text{ lb},$$

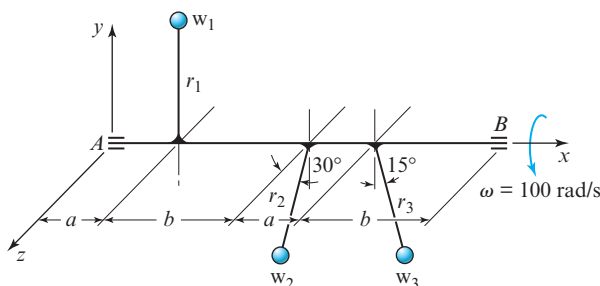


Figure 15.10 $a = 1$ in, $b = 2$ in, $r_1 = 3$ in, $r_2 = 2$ in, and $r_3 = 2.5$ in.

and

$$m_3 r_3 \omega^2 = \frac{(1.5 \text{ oz})(2.5 \text{ in})(100 \text{ rad/s})^2}{(386 \text{ in/s}^2)(16 \text{ oz/lb})} = 6.07 \text{ lb}.$$

These three forces are each parallel to the yz plane and can be written in vector form as

$$\mathbf{F}_1 = m_1 r_1 \omega^2 \angle \theta_1 = 9.72 \text{ lb} \angle 0^\circ = 9.72 \hat{\mathbf{j}} \text{ lb},$$

$$\mathbf{F}_2 = m_2 r_2 \omega^2 \angle \theta_2 = 3.24 \text{ lb} \angle 120^\circ = -1.62 \hat{\mathbf{j}} + 2.81 \hat{\mathbf{k}} \text{ lb},$$

and

$$\mathbf{F}_3 = m_3 r_3 \omega^2 \angle \theta_3 = 6.07 \text{ lb} \angle 195^\circ = -5.86 \hat{\mathbf{j}} - 1.57 \hat{\mathbf{k}} \text{ lb},$$

where the angles θ_1, θ_2 , and θ_3 are measured counterclockwise from the y axis when viewed from the positive x axis. The moments of these forces taken about the bearing at A must be balanced by the moment of the bearing reaction at B . Therefore,

$$\begin{aligned} \sum \mathbf{M}_A = & (1 \hat{\mathbf{i}} \text{ in}) \times (9.72 \hat{\mathbf{j}} \text{ lb}) + (3 \hat{\mathbf{i}} \text{ in}) \times (-1.62 \hat{\mathbf{j}} + 2.81 \hat{\mathbf{k}} \text{ lb}) \\ & + (4 \hat{\mathbf{i}} \text{ in}) \times (-5.86 \hat{\mathbf{j}} - 1.57 \hat{\mathbf{k}} \text{ lb}) + (6 \hat{\mathbf{i}} \text{ in}) \times (\mathbf{F}_B) = \mathbf{0}. \end{aligned}$$

Solving this equation, the bearing reaction at B is

$$\mathbf{F}_B = 3.10 \hat{\mathbf{j}} - 0.36 \hat{\mathbf{k}} \text{ lb} = 3.12 \text{ lb} \angle -6.6^\circ. \quad \text{Ans.}$$

To find the bearing reaction at A , we repeat the analysis. Taking moments about B gives

$$\begin{aligned} \sum \mathbf{M}_B = & (-2 \hat{\mathbf{i}} \text{ in}) \times (-5.86 \hat{\mathbf{j}} - 1.57 \hat{\mathbf{k}} \text{ lb}) + (-3 \hat{\mathbf{i}} \text{ in}) \times (-1.62 \hat{\mathbf{j}} + 2.81 \hat{\mathbf{k}} \text{ lb}) \\ & + (-5 \hat{\mathbf{i}} \text{ in}) \times (9.72 \hat{\mathbf{j}} \text{ lb}) + (-6 \hat{\mathbf{i}} \text{ in}) \times (\mathbf{F}_A) = \mathbf{0}. \end{aligned}$$

Then solving this equation, the bearing reaction at A is

$$\mathbf{F}_A = -5.34 \hat{\mathbf{j}} - 0.88 \hat{\mathbf{k}} \text{ lb} = 5.41 \text{ lb} \angle 189.4^\circ. \quad \text{Ans.}$$

Note that these are rotating reactions and that static (stationary) components caused by gravity are not included.

EXAMPLE 15.3

The angular speed of the system shown in Fig. 15.11 is 750 rev/min. Determine (a) the magnitudes and orientations of the bearing reactions at A and B , and (b) the magnitude and location of a correcting (or balancing) mass to be added at a radius of 0.25 m.

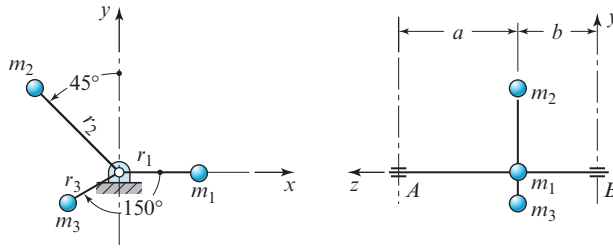


Figure 15.11 $a = 0.3$ m, $b = 0.2$ m, $r_1 = 0.2$ m, $r_2 = 0.3$ m, $r_3 = 0.15$ m, $m_1 = 12$ kg, $m_2 = 3$ kg, and $m_3 = 10$ kg.

SOLUTION

- (a) The angular velocity of this system is

$$\omega = (750 \text{ rev/min}) (2\pi \text{ rad/rev}) / (60 \text{ s/min}) = 78.54 \text{ rad/s}.$$

The inertia forces caused by the rotating masses are

$$F_1 = m_1 r_1 \omega^2 = (12 \text{ kg}) (0.2 \text{ m}) (78.54 \text{ rad/s})^2 = 14.80 \text{ kN},$$

$$F_2 = m_2 r_2 \omega^2 = (3 \text{ kg}) (0.3 \text{ m}) (78.54 \text{ rad/s})^2 = 5.55 \text{ kN},$$

and

$$F_3 = m_3 r_3 \omega^2 = (10 \text{ kg}) (0.15 \text{ m}) (78.54 \text{ rad/s})^2 = 9.25 \text{ kN}.$$

These forces can be written in vector form as

$$\mathbf{F}_1 = 14.80 \text{ kN} \angle 0^\circ = 14.80 \hat{\mathbf{i}} \text{ kN},$$

$$\mathbf{F}_2 = 5.55 \text{ kN} \angle 135^\circ = -3.93 \hat{\mathbf{i}} + 3.93 \hat{\mathbf{j}} \text{ kN},$$

$$\mathbf{F}_3 = 9.25 \text{ kN} \angle -150^\circ = -8.01 \hat{\mathbf{i}} - 4.63 \hat{\mathbf{j}} \text{ kN}.$$

To find the bearing reaction at B we take moments about bearing A . This equation is written as

$$\begin{aligned} \sum \mathbf{M}_A = & (-0.3 \hat{\mathbf{k}} \text{ m}) \times \left[(14.80 \hat{\mathbf{i}} \text{ kN}) + (-3.93 \hat{\mathbf{i}} + 3.93 \hat{\mathbf{j}} \text{ kN}) \right. \\ & \left. + (-8.01 \hat{\mathbf{i}} - 4.63 \hat{\mathbf{j}} \text{ kN}) \right] + (-0.5 \hat{\mathbf{k}} \text{ m}) \times \mathbf{F}_B = \mathbf{0}. \end{aligned}$$

Taking the cross products and rearranging gives

$$(0.5 \hat{\mathbf{k}} \text{ m}) \times \mathbf{F}_B = -0.21 \hat{\mathbf{i}} - 0.86 \hat{\mathbf{j}} \text{ kN} \cdot \text{m}.$$

Solving this equation for $\mathbf{F}_B$ gives

$$\mathbf{F}_B = -1.72\hat{\mathbf{i}} + 0.42\hat{\mathbf{j}} \text{ kN} = 1.77 \text{ kN} \angle 166.28^\circ. \quad \text{Ans.}$$

Summing forces gives

$$\mathbf{F}_A + \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 + \mathbf{F}_B = \mathbf{0}.$$

Therefore, the reaction at A is

$$\begin{aligned} \mathbf{F}_A = & -\left(14.80\hat{\mathbf{i}} \text{ kN}\right) - \left(-3.93\hat{\mathbf{i}} + 3.93\hat{\mathbf{j}} \text{ kN}\right) - \left(-8.00\hat{\mathbf{i}} - 4.63\hat{\mathbf{j}} \text{ kN}\right) \\ & - \left(-1.72\hat{\mathbf{i}} + 0.42\hat{\mathbf{j}} \text{ kN}\right) \end{aligned}$$

or

$$\mathbf{F}_A = -1.14\hat{\mathbf{i}} + 0.28\hat{\mathbf{j}} \text{ kN} = 1.17 \angle 166.20^\circ \text{ kN}. \quad \text{Ans.}$$

(b) Let $\mathbf{F}_C$ be the correcting force. Then, for zero bearing reactions

$$\sum \mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 + \mathbf{F}_C = \mathbf{0}.$$

Thus,

$$\begin{aligned} \mathbf{F}_C = & -\left(14.80\hat{\mathbf{i}} \text{ kN}\right) - \left(-3.93\hat{\mathbf{i}} + 3.93\hat{\mathbf{j}} \text{ kN}\right) - \left(-8.00\hat{\mathbf{i}} - 4.63\hat{\mathbf{j}} \text{ kN}\right) \\ = & -2.86\hat{\mathbf{i}} + 0.70\hat{\mathbf{j}} \text{ kN} = 2.94 \text{ kN} \angle 166.25^\circ. \end{aligned}$$

Therefore, the correcting mass should be added at 166.25° .

Ans.

The correcting mass at a radius of 0.25 m is

$$m_C = \frac{F_C}{r_C \omega^2} = \frac{2.94 (10)^3 \text{ N}}{0.25 \text{ m} (78.54 \text{ rad/s})^2} = 1.91 \text{ kg}. \quad \text{Ans.}$$

Scalar Equations For a rotating system with n discrete mass particles and two correcting planes ($j = 1, 2$), we can write the following four scalar equations:

$$\sum_{i=1}^n m_i r_i \cos \phi_i + \sum_{j=1}^2 m_{cj} r_{cj} \cos \phi_{cj} = 0, \quad (15.6a)$$

$$\sum_{i=1}^n m_i r_i \sin \phi_i + \sum_{j=1}^2 m_{cj} r_{cj} \sin \phi_{cj} = 0, \quad (15.6b)$$

$$\sum_{i=1}^n z_i m_i r_i \cos \phi_i + \sum_{j=1}^2 z_{cj} m_{cj} r_{cj} \cos \phi_{cj} = 0, \quad (15.7a)$$

$$\sum_{i=1}^n z_i m_i r_i \sin \phi_i + \sum_{j=1}^2 z_{cj} m_{cj} r_{cj} \sin \phi_{cj} = 0. \quad (15.7b)$$

Note that the equations do not depend on the angular velocity of the rotating system; that is, if the system is balanced for one speed, then it is balanced for all speeds.

For a distributed mass system and two correcting planes, we can write the following four scalar equations:

$$F_{A21}^x + F_{B21}^x + \omega^2 \sum_{j=1}^2 m_{cj} r_{cj} \cos \phi_{cj} = 0, \quad (15.8a)$$

$$F_{A21}^y + F_{B21}^y + \omega^2 \sum_{j=1}^2 m_{cj} r_{cj} \sin \phi_{cj} = 0, \quad (15.8b)$$

$$z_A F_{A21}^x + z_B F_{B21}^x + \omega^2 \sum_{j=1}^2 z_j m_{cj} r_{cj} \cos \phi_{cj} = 0, \quad (15.9a)$$

$$z_A F_{A21}^y + z_B F_{B21}^y + \omega^2 \sum_{j=1}^2 z_j m_{cj} r_{cj} \sin \phi_{cj} = 0. \quad (15.9b)$$

A Direct Method of Dynamic Balancing To avoid solving four equations in four unknowns, that is Eqs. (15.6) and (15.7) or Eqs. (15.8) and (15.9), we can take moments about one correction plane and solve for two unknowns. Then, we can take moments about the other correction plane and solve for the remaining two unknowns. Recall that if a rotating system is in dynamic balance, then it is automatically in static balance. However, if the system is in static balance, there is no guarantee that it is in dynamic balance (Sec. 15.4).

The following example illustrates the direct method of dynamic balancing of a distributed mass system.

EXAMPLE 15.4

The distributed mass system shown in Fig. 15.12 has been tested for unbalance by rotating the rotor at an angular velocity of 100 rad/s. The bearing reactions on the frame at two bearings *A* and *B* are $F_A = 40$ lb and $F_B = 35$ lb, respectively. Determine the magnitudes and orientations of the correcting masses to be removed in the specified planes 1 and 2 to achieve dynamic balance. The correcting masses are to be located at radii $r_1 = r_2 = r = 4$ in.

SOLUTION

The *x* and *y* components of the forces acting on frame at bearings *A* and *B* are

$$F_{A21}^x = 40 \cos 240^\circ \text{ lb} = -20 \text{ lb}, \quad (1a)$$

$$F_{A21}^y = 40 \sin 240^\circ \text{ lb} = -20\sqrt{3} \text{ lb}, \quad (1b)$$

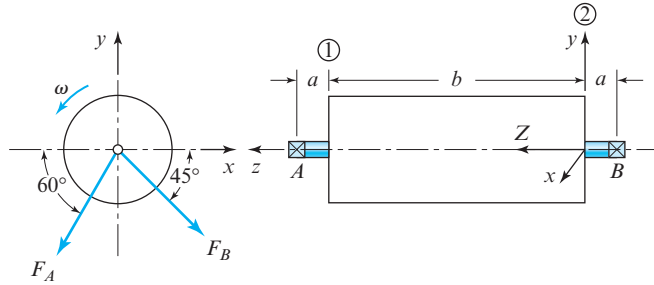


Figure 15.12 $a = 5$ in and $b = 40$ in.

$$F_{B21}^x = 35 \cos(-45^\circ) \text{ lb} = 17.5\sqrt{2} \text{ lb} \quad (1c)$$

$$F_{B21}^y = 35 \sin(-45^\circ) \text{ lb} = -17.5\sqrt{2} \text{ lb.} \quad (1d)$$

Equations (15.8) and (15.9) can be written as

$$m_1 r \omega^2 \cos \phi_1 + m_2 r \omega^2 \cos \phi_2 = -F_{A21}^x - F_{B21}^x, \quad (2a)$$

$$m_1 r \omega^2 \sin \phi_1 + m_2 r \omega^2 \sin \phi_2 = -F_{A21}^y - F_{B21}^y, \quad (2b)$$

$$z_1 m_1 r \omega^2 \cos \phi_1 + z_2 m_2 r \omega^2 \cos \phi_2 = -z_A F_{A21}^x - z_B F_{B21}^x, \quad (3a)$$

$$z_1 m_1 r \omega^2 \sin \phi_1 + z_2 m_2 r \omega^2 \sin \phi_2 = -z_A F_{A21}^y - z_B F_{B21}^y. \quad (3b)$$

Using the direct method, we take moments about one of the two balancing planes. Here we take moments about plane 2 and note that the distances are

$$z_1 = +40 \text{ in}, z_2 = 0, z_A = +45 \text{ in}, \text{ and } z_B = -5 \text{ in.} \quad (4)$$

Substituting Eqs. (4) and the known data into Eqs. (2) and (3) gives

$$(4 \text{ in}) (100 \text{ rad/s})^2 (m_1 \cos \phi_1 + m_2 \cos \phi_2) = -4.749 \text{ lb}, \quad (5a)$$

$$(4 \text{ in}) (100 \text{ rad/s})^2 (m_1 \sin \phi_1 + m_2 \sin \phi_2) = 59.390 \text{ lb}, \quad (5b)$$

$$(40 \text{ in}) (4 \text{ in}) (100 \text{ rad/s})^2 m_1 \cos \phi_1 = 1\,023.744 \text{ in} \cdot \text{lb}, \quad (6a)$$

$$(40 \text{ in}) (4 \text{ in}) (100 \text{ rad/s})^2 m_1 \sin \phi_1 = 1\,435.102 \text{ in} \cdot \text{lb}. \quad (6b)$$

Dividing Eq. (6b) by Eq. (6a), the location of the first correcting mass is

$$\phi_1 = \tan^{-1} \left(\frac{1\,435.102 \text{ in} \cdot \text{lb}}{1\,023.744 \text{ in} \cdot \text{lb}} \right) = 54.5^\circ. \quad \text{Ans. (7a)}$$

Substituting Eq. (7a) into Eq. (6a) or (6b), the first correcting mass is

$$m_1 = \frac{(1\,435.102 \text{ in} \cdot \text{lb}) (386 \text{ in/s}^2)}{(1\,600\,000 \text{ in}^2/\text{s}^2) \sin 54.5^\circ} = 0.425 \text{ lb.} \quad \text{Ans. (7b)}$$

Substituting Eqs. (7) into Eqs. (5), rearranging, and dividing Eq. (5b) by Eq. (5a), the location of the second correcting mass is

$$\phi_2 = \tan^{-1} \left(\frac{23.535 \text{ lb}}{-30.324 \text{ lb}} \right) = 142.2^\circ. \quad \text{Ans. (8a)}$$

Substituting Eq. (8a) into Eq. (5a) or (5b), the second correcting mass is

$$m_2 = \frac{(-30.342 \text{ lb}) (386 \text{ in/s}^2)}{(40\,000 \text{ in/s}^2) \cos 142.2^\circ} = 0.370 \text{ lb}. \quad \text{Ans. (8b)}$$

Note that the answers given by Eq. (7a) and (8a) are for adding mass to the system.

To balance the rotating system by removing mass, we use:

$$\phi_{\text{remove}} = \phi_{\text{add}} \pm 180^\circ.$$

Therefore, using this equation the mass to be removed from plane 1 is

$$m_1 = 0.425 \text{ lb at the angle } 54.5^\circ + 180^\circ = 234.5^\circ = -125.5^\circ. \quad \text{Ans.}$$

Also, the mass to be removed from plane 2 is

$$m_2 = 0.370 \text{ lb at the angle } 142.2^\circ + 180^\circ = 322.2^\circ = -37.8^\circ. \quad \text{Ans.}$$

Numeric Analysis For this approach, it is convenient to choose the xy plane as the plane of rotation with z as the axis of rotation, as shown in Fig. 15.13. In this manner, the unbalance vectors $m_i \mathbf{R}_i$ and the two correction vectors, $m_L \mathbf{R}_L$ in the left plane and $m_R \mathbf{R}_R$ in the right plane, can be expressed in the two-dimensional polar notation $m\mathbf{R} = mR \angle \theta$. This makes it easy to use the polar-rectangular conversion feature and its inverse, found on many calculators.

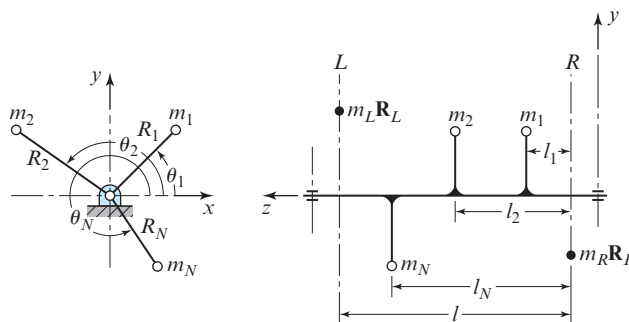


Figure 15.13 Notation for numeric solution; correction vectors are not shown in end view.

Note that Fig. 15.13 has $m_1, m_2, \dots, m_N$ unbalances. Solving Eqs. (b) and (c), we have

$$m_L \mathbf{R}_L = - \sum_{i=1}^N \frac{m_i l_i}{l} \mathbf{R}_i, \quad (15.10)$$

$$m_R \mathbf{R}_R = -m_L \mathbf{R}_L - \sum_{i=1}^N m_i \mathbf{R}_i. \quad (15.11)$$

These two equations can easily be solved. If a calculator is used, it is suggested that the summation key be employed with each term of the summation entered.

15.6 DYNAMIC BALANCING

The units in which each unbalance is measured have customarily been the ounce-inch (oz·in), the gram-centimeter (g·cm), or the inconsistent unit of gram-inch (g·in). If correct practice is followed in the use of SI units, however, the most appropriate unit of unbalance is the milligram-meter (mg·m), since prefixes in multiples of 1 000 are preferred in SI; thus, the prefix centi- is not recommended. Furthermore, only one prefix should be used in a compound unit; preferably, the first-named quantity should be prefixed. Thus, although both are acceptable in size, neither the gram-centimeter nor the kilogram-millimeter should be used. In this text we use the ounce-inch (oz·in) and the milligram-meter (mg·m) for units of unbalance.

We have seen that static balancing is sufficient for rotating disks, wheels, gears, and the like, where the mass can be assumed to exist in a single rotating plane. In the case of longer-axis machine elements, such as turbine rotors or motor armatures, the unbalanced inertia forces result in a couple whose effect tends to cause the rotor to turn end over end. The purpose of balancing is to determine this unbalanced couple and to add a new couple of equal magnitude in the opposite direction. The new couple is introduced by the addition of masses in two preselected correction planes or by subtracting (drilling out) masses from these two planes. To require balancing, a rotor usually has both static and dynamic unbalance; consequently, the correction masses, their radial orientations, or both are not the same for the two correction masses; also their radial orientations need not be the same for the two correction planes. This means that the angular separation of the correction masses on the two planes is usually not 180° . Thus, to balance a rotor, we must measure the magnitude and the angular orientation of the correction mass for each of the two correction planes.

Three methods of measuring the corrections for two planes are in general use, the *pivoted-cradle*, the *nodal-point*, and the *mechanical-compensation* methods.

Figure 15.14 schematically illustrates a pivoted-cradle balancing machine with a rotor to be balanced, mounted on half-bearings or rollers attached to a cradle. The right end of the rotor is connected to a drive motor through a universal joint. The cradle can be rocked about either of two points that are adjusted to coincide with the two correction planes on the rotor. The left pivot is shown in the released position and the cradle and rotor are free to rock or oscillate about the right pivot, which is in the locked position. Springs and dashpots are secured at each end of the cradle to provide a single-degree-of-freedom

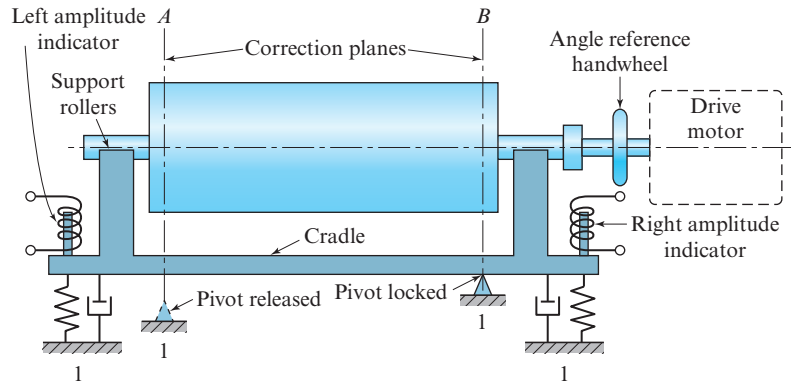


Figure 15.14 Schematic drawing of a pivoted-cradle balancing machine.

vibrating system. Often, these are made adjustable so that the motor speed can be tuned to match the natural frequency. Amplitude indicators at each end of the cradle are differential transformers, or they may consist of a permanent magnet mounted on the cradle that moves relative to a stationary coil to generate a voltage proportional to the unbalance.

With the pivots located in the two correction planes, the operator can lock either pivot and take readings of the magnitude and orientation angle of the correction in the unlocked plane. The readings obtained are completely independent of the measurements taken in the other correction plane, since an unbalance in the plane of the locked pivot has no moment about that pivot. With the right pivot locked, an unbalance correctable in the left correction plane causes vibration whose amplitude is measured by the left amplitude indicator. Once this correction is measured, the right pivot is released, the left pivot is locked, and another set of measurements is made for the right correction plane using the right amplitude indicator.

Not shown in the balancing machine of Fig. 15.14 is a sine-wave signal generator that is attached to the drive shaft. If the resulting sine-wave signal generator is compared on a dual-beam oscilloscope with the wave generated by one of the amplitude indicators, a phase difference is found. The angular phase difference is the angular orientation of the unbalance. In a balancing machine, an electronic phasemeter measures the phase angle and gives the result on another meter calibrated in degrees. To locate the correction on the rotor, the angular reference handwheel is turned by hand until the indicated angle is in line with a reference pointer. This places the heavy side of the rotor in a preselected posture and permits the correction to be made.

The relation between the amount of unbalance and the measured amplitude is given by Eq. (15.4). Rearranging that equation, and substituting r for e , the amplitude of the motion can be written as

$$X = \frac{m_u r (\omega/\omega_n)^2}{m \sqrt{(1 - \omega^2/\omega_n^2)^2 + (2\zeta\omega/\omega_n)^2}}, \quad (15.12)$$

where $m_u r$ is the unbalance, and m is the mass of the cradle and the specimen. This equation demonstrates that the amplitude X is directly proportional to the unbalance,

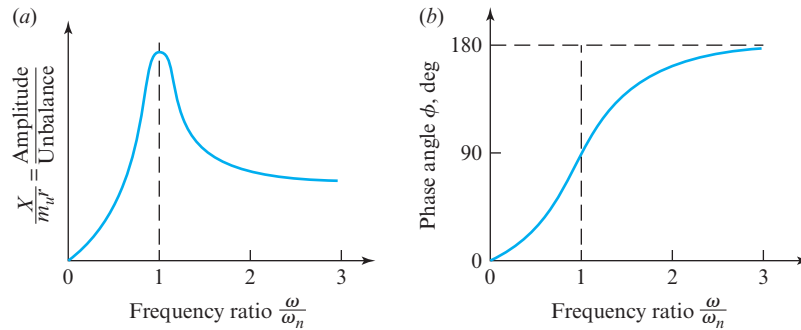


Figure 15.15

$m_u r$. Figure 15.15a shows a plot of this equation for a particular damping ratio, ζ , and demonstrates that the machine is most sensitive near resonance ($\omega = \omega_n$), since, in this region, the greatest amplitude is recorded for a given unbalance. Damping is deliberately introduced in balancing machines to filter noise and other vibrations that might affect the results. Damping also helps to maintain calibration against effects of temperature and other environmental conditions.

Substituting Eqs. (15.1) and (15.3) into Eq. (c) of Sec. 15.2 and rearranging, the phase angle can be written in parametric form as

$$\phi = \tan^{-1} \frac{2\zeta \omega / \omega_n}{1 - \omega^2 / \omega_n^2}. \quad (15.13)$$

A plot of this equation for a single damping ratio and for various frequency ratios is shown in Fig. 15.15b. This curve indicates that, at resonance, when the speed, ω , matches the natural frequency, ω_n , the displacement lags the unbalance by the angle $\phi = 90^\circ$. If the top of the rotor is turning away from the operator, the unbalance is horizontal and directly in front of the operator when the displacement is maximum downward. The angular orientation approaches 180° as ω is increased above resonance.

15.7 DYNAMIC BALANCING MACHINES

A dynamic balancing machine for high-speed production is shown in Fig. 15.16.

Nodal-Point Balancing Plane separation using a point of zero or minimum vibration is called the *nodal-point method of balancing*. To see how this method is used, examine Fig. 15.17. Here, the specimen to be balanced is shown mounted on bearings that are fastened to a nodal bar that experiences the same vibration as the specimen. We assume that the specimen is already balanced in the left correction plane and that an unbalance still exists in the right plane, as shown. Because of this unbalance, a vibration of the entire assembly takes place, causing the nodal bar to oscillate about some point O , occupying first (exaggerated) posture CC and then DD . Point O is easily located by sliding a dial indicator along the nodal bar; a point of zero motion or minimum motion is readily determined. This

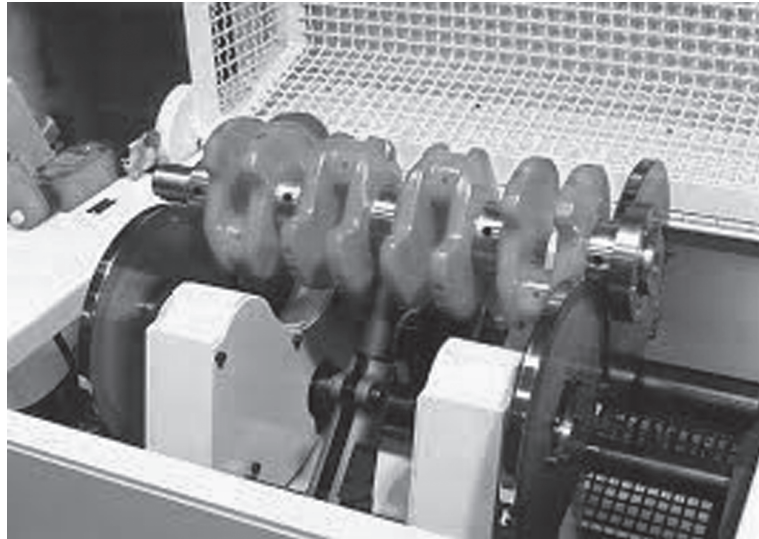


Figure 15.16 Engine crankshaft dynamic balancing machine. (Courtesy of Schenck Ro Tec Corporation, Auburn Hills, MI.)

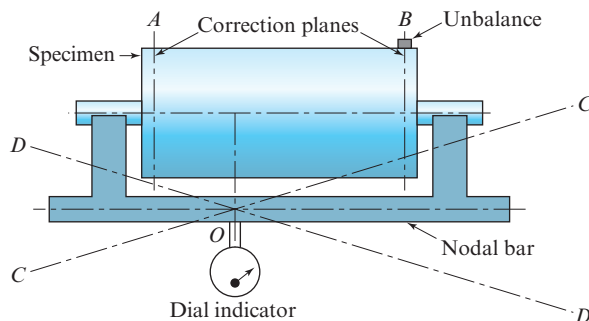


Figure 15.17 Plane separation by the nodal-point method.

is the null or nodal point. Its location is the center of oscillation for a center of percussion (Sec. 12.6) in the right correction plane.

We assumed, at the beginning of this discussion, that no unbalance existed in the left correction plane. However, if unbalance is present, its magnitude is given by the dial indicator located at the nodal point just determined. Thus, by locating the dial indicator at this nodal point, we measure the unbalance in the left plane without any interference from that in the right plane. In a similar manner, another nodal point can be found that will measure only the unbalance in the right correction plane without any interference from that in the left plane.

In commercial balancing machines employing the nodal-point principle, the plane separation is accomplished in electronic networks. Typical of these is the Micro Dynamic

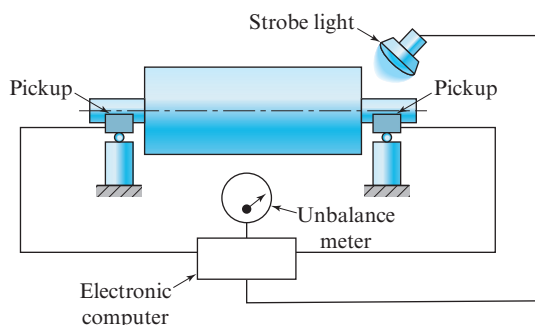


Figure 15.18 Diagram of the electronic circuit in a Micro Dynamic Balancer. (Courtesy of Schenck Ro Tec Corporation, Auburn Hills, MI.)

Balancer; a schematic is shown in Fig. 15.18. On this machine, a switching knob selects either correction plane and displays the unbalance on a voltmeter, which is calibrated in appropriate unbalance units. The computer contains a filter that eliminates bearing noise and other frequencies not related to the unbalance. A multiplying network is used to give the sensitivity desired and to cause the meter to read in preselected balance units. The strobe light is driven by an oscillator that is synchronized to the rotor speed.

The rotor is driven at a speed that is much greater than the natural frequency of the system, and, since the damping is quite small, the direction of the vibration is horizontal, and the phase angle is approximately 180° . Marked on the right end of the rotor are degrees or numbers that are readable and stationary under the strobe light during rotation of the rotor. Thus, it is only necessary to observe the particular station number of the degree marking under the strobe light to locate the heavy spot. When the switch is shifted to the other correction plane, the meter again reads the amount, and the strobe light illuminates the station. Sometimes as few as five station numbers distributed uniformly around the periphery are adequate for balancing. Rotation such that the top of the rotor moves away from the operator causes the heavy spot to be in a horizontal plane and on the near side of the axis when illuminated by the strobe lamp. A pointer is usually placed here to indicate its location. If, during production balancing, it is found that the phase angle is less than 180° , the pointer can be shifted slightly to indicate the proper position to observe.

Mechanical Compensation An unbalanced rotating rotor mounted in a balancing machine produces a vibration. Counterforces can be introduced in each correction plane in the balancing machine that exactly balance the forces causing the vibration. The result of introducing these forces is a smooth running system. Upon stopping, the location and amount of each counterforce can be measured to give the exact correction required. This is called *mechanical compensation*.

When mechanical compensation is used, the speed of the rotor during balancing is not important since the equipment is in calibration for all speeds. The rotor may be driven by a belt, through a universal joint, or it may be self-driven if, for example, it is a gasoline engine. The electronic equipment is simple, no built-in damping is necessary, and the machine is easy to operate, since the unbalances in both correction planes are measured simultaneously and the magnitudes and orientations are read directly.

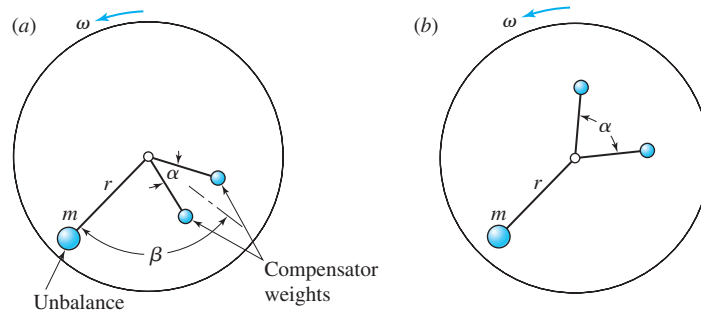


Figure 15.19 (a) Postures of compensator weights; (b) after compensation

We can understand how mechanical compensation is applied by examining Fig. 15.19a. Looking at the end of the rotor, we see one of the correction planes with the unbalance to be corrected (represented by mr). The correction plane, viewed along the axis of rotation to indicate the unbalance, and the two compensator weights are shown in Fig. 15.19a. All three of these weights are to rotate with the same angular velocity ω , but the postures of the compensator weights relative to one another and their posture relative to the unbalanced weight can be varied by two controls. One of these controls changes the angle α , that is, the angle between the compensator weights. The other control changes the angular posture of the pair of compensator weights relative to the unbalance, that is, the angle β . The knob that changes the angle β is the *orientation control*, and, when the rotor is compensated (balanced) in this plane, a pointer on the knob indicates the exact angular orientation of the unbalance. The knob that changes the angle α is the *amount control*, and it also gives a direct reading when the rotor unbalance is compensated as shown in Fig. 15.19b. The magnitude of the vibration is measured electrically and displayed on a voltmeter. Thus, compensation is found when the controls are manipulated to make the voltmeter read zero.

15.8 FIELD BALANCING WITH A PROGRAMMABLE CALCULATOR*

Field balancing is necessary for very large rotors for which balancing machines are impractical. Despite the fact that high-speed rotors are balanced in the shop during manufacture, it is frequently necessary to rebalance them in the field because of slight deformations brought on by shipping, creep, or high operating temperatures.

It is possible to balance a machine in the field by balancing a single plane at a time. But cross-coupling effects and correction-plane interference often require balancing each end of a rotor two or three times to obtain satisfactory results. Some machines may require as much as an hour to bring them up to full speed, resulting in even more delays in the balancing procedure.

*The authors are grateful to Prof. W. B. Fagerstrom, University of Delaware, for contributing some of the ideas of this section.

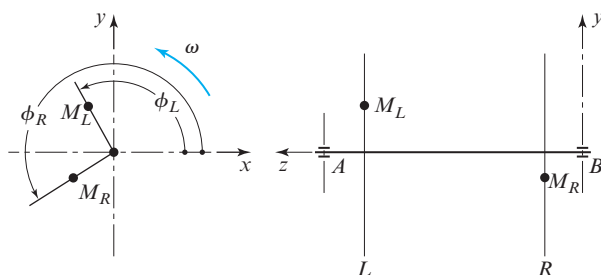


Figure 15.20 Notation for two-plane field balancing.

Both Rathbone [5] and Thearle [8] have developed methods of two-plane field balancing that can be expressed in complex-number notation and solved using a programmable calculator. The time saved is several hours when compared with graphic methods or analysis with complex numbers using an ordinary scientific calculator.

In the analysis that follows, boldface letters are used to represent complex numbers; for example

$$\mathbf{R} = R \angle \theta = R e^{j\theta} = x + jy.$$

In Fig. 15.20, unknown unbalances $\mathbf{M}_L$ and $\mathbf{M}_R$ are assumed to exist in the left and right correction planes, respectively. The magnitudes of these unbalances are M_L and M_R , and they are located at angles ϕ_L and ϕ_R from the rotating reference, denoted as the xy system. When these unbalances have been found, their negatives are located in the left and right planes to achieve balance.

The rotating unbalances $\mathbf{M}_L$ and $\mathbf{M}_R$ produce disturbances at bearings A and B . Using commercial field-balancing equipment, it is possible to measure the amplitudes and the angular locations of these disturbances. The notation $\mathbf{X} = X \angle \phi$, with appropriate subscripts, is used to designate these quantities.

In field balancing, three runs or tests are made, as follows:

- *First run:* Measure vector $\mathbf{X}_A = X_A \angle \phi_A$ at bearing A and vector $\mathbf{X}_B = X_B \angle \phi_B$ at bearing B due only to the original unbalances, $\mathbf{M}_L = M_L \angle \phi_L$ and $\mathbf{M}_R = M_R \angle \phi_R$.
- *Second run:* Add trial mass $\mathbf{m}_L = m_L \angle \theta_L$ to the left correction plane, and measure the amplitudes $\mathbf{X}_{AL} = X_{AL} \angle \phi_{AL}$ and $\mathbf{X}_{BL} = X_{BL} \angle \phi_{BL}$ at the left and right bearings (A and B), respectively.
- *Third run:* Remove trial mass $\mathbf{m}_L = m_L \angle \theta_L$. Add trial mass $\mathbf{m}_R = m_R \angle \theta_R$ to the right correction plane and again measure the bearing amplitudes. These results are designated $\mathbf{X}_{AR} = X_{AR} \angle \phi_{AR}$ for bearing A and $\mathbf{X}_{BR} = X_{BR} \angle \phi_{BR}$ for bearing B .

Note in the above runs that the term “trial mass” means the same as a trial unbalance, and a unit distance from the axis of rotation is used.

To develop the equations for the unbalance being sought, we first define *complex stiffness* as the amplitude that would result at either bearing caused by a unit unbalance located at the intersection of the rotating reference mark and one of the correction planes. Thus, we must find the complex stiffnesses $\mathbf{A}_L$ and $\mathbf{B}_L$ caused by a unit unbalance located at the intersection of the rotating reference mark and plane L . In addition, we require the

complex stiffnesses $\mathbf{A}_R$ and $\mathbf{B}_R$ caused by a unit unbalance located at the intersection of the rotating reference mark and plane R .

If these stiffnesses were available, we could write the following sets of complex equations:

$$\mathbf{X}_{AL} = \mathbf{X}_A + \mathbf{A}_L \mathbf{m}_L \quad \mathbf{X}_{BL} = \mathbf{X}_B + \mathbf{B}_L \mathbf{m}_L, \quad (a)$$

$$\mathbf{X}_{AR} = \mathbf{X}_A + \mathbf{A}_R \mathbf{m}_R \quad \mathbf{X}_{BR} = \mathbf{X}_B + \mathbf{B}_R \mathbf{m}_R. \quad (b)$$

After the three runs are made, the stiffnesses are the only unknowns in these equations. Therefore,

$$\begin{aligned} \mathbf{A}_L &= \frac{\mathbf{X}_{AL} - \mathbf{X}_A}{\mathbf{m}_L}, \quad \mathbf{B}_L = \frac{\mathbf{X}_{BL} - \mathbf{X}_B}{\mathbf{m}_L}, \\ \mathbf{A}_R &= \frac{\mathbf{X}_{AR} - \mathbf{X}_A}{\mathbf{m}_R}, \quad \mathbf{B}_R = \frac{\mathbf{X}_{BR} - \mathbf{X}_B}{\mathbf{m}_R}. \end{aligned} \quad (15.14)$$

Then, from the definitions of these stiffnesses, we have from the first run

$$\mathbf{X}_A = \mathbf{A}_L \mathbf{M}_L + \mathbf{A}_R \mathbf{M}_R, \quad \mathbf{X}_B = \mathbf{B}_L \mathbf{M}_L + \mathbf{B}_R \mathbf{M}_R. \quad (c)$$

Solving this pair of equations simultaneously gives

$$\mathbf{M}_L = \frac{\mathbf{X}_A \mathbf{B}_R - \mathbf{A}_R \mathbf{X}_B}{\mathbf{A}_L \mathbf{B}_R - \mathbf{A}_R \mathbf{B}_L}, \quad \mathbf{M}_R = \frac{\mathbf{A}_L \mathbf{X}_B - \mathbf{X}_A \mathbf{B}_L}{\mathbf{A}_L \mathbf{B}_R - \mathbf{A}_R \mathbf{B}_L}. \quad (15.15)$$

These equations can be programmed, either in complex polar form or in complex rectangular form. The suggestions that follow were formed assuming a complex rectangular form for the solution.

Since the original data are formulated in polar coordinates, a subprogram should be written to transform the data into rectangular coordinates before storage.

The equations reveal that complex subtraction, division, and multiplication are used often. These operations can be set up as subprograms to be called from the main program. If $\mathbf{A} = a + jb$ and $\mathbf{B} = c + jd$, the formula for complex subtraction is

$$\mathbf{A} - \mathbf{B} = (a - c) + j(b - d). \quad (15.16)$$

For complex multiplication, the formula is

$$\mathbf{AB} = (ac - bd) + j(bc + ad) \quad (15.17)$$

and for complex division, the formula is

$$\frac{\mathbf{A}}{\mathbf{B}} = \frac{(ac + bd) + j(bc - ad)}{c^2 + d^2}. \quad (15.18)$$

With these subprograms, it is an easy matter to program Eqs. (15.14) and (15.15).

As a check on your programming, you may wish to use the following data: $\mathbf{X}_A = 8.6 \angle 63^\circ$, $\mathbf{X}_B = 6.5 \angle 206^\circ$, $\mathbf{m}_L = 10 \angle 270^\circ$, $\mathbf{m}_R = 12 \angle 180^\circ$, $\mathbf{X}_{AL} = 5.9 \angle 123^\circ$,

$\mathbf{X}_{BL} = 4.5\angle 228^\circ$, $\mathbf{X}_{AR} = 6.2\angle 36^\circ$, $\mathbf{X}_{BR} = 10.4\angle 162^\circ$. The correct results are $\mathbf{M}_L = 10.76\angle 146.6^\circ$ and $\mathbf{M}_R = 6.20\angle 245.4^\circ$.

According to Fagerstrom, the vibration angles used can be expressed in two different systems. The first is the rotating-protractor stationary-mark system (RPSM). This is the system used in the preceding analysis and is the one a theoretician would prefer. In actual practice, however, it is usually easier to have the protractor stationary and use a rotating mark like a key or keyway. This is called the rotating-mark stationary-protractor system (RMSP). The only difference between the two systems is in the sign of the vibration angle, but there is no sign change on the trial or correction masses.

15.9 BALANCING A SINGLE-CYLINDER ENGINE

The rotating masses in a single-cylinder engine can be balanced using the methods already discussed in this chapter. The reciprocating masses, however, cannot be completely balanced, and so our studies in this section are really concerned with minimizing the unbalance. Although the reciprocating masses cannot be totally balanced using a simple counterweight, it is possible to modify the shaking forces (Sec. 14.9) by unbalancing the rotating masses. As an example of this, let us add a counterweight opposite the crankpin whose mass exceeds the rotating mass by one-half of the reciprocating mass; (from one-half to two-thirds of the reciprocating mass is usually added to the counterweight to alter the balance characteristics in a single-cylinder engine). We designate the mass of the counterweight m_C , substitute this mass into Eq. (14.24), and use a negative sign, since the counterweight is opposite the crankpin; then, the inertia force caused by this counterweight is

$$\mathbf{F}_C = -m_C r \omega^2 \cos \omega t \hat{\mathbf{i}} - m_C r \omega^2 \sin \omega t \hat{\mathbf{j}}. \quad (a)$$

Note that the balancing mass and the crankpin both have the same radius. Designating by m_A and m_B the masses of the rotating and the reciprocating parts, respectively, as in Chap. 14, then, according to the supposition above, we have

$$m_C = m_A + \frac{1}{2}m_B. \quad (b)$$

Equation (a) can now be written

$$\mathbf{F}_C = -\left(m_A + \frac{1}{2}m_B\right)r\omega^2 \cos \omega t \hat{\mathbf{i}} - \left(m_A + \frac{1}{2}m_B\right)r\omega^2 \sin \omega t \hat{\mathbf{j}}. \quad (c)$$

From Eqs. (14.24) and (14.25), the inertia force caused by the rotating and reciprocating masses can be written as

$$\mathbf{F}_{A,B} = F^x \hat{\mathbf{i}} + F^y \hat{\mathbf{j}} = \left[(m_A + m_B)r\omega^2 \cos \omega t + \left(m_B \frac{r}{l}\right)r\omega^2 \cos 2\omega t\right] \hat{\mathbf{i}} + m_A r \omega^2 \sin \omega t \hat{\mathbf{j}}. \quad (d)$$

Adding Eqs. (c) and (d), the resultant inertia force can be written as

$$\mathbf{F} = \left[\frac{1}{2}m_B r \omega^2 \cos \omega t + \left(m_B \frac{r}{l}\right)r\omega^2 \cos 2\omega t\right] \hat{\mathbf{i}} - \frac{1}{2}m_B r \omega^2 \sin \omega t \hat{\mathbf{j}}. \quad (15.19)$$

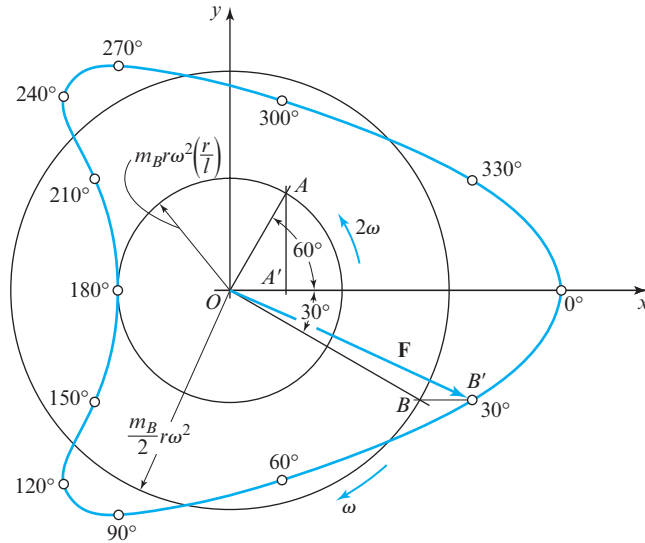


Figure 15.21 Polar diagram of inertia forces.

The first term is called the *primary component*. This component has a magnitude $\frac{1}{2}m_B r \omega^2$ and can be represented as a backward- (clockwise-) rotating vector with angular velocity ω . The second term is called the *secondary component*. This component is the x projection of a vector of length $m_B (r/l) r \omega^2$ rotating forward (counterclockwise) with an angular velocity of 2ω .

The maximum inertia force occurs when $\omega t = 0$. Substituting this condition into Eq. (15.19) gives

$$F_{\max} = m_B r \omega^2 \left(\frac{r}{l} + \frac{1}{2} \right), \quad (e)$$

since $\cos \omega t = \cos 2\omega t = 1$ and $\sin \omega t = 0$ when $\omega t = 0$. Before the extra counterweight is added, the maximum inertia force is

$$F_{\max} = m_B r \omega^2 \left(\frac{r}{l} + 1 \right). \quad (f)$$

Therefore, in this instance, the effect of the added counterweight is to reduce the maximum shaking force by 50% of the primary component and to add axial inertia forces where formerly none existed. Equation (15.19) is plotted as a polar diagram in Fig. 15.21 for $r/l = 1/4$. Here, vector OA rotates counterclockwise at angular velocity of 2ω . The horizontal projection of this vector, OA' , is the secondary component. Vector OB , the primary component, rotates clockwise at an angular velocity ω . The total shaking force, F , is shown for the 30° orientation and is the sum of vectors OB and $BB' = OA'$.

Imaginary-Mass Approach Stevensen has refined and extended a method of engine balancing that here is called the *imaginary-mass approach*.^{*} It is possible that the method is known in some circles as the *virtual-rotor approach*, since it uses what might be called a virtual rotor that counter-rotates to accommodate part of the piston effect in a reciprocating engine.

Before going into details, it is necessary to explain a change in the method of viewing the crank circle of an engine. In developing the imaginary-mass approach in this and the following section, we use the right-handed coordinate system of Fig. 15.22a. This might appear to be a left-handed system, since the y axis is located clockwise from the x axis, and since positive rotation is shown as clockwise. If you prefer, you can think of this system as a right-handed three-dimensional system viewed from the negative z axis. We adopt this notation, since it has been used for so long by the automotive industry.[†]

The imaginary-mass approach uses two fictitious masses, each equal to half the equivalent reciprocating mass at the particular harmonic frequency studied. The purpose of these fictitious masses is to replace the effects of the reciprocating mass. These imaginary masses rotate about the crank center in opposite directions and with equal velocities. They are arranged so that they come together at both the top dead center (TDC) and bottom dead center (BDC), as shown in Fig. 15.22a. The mass, $+\frac{1}{2}m_B$, rotates *with* the crank motion; the other mass rotates opposite to the crank motion. The mass rotating with the crank motion is designated in Fig. 15.22a by a plus sign, and the mass rotating in the opposite direction is designated by a minus sign. The center of mass of these two rotating masses always lies on the cylinder axis, x .

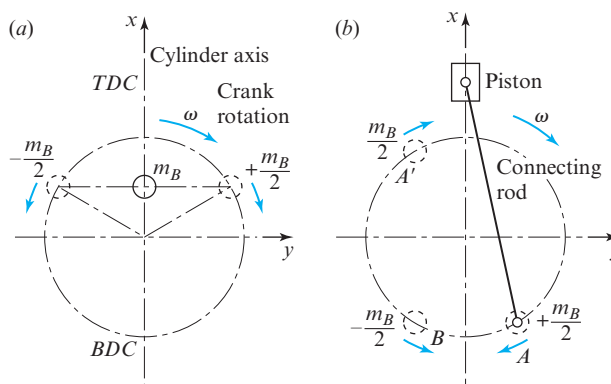


Figure 15.22 View of the crank circle from the negative z axis.

^{*} The presentation here is from Prof. E. N. Stevensen, University of Hartford, class notes [7], with his permission. Although some changes have been made to conform to the notation of this text, the material is all Stevensen's. He refers to Maleev [4] and Lichty [2] and states that the method first came to his attention in both of those books.

[†] Readers who are antique automobile buffs will understand this convention, since it is the direction in which an antique engine is hand cranked.

The imaginary-mass approach was conceived remembering that the piston motion and the resulting inertia force can always be represented by a Fourier series. Such a series has an infinite number of terms, each term representing a simple harmonic motion having a known frequency and amplitude. The higher-frequency amplitudes are so small that they can be neglected, and hence, only a small number of the lower-frequency amplitudes are needed. Also, the odd harmonics (third, fifth, etc.) are not present because of the symmetry of the piston motion.

Each harmonic (the first, second, fourth, and so on) is represented by a pair of imaginary masses. The angular velocities of these masses are $\pm\omega$ for the first harmonic, $\pm 2\omega$ for the second, $\pm 4\omega$ for the fourth, and so on. It is rarely necessary to consider the sixth or higher harmonics.

Stevensen gives the following rule for placing the imaginary masses:

For a given orientation of the cranks the orientations of the imaginary masses are found, first, by determining the angles of travel of each crank from its top dead center and, second, by moving their imaginary masses, one clockwise and the other counterclockwise, by angles equal to the crank angle times the number of the harmonic.

All of these angles must be measured from the same dead-center crank orientation.

Let us apply this approach to the single-cylinder engine, considering only the first harmonic. In Fig. 15.22*b*, the mass $+\frac{1}{2}m_B$ at *A* rotates at the angular velocity ω with the crank, whereas mass $-\frac{1}{2}m_B$ at *B* rotates at the angular velocity $-\omega$ opposite to crank rotation. The imaginary mass at *A* can be balanced by adding an equal mass at *A'* to rotate with the crankshaft. However, the mass at *B* can be balanced neither by the addition nor by the subtraction of mass from any part of the crankshaft, since it is rotating in the opposite direction. When half the mass of the reciprocating parts is balanced in this manner—that is, by adding the mass at *A'*—the unbalanced part of the first harmonic, caused by the mass at *B*, causes the engine to vibrate in the plane of rotation equally in all directions like a true unbalanced rotating mass.

It is interesting to find that in a one-cylinder motorcycle engine, a fore-and-aft unbalance is less objectionable than an up-and-down unbalance. For this reason, such engines are overbalanced by using a counterweight whose mass is more than half the reciprocating mass.

It is impossible to balance the second and higher harmonics with masses rotating at crankshaft speeds, since the frequency of the unbalance is higher than that of the crankshaft rotation. Balancing of second harmonics has been accomplished by using shafts geared to run at twice the engine crankshaft speed, as in the case of the 1976 Plymouth Arrow engine, but at the cost of complexity. It is not usually done.

For a convenient reference, we summarize the inertia forces in a single-cylinder engine with balancing masses; see Table 15.1. Note that the expressions in the final two columns of the table are obtained from Eqs. (14.24) and (14.27). Note that the subscript *C* is used to designate counterweights (balancing masses) and their radii. Also, note that the effect of the second harmonic is presented as a mass equal to $\frac{1}{4}m_B(r/l)$ reciprocating at a speed of 2ω ; see column 2. Since the inertia balancing masses are selected and placed to counterbalance the rotating inertia forces, then the only unbalance that results along the cylinder axis is due to the harmonics of the reciprocating masses. Similarly, the only unbalance across the

Table 15.1 One-cylinder Engine Inertia Forces

Type	Equivalent mass	Radius	Along (x) cylinder axis	Across (y) cylinder axis
Rotating	m_A	r	$m_A r \omega^2 \cos \omega t$	$m_A r \omega^2 \sin \omega t$
	m_{AC}	r_C	$m_{AC} r_C \omega^2 \cos(\omega t + \pi)$	$m_{AC} r_C \omega^2 \sin(\omega t + \pi)$
Reciprocating	m_B	r	$m_B r \omega^2 \cos \omega t$	0
First harmonic	m_{BC}	r_C	$m_{BC} r_C \omega^2 \cos(\omega t + \pi)$	$m_{BC} r_C \omega^2 \sin(\omega t + \pi)$
Reciprocating				
Second harmonic	$\frac{m_B r}{4l}$	r	$\frac{m_B r}{4l} (r)(2\omega)^2 \cos 2\omega t$	0

cylinder axis is due to the first harmonic of the reciprocating masses. The maximum values of the unbalance in these two directions can be predetermined in any desired ratio to each other, as indicated previously, and a solution can be obtained for the equivalent mass, m_{BC} , at radius r_C .

If this approach is used to also include the effect of the fourth harmonic, there results an added mass of $(1/16)m_B(r/l)^3$ reciprocating at the speed of 2ω and another mass of $-(1/64)m_B(r/l)^3$ reciprocating at a speed of 4ω , illustrating the decreasing significance of higher harmonics.

15.10 BALANCING MULTI-CYLINDER ENGINES

To obtain a basic understanding of the balancing problem in multi-cylinder engines, let us consider the two-cylinder in-line engine with three main bearings shown in Fig. 15.23*a*. The cranks (numbered 1 and 2) are 180° apart, and the rotating parts are already balanced by counterweights. Applying the imaginary-mass approach for the first harmonic results

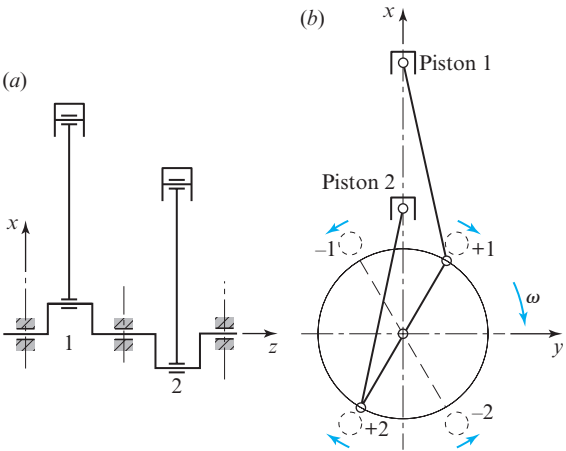


Figure 15.23 (a) Two-cylinder engine; (b) first harmonics.

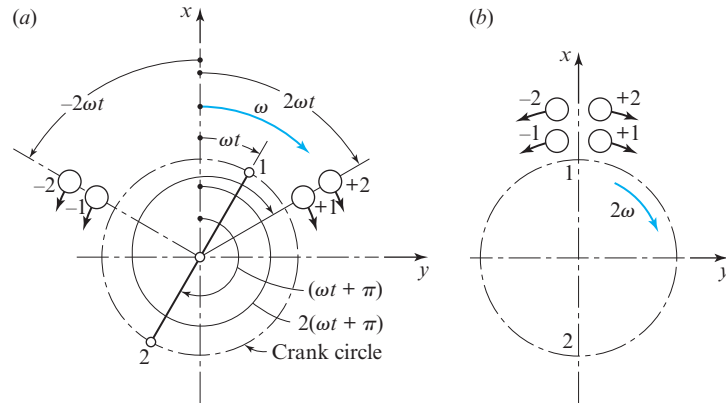


Figure 15.24 (a) Crank 1 at TDC; (b) extreme, or dead-center, postures.

in the diagram of Fig. 15.23b. This diagram shows that masses +1 and +2, rotating clockwise, balance each other, as do masses -1 and -2, rotating counterclockwise. Thus, the first harmonic forces are inherently balanced for this crank arrangement. These forces, however, are not in the same plane; therefore, unbalanced couples will be set up that tend to rotate the engine about the y axis. The values of these couples can be determined using the force expressions in Table 15.1 together with the coupling distance, since the equations can be applied to each cylinder separately. It is possible to balance the couple resulting from the real rotating masses as well as the imaginary half-masses that rotate with the engine; however, the couple resulting from the half-mass of the first harmonic that is counter-rotating cannot be balanced.

The locations of the imaginary masses for the second harmonic using Stevensen's rule are shown in Fig. 15.24a. This diagram shows that the second-harmonic forces due to the imaginary masses are not balanced. Since the greatest unbalance occurs at the dead-center orientations, the diagrams are usually drawn for this extreme orientation, with crank 1 at TDC, as in Fig. 15.24b. This unbalance causes a vibration in the xz plane having the frequency 2ω . The diagram for the fourth harmonics, not shown, is the same as in Fig. 15.24b, but, of course, the speed is 4ω .

Four-Cylinder Engine A four-cylinder in-line engine with cranks spaced 180° apart is shown in Fig. 15.25c. This engine can be treated as two two-cylinder engines placed back to back. Thus, the first harmonic forces still balance, and, in addition, from Figs. 15.25a and 15.25c, the first-harmonic couples also balance. These couples tend, however, to deflect the center bearing of a three-bearing crankshaft up and down and to bend the center of a two-bearing shaft in the same manner.

Figure 15.25b shows that when cranks 1 and 4 are at TDC, all the masses representing the second harmonic traveling in both directions accumulate at TDC, giving an unbalanced force. The center of mass of all the masses is always on the x axis, and so the unbalanced second harmonics cause a vertical vibration with a frequency of twice the engine speed.

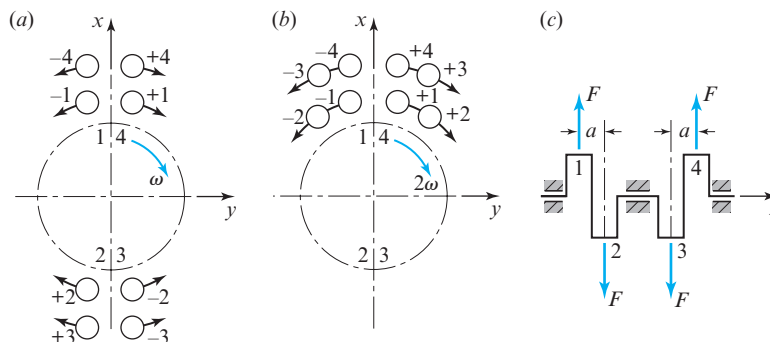


Figure 15.25 Postures for: (a) first harmonics; (b) second harmonics. (c) Crankshaft with first-harmonic couples added.

This characteristic is typical of all four-cylinder engines with this crank arrangement. Since the masses and forces all act in the same direction, there is no coupling action.

A diagram of the fourth harmonics is identical to that of Fig. 15.25b, and the effects are the same, but they do have a higher frequency and exert less force.

Three-Cylinder Engine A three-cylinder in-line engine with cranks spaced 120° apart is shown in Fig. 15.26. Note that the cylinders are numbered according to the order in which they arrive at TDC. The analysis, using imaginary masses, of the couples of the first harmonic forces indicates that when crank 1 is at TDC (Fig. 15.26), there is a vertical component of the forces on cranks 2 and 3 equal in magnitude to half of the force on crank 1. The resultant of these two downward components is equivalent to a force downward, equal in magnitude to the force on crank 1 and located halfway between cranks 2 and 3. Thus, a couple is set up with an arm equal to the distance between the center of crank 1 and the centerline between cranks 2 and 3.

At the same time, the horizontal components of the $+2$ and -2 forces cancel each other, as do the horizontal components of the $+3$ and -3 forces as shown in Fig. 15.27. Therefore,

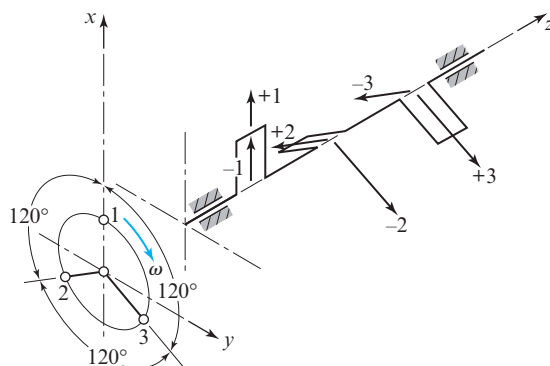


Figure 15.26 First-harmonic forces.

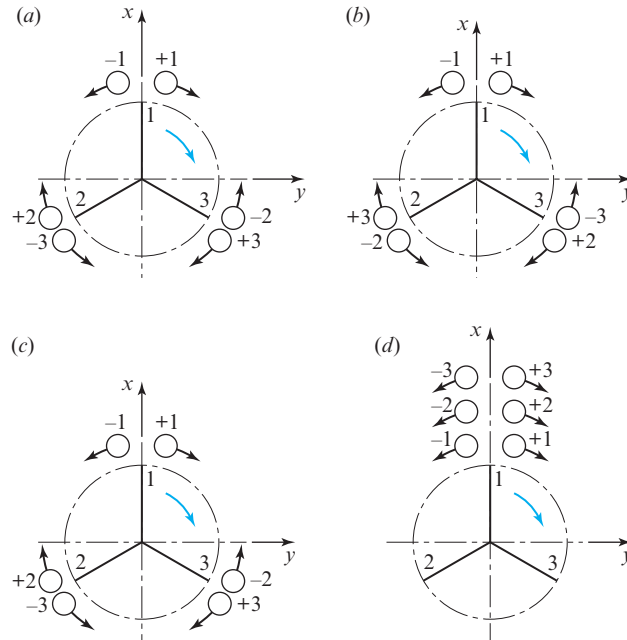


Figure 15.27 Postures for: (a) first harmonics; (b) second harmonics; (c) fourth harmonics; (d) sixth harmonics.

no horizontal couple exists. Similar couples are found for both the second and fourth harmonics. Thus, a three-cylinder engine is inherently balanced for forces in the first, second, and fourth harmonics; only the sixth harmonic forces are completely unbalanced. These unbalanced forces tend to create a vibration in the plane of the centerlines of the cylinders, but the magnitude of the forces is very small and can be neglected as far as vibration is concerned.

Six-Cylinder Engine If a six-cylinder in-line engine is conceived as a combination of two three-cylinder engines placed back to back with parallel cylinders, it has the same inherent balance of the first, second, and fourth harmonics. By virtue of symmetry, the couples of each three-cylinder engine act in opposite directions and balance the other. These couples, although perfectly balanced, tend to bend the crankshaft and crankcase and necessitate the use of rigid construction for high-speed operation. As before, the sixth harmonic forces are completely unbalanced and tend to create a vibration in the vertical plane with a frequency of 6ω . The magnitude of these forces, however, is very small and practically negligible as a source of vibration.

Other Engines Taking into consideration the cylinder arrangement and crank spacing permits a great many configurations. For any combination, the balancing situation can be investigated to any harmonic desired by the methods outlined in this section. Particular attention must be paid when analyzing the part of Stevensen's rule that calls for

determining the angle of travel from the TDC of the cylinder under consideration and moving the imaginary masses through the appropriate angles from that same TDC. This is especially important when radial and opposed-piston engines are investigated.

As practice problems you may wish to use these methods to confirm the following facts:

1. In a three-cylinder radial engine with one crank and three connecting rods having the same crankpin, the negative masses are inherently balanced for the first harmonic forces, whereas the positive masses are always located at the crankpin. These two findings are inherently true for all radial engines. Also, since the radial engine has its cylinders in a single plane, unbalanced couples do not occur. The three-cylinder engine has unbalanced forces in the second and higher harmonics.
2. A two-cylinder opposed-piston engine with a crank spacing of 180° is balanced for forces in the first, second, and fourth harmonics but unbalanced for couples.
3. A four-cylinder in-line engine with cranks at 90° is balanced for forces in the first harmonic but unbalanced for couples. In the second harmonic, it is balanced for both forces and couples.
4. An eight-cylinder in-line engine with the cranks at 90° is inherently balanced for both forces and couples in the first and second harmonics but unbalanced in the fourth harmonic.
5. An eight-cylinder V-engine with cranks at 90° is inherently balanced for forces in the first and second harmonics and for couples in the second harmonic. The unbalanced couples in the first harmonic can be balanced by counterweights that introduce an equal and opposite couple. Such an engine is unbalanced for forces in the fourth harmonic.

15.11 ANALYTIC TECHNIQUE FOR BALANCING MULTI-CYLINDER ENGINES

First-Harmonic Forces The x and y components of the resultant of the first harmonic forces for any multi-cylinder reciprocating engine can be written in the form

$$\begin{aligned} F_p^x &= A \cos \theta + B \sin \theta, \\ F_p^y &= C \cos \theta + D \sin \theta, \end{aligned} \quad (15.20)$$

where

$$A = \sum_{i=1}^n P_i \cos(\psi_i - \phi_i) \cos \psi_i, \quad (15.21a)$$

$$B = \sum_{i=1}^n P_i \sin(\psi_i - \phi_i) \cos \psi_i, \quad (15.21b)$$

$$C = \sum_{i=1}^n P_i \cos(\psi_i - \phi_i) \sin \psi_i, \quad (15.21c)$$

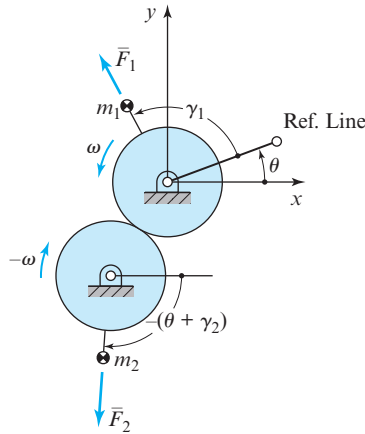


Figure 15.28 General arrangement of counter-rotating masses for balancing the primary forces.

and

$$D = \sum_{i=1}^n P_i \sin(\psi_i - \phi_i) \sin \psi_i, \quad (15.21d)$$

where n is the number of cylinders, $P_i = m_i r_i \omega^2$ is the inertia force of piston i , m_i is the effective mass of piston i (Sec. 14.5), r_i is the length of crank i , ω is the constant angular velocity of the crankshaft, ψ_i is the angle from the x axis to the line of sliding of piston i , and ϕ_i is the angle from the reference line on crank 1 to crank i (by definition, $\phi_1 = 0$). Note that, if the coefficients A , B , C , and D are all zero, then the x and y components of the resultant of the first-harmonic forces are zero; that is, there is no primary shaking force. This is the case, for example, with a six-cylinder in-line engine (previous section).

If the coefficients A , B , C , and D are not all zero, then the first-harmonic forces can always be balanced by a pair of rotating masses or in some special cases by a single mass, as shown in Fig. 15.28. The correcting masses are rotating with the same speed as the crankshaft. The correcting mass m_1 creates a correcting inertia force F_1 at an angle of $(\theta + \gamma_1)$ from the x axis, and the correcting mass m_2 creates a correcting inertia force F_2 at an angle of $-(\theta + \gamma_2)$ from the x axis. For balance, these two correcting forces plus the resultant of the first harmonic forces (the primary shaking force) must be equal to zero; that is,

$$\begin{aligned} A \cos \theta + B \sin \theta + F_1 \cos(\theta + \gamma_1) + F_2 \cos[-(\theta + \gamma_2)] &= 0, \\ C \cos \theta + D \sin \theta + F_1 \sin(\theta + \gamma_1) + F_2 \sin[-(\theta + \gamma_2)] &= 0. \end{aligned} \quad (15.22)$$

Expanding these two equations, in terms of functions of θ , γ_1 , and γ_2 , and rearranging, gives

$$\begin{aligned} (F_1 \cos \gamma_1 + F_2 \cos \gamma_2) \cos \theta - (F_1 \sin \gamma_1 + F_2 \sin \gamma_2) \sin \theta &= -A \cos \theta - B \sin \theta, \\ (F_1 \sin \gamma_1 - F_2 \sin \gamma_2) \cos \theta + (F_1 \cos \gamma_1 - F_2 \cos \gamma_2) \sin \theta &= -C \cos \theta - D \sin \theta. \end{aligned} \quad (15.23)$$

To satisfy Eqs. (15.23), for all values of the crank angle θ , the necessary conditions are

$$\begin{aligned} F_1 \cos \gamma_1 + F_2 \cos \gamma_2 &= -A, \\ F_1 \sin \gamma_1 + F_2 \sin \gamma_2 &= B, \\ F_1 \sin \gamma_1 - F_2 \sin \gamma_2 &= -C, \\ F_1 \cos \gamma_1 - F_2 \cos \gamma_2 &= -D. \end{aligned} \quad (15.24)$$

Solving Eqs. (15.24), the forces are

$$\begin{aligned} F_1 &= \frac{1}{2} \sqrt{(D+A)^2 + (C-B)^2}, \\ F_2 &= \frac{1}{2} \sqrt{(D-A)^2 + (C+B)^2}, \end{aligned} \quad (15.25)$$

and the orientation angles are given by

$$\begin{aligned} \tan \gamma_1 &= (C-B)/(D+A) \\ \tan \gamma_2 &= (C+B)/(D-A) \end{aligned} \quad (15.26)$$

Second Harmonic Forces The x and y components of the resultant of the second harmonic forces can be written in the form

$$\begin{aligned} F_S^x &= A' \cos 2\theta + B' \sin 2\theta, \\ F_S^y &= C' \cos 2\theta + D' \sin 2\theta, \end{aligned} \quad (15.27)$$

where

$$\begin{aligned} A' &= \sum_{i=1}^n Q_i \cos 2(\psi_i - \phi_i) \cos \psi_i, \\ B' &= \sum_{i=1}^n Q_i \sin 2(\psi_i - \phi_i) \cos \psi_i, \\ C' &= \sum_{i=1}^n Q_i \cos 2(\psi_i - \phi_i) \sin \psi_i, \end{aligned}$$

and

$$D' = \sum_{i=1}^n Q_i \sin 2(\psi_i - \phi_i) \sin \psi_i,$$

where the inertia force $Q_i = m_i(r_i/l_i)r_i\omega_i^2 = (r_i/l_i)P_i$. We have stated previously that, in general, $r_i/l_i < 1/4$; therefore, $Q_i < P_i$, which indicates that the magnitudes of the second harmonic forces are much smaller than the magnitudes of the first harmonic forces.

The second harmonic forces can always be balanced by a pair of gears rotating at twice the crankshaft speed, as shown in Fig. 15.29. Mass m'_1 creates force F'_1 at an orientation angle of $(2\theta + \lambda_1)$ from the x axis, and mass m'_2 creates force F'_2 at an orientation angle of

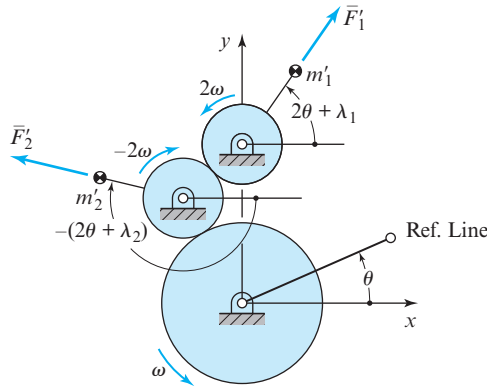


Figure 15.29 General arrangement of counter-rotating masses for balancing the secondary forces.

$-(2\theta + \lambda_2)$ from the x axis. For balance, these two forces plus the resultant of the second harmonic forces must be equal to zero.

Note that the analysis for the second harmonic forces is exactly the same as for the first harmonic forces, with A , B , C , and D replaced by A' , B' , C' , and D' , respectively, θ replaced by 2θ , and the orientation angles γ_1 and γ_2 replaced by orientation angles λ_1 and λ_2 , respectively. The two correcting forces can be written as

$$F'_1 = \frac{1}{2} \sqrt{(D' + A')^2 + (C' - B')^2}$$

and

$$F'_2 = \frac{1}{2} \sqrt{(D' - A')^2 + (C' + B')^2}.$$

The two orientation angles can be written as

$$\tan \lambda_1 = \frac{(C' - B')}{(D' + A')}$$

and

$$\tan \lambda_2 = \frac{(C' + B')}{(D' - A')}.$$

EXAMPLE 15.5

Determine the magnitudes and orientations of the forces created by the correcting masses needed to balance the first and second harmonic shaking forces of the three-cylinder engine shown in Fig. 15.30.

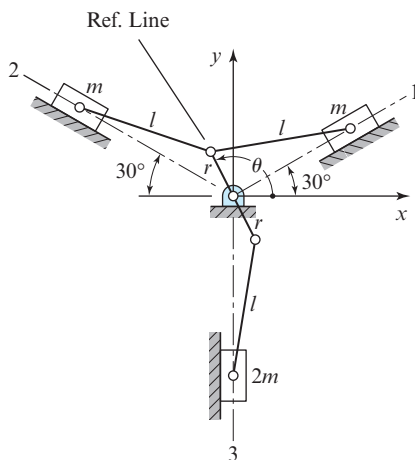


Figure 15.30
Three-cylinder
arrangement.

SOLUTION

The resultant first harmonic force vector is

$$\mathbf{F}_P = 1.5P \cos \theta \hat{\mathbf{i}} - 1.5P \sin \theta \hat{\mathbf{j}}, \quad (1)$$

where

$$P = mr\omega^2.$$

Comparing Eq. (1) with Eqs. (15.20), the coefficients are

$$A = 1.5P, \quad B = 0, \quad C = 0, \quad \text{and} \quad D = -1.5P.$$

Therefore, from Eqs. (15.25), the forces are

$$F_1 = \frac{1}{2} \sqrt{(D+A)^2 + (C-B)^2} = 0$$

and

$$F_2 = \frac{1}{2} \sqrt{(D-A)^2 + (C+B)^2} = 1.5P.$$

Since the first force is zero, there is only one correcting mass, that is, m_2 . The orientation angle for this mass, from the second of Eqs. (15.26), is

$$\tan \gamma_2 = (C+B)/(D-A) = 0/-3P = 0.$$

Therefore, since the denominator (the cosine of the angle) is negative, the result is

$$\gamma_2 = 180^\circ. \quad \text{Ans.}$$

The conclusion is that the correcting mass m_2 is rotating opposite to the crankshaft (at crankshaft speed) and oriented at angle $-(\theta + 180^\circ)$. Figure 15.31 shows the orientation of the correcting mass for $\theta = 0^\circ$, that is, the reference line is coincident with the x axis. If the correcting mass m_2 is placed at a radius r_2 , then the inertia force is

$$F_2 = m_2 r_2 \omega^2 = 1.5mr\omega^2.$$

Therefore, the product is

$$m_2 r_2 = 1.5mr.$$

Ans.

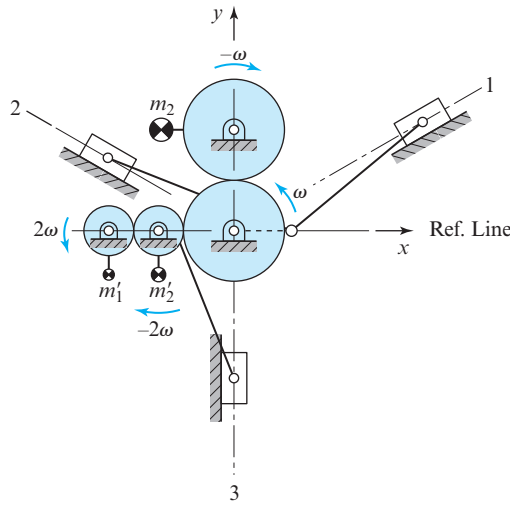


Figure 15.31 Balancing arrangement for $\theta = 0^\circ$.

The second harmonic force vector is

$$\mathbf{F}_s = 1.5Q \sin 2\theta \hat{\mathbf{i}} + 2.5Q \cos 2\theta \hat{\mathbf{j}}, \quad (2)$$

where

$$Q = \frac{mr^2\omega^2}{l}.$$

Comparing Eq. (2) with Eqs. (15.23), the coefficients are

$$A' = 0, \quad B' = 1.5Q, \quad C' = 2.5Q, \quad \text{and} \quad D' = 0.$$

Therefore, the inertia forces, from Eqs. (15.25), are

$$F'_1 = \frac{1}{2} \sqrt{(D' + A')^2 + (C' - B')^2} = 0.5Q$$

and

$$F'_2 = \frac{1}{2} \sqrt{(D' - A')^2 + (C' + B')^2} = 2.0Q.$$

The forces can be expressed as

$$F'_1 = m'_1 r'_1 (2\omega)^2 = 0.5m \frac{r^2}{l} \omega^2$$

and

$$F'_2 = m'_2 r'_2 (2\omega)^2 = 2m \frac{r^2}{l} \omega^2.$$

Therefore, the correcting mass m'_1 can be placed at a radius r'_1 such that

$$m'_1 r'_1 = 0.125mr^2/l, \quad \text{Ans.}$$

and the correcting mass m'_2 can be placed at radius r'_2 such that

$$m'_2 r'_2 = 0.5mr^2/l. \quad \text{Ans.}$$

From Eq. (15.26a), the first orientation angle is

$$\tan \lambda_1 = (C' - B) / (D' + A') = -1.0Q/0 = \infty.$$

Therefore, $\lambda_1 = 270^\circ$, since the numerator (the sine of the angle) is negative.

From Eq. (15.25b), the second orientation angle is

$$\tan \lambda_2 = (C' + B') / (D' - A') = +4.0Q/0 = \infty.$$

Thus, $\lambda_2 = 90^\circ$, since the numerator (the sine of the angle) is positive.

The conclusion is that we need an arrangement such as that shown in Fig. 15.31. Mass m'_1 is located on a shaft turning in the same direction as the crankshaft at twice the crankshaft speed. The orientation angle is $2\theta + 270^\circ$. Mass m'_2 is located on a shaft turning opposite to the crankshaft at twice the crankshaft speed. The orientation angle is $-(2\theta + 90^\circ)$.

15.12 BALANCING LINKAGES*

The two problems that arise in balancing linkages are balancing the shaking force and balancing the shaking moment. Lowen and Berkof [3] note that very few studies have been reported on the problem of balancing the shaking moment. This problem is discussed further in Sec. 15.13.

In force balancing a linkage, we must concern ourselves with the position of the total center of mass. If a way can be found to cause this total center of mass to remain stationary, the vector sum of all the frame forces will always be zero. Berkof and Lowen [1] have listed five methods of force balancing:

*Those who wish to investigate this topic in detail should begin with reference [3], in which the entire issue is devoted to the subject of linkage balancing. This issue contains 11 translations on the subject from the German and Russian literature.

1. The method of static balancing, in which concentrated link masses are replaced by systems of masses that are statically equivalent.
2. The method of principal vectors, in which an analytic expression is obtained for the center of mass and then manipulated to learn how its trajectory can be influenced.
3. The method of linearly independent vectors, in which the center of mass of a mechanism is made stationary, causing the coefficients of the time-dependent terms of the equation describing the trajectory of the total center of mass to vanish.
4. The use of cam-driven masses to keep the total center of mass stationary.
5. The addition of an axially symmetric duplicate mechanism by which the new combined total center of mass is made stationary.

Here we present only the Berkof-Lowen method [1], which employs the method of linearly independent vectors. The method is developed completely for the four-bar linkage, but the final results alone are given for a typical six-bar linkage. The procedure is as follows. First, find the equation that describes the trajectory of the total center of mass of the linkage. This equation will contain certain terms whose coefficients are time dependent. Then the total center of mass is made stationary by changing the positions of the individual link masses so that the coefficients of all time-dependent terms vanish. To accomplish this, it is necessary to write the equation in such a form that the time-dependent unit vectors contained in the equation are independent.

In Fig. 15.32, a general four-bar linkage is shown having link mass m_2 located at G_2 , m_3 located at G_3 , and m_4 located at G_4 . The total mass M of the four-bar linkage is

$$M = m_2 + m_3 + m_4. \quad (a)$$

We begin by defining the position of the total center of mass of the linkage by the vector $\mathbf{r}_s$,

$$\mathbf{r}_s = \frac{1}{M} (m_2 \mathbf{r}_{s_2} + m_3 \mathbf{r}_{s_3} + m_4 \mathbf{r}_{s_4}), \quad (b)$$

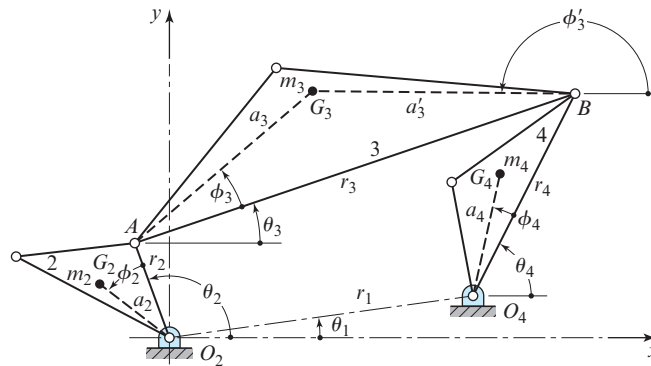


Figure 15.32 Four-bar linkage showing arbitrary postures of the link masses.

where $\mathbf{r}_{s_2}$, $\mathbf{r}_{s_3}$, and $\mathbf{r}_{s_4}$ are the vectors that describe the positions of m_2 , m_3 , and m_4 , respectively, in the global xy coordinate system. Coordinates a_i , ϕ_i describe the positions of the mass centers within each link. Thus, from Fig. 15.32,

$$\begin{aligned}\mathbf{r}_{s_2} &= a_2 e^{j(\theta_2 + \phi_2)}, \\ \mathbf{r}_{s_3} &= r_2 e^{j\theta_2} + a_3 e^{j(\theta_3 + \phi_3)}, \\ \mathbf{r}_{s_4} &= r_1 e^{j\theta_1} + a_4 e^{j(\theta_4 + \phi_4)}.\end{aligned}\quad (c)$$

Substituting Eqs. (c) into (b) gives

$$M\mathbf{r}_s = (m_2 a_2 e^{j\phi_2} + m_3 r_2) e^{j\theta_2} + (m_3 a_3 e^{j\phi_3}) e^{j\theta_3} + (m_4 a_4 e^{j\phi_4}) e^{j\theta_4} + m_4 r_1 e^{j\theta_1}, \quad (d)$$

where we have used the identity $e^{j\alpha} e^{j\beta} = e^{j(\alpha + \beta)}$.

For a four-bar linkage the loop-closure equation can be written as

$$r_2 e^{j\theta_2} + r_3 e^{j\theta_3} - r_4 e^{j\theta_4} - r_1 e^{j\theta_1} = 0. \quad (e)$$

Thus, the time-dependent terms $e^{j\theta_2}$, $e^{j\theta_3}$, and $e^{j\theta_4}$ in Eq. (d) are not independent. To make them so, we solve Eq. (e) for one of the unit vectors, say $e^{j\theta_3}$, and substitute the result into Eq. (d). Thus,

$$e^{j\theta_3} = \frac{1}{r_3} (r_1 e^{j\theta_1} - r_2 e^{j\theta_2} + r_4 e^{j\theta_4}), \quad (f)$$

and Eq. (d) now becomes

$$\begin{aligned}M\mathbf{r}_s &= (m_2 a_2 e^{j\phi_2} + m_3 r_2 - m_3 a_3 \frac{r_2}{r_3} e^{j\phi_3}) e^{j\theta_2} + (m_4 a_4 e^{j\phi_4} + m_3 a_3 \frac{r_4}{r_3} e^{j\phi_3}) e^{j\theta_4} \\ &\quad + (m_4 r_1 + m_3 a_3 \frac{r_1}{r_3} e^{j\phi_3}) e^{j\theta_1}.\end{aligned}\quad (g)$$

Equation (g) shows that the center of mass will be stationary at the position

$$\mathbf{r}_s = \frac{r_1}{r_3 M} (m_4 r_3 + m_3 a_3 e^{j\phi_3}) e^{j\theta_1} \quad (15.28)$$

if we make the following coefficients of the time-dependent terms vanish:

$$m_2 a_2 e^{j\phi_2} + m_3 r_2 - m_3 a_3 \frac{r_2}{r_3} e^{j\phi_3} = 0, \quad (h)$$

$$m_4 a_4 e^{j\phi_4} + m_3 a_3 \frac{r_4}{r_3} e^{j\phi_3} = 0. \quad (i)$$

But, Eq. (h) can be simplified by locating G_3 from point B instead of point A (Fig. 15.32). Thus,

$$a_3 e^{j\phi_3} = r_3 + a'_3 e^{j\phi'_3}.$$

With this substitution, Eq. (h) becomes

$$m_2 a_2 e^{j\phi_2} - m_3 a'_3 \frac{r_2}{r_3} e^{j\phi'_3} = 0. \quad (j)$$

Equations (i) and (j) must be satisfied to obtain total force balance. These equations yield the two sets of conditions:

$$\begin{aligned} m_2 a_2 &= m_3 a'_3 \frac{r_2}{r_3} & \text{and} & \quad \phi_2 = \phi'_3, \\ m_4 a_4 &= m_3 a_3 \frac{r_4}{r_3} & \text{and} & \quad \phi_4 = \phi_3 + \pi. \end{aligned} \quad (15.29)$$

A study of these conditions indicates that the mass and its location can be specified in advance for any single link; then, full balance can be obtained by rearranging the mass of the other two links.

The usual problem in balancing a four-bar linkage is that the link lengths r are specified in advance because of the functional requirements. For this situation, counterweights can be added to the input and output links to redistribute their masses, while the geometry of the coupler link is undisturbed.

When adding counterweights, the following relations must be satisfied:

$$m_i a_i \angle \phi_i = m_i^\circ a_i^\circ \angle \phi_i^\circ + m_i^* a_i^* \angle \phi_i^*, \quad (15.30)$$

where $m_i^\circ, a_i^\circ, \phi_i^\circ$ are the parameters of the unbalanced linkage, m_i^*, a_i^*, ϕ_i^* are the parameters of the counterweights, and m_i, a_i, ϕ_i are the parameters that result from Eqs. (15.29). A second condition that must generally be satisfied is

$$m_i = m_i^\circ + m_i^*. \quad (15.31)$$

If the solution to a balancing problem can remain as the mass distance product $m_i^* a_i^*$, Eq. (15.31) need not be used, and Eq. (15.30) can be solved to yield

$$m_i^* a_i^* = \sqrt{(m_i a_i)^2 + (m_i^\circ a_i^\circ)^2 - 2(m_i a_i)(m_i^\circ a_i^\circ) \cos(\phi_i - \phi_i^\circ)}, \quad (15.32)$$

$$\phi_i^* = \tan^{-1} \left(\frac{m_i a_i \sin \phi_i - m_i^\circ a_i^\circ \sin \phi_i^\circ}{m_i a_i \cos \phi_i - m_i^\circ a_i^\circ \cos \phi_i^\circ} \right). \quad (15.33)$$

Figure 15.33 shows a typical six-bar linkage and the notation. For this, the Berkof-Lowen conditions for total balance are:

$$\begin{aligned} m_2 \frac{a_2}{r_2} e^{j\phi_2} &= m_5 \frac{a'_5}{r_5} \frac{b_2}{r_2} e^{j(\phi_1 + \alpha_2)} + m_3 \frac{a'_3}{r_3} e^{j\phi_1}, \\ m_4 \frac{a_4}{r_4} e^{j\phi_4} &= m_6 \frac{a'_6}{r_6} \frac{b_4}{r_4} e^{j\phi'_6} - m_3 \frac{a_3}{r_3} e^{j(\phi_3 + \alpha_4)}, \\ m_5 \frac{a_5}{r_5} e^{j\phi_5} &= -m_6 \frac{a_6}{r_6} e^{j\phi_6}. \end{aligned} \quad (15.34)$$

Similar relations can be devised for other six-bar linkages. For total balance, Eqs. (15.34) demonstrate that a certain mass-geometry relation between links 5 and 6 must be satisfied,

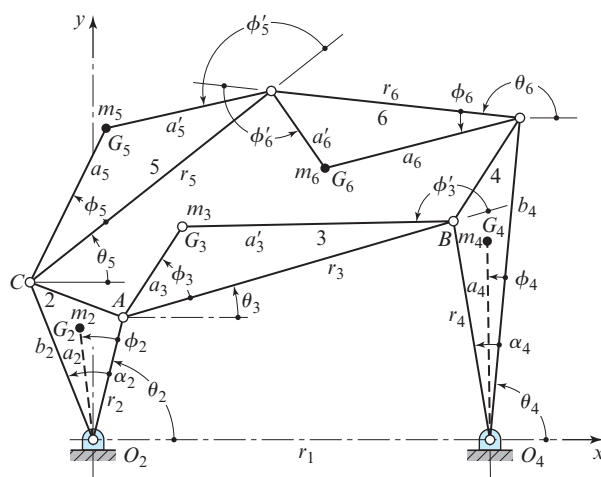


Figure 15.33 Notation for a six-bar linkage.

after which the masses of any two links and their locations can be specified. Balance is then achieved by a redistribution of the masses of the remaining three movable links.

It is important to note that the addition of counterweights to balance the shaking forces will probably increase the internal bearing forces as well as the shaking moment. Thus, only a partial balance may represent the best compromise between these three effects.

EXAMPLE 15.6

Table 15.2 is a tabulation of the dimensions, masses, and the locations of the mass centers of a four-bar linkage having link 2 as the input and link 4 as the output. Complete force balancing is to be achieved by adding counterweights to the input and the output links. Find the mass-distance product and the angular orientation of each counterweight.

Table 15.2 Parameters of An Unbalanced Four-Bar Linkage

Link <i>i</i>	1	2	3	4
r_i , mm	140	50	150	75
a_i^o , mm	—	25	80	40
ϕ_i^o	—	0°	15°	0°
$a_i'^o$, mm	—	—	75.6	—
$\phi_i'^o$	—	—	164.1°	—
m_i^o , kg	—	0.046	0.125	0.054

SOLUTION

From Eqs. (15.29) we first find

$$m_2 a_2 = m_3^o a_3'^o \frac{r_2}{r_3} = (0.125 \text{ kg})(75.6 \text{ mm}) \frac{50 \text{ mm}}{150 \text{ mm}} = 3.150 \text{ g} \cdot \text{m},$$

$$\phi_2 = \phi_3' = 164.1^\circ,$$

$$m_4 a_4 = m_4^o a_3^o \frac{r_4}{r_3} = (0.125 \text{ kg})(80 \text{ mm}) \frac{75 \text{ mm}}{150 \text{ mm}} = 5.000 \text{ g} \cdot \text{m},$$

$$\phi_4 = \phi_4^o + 180^\circ = 15^\circ + 180^\circ = 195^\circ.$$

Note that $m_2 a_2$ and $m_4 a_4$ are the mass-distance products *after* the counterweights have been added. Also, note that the link 3 parameters are altered. We next compute

$$m_2^o a_2^o = (0.046 \text{ kg})(25 \text{ mm}) = 1.150 \text{ g} \cdot \text{m} \text{ and}$$

$$m_4^o a_4^o = (0.054 \text{ kg})(40 \text{ mm}) = 2.160 \text{ g} \cdot \text{m}.$$

Using Eq. (15.32), we compute the mass-distance product for the link 2 counterweight as

$$\begin{aligned} m_2^* a_2^* &= \sqrt{(m_2 a_2)^2 + (m_2^o a_2^o)^2 - 2(m_2 a_2)(m_2^o a_2^o) \cos(\phi_2 - \phi_2^o)} \\ &= \sqrt{(3.150 \text{ g} \cdot \text{m})^2 + (1.150 \text{ g} \cdot \text{m})^2 - 2(3.150 \text{ g} \cdot \text{m})(1.150 \text{ g} \cdot \text{m}) \cos(164.1^\circ - 0^\circ)} \\ &= 4.268 \text{ g} \cdot \text{m} \end{aligned} \quad \text{Ans.}$$

From Eq. (15.33), we find the orientation of this counterweight to be

$$\begin{aligned} \phi_2^* &= \tan^{-1} \left(\frac{m_2 a_2 \sin \phi_2 - m_2^o a_2^o \sin \phi_2^o}{m_2 a_2 \cos \phi_2 - m_2^o a_2^o \cos \phi_2^o} \right) \\ &= \tan^{-1} \left(\frac{(3.150 \text{ g} \cdot \text{m}) \sin 164.1^\circ - (1.150 \text{ g} \cdot \text{m}) \sin 0^\circ}{(3.150 \text{ g} \cdot \text{m}) \cos 164.1^\circ - (1.150 \text{ g} \cdot \text{m}) \cos 0^\circ} \right) = 168.3^\circ \end{aligned} \quad \text{Ans.}$$

Using the same procedure for link 4 yields

$$m_4^* a_4^* = 7.11 \text{ g} \cdot \text{m} \quad \text{at} \quad \phi_4^* = 190.5^\circ. \quad \text{Ans.}$$

Figure 15.34 is a scale drawing of the complete linkage with the two counterweights added to achieve complete force balance.

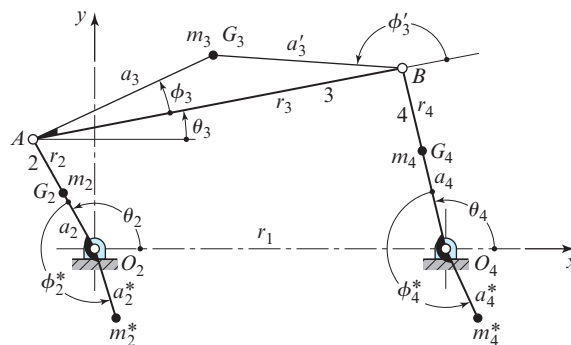


Figure 15.34 Four-bar crank-rocker linkage showing counterweights added to input and output links.

15.13 BALANCING OF MACHINES*

In Sec. 15.12, we learned how to force balance a simple linkage using two or more counterweights, depending upon the number of links composing the linkage. Unfortunately, this does not balance the shaking moment and, in fact, may even make it worse because of the addition of the counterweights. If a machine is imagined to be composed of several mechanisms, one might consider balancing the machine by balancing each mechanism separately, but this may not result in the best balance for the machine, since the addition of a large number of counterweights may cause the inertia torque to be completely unacceptable. Furthermore, unbalance of one mechanism may counteract the unbalance in another, eliminating the need for some of the counterweights in the first place.

Stevensen demonstrates that any single harmonic of unbalanced forces, moments of forces, and torques in a machine can be balanced by the addition of six counterweights. They are arranged on three shafts, two per shaft, driven at the constant speed of the harmonic, and have axes parallel, respectively, to each of the three mutually perpendicular axes through the center of mass of the machine. The method is too complex to be included here, but it is worthwhile to look at the overall approach.

Using the methods of this text together with a computer, the rectilinear and angular accelerations of each of the moving mass centers of a machine are computed for postures throughout a cycle of motion. The masses and mass moments of inertia of the machine must also be computed or determined experimentally. Then, the inertia forces, the inertia torques, and the moments of the forces are computed with reference to the three mutually perpendicular coordinate axes through the center of mass of the machine. When these are summed for each posture in the cycle, six functions of time are the result, three for the forces and three for the moments. With the aid of a computer, it is then possible to use numeric harmonic analysis to define the component harmonics of the unbalanced forces parallel to the three axes and of the unbalanced moments about the three axes.

To balance a single harmonic, each component of the unbalance of the machine is represented in the form $A \cos \omega t + B \sin \omega t$ with appropriate subscripts. Six equations of equilibrium are then written, which include the unbalances as well as the effects of the six unknown counterweights. These equations are arranged so that each of the $\sin \omega t$ and $\cos \omega t$ terms is multiplied by parenthetic coefficients. Balance is then achieved by setting the parenthetic terms in each equation equal to zero, much in the same manner as in the preceding section. This results in 12 equations, all linear, in 12 unknowns. With the locations for the balancing counterweights on the three shafts specified, the 12 equations can be solved for the six mr products, and the six phase angles needed for the six balancing weights. Stevenson goes on to demonstrate that when fewer than the necessary three shafts are available, it becomes necessary to optimize some effect of the unbalance, such as the motion of a point on the machine.

* The material for this section is from Stevenson [6]. It is included with the advice and consent of Professor E. N. Stevenson.

15.14 REFERENCES

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PROBLEMS

- 15.1** Determine the bearing reactions at A and B if the speed of the shaft is 300 rev/min. Also determine the magnitude and orientation of the balancing mass if it is located at a radius of 50 mm.

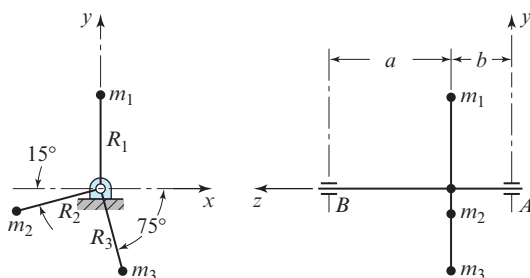


Figure P15.1 $a = 800$ mm, $b = 200$ mm, $R_1 = 25$ mm, $R_2 = 35$ mm, $R_3 = 40$ mm, $m_1 = 2$ kg, $m_2 = 1.5$ kg, and $m_3 = 3$ kg.

- 15.2** Three weights are connected to a shaft that rotates in bearings at A and B. Determine the magnitudes and orientations of the bearing reactions if the speed of the shaft is 300 rev/min. Also, determine the magnitude and orientation of a counterweight that is to be located at a radius of 10 in.

- †15.3** Two weights are connected to a rotating shaft and mounted outboard of bearings A and B. If the speed of the shaft is 120 rev/min, what are the magnitudes and orientations of the bearing reactions at A and B?

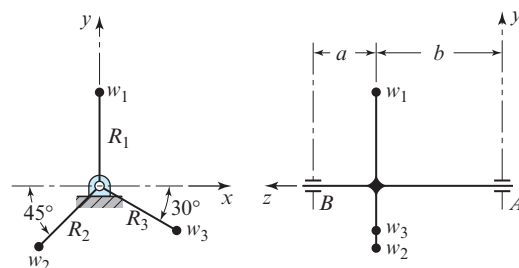


Figure P15.2 $a = 6$ in, $b = 12$ in, $R_1 = 8$ in, $R_2 = 12$ in, $R_3 = 6$ in, $w_1 = 2$ oz, $w_2 = 1.5$ oz, and $w_3 = 3$ oz.

Suppose the system is to be balanced by removing weight at a radius of 5 in. Determine the magnitude and orientation of the weight to be removed.

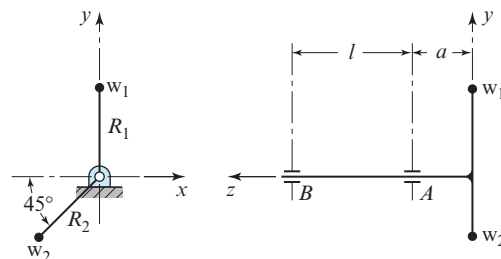


Figure P15.3 $l = 4$ in, $a = 2$ in, $R_1 = 4$ in, $R_2 = 6$ in, $w_1 = 4$ lb, and $w_2 = 3$ lb.

- 15.4** If the speed of the shaft is 220 rev/min, calculate the magnitudes and orientations of the bearing reactions at A and B for the two-mass system.

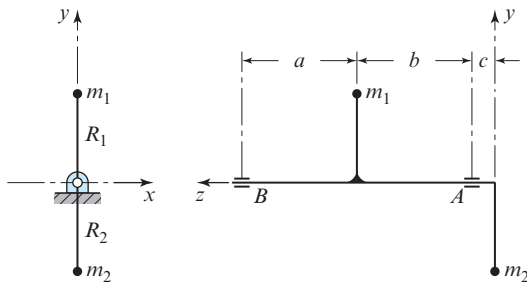


Figure P15.4 $a = b = 250$ mm, $c = 50$ mm, $R_1 = 60$ mm, $R_2 = 40$ mm, $m_1 = 2$ kg, and $m_2 = 1.5$ kg.

- 15.5** Determine the bearing reactions at A and B and their orientations if the shaft speed is 100 rev/min.

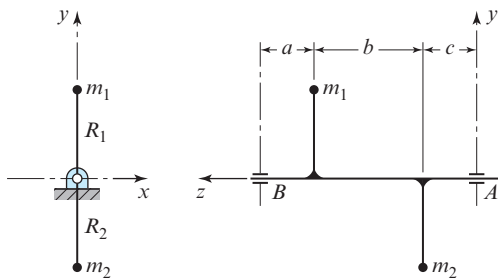


Figure P15.5 $a = c = 300$ mm, $b = 600$ mm, $R_1 = R_2 = 60$ mm, $m_1 = 1$ kg, and $m_2 = 3$ kg.

- 15.6** The rotating shaft shown in Fig. P15.5 supports two masses m_1 and m_2 , whose weights are 4 lb and 5 lb, respectively. The dimensions are $a = 2$ in, $b = 8$ in, $c = 3$ in, $R_1 = 4$ in, and $R_2 = 3$ in. Find the magnitudes of the bearing reactions at A and B and their orientations if the shaft speed is 360 rev/min.
- 15.7** The shaft is to be balanced by placing masses in correction planes L and R. Calculate the magnitudes and orientations of the correction masses.
- 15.8** The shaft of Problem 15.7 is to be balanced by removing weight from the two correction planes. Determine the correction masses and their orientations.

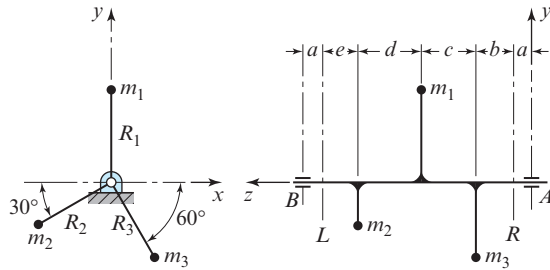


Figure P15.7 $a = 1$ in, $b = e = 8$ in, $c = 10$ in, $d = 9$ in, $R_1 = R_3 = 5$ in, $R_2 = 4$ in, $w_1 = w_3 = 4$ oz, and $w_2 = 3$ oz.

- †15.9** The shaft shown in Fig. P15.7 is to be balanced by removing masses in the two correction planes, L and R. The three masses are $m_1 = 6$ g, $m_2 = 7$ g, and $m_3 = 5$ g. The dimensions are $a = 25$ mm, $b = 300$ mm, $c = 600$ mm, $d = 150$ mm, $e = 75$ mm, $R_1 = 125$ mm, $R_2 = 150$ mm, and $R_3 = 100$ mm. Calculate the magnitudes and orientations of the correction masses.

- 15.10** Repeat Problem 15.9 if masses are to be added in the two correction planes.

- 15.11** Solve the two-plane balancing problem as stated in Sec. 15.8.

- 15.12** A rotor to be balanced in the field yields an amplitude of 5 at an angle of 142° at the left bearing and an amplitude of 3 at an angle of -22° at the right bearing because of unbalance. To correct this, a trial mass of 12 is added to the left correction plane at an angle of 210° from the rotating reference. A second run then gives left and right responses of $8 \angle 160^\circ$ and $4 \angle 260^\circ$, respectively. The first trial mass is then removed and a second mass of 6 is added to the right correction plane at an angle of -70° . The responses to this are $2 \angle 74^\circ$ and $4.5 \angle -80^\circ$ for the left and right bearings, respectively. Determine the original unbalances.

- 15.13** A rotor is rotating with a constant angular velocity $\omega = 50$ rad/s, and is dynamically balanced by the two correcting masses. A decision has been made to use planes 1 and 3 instead of planes 1 and 2. Determine the magnitudes and orientations of the new masses $(m_{C1})_{\text{new}}$ and $(m_{C3})_{\text{new}}$ at the same radii $R_{C1} = 50$ mm and $R_{C3} = 150$ mm.

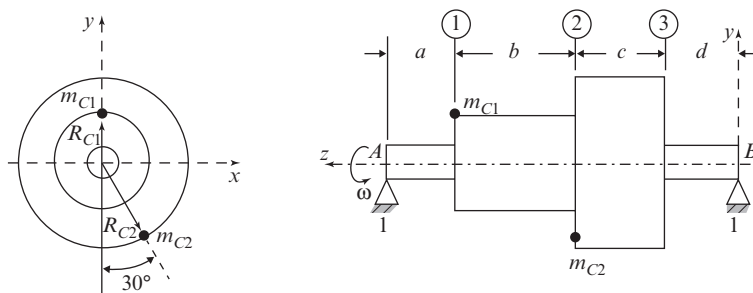


Figure P15.13 $a = d = 60$ mm, $b = 100$ mm, $c = 80$ mm, $R_{C1} = 50$ mm, $R_{C2} = 150$ mm, $m_{C1} = 3$ kg, and $m_{C2} = 2$ kg.

- 15.14** The shaft is rotating with a constant angular velocity $\omega = 80$ rad/s. For dynamic balance, determine the magnitudes and orientations of the correcting masses to be removed in planes 1 and 2 at radii $R_{C1} = R_{C2} = 2.6$ in.

- 15.15** The shaft is simply supported by the bearings at A and B and is rotating with a constant angular

velocity $\omega = 50$ rad/s. Determine the magnitudes and orientations of the correcting masses that must be removed in the correcting planes 1 and 2 to dynamically balance the system. The radial distances from the shaft axis are $R_{C1} = R_{C2} = 70$ mm. Show the orientations of the correcting masses on the right figure.

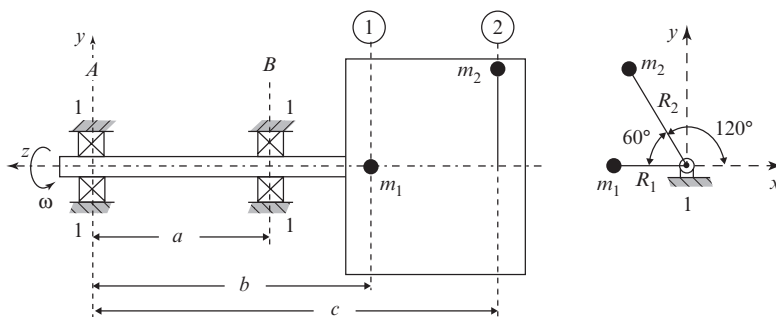


Figure P15.14 $a = 10$ in, $b = 16$ in, $c = 24$ in, $m_1 = 8.8$ lb, and $m_2 = 4.4$ lb.

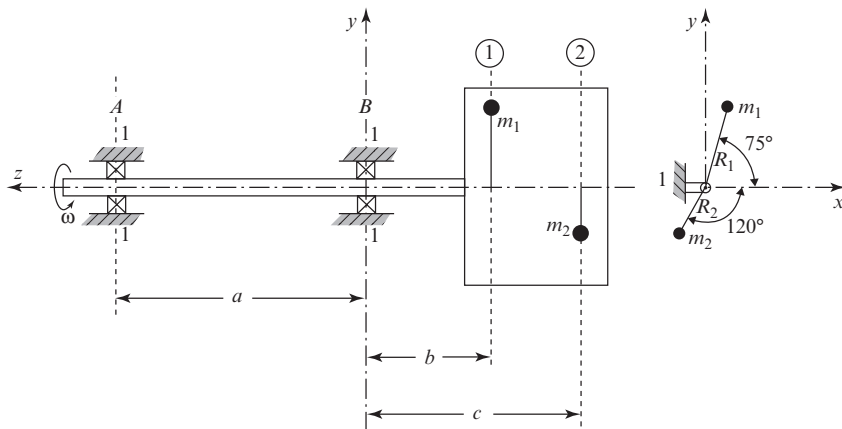


Figure P15.15 $a = 400$ mm, $b = 150$ mm, $c = 300$ mm, $R_1 = 60$ mm, $R_2 = 35$ mm, $m_1 = 12$ kg, and $m_2 = 15$ kg.

- 15.16** The shaft is simply supported by the bearings at A and B and is rotating with a constant angular velocity $\omega = 75 \text{ rad/s}$. To dynamically balance the system, determine the magnitudes and orientations of the correcting masses that must be removed in the correcting planes 1 and 2 at radial distances $R_{C1} = R_{C2} = 2.2 \text{ in}$.
- 15.17** Three mass particles are rigidly attached to a simply supported shaft that is rotating with a constant angular velocity $\omega = 15 \text{ rad/s}$. Determine the magnitudes and orientations of the bearing reaction forces at A and B . Then determine the magnitudes and orientations of the correcting masses that must

be added in the two correcting planes, 1 and 2, to dynamically balance the system. The correcting masses are to be placed at radial distances $R_{C1} = R_{C2} = 25 \text{ mm}$ from the shaft axis.

- 15.18** Three mass particles are rigidly attached to the shaft, which is rotating with a constant angular velocity $\omega = 350 \hat{k} \text{ rev/min}$. Determine the magnitudes and orientations of the reaction forces at the bearings A and B . Then, determine the correcting masses that must be removed in the correction planes 1 and 2 to dynamically balance the system if the radial distances of the correcting masses are specified as $R_{C1} = R_{C2} = 3 \text{ in}$.

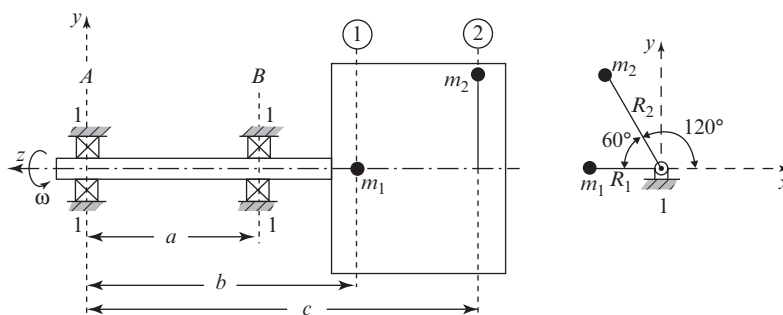


Figure P15.16 $a = 10 \text{ in}$, $b = 16 \text{ in}$, $c = 24 \text{ in}$, $R_1 = 1 \text{ in}$, $R_2 = 1.6 \text{ in}$, $m_1 = 11 \text{ lb}$, and $m_2 = 15.4 \text{ lb}$.

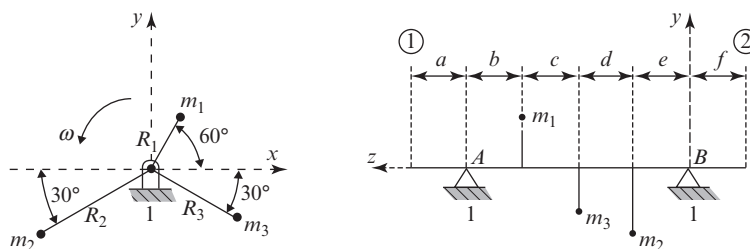


Figure P15.17 $a = 50 \text{ mm}$, $b = 215 \text{ mm}$, $c = 275 \text{ mm}$, $d = 100 \text{ mm}$, $e = 150 \text{ mm}$, $f = 125 \text{ mm}$, $R_1 = 15 \text{ mm}$, $R_2 = 35 \text{ mm}$, $R_3 = 20 \text{ mm}$, $m_1 = 3 \text{ kg}$, $m_2 = 1 \text{ kg}$, and $m_3 = 2 \text{ kg}$.

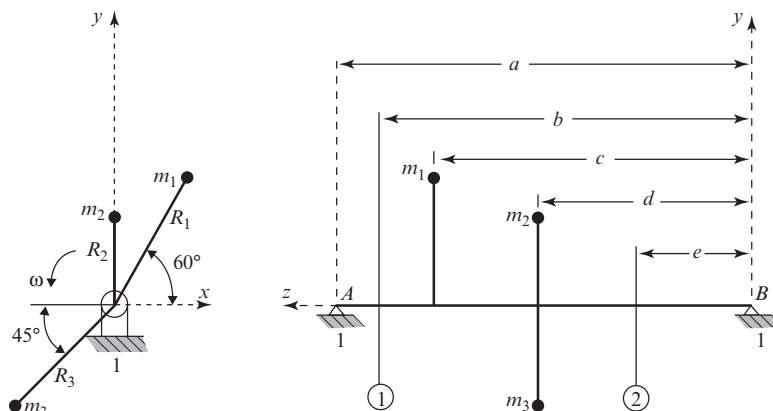


Figure P15.18 $a = 10.6$ in, $b = 9.4$ in, $c = 8$ in, $d = 5.4$ in, $e = 3$ in, $R_1 = 3.8$ in, $R_2 = 2.2$ in, $R_3 = 3.6$ in, $m_1 = 8.8$ lb, $m_2 = 7$ lb, and $m_3 = 1.1$ lb.

15.19 The shaft of the distributed mass system is simply supported by the bearings at A and B and has a constant speed of 550 rev/min. The reaction forces acting on the bearings at A and B are $\mathbf{F}_A = -17.5\hat{\mathbf{i}} + 30.3\hat{\mathbf{j}}$ N and $\mathbf{F}_B = -25.0\hat{\mathbf{i}} - 43.3\hat{\mathbf{j}}$ N, respectively. Using the graphic approach determine the magnitudes and angular orientations of the reaction forces at bearings A and B.

15.20 The distributed mass system, denoted as body 2, is simply supported bearings at A and B and has a constant speed of 240 rev/min. The forces from the system acting on the ground at bearings A and B are $(\mathbf{F}_{21})_A = 250\hat{\mathbf{i}} + 75\hat{\mathbf{j}}$ N and $(\mathbf{F}_{21})_B = 80\hat{\mathbf{i}} - 125\hat{\mathbf{j}}$ N, respectively. Determine the magnitudes and orientations of the correcting masses that must be removed in the correction planes 1 and 2 to ensure moment (dynamic) balance of the system.

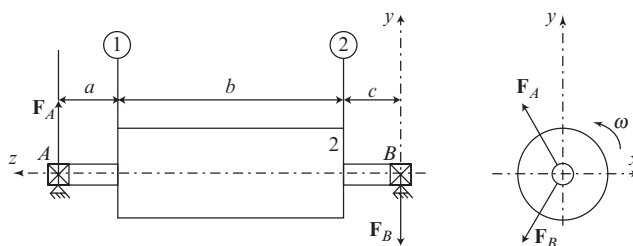


Figure P15.19 $a = 1.2$ in, $b = c = 1.0$ in, $d = 0.8$ in, $R_1 = 0.80$ in, $R_2 = 0.80$ in, $R_3 = 0.60$ in, $m_1 = 15.4$ lb, $m_2 = 26.5$ lb, and $m_3 = 17.6$ lb.

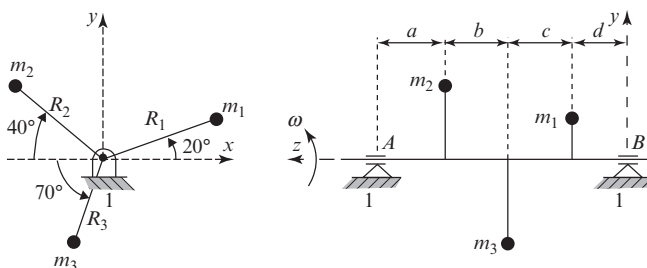


Figure P15.20 $a = 250$ mm, $b = 200$ mm, $c = 75$ mm, and $R_{C1} = R_{C2} = 30$ mm.

15.21 The shaft, simply supported by bearings at A and B , has a constant speed of 300 rev/min. Using the graphic approach determine the magnitudes and orientations of the reaction forces at bearings A and B .

15.22 The angular speed of the continuous mass system, denoted as 2, in the simply supported bearings at A and B is a constant 360 rev/min. The forces from the system acting on the ground at bearings A and B are specified as $(\mathbf{F}_{21})_A = 73\hat{\mathbf{i}} - 79\hat{\mathbf{j}}$ lb and $(\mathbf{F}_{21})_B = 63\hat{\mathbf{i}} + 118\hat{\mathbf{j}}$ lb, respectively. Determine the magnitudes and orientations of the masses that

must be removed in the correcting planes 1 and 2 to ensure dynamic balance of the system.

15.23 The constant angular velocity of the two-cylinder engine crankshaft is $\omega = 200$ rad/s counterclockwise. Determine the x and y components of the primary shaking force acting on the crankshaft bearing in terms of the crank angle θ . Then, determine the magnitudes and orientations of the correcting masses that must be added at the radial distance $R_C = 40$ mm from the crankshaft axis. Determine the answers when the reference line (attached to crank 1) is specified at the crank angle $\theta = 30^\circ$, as shown on the figure to the right.

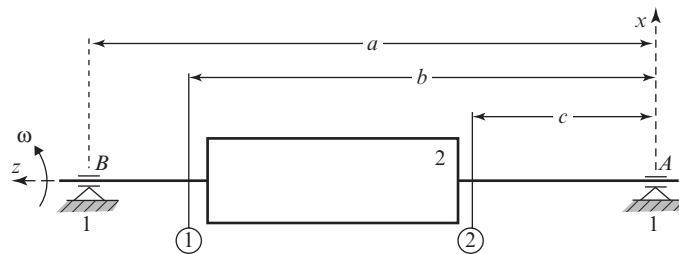


Figure P15.21 $a = 40$ mm, $b = c = 15$ mm, $d = 25$ mm, $R_1 = 10$ mm, $R_2 = 35$ mm, $R_3 = 15$ mm, $m_1 = 17$ kg, $m_2 = 4$ kg, and $m_3 = 8$ kg.

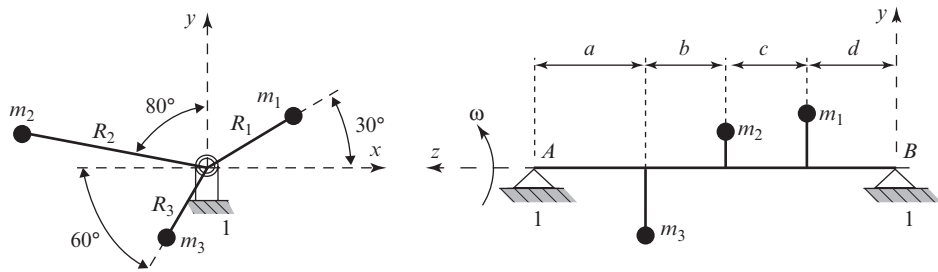


Figure P15.22 $a = 30$ in, $b = 27$ in, $c = 7$ in, and $R_{C1} = R_{C2} = 2$ in.

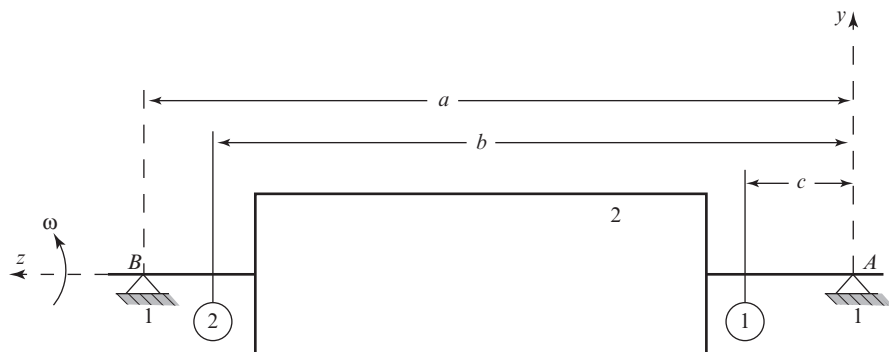


Figure P15.23 $R_1 = R_2 = R = 80$ mm, $L_1 = L_2 = L = 160$ mm, and $m_1 = m_2 = m = 15$ kg.

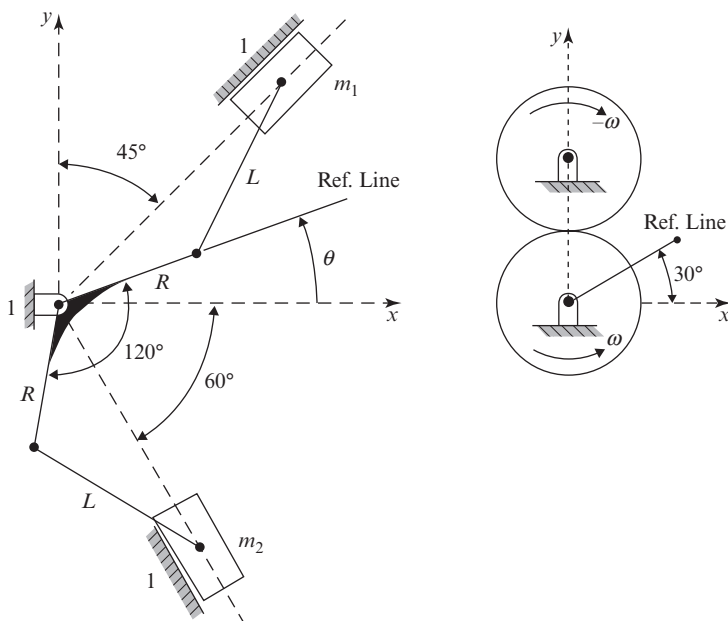


Figure P15.24 $R_1 = R_2 = R = 3$ in, $L_1 = L_2 = L = 24$ in, and $m_1 = m_2 = m = 13.25$ lb.

- 15.24** The two-cylinder engine crankshaft is rotating counterclockwise with a constant angular velocity $\omega = 45$ rad/s. Determine the magnitude and direction of the primary shaking force in terms of crank angle θ . If correcting masses are required to balance the primary shaking force, then determine: (a) the magnitudes and orientations of the inertial forces created by these correcting masses, and (b) the magnitudes and orientations of the correcting masses if $R_{C1} = R_{C2} = 6$ in. Determine the answers for parts (a) and (b) when the reference line attached to crank 1 is at crank angle $\theta = 210^\circ$.
- 15.25** The crankshaft of the two-cylinder engine (Fig. P15.25, next page) is rotating counterclockwise with a constant angular velocity $\omega = \dot{\theta} = 45$ rad/s. Determine the magnitude and orientation of the primary shaking force in terms of crank angle θ . If correcting masses are required to balance the primary shaking force, then determine (a) the magnitudes and orientations of the inertial forces created by these correcting masses, (b) the magnitudes and orientations of the correcting masses if $R_{C1} = R_{C2} = 200$ mm, and (c) determine the answers when the reference line (attached to crank 1) is specified at crank angle $\theta = 0^\circ$.
- 15.26** The crankshaft of the three-cylinder engine (Fig. P15.26, next page) is rotating with a constant angular velocity $\omega = 50\hat{k}$ rad/s. Determine the x and y components of the primary shaking force, the magnitude(s) of the correcting force (or forces) created by the correcting mass (or masses), and the orientation(s) of the correcting force (or forces). Determine the answers when the reference line (attached to crank 1) is specified at crank angle $\theta = 0^\circ$.
- 15.27** The crankshaft of the two-cylinder engine (Fig. P15.27, page 883) is rotating counterclockwise with a constant angular speed $\omega = 330$ rev/min. Determine the x and y components of the primary shaking force on the crankshaft bearing, and the magnitudes and orientations of the inertial forces created by the correcting masses that balance the primary shaking force. Determine the magnitude and orientation of the primary shaking force when the reference line (attached to crank 1) is specified at crank angle $\theta = 0^\circ$.

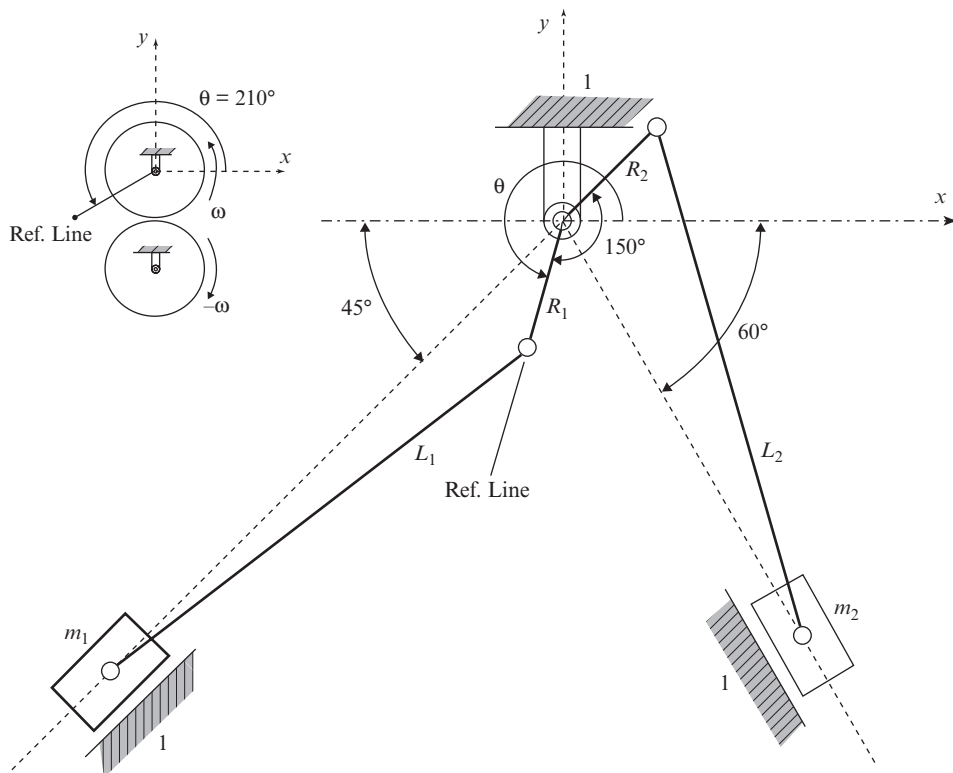


Figure P15.25 $R_1 = R_2 = R = 100$ mm, $L_1 = L_2 = L = 550$ mm, and $m_1 = m_2 = m = 5$ kg.

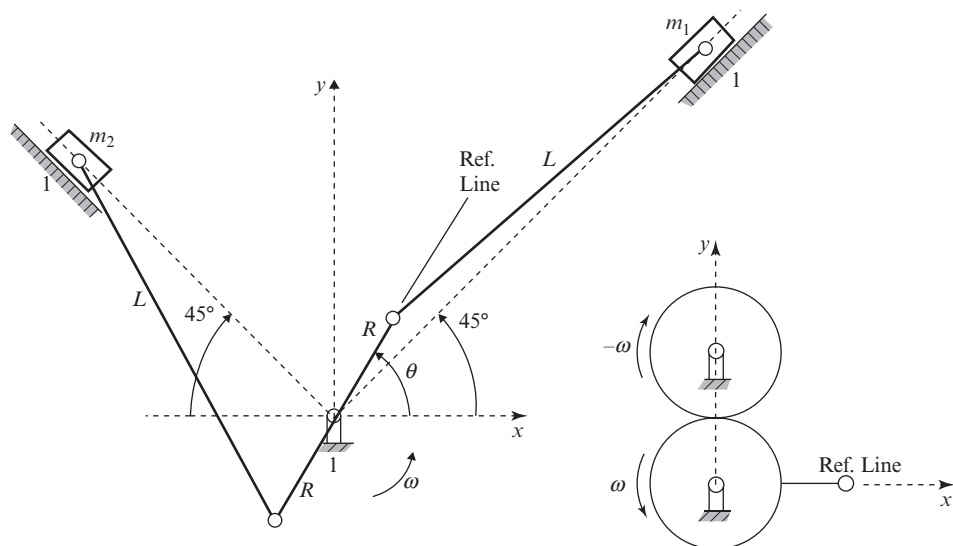


Figure P15.26 $R_1 = R_2 = R_3 = R = 6$ in, $L_1 = L_2 = L_3 = L = 30$ in, $m_1 = 2m = 22$ lb, and $m_2 = m_3 = m = 11$ lb.

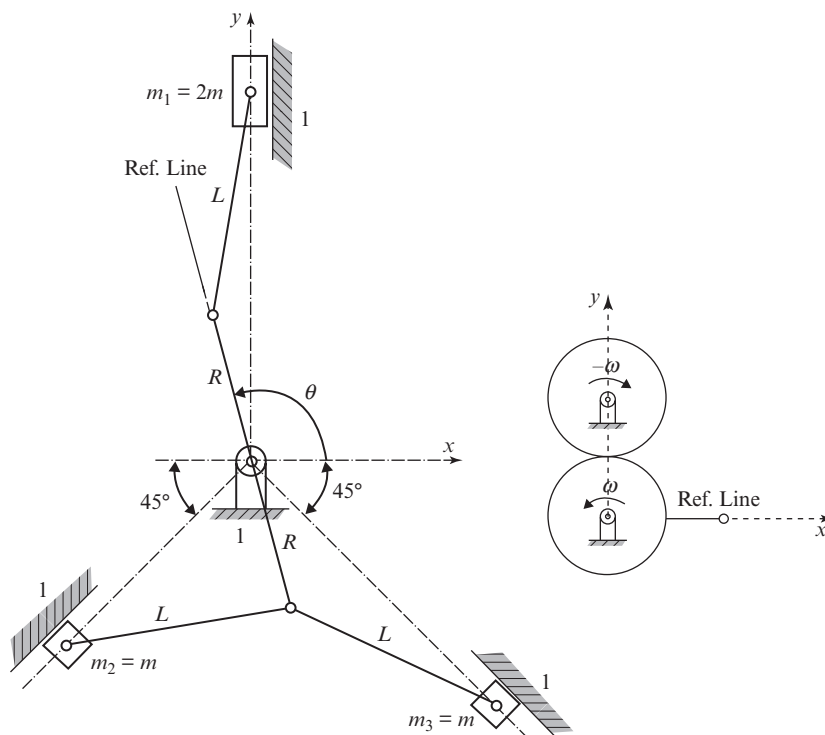


Figure P15.27 $R_1 = R_2 = R = 120$ mm, $L_1 = L_2 = L = 450$ mm, $m_1 = 3m = 60$ kg, and $m_2 = m = 20$ kg.

15.28 The two cranks of the two-cylinder engine (Fig. P15.28, next page) are oriented at 240° to each other, and the crankshaft is rotating counter-clockwise with a constant angular speed $\omega = 690$ rev/min. Both pistons are in the same xy plane. Determine the x and y components of the primary shaking force acting on the ground bearing, and the magnitudes and orientations of the correcting masses that balance the primary shaking force. The radial distances of the correcting masses are $R_{C1} = R_{C2} = 16$ in. Determine the answers when the reference line (attached to crank 1) is specified at the crank angle $\theta = 60^\circ$.

15.29 The constant angular velocity of the crankshaft of the two-cylinder engine (Fig. P15.29, next page) is $\omega = 250\hat{k}$ rad/s. Determine the x and y components of the primary shaking force in terms of the crank angle θ , and determine the magnitudes and orientations of the correction masses. The correction masses are to be added at a radial distance $R_C = 50$ mm from the crankshaft axis. Determine the answers when the reference line (attached to crank 1) is specified at the crank angle $\theta = 30^\circ$.

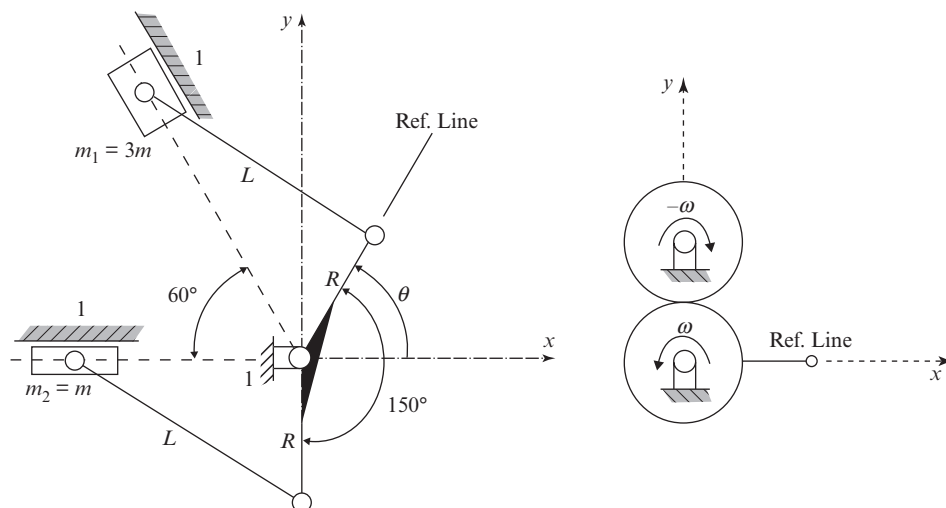


Figure P15.28 $R_1 = 12$ in, $R_2 = 6$ in, $L_1 = L_2 = L = 26$ in, $m_1 = 22$ lb, and $m_2 = 110$ lb.

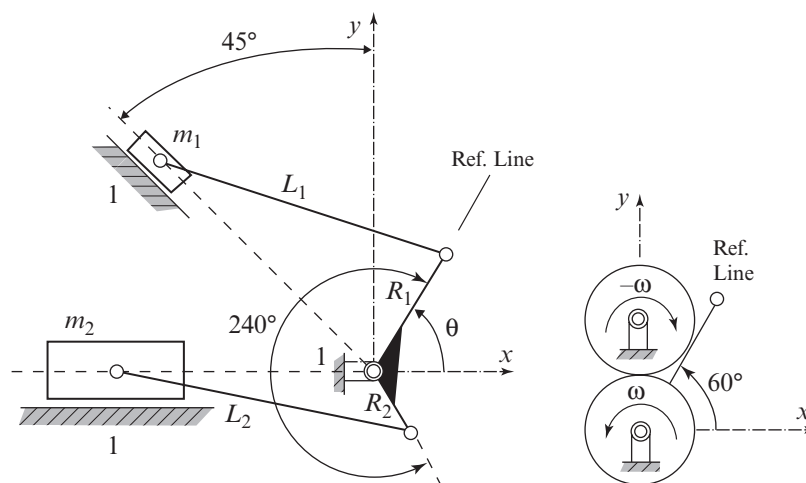
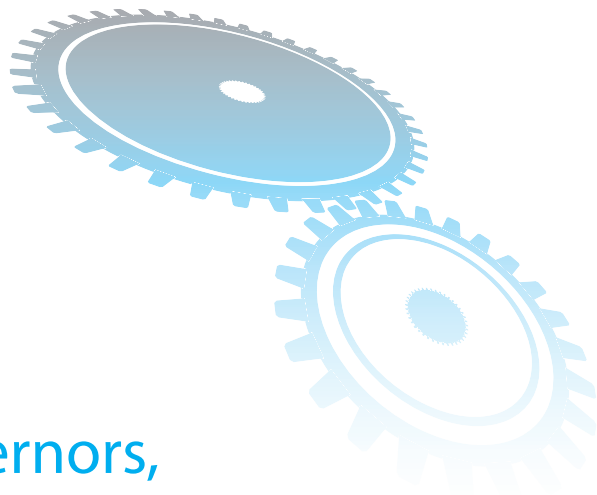


Figure P15.29 $R_1 = R_2 = R = 150$ mm, $L_1 = L_2 = L = 300$ mm, and $m_1 = m_2 = m = 20$ kg.

16 Flywheels, Governors, and Gyroscopes



16.1 DYNAMIC THEORY OF FLYWHEELS

A flywheel is an energy storage device. It absorbs mechanical energy by increasing its angular velocity and delivers energy by decreasing its angular velocity. Commonly, a flywheel is used to smooth the flow of energy between a power source and its load. If the load happens to be a punch press, for example, the actual punching operation requires energy for only a small fraction of its motion cycle. As another example, if the power source happens to be a two-cylinder, four-stroke engine, the engine delivers energy during only about half of its motion cycle. Other applications involve using flywheels to absorb braking energy and deliver acceleration energy for automobiles, or to act as energy-smoothing devices for electric utilities as well as solar- and wind-power-generating facilities. Electric railways have long used regenerative braking by absorbing braking energy back into power lines, but newly introduced and stronger materials now make the flywheel feasible for such purposes.

Figure 16.1 is a schematic representation of a flywheel. The flywheel, whose motion is measured by the angular coordinate θ has a mass moment of inertia I . An input torque T_i , corresponding to coordinate θ_i , causes the flywheel speed to increase. A load or output torque, T_o , with corresponding coordinate θ_o , absorbs energy from the flywheel and causes it to slow down. If the work entering into the system is considered positive and work output is negative, the equation of motion of the flywheel is

$$\sum M = T_i(\theta_i, \dot{\theta}_i) - T_o(\theta_o, \dot{\theta}_o) - I\ddot{\theta} = 0,$$

which can be written as

$$I\alpha = T_i(\theta_i, \omega_i) - T_o(\theta_o, \omega_o). \quad (a)$$

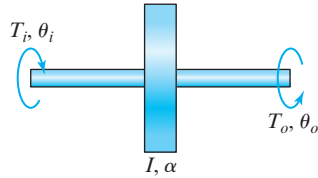


Figure 16.1 Representation of a flywheel.

Note that both T_i and T_o may depend for their values on their angular positions, θ_i and θ_o , as well as their angular velocities, ω_i and ω_o . Typically, the torque characteristic depends upon only one variable, either position or velocity. For example, the torque delivered by an induction motor depends on the speed of the motor. In fact, it is common practice for manufacturers of electric motors to publish charts detailing the torque-speed characteristics of their motors.

When the input and output torque functions are known, Eq. (a) can be solved for the motion of the flywheel using well-known techniques for solving linear and nonlinear differential equations. The resulting equations can easily be solved using the methods of Chap. 13. In Chap. 16, we assume rigid shafting, giving $\theta_i = \theta_o = \theta$. Thus, Eq. (a) becomes

$$I\alpha = T_i(\theta, \omega) - T_o(\theta, \omega). \quad (b)$$

When the two torque functions are known, and the starting values of displacement θ and velocity ω are given, Eq. (b) can be solved for θ , ω , and α as functions of time. However, typically, we are usually not interested in the instantaneous values of the kinematic parameters. Primarily, we want to know the overall performance of the flywheel. Important questions include: what is the mass moment of inertia; how do we match the power source to the load to get an optimum size of motor or engine; and what are the resulting performance characteristics of the system?

To gain insight into the problem, consider a hypothetical situation where an input power source subjects a flywheel to a constant torque T_i while the output shaft rotates from θ_1 to θ_2 . This is positive torque and is plotted upward in Fig. 16.2. Equation (b) indicates that a positive acceleration, α , is the result, and so the shaft angular velocity increases from ω_1 to ω_2 . As shown in the diagram, the shaft now rotates from θ_2 to θ_3 with zero torque and hence, from Eq. (b), with zero angular acceleration; therefore, $\omega_3 = \omega_2$. From θ_3 to θ_4 , a load, or output torque, of constant magnitude is applied, causing the shaft to slow from ω_3 to ω_4 . Note that the output torque is plotted in the negative direction in accordance with Eq. (b).

The energy input to the flywheel is the area of the rectangle between θ_1 and θ_2 , or

$$U_i = T_i(\theta_2 - \theta_1). \quad (c)$$

The energy output from the flywheel is the area of the rectangle between θ_3 and θ_4 , or

$$U_o = T_o(\theta_4 - \theta_3). \quad (d)$$

If U_o is greater than U_i , the load uses more energy than has been entered into the flywheel, and so ω_4 will be less than ω_1 . If U_o is equal to U_i , then ω_4 will equal ω_1 , since the gain

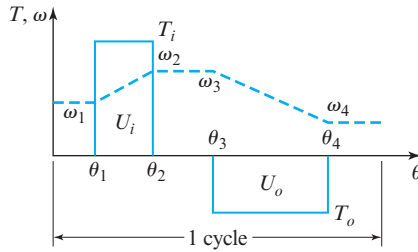


Figure 16.2

and loss are equal; we are assuming no friction losses. Finally, if U_i is greater than U_o , then ω_4 will be greater than ω_1 .

We can also write these relations in terms of kinetic energy. At $\theta = \theta_1$, the flywheel has an angular velocity of ω_1 rad/s, and so its kinetic energy is

$$U_1 = \frac{1}{2} I \omega_1^2. \quad (e)$$

At $\theta = \theta_2$, the angular velocity is ω_2 , and so its kinetic energy is

$$U_2 = \frac{1}{2} I \omega_2^2. \quad (f)$$

Therefore, the change in kinetic energy is

$$U_2 - U_1 = \frac{1}{2} I (\omega_2^2 - \omega_1^2). \quad (16.1)$$

16.2 INTEGRATION TECHNIQUE

Many of the torque-displacement functions encountered in practical engineering situations are so complex that they must be integrated by approximate methods. Figure 16.3, for example, is a plot of the engine torque for one cycle of motion of a single-cylinder, four-stroke internal combustion engine. Since part of the torque curve is negative, it is clear that the flywheel returns part of its energy back to the engine. Approximate integration of this curve for a cycle of 4π rad yields the mean torque, T_m , available to drive the load.

A simple integration routine is Simpson's rule; this approximation can be programmed for any computer and is simple enough for even the crudest programmable calculators. In fact, this routine is often found as a part of the library for many calculators and personal computers. The equation used is

$$\int_{x_0}^{x_n} f(x) dx = \frac{h}{3} (f_0 + 4f_1 + 2f_2 + 4f_3 + 2f_4 + \dots + 2f_{n-2} + 4f_{n-1} + f_n), \quad (16.2)$$

where

$$h = (x_n - x_0)/n \quad \text{with} \quad x_n > x_0,$$

and n is the number of subintervals used, 2, 4, 6, ..., which must be even. If memory is limited, Eq. (16.2) can be solved in two or more steps, say from 0 to $n/2$ and then from $n/2$ to n .

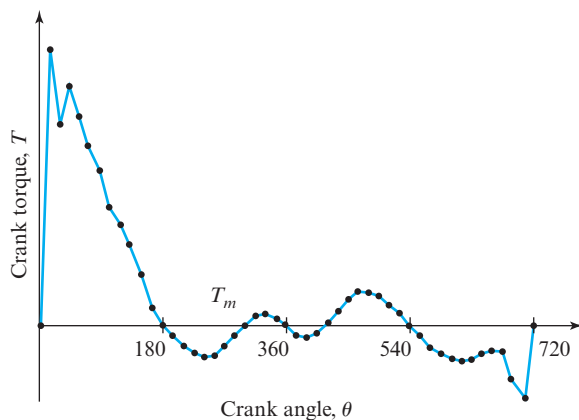


Figure 16.3 Relation between torque and crank angle for a single-cylinder, four-stroke engine.

For flywheel applications, it is convenient to define a *coefficient of speed fluctuation* as

$$C_s = \frac{\omega_2 - \omega_1}{\omega}, \quad (16.3)$$

where ω is the nominal or average angular velocity, given by

$$\omega = \frac{1}{2} (\omega_2 + \omega_1). \quad (16.4)$$

Equation (16.1) can be factored to give the change in kinetic energy as

$$U_2 - U_1 = \frac{1}{2} I (\omega_2 - \omega_1) (\omega_2 + \omega_1).$$

Substituting Eqs. (16.3) and (16.4) into this equation gives

$$U_2 - U_1 = C_s I \omega^2, \quad (16.5)$$

which can be used to obtain an appropriate flywheel inertia corresponding to the change in the kinetic energy.

EXAMPLE 16.1

Table 16.1 lists numeric values of the torque plotted in Fig. 16.3. The nominal speed of the engine is to be 250 rad/s.

- Integrate the torque-displacement function for one cycle, and find the energy that can be delivered to the load during the cycle.
- Determine the mean torque T_m (Fig. 16.3).

Table 16.1 Torque data for Fig. 16.3

θ_i deg	T_i in · lb	θ_i deg	T_i in · lb	θ_i deg	T_i in · lb	θ_i deg	T_i in · lb
0	0	180	0	360	0	540	0
15	2 800	195	−107	375	−85	555	−107
30	2 090	210	−206	390	−125	570	−206
45	2 430	225	−280	405	−89	585	−292
60	2 160	240	−323	420	8	600	−355
75	1 840	255	−310	435	126	615	−371
90	1 590	270	−242	450	242	630	−362
105	1 210	285	−126	465	310	645	−312
120	1 066	300	−8	480	323	660	−272
135	803	315	89	495	280	675	−274
150	532	330	125	510	206	690	−548
165	184	345	85	525	107	705	−760

- (c) The greatest energy fluctuation occurs approximately between $\theta = 15^\circ$ and $\theta = 150^\circ$ on the T diagram (Fig. 16.3). Using a coefficient of speed fluctuation of $C_S = 0.1$, find a suitable value for the flywheel inertia.
- (d) Find ω_2 and ω_1 .

SOLUTION

- (a) Using $n = 48$ and $h = 4\pi/48$, we enter the data of Table 16.1 into our calculator and compute the integral by Simpson's rule as defined in Eq. (16.2); the amount of energy that can be delivered to the load during the cycle is

$$U = 3\,490 \text{ in} \cdot \text{lb.} \quad \text{Ans.}$$

- (b) The mean torque is

$$T_m = 3\,490 \text{ in} \cdot \text{lb} / 4\pi \text{ rad} = 278 \text{ in} \cdot \text{lb.} \quad \text{Ans.}$$

The largest positive loop in the torque-displacement diagram occurs between $\theta = 0^\circ$ and $\theta = 180^\circ$. We select this loop as yielding the largest speed change. Subtracting 278 in · lb from the values in Table 16.1 for this loop gives, respectively, −278, 2 522, 1 812, 2 152, 1 882, 1 562, 1 312, 932, 788, 535, 254, −94, and −278 in · lb.

Entering the data into Simpson's rule again, using $n = 12$ and $h = \pi/12$, the change in kinetic energy is

$$U_2 - U_1 = 3\,663 \text{ in} \cdot \text{lb.}$$

Rearranging Eq. (16.5), the mass moment of inertia of the flywheel can be written as

$$I = \frac{U_2 - U_1}{C_S \omega^2}.$$

Substituting the known information into this equation, a suitable value for the flywheel inertia is

$$I = \frac{3\,663 \text{ in} \cdot \text{lb}}{0.1 (250 \text{ rad/s})^2} = 0.586 \text{ in} \cdot \text{lb} \cdot \text{s}^2. \quad \text{Ans.}$$

- (c) Equations (16.3) and (16.4) can now be solved simultaneously for ω_2 and ω_1 . Substituting appropriate values into these two equations yields

$$\omega_2 = \frac{1}{2} (2 + C_S) \omega = \frac{1}{2} (2 + 0.1) 250 \text{ rad/s} = 262.5 \text{ rad/s}, \quad \text{Ans.}$$

$$\omega_1 = 2\omega - \omega_2 = 2(250 \text{ rad/s}) - 262.5 \text{ rad/s} = 237.5 \text{ rad/s}. \quad \text{Ans.}$$

These two speeds occur at $\theta = 180^\circ$ and $\theta = 0$, respectively.

16.3 MULTI-CYLINDER ENGINE TORQUE SUMMATION

Once the torque-displacement relation has been defined for a single cylinder, it is easy to assume these for multi-cylinder engines. If, for example, we wish to find the torque function for a three-cylinder engine, then we examine the firing order shown in Fig. 14.1 and write

$$T_{\text{total}} = T_\theta + T_{\theta+240^\circ} + T_{\theta+480^\circ}, \quad (a)$$

since the torque events are spaced 240° apart. Here, θ is the crank angle of the first cylinder.

Applying this same procedure for a four-cylinder, four-stroke internal combustion engine, along with using the torque values in Table 16.1, gives the results in Table 16.2. Figure 16.4 shows how this torque varies during each 180° of crank rotation.

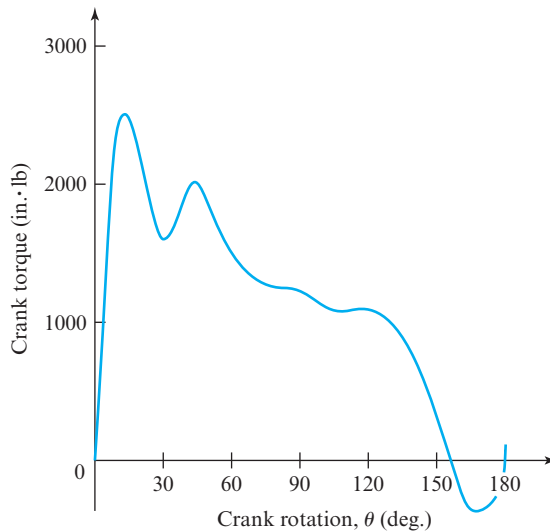
16.4 CLASSIFICATION OF GOVERNORS

The availability today of a wide variety of low-priced solid-state electronic devices and transducers makes it possible to regulate the speed of mechanical systems to a finer degree and at less cost than was possible with the older all-mechanical governors. The electronic governor also has the advantage that the speed to be regulated can be changed quite easily and at will.

So far in Chap. 16, we have we learned that flywheels are used to regulate speed over short intervals of time, such as a single revolution or the duration of an engine cycle. *Governors* are also devices that are used to regulate speed. In contrast to flywheels, however, governors are used to regulate speed over much longer intervals of time; in fact, they are intended to maintain a balance between the energy supplied to a moving system

Table 16.2 Torque data for the Four-Cylinder, Four-Stroke Engine

θ_i deg	T_θ in · lb	$T_{\theta+180}$ in · lb	$T_{\theta+360}$ in · lb	$T_{\theta+540}$ in · lb	T_{total} in · lb
0	0	0	0	0	0
15	2 800	−107	−85	−107	2 501
30	2 090	−206	−125	−206	1 553
45	2 430	−280	−89	−292	1 769
60	2 160	−323	8	−355	1 490
75	1 840	−310	126	−371	1 285
90	1 590	−242	242	−362	1 228
105	1 210	−126	310	−312	1 082
120	1 066	−8	323	−272	1 109
135	803	89	280	−274	898
150	532	125	206	−548	315
165	184	85	107	−760	−384

**Figure 16.4** Relation between torque and crank angle for the four-cylinder, four-stroke engine.

and the external load or resistance applied to that system. When the speed of a machine must, over its lifetime, always be kept at the same level, or approximately so, then a shaft-mounted mechanical device may be an appropriate speed regulator. Such governors may be classified as either centrifugal governors or inertia governors.

16.5 CENTRIFUGAL GOVERNORS

Figure 16.5 shows a simple spring-controlled shaft governor. Masses attached to the bell-crank levers are driven outward by inertia force against springs. The motion of the shaft-mounted sleeve is dependent on the motions of the masses and the ratio of the bell-crank lengths.

Defining the nomenclature for Fig. 16.6, we let

k = spring rate,

P = spring force,

W = sleeve weight acting along the vertical shaft,

r = instantaneous radial position of mass center, and

r_0 = position of mass center at zero spring force.

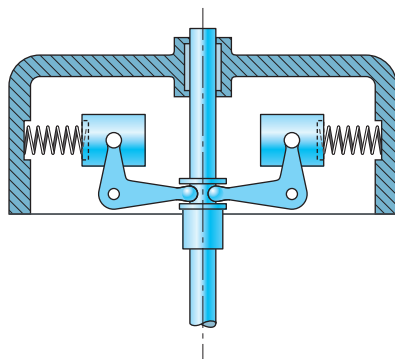


Figure 16.5 Centrifugal shaft governor.

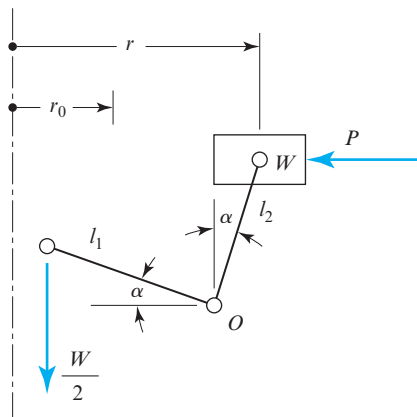


Figure 16.6

Then the spring force at any position r is

$$P = k(r - r_0). \quad (a)$$

Taking moments about pivot O gives

$$\sum M_O = Pl_2 \cos \alpha + \frac{W}{2} l_1 \cos \alpha + (-m\ddot{r}) l_2 \cos \alpha = 0. \quad (b)$$

Now, with $\ddot{r} = r\omega^2$ and $m = W/g$, this gives

$$\frac{W}{g} r\omega^2 l_2 = Pl_2 + \frac{W}{2} l_1. \quad (c)$$

The controlling force can be written as

$$F = \frac{W}{g} r\omega^2. \quad (d)$$

Then substituting this expression into Eq. (c), and using Eq. (a), the controlling force is

$$F = P + \frac{W}{2} \frac{l_1}{l_2} = k(r - r_0) + \frac{W}{2} \frac{l_1}{l_2}. \quad (16.6)$$

Equating Eqs. (d) and (16.6) gives

$$\frac{W}{g} r\omega^2 = k(r - r_0) + \frac{W}{2} \frac{l_1}{l_2}.$$

Then, rearranging this equation, the corresponding angular velocity can be written as

$$\omega^2 = \frac{kg}{Wr} \left[r - \left(r_0 - \frac{W}{2k} \frac{l_1}{l_2} \right) \right]. \quad (16.7)$$

16.6 INERTIA GOVERNORS

In a centrifugal governor, an increase in speed causes the rotating masses to move radially outward. Thus, it is the radial (normal) component of the acceleration that is primarily responsible for creating the controlling force. In an inertia shaft governor, as shown in Fig. 16.7, the mass pivot is located at point A very close to the shaft center at O . Thus, the radial component of acceleration is small and much less effective, but a sudden change of speed, producing an angular acceleration, causes the masses to lag and produces tension in the spring. This spring tension produces a transverse (tangential) acceleration force. Consequently, it is the angular acceleration that determines the positions of the masses.

Compared with a centrifugal governor, an inertia governor is more sensitive, since it acts at the very beginning of a speed change.

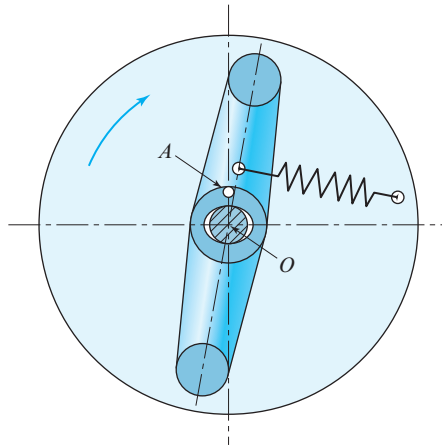


Figure 16.7 Inertia shaft governor.

16.7 MECHANICAL CONTROL SYSTEMS

Many mechanical control systems are represented schematically as shown in Fig. 16.8. Here, θ_i and θ_o represent any set of input and output signals. In the case of a governor, θ_i represents the desired speed and θ_o the actual speed. For control systems in general, the input and output functions can represent force, torque, or displacement, either rectilinear or angular.

The system shown in Fig. 16.8 is called a *closed-loop* or *feedback control system*, since the output value, θ_o , is fed back to the detector at the input to indicate error ε , which is the difference between the input and the output. The objective of the controller is to cause this error to become as close to zero as possible. The mechanical characteristics of the system, that is, the mechanical clearances, friction, inertias, and stiffnesses, sometimes cause the output to differ somewhat from the input, and so it is the designer's responsibility to examine these mechanical effects in an effort to minimize the error for all operating conditions.

The differential equation for a control system is always written as an expression of the dynamic equilibrium of the elements of the system. Thus, for the rotational system of Fig. 16.8, we express mathematically that the inertia torque is equal to the sum of all other torques acting upon the system. If the load has both inertia and viscous damping, then the

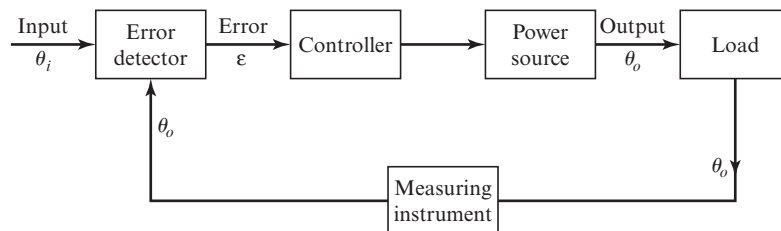


Figure 16.8 Block diagram of a closed-loop control system.

expression is

$$I\ddot{\theta}_o = -c\dot{\theta}_o + k'f(\varepsilon). \quad (a)$$

The function $f(\varepsilon)$ depends upon the characteristics of the controller and the power source. A very simple system would result if the components had characteristics such that the relation between $f(\varepsilon)$ and ε is linear. For this condition we can make the substitution

$$k'f(\varepsilon) = k\varepsilon. \quad (b)$$

Substituting Eq. (b) into Eq. (a), and rearranging gives

$$I\ddot{\theta}_o + c\dot{\theta}_o = k\varepsilon. \quad (c)$$

By definition, we have

$$\varepsilon = \theta_i - \theta_o. \quad (d)$$

Therefore, Eq. (c) can be written as

$$I\ddot{\theta}_o + c\dot{\theta}_o + k\theta_o = k\theta_i. \quad (16.8)$$

This second-order differential equation can be simplified by the following substitutions

$$\omega_n = \sqrt{\frac{k}{I}} \quad (16.9)$$

and

$$\zeta = \frac{c}{2\sqrt{kI}}, \quad (16.10)$$

where I = mass moment of inertia, c = torsional damping coefficient, k = torsional stiffness, ω_n = natural frequency, and ζ = damping ratio, c/c_{cr} .

After a review of this notation for linear systems in Chap. 13, Eq. (16.8) can be written in the form

$$\ddot{\theta}_o + 2\zeta\omega_n\dot{\theta}_o + \omega_n^2\theta_o = \omega_n^2\theta_i. \quad (16.11)$$

16.8 STANDARD INPUT FUNCTIONS

There is a great deal of useful information that can be gained from a mathematical analysis as well as a laboratory analysis of feedback control systems when standard input functions are used to study their performance. Standard input functions result in differential equations that are easier to analyze mathematically than they are if actual

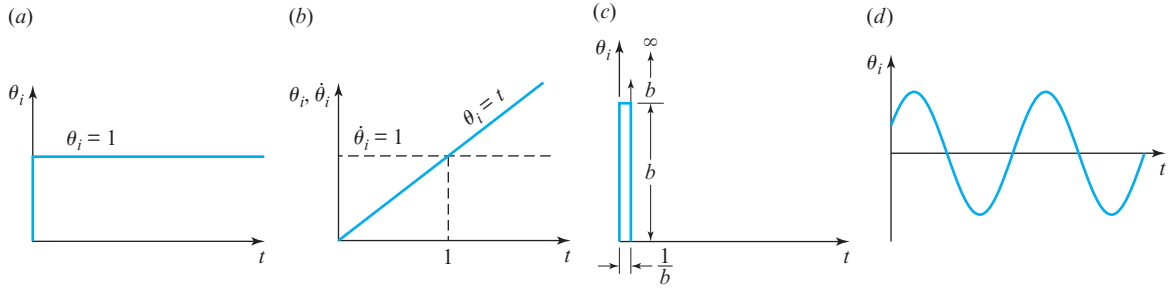


Figure 16.9 (a) Unit-step function; (b) unit-step velocity function; (c) unit-impulse function; (d) harmonic input function.

operating conditions are used as input. Furthermore, the use of standard inputs makes it possible to compare the performance of different control systems.

One of the most useful standard input functions is the *unit-step function* shown in Fig. 16.9a. This function is not continuous, and, consequently, we cannot define initial conditions at $t = 0$. It is customary to specify the conditions at $t = 0+$ and $t = 0-$, where the signs indicate the conditions for values of time slightly greater than or slightly less than zero. Thus, for the unit-step function we have

$$\begin{aligned} \theta_i &= 0, & \dot{\theta}_i &= 0 & \text{when } t &= 0-, \\ \theta_i &= 1, & \dot{\theta}_i &= 0 & \text{when } t &= 0+. \end{aligned}$$

The performance resulting from the application of these input conditions is called the *unit-step response*.

Another standard input function is the *unit-step velocity function*, shown in Fig. 16.9b. This function is sometimes called the *unit-slope ramp function* because of the form of its displacement diagram. This function is defined as follows:

$$\begin{aligned} \theta_i &= 0, & \dot{\theta}_i &= 0 & \text{when } t &= 0-, \\ \theta_i &= t, & \dot{\theta}_i &= 1 & \text{when } t &= 0+. \end{aligned}$$

Also shown in Figs. 16.9c and 16.9d are the unit-impulse function and the harmonic input function, both of which are useful in studying a system's response.

It sometimes happens in control systems that the input function is relatively constant, but the output is subjected to abruptly varying applied forces or torques. For example, an automatically controlled machine tool may encounter a deeper cut, causing a greater load torque to be exerted on the machine, thus tending to slow it down. The differential equation for this condition is written as

$$I\ddot{\theta}_o + c\dot{\theta}_o + k\theta_o = k\theta_i - T. \quad (a)$$

When this equation is simplified by substituting the values from Eqs. (16.9) and (16.10) and rearranged, it becomes

$$\ddot{\theta}_o + 2\zeta\omega_n\dot{\theta}_o + \omega_n^2\theta_o = \omega_n^2\theta_i - \frac{T}{I}. \quad (16.12)$$

When the solution to Eq. (16.11) is obtained for a given function θ_i , the solution to Eq. (16.12) can be obtained by superposition.

16.9 SOLUTION OF LINEAR DIFFERENTIAL EQUATIONS

In the analysis of feedback control systems, n th-order linear differential equations are frequently encountered. The general form of these equations is

$$a_n \frac{d^n \theta}{dt^n} + a_{n-1} \frac{d^{n-1} \theta}{dt^{n-1}} + \cdots + a_1 \frac{d\theta}{dt} + a_0 \theta = f(t). \quad (16.13)$$

The function $f(t)$ is the driving or forcing function, and so t and θ are the independent and dependent variables, respectively. The coefficients $a_0, a_1, \dots, a_n$ are constants and are independent of t and θ .

The solution to equations of the form of Eq. (16.13) is composed of two parts (Chap. 13). The first part is called the *complementary solution*, and it is the solution to the homogeneous equation

$$a_n \frac{d^n \theta}{dt^n} + a_{n-1} \frac{d^{n-1} \theta}{dt^{n-1}} + \cdots + a_1 \frac{d\theta}{dt} + a_0 \theta = 0. \quad (16.14)$$

The complementary solution is also called the *transient solution* in the literature of automatic controls since, for a damped stable system, this part of the solution fades away. If the control system should happen to be unstable, then the controlled quantity increases without limit. Note that the transient solution is obtained by solving with a forcing function of zero.

The other part of the solution is called the *particular solution* by mathematicians and the *steady-state solution* by control engineers. It is any specific solution of Eq. (16.13).

The complete solution is the sum of the transient and the steady-state solutions. The transient part contains n arbitrary constants that are evaluated from the initial conditions. The amplitude of the transient solution depends upon both the forcing function and the initial conditions, but all other characteristics of the transient solution are completely independent of the forcing function. We find, for linear systems, that neither the forcing function nor the amplitude have any effect on system stability, this being dependent only on the transient portion of the solution.

The Transient Solution The following steps are used to obtain the transient solution:

1. Set the forcing function equal to zero and arrange the equation in the form of Eq. (16.14).
2. Assume a solution of the form

$$\theta = Ae^{st}. \quad (16.15)$$

3. Substitute Eq. (16.15) and its derivatives into the differential equation, and simplify to obtain the characteristic equation.
4. Solve the characteristic equation for its roots.
5. Obtain the transient solution by substituting the roots into the assumed solution.

As an example of this procedure let us solve Eq. (16.11) for the transient solution. Following Step 1 we write

$$\ddot{\theta}_o + 2\zeta\omega_n\dot{\theta}_o + \omega_n^2\theta_o = 0, \quad (a)$$

and for Step 2 we have

$$\theta_o = Ae^{st}, \quad \dot{\theta}_o = Ase^{st}, \quad \ddot{\theta}_o = As^2e^{st}. \quad (b)$$

For Step 3, we substitute Eqs. (b) into Eq. (a), which gives

$$As^2e^{st} + 2\zeta\omega_nAse^{st} + \omega_n^2Ae^{st} = 0. \quad (c)$$

The characteristic equation is then obtained by dividing out common factors, that is,

$$s^2 + 2\zeta\omega_ns + \omega_n^2 = 0. \quad (d)$$

Step 4 is to solve this equation and obtain its roots, namely

$$\begin{aligned} s_1 &= -\zeta\omega_n + \sqrt{\zeta^2 - 1}\omega_n, \\ s_2 &= -\zeta\omega_n - \sqrt{\zeta^2 - 1}\omega_n. \end{aligned} \quad (e)$$

We shall not consider the situation in which $\zeta > 1$. Consequently, for $\zeta \leq 1$, the radical in Eq. (e) has an imaginary value, and we prefer to write the roots in the form

$$\begin{aligned} s_1 &= -\zeta\omega_n + j\sqrt{1 - \zeta^2}\omega_n, \\ s_2 &= -\zeta\omega_n - j\sqrt{1 - \zeta^2}\omega_n, \end{aligned} \quad (f)$$

where $j = \sqrt{-1}$.

Finally, for Step 5, we substitute the roots back into the assumed solution. Then, after factoring, the transient solution becomes

$$\theta_{o,t} = e^{-\zeta\omega_nt} \left(Ce^{j\sqrt{1-\zeta^2}\omega_nt} + De^{-j\sqrt{1-\zeta^2}\omega_nt} \right), \quad (g)$$

where C and D are the constants of integration and are two in number, the same as the order of the highest derivative in the differential equation. These cannot be evaluated until the complete solution is found. At that time, they will be determined from the initial conditions. We can, however, transform Eq. (g) into trigonometric form by using Euler's equation, [Eq. (2.40)]. The result of this transformation is

$$\theta_{o,t} = e^{-\zeta\omega_nt} \left(A \cos \sqrt{1 - \zeta^2}\omega_nt + B \sin \sqrt{1 - \zeta^2}\omega_nt \right). \quad (16.16)$$

The coefficient $e^{-\zeta\omega_n t}$ of Eq. (16.16) indicates its transient nature, since this term becomes approximately zero for large values of t .

It can sometimes happen that two or more roots of the characteristic equation are identical. For the case of two identical roots, $s_1 = s_2$, the transient solution is

$$\theta_{o,t} = Ee^{s_1 t} + Fte^{s_1 t}. \quad (h)$$

For other special cases, the reader should refer to a text such as [2] that focuses on the analysis of control systems.

The Steady-State Solution Here we shall solve Eq. (16.11) for various standard inputs.

For the unit-step function, $\theta_i = 1$. Substituting this function into Eq. (16.11) gives

$$\ddot{\theta}_o + 2\zeta\omega_n\dot{\theta}_o + \omega_n^2\theta_o = \omega_n^2. \quad (i)$$

The solution is

$$\theta_{o,s} = 1, \quad (16.17)$$

which can be verified by substituting Eq. (16.17) and its derivatives into Eq. (i).

For the unit-slope ramp function, $\theta_i = t$, substituting this function into Eq. (16.11) gives

$$\ddot{\theta}_o + 2\zeta\omega_n\dot{\theta}_o + \omega_n^2\theta_o = \omega_n^2 t. \quad (j)$$

We assume a solution of the form

$$\theta_{o,s} = At + B. \quad (k)$$

The successive derivatives are

$$\dot{\theta}_{o,s} = A, \quad \ddot{\theta}_{o,s} = 0.$$

Substituting the assumed solution and its derivatives into Eq. (j) gives

$$2\zeta\omega_n A + \omega_n^2(At + B) = \omega_n^2 t.$$

We now arrange the equation into the form

$$(2\zeta\omega_n A + \omega_n^2 B) + (\omega_n^2 A)t = \omega_n^2 t,$$

and solve for A and B . The solutions are

$$A = 1, \quad B = -\frac{2\zeta}{\omega_n}.$$

Substituting these into Eq. (k), the steady-state solution for a unit-slope ramp function is

$$\theta_{o,s} = t - \frac{2\zeta}{\omega_n}. \quad (16.18)$$

The Complete Solution The complete solution is the sum of the transient and the steady-state solutions. Thus,

$$\theta_o = \theta_{o,t} + \theta_{o,s}.$$

Therefore, adding Eqs. (16.16) and (16.17) gives

$$\theta_o = e^{-\zeta \omega_n t} \left(C_1 \cos \sqrt{1 - \zeta^2} \omega_n t + C_2 \sin \sqrt{1 - \zeta^2} \omega_n t \right) + 1. \quad (16.19)$$

The initial conditions for $t = 0+$ can now be applied in order to evaluate the constants C_1 and C_2 . Imposing the condition that $\theta_o = 0$ before the step ($t = 0-$) gives

$$0 = 1 (C_1 \cos 0 + C_2 \sin 0) + 1$$

or

$$C_1 = -1.$$

The second condition to be imposed is that $\dot{\theta}_o = 0$ at $t = 0+$. To apply this condition, it is necessary to take the derivative of Eq. (16.19). This is

$$\begin{aligned} \dot{\theta}_o = & -\zeta \omega_n e^{-\zeta \omega_n t} \left(C_1 \cos \sqrt{1 - \zeta^2} \omega_n t + C_2 \sin \sqrt{1 - \zeta^2} \omega_n t \right) \\ & + e^{-\zeta \omega_n t} \left(-C_1 \sqrt{1 - \zeta^2} \omega_n \sin \sqrt{1 - \zeta^2} \omega_n t + C_2 \sqrt{1 - \zeta^2} \omega_n \cos \sqrt{1 - \zeta^2} \omega_n t \right). \end{aligned}$$

Substituting $\dot{\theta}_o = 0$ and $t = 0$ gives

$$0 = -\zeta \omega_n (C_1 \cos 0 + C_2 \sin 0) + 1 \left(-C_1 \sqrt{1 - \zeta^2} \omega_n \sin 0 + C_2 \sqrt{1 - \zeta^2} \omega_n \cos 0 \right).$$

Substituting the value of C_1 and solving yields

$$C_2 = -\frac{\zeta}{\sqrt{1 - \zeta^2}}.$$

Substituting the values of C_1 and C_2 into Eq. (16.19) and rearranging, the complete solution can be written as

$$\theta_o = 1 - e^{-\zeta \omega_n t} \left(\cos \sqrt{1 - \zeta^2} \omega_n t + \sin \sqrt{1 - \zeta^2} \omega_n t \right). \quad (16.20)$$

If transformed to a single trigonometric term with a phase angle as demonstrated in Chap. 13, the complete solution is

$$\theta_o = 1 - \frac{e^{-\zeta \omega_n t}}{\sqrt{1 - \zeta^2}} \cos \left(\sqrt{1 - \zeta^2} \omega_n t - \phi \right), \quad (16.21)$$

where

$$\phi = \tan^{-1} \left(\frac{\zeta}{\sqrt{1-\zeta^2}} \right).$$

The complete solution for the unit-slope ramp function input is obtained in a similar manner. The initial conditions to be applied in order to evaluate the constants of integration are

$$\theta_o = 0 \quad \text{and} \quad \dot{\theta}_o = 0 \quad \text{when} \quad t = 0^-.$$

The equation to be solved for the integration constants is obtained by adding Eqs. (16.16) and (16.18); it is

$$\theta_o = t - \frac{2\zeta}{\omega_n} + e^{-\zeta\omega_n t} \left(C_1 \cos \sqrt{1-\zeta^2}\omega_n t + C_2 \sin \sqrt{1-\zeta^2}\omega_n t \right). \quad (16.22)$$

The two constants are found to be

$$C_1 = \frac{2\zeta}{\omega_n} \quad \text{and} \quad C_2 = -\frac{1-2\zeta^2}{\sqrt{1-\zeta^2}\omega_n},$$

and the complete solution is

$$\theta_o = t - \frac{2\zeta}{\omega_n} + e^{-\zeta\omega_n t} \left(\frac{2\zeta}{\omega_n} \cos \sqrt{1-\zeta^2}\omega_n t - \frac{1-2\zeta^2}{\sqrt{1-\zeta^2}\omega_n} \sin \sqrt{1-\zeta^2}\omega_n t \right) \quad (16.23)$$

or

$$\theta_o = t - \frac{2\zeta}{\omega_n} + \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}\omega_n} \cos \left(\sqrt{1-\zeta^2}\omega_n t - \phi \right), \quad (16.24)$$

where the phase angle is

$$\phi = \tan^{-1} \left(\frac{2\zeta^2 - 1}{2\zeta\sqrt{1-\zeta^2}} \right).$$

16.10 ANALYSIS OF PROPORTIONAL-ERROR FEEDBACK SYSTEMS

As the name implies, a proportional-error feedback control system is one that operates by applying a correction that is directly proportional to any error that might exist. Such a system must be analyzed in order to determine the following:

1. The response time, or the time required to reach steady-state operation after the application of a disturbance to the system; since a system with viscous damping never reaches a steady-state condition, this is usually defined as the time required for the transient to decay to 5% of its initial value.
2. The natural frequency.

3. The steady-state error, that is, the difference between the output and input during steady-state operation.
4. The maximum overshoot, or the maximum deviation between output and input during transient conditions.

Output response of the system to a unit-step input is given by Eq. (16.21) and has been plotted in Fig. 16.10 for various values of the damping ratio. The abscissa is measured in dimensionless time, $\omega_n t/2\pi$, so the curves can be applied to any physical system. Thus, if the undamped natural frequency, ω_n , is known for a given system, then the response time can be calculated simply by multiplying dimensionless time by the quantity $2\pi/\omega_n$. The response for zero damping is included for its academic interest; an automatic control system with no damping at all would give completely unsatisfactory performance.

The response for critical damping, $\zeta = 1$, shows no overshoot, and the steady-state condition is reached at about $\omega_n t/2\pi = 1.0$.

It is convenient to define an *overshoot ratio* as the ratio of overshoot with damping to the overshoot that would exist if no damping were present. Figure 16.11 shows a plot of this ratio versus the damping ratio. For $\zeta = 0.40$, the overshoot ratio is approximately 0.25; this quantity can also be read from Fig. 16.10. Note that the steady-state condition is not reached for $\zeta = 0.40$ until about $\omega_n t/2\pi = 2.0$. The response time, therefore, is twice that for critical damping.

Since the response time is proportional to $\omega_n t/2\pi$, we can make this time short simply by making the undamped natural frequency, ω_n , large.

In practice, ζ is usually between 0.4 and 0.7 in a physical system. If a certain deviation from steady state is permitted, say 5%, then a system with $\zeta = 0.7$ will reach steady state before one having $\zeta = 1.0$, since the early portion of its response curve is steeper.

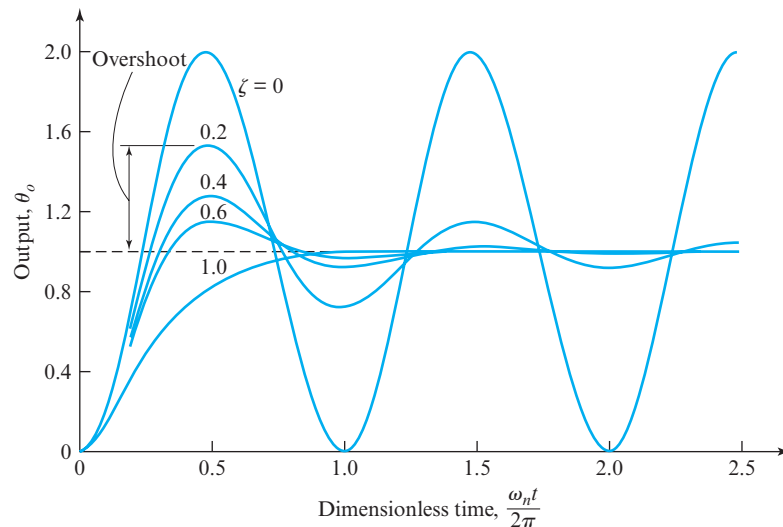


Figure 16.10 Response to a unit-step input.

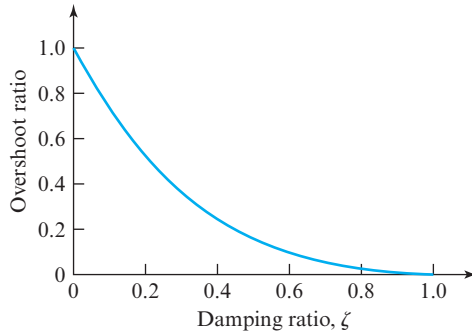


Figure 16.11

Figure 16.10 demonstrates that the steady-state value of the output is unity. Consequently, there is no steady-state error for step-input functions when applied to proportional-error feedback systems.

Local Disturbance Another look at the behavior of a system can be obtained by considering that the input signal is constant or zero and that the load is subjected to a disturbance. Selecting the zero input condition, for simplicity, Eq. (16.12) then becomes

$$\ddot{\theta}_o + 2\zeta\omega_n\dot{\theta}_o + \omega_n^2\theta_o = -\frac{T}{I}, \quad (16.25)$$

where T represents a constant torque suddenly applied to the input. The steady-state component of the solution is

$$\theta_{o,s} = -\frac{T}{I\omega_n^2}, \quad (a)$$

so that the complete solution must be

$$\theta_o = e^{-\zeta\omega_n t} \left(C_1 \cos \sqrt{1-\zeta^2}\omega_n t + C_2 \sin \sqrt{1-\zeta^2}\omega_n t \right) - \frac{T}{I\omega_n^2}. \quad (b)$$

The initial conditions are $\theta_o = 0$ and $\dot{\theta}_o = 0$ at $t = 0$. Using these conditions, we can solve Eq. (b) for the two constants

$$C_1 = \frac{T}{I\omega_n^2} \text{ and } C_2 = \frac{\zeta T}{\sqrt{1-\zeta^2}I\omega_n^2}.$$

Substituting these into Eq. (b) and transforming to a single trigonometric term, the result is

$$\theta_o = \frac{T}{I\omega_n^2} \left[\frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \cos \left(\sqrt{1-\zeta^2}\omega_n t - \phi \right) - 1 \right], \quad (16.26)$$

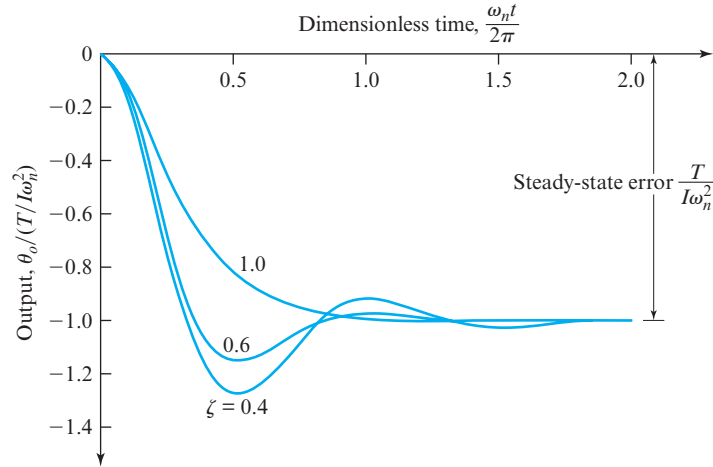


Figure 16.12 Response of proportional-error feedback control.

where

$$\phi = \tan^{-1} \left(\frac{\zeta}{\sqrt{1 - \zeta^2}} \right).$$

Equation (16.26) is plotted in Fig. 16.12 for three values of the damping ratio to show what happens. As shown, the disturbance decays at a rate that is dependent on the amount of damping. It finally reaches a steady-state condition that is not zero but is of an amount

$$\varepsilon_s = \frac{T}{I\omega_n^2}. \quad (16.27)$$

This is the steady-state error. If we substitute $\omega_n^2 = k/I$, the equation becomes

$$\varepsilon_s = \frac{T}{k}. \quad (16.28)$$

Hence, the only method of reducing the magnitude of this error is to increase the gain, k , of the system.

We have seen that the control system is defined by parameters I , c , and k . Of these three, the gain, k , is usually the easiest to change. Some variation in the damping, c , is usually possible, but the employment of dashpots or friction dampers is not often a good solution. Variation in the inertia, I , is the most difficult change to make, since this is fixed by the design of the driven element, and, furthermore, improvement always requires a decrease in the inertia.

Unit-Step Velocity Input For the unit-step velocity input, we have $\theta_i = t$, and from Eq. (16.18),

$$\theta_{o,s} = t - \frac{2\zeta}{\omega_n}$$

after the transient decays. Therefore, the steady-state error is

$$\varepsilon_s = \frac{2\zeta}{\omega_n} = \frac{c}{k}, \quad (16.29)$$

which is obtained by substitution of the value of ζ from Eq. (16.10). This error persists only as long as the input function signals for a constant velocity. Again, we see that the magnitude can be reduced by increasing the gain, k .

The widely used automotive cruise-control system is an excellent example of an electromechanical governor. A transducer is attached to the speedometer cable, and the electrical output of this transducer is the signal, θ_o , fed to the error detector of Fig. 16.8. In some cases, magnets are mounted on the driveshaft of the car to activate the transducer. In the cruise-control system, the error detector is an electronic regulator, usually mounted under the dashboard. The regulator is turned on by an engagement switch under, or near, the steering wheel. A power unit is connected to the carburetor throttle linkage; the power unit is controlled by the regulator and gets its power from a vacuum port on the engine. Such systems have one or two brake-release switches as well as the engagement switch. The accelerator pedal can also be used to override the system.

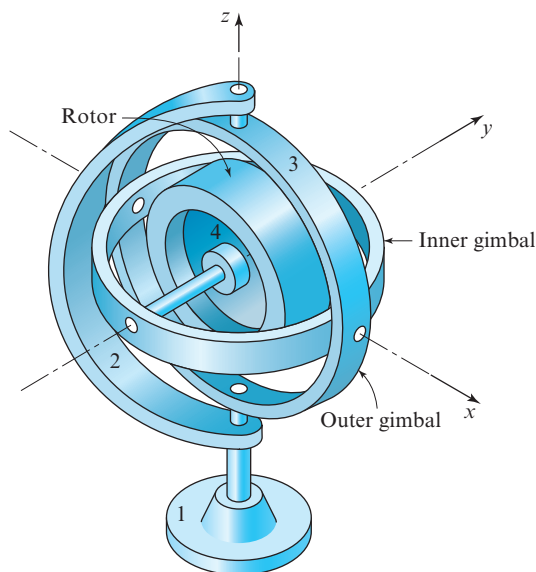
16.11 INTRODUCTION TO GYROSCOPES

A gyroscope is defined as a rigid body capable of three-dimensional rotation with high angular velocity about any axis that passes through a fixed point called the center, which may or may not be its center of mass. A child's toy top fits this definition and is a form of gyroscope; its fixed point is the point of contact of the top with the floor or table on which it spins.

The usual form of a gyroscope is a mechanical device in which the essential part is a rotor having a heavy rim spinning at high speed about a fixed point on the rotor axis. The rotor is then mounted so as to turn freely about its center of mass by means of double gimbals called a Cardan suspension; an example is pictured in Fig. 16.13.

The gyroscope has fascinated students of mechanics and applied mathematics for many years. Indeed, once the rotor is set spinning, a gyroscope seems to act like a device possessing intelligence. If we attempt to move some of its parts, it seems not only to resist this motion but even to evade it. It seemingly fails to conform to the laws of static equilibrium and of gravitation.

The early history of the gyroscope is rather obscure. Probably the earliest gyroscope of the type now in use was constructed by Bohnenburger in Germany in 1817 [1]. In 1852, Leon Foucault, of Paris, constructed a very refined version to demonstrate the rotation of the earth; it was Foucault who named the instrument the gyroscope from the Greek words *gyros*, circle or ring, and *skopien*, to view [4]. The mathematical foundations of gyroscopic theory were laid by Leonhard Euler in 1765 in his work on the dynamics of rigid bodies [3]. Gyroscopes were not put to practical and industrial use until the beginning of the twentieth century in the United States, at which time the gyroscopic compass, the ship stabilizer, and the monorail car were all invented. Subsequent uses of the gyroscope as a turn-and-bank indicator, artificial horizon, and automatic pilot in aircraft and missiles are well known.

Figure 16.13 Laboratory gyroscope.

We may also become concerned with gyroscopic effects in the design of machines, although not always intentionally. Such effects are present in riding a motorcycle or bicycle; they are also present, owing to the rotating masses, when an airplane or automobile makes a turn. Sometimes these gyroscopic effects are desirable, but more often they are undesirable, and designers must account for them in their selection of bearings and rotating parts. It is certainly true that, as machine speeds increase and as factors of safety decrease, we must stop neglecting gyroscopic forces in our design calculations, since their values will become more significant.

16.12 MOTION OF A GYROSCOPE

Although we have noted above that a gyroscope seems to have intelligence and appears to avoid compliance with the fundamental laws of mechanics, this is not truly the case. In fact, we have thoroughly covered the basic theory involved in Chap. 12, where we studied dynamic forces with spatial motion. Still, since gyroscopic forces are of increasing importance in high-speed machines, we will look at them again and try to explain the apparent paradoxes they seem to raise.

To provide a vehicle for the explanation of the simpler motions of a gyroscope, we will consider a series of experiments to be performed on the one pictured in Fig. 16.13. In the following discussion, we assume that the rotor is already spinning rapidly and that all pivot friction is negligible.

As a first experiment, we transport the entire gyroscope about the table or around the room. We find that even though we travel along a curved path, the orientation of the rotor's axis of rotation does not change as we move. This is a consequence of the law of conservation of angular momentum. If the orientation of the axis of rotation is to change,

then the orientation of the angular momentum vector must also change. However, this requires an externally applied moment that, with the three sets of frictionless bearings, has not been supplied for this experiment. Therefore, the orientation of the rotation axis does not change.

As a second experiment, while the rotor is still spinning, we lift the inner gimbal out of its bearings and move it about. We again find that it can be translated anywhere but that we meet with definite resistance if we attempt to rotate the axis of spin. In other words, the rotor persists in maintaining the orientation of its plane of rotation.

As a third experiment, we replace the inner gimbal back into its bearings with the axis of *spin* horizontal, as shown in Fig. 16.13. If we now apply a steady downward force to the inner gimbal at one end of the spin axis, say by pushing on it with a pencil, we find that the end of the spin axis does not move downward as we might expect. Not only do we meet with resistance to the force of the pencil, but the outer gimbal begins to rotate about the vertical axis, causing the rotor to skew around in the horizontal plane, and it continues this rotation until we remove the force of the pencil. This skewing motion of the spin axis is called *precession*.

Although it may seem strange and unexpected, this precession motion is in strict obedience to the laws of dynamic equilibrium as expressed by the Euler equations of motion in Eqs. (12.111). Yet, it is easier to understand and explain this phenomena if we again think in terms of the angular momentum vector. The downward force applied by the pencil to the inner gimbal produces a net external moment, $\mathbf{M}$, on the rotor shaft through a pair of equal and opposite forces at the bearings and results in a time rate of change of the angular momentum vector, $\mathbf{H}$. As we demonstrated in Eq. (12.121), the time rate of change of the angular momentum can be written as

$$\frac{d\mathbf{H}}{dt} = \sum \mathbf{M}. \quad (16.30)$$

Since this applied external moment cannot change the rate of spin of the rotor about its own axis, it changes the angular momentum by making the rotor rotate about the vertical axis as well, thus causing the precession.

As our next set of experiments, we repeat the previous experiment, watching carefully the directions involved. We first cause the rotor to spin in the positive direction, that is, with its angular velocity vector in the positive y direction. If we next apply a positive torque (moment vector in the positive x direction) using downward force of our pencil on the negative y end of the rotor axis, the precession (rotation) of the outer gimbal is found to be in the negative z direction. Further experiments indicate that either a negative spin velocity or a negative moment caused by the pencil result in a positive direction for the precession.

For a final experiment, we apply a moment to the outer gimbal in an attempt to cause the rotor to rotate about the z axis. Such an attempt meets with definite resistance and causes the inner gimbal and the spin axis to rotate. Note again in this case that the angular momentum vector is changing as the result of the application of external torque. If the spin axis starts in the vertical direction (aligned along z), however, then the gyroscope is in stable equilibrium, and the outer and inner gimbals can be turned together quite freely.

16.13 STEADY OR REGULAR PRECESSION

Consider the heavy rotor shown in Fig. 16.14 that is spinning with a constant angular velocity ω_s about a spin axis that is tipped at a constant angle, θ , from the vertical. At the same time, assume that the spin axis has a constant angular velocity, ω_p . The state of motion just described is called *steady* or *regular* precession.

We should take careful note of several things in Fig. 16.14. First, we have shown the rotor at a posture for which the inner gimbal is rotated to quite a different angle than in Fig. 16.13; yet the xyz coordinate system is still oriented with x along the pivot axis between the inner and outer gimbals, and z is still along the true axis of precession. Second, we have designated a new coordinate system $x'y'z'$ with the x' axis coincident with x and the z' axis aligned along the axis of the rotor, the spin axis. Third, we should note that neither of these two coordinate systems is stationary and, similarly, that $x'y'z'$ is not attached to the rotor and does not experience the angular velocity ω_s . The $x'y'z'$ coordinate system remains fixed to the inner gimbal, whereas xyz remains fixed to the outer gimbal.

Nevertheless, we see that, for a disk-shaped rotor mounted as described, the $x'y'z'$ axes are the principal axes of inertia of the rotor. We designate the corresponding principal mass moments of inertia $I^{z'z'} = I^s$ and $I^{x'x'} = I^{y'y'} = I$.

Recognize that the constant angular spin velocity, ω_s , is not truly an absolute angular velocity; it is the “relative” or *apparent* angular velocity of the rotor with respect to the inner gimbal. If we designate the rotor as body 4, with the inner and outer gimbals being links 3 and 2, respectively, then

$$\omega_{4/3} = \omega_s = \omega_s \hat{\mathbf{k}}'. \quad (a)$$

Similarly, the angular velocity of precession is really the angular velocity of the outer gimbal:

$$\omega_2 = \omega_p = \omega_p \hat{\mathbf{k}} = (\omega_p \sin \theta) \hat{\mathbf{j}}' + (\omega_p \cos \theta) \hat{\mathbf{k}}'. \quad (b)$$

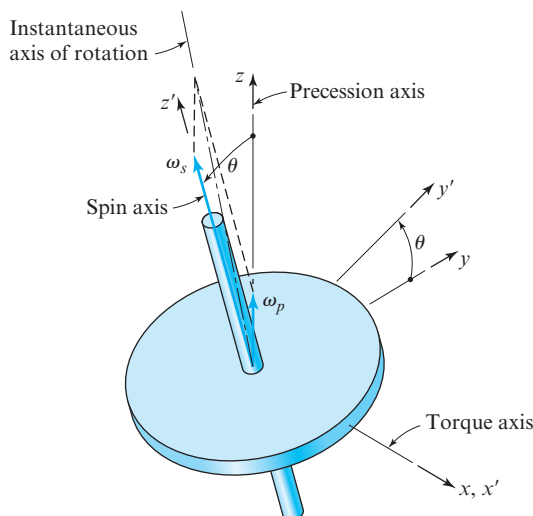


Figure 16.14 Angular velocity components.

Since we are interested in steady precession here, the angle θ has been assumed constant; thus, there is no rotation between the inner and outer gimbals:

$$\boldsymbol{\omega}_{3/2} = \dot{\theta} \hat{\mathbf{i}}' = 0. \quad (c)$$

We can now use Eqs. (a) through (c) to find the absolute angular velocity of the rotor:*

$$\begin{aligned} \boldsymbol{\omega}_4 &= \boldsymbol{\omega}_2 + \boldsymbol{\omega}_{3/2} + \boldsymbol{\omega}_{4/3} = \boldsymbol{\omega}_p + \boldsymbol{\omega}_s \\ &= (\omega_p \sin \theta) \hat{\mathbf{j}}' + (\omega_s + \omega_p \cos \theta) \hat{\mathbf{k}}'. \end{aligned} \quad (16.31)$$

If we assume that the masses of the gimbals are negligible in comparison with that of the rotor, then the angular momentum of the system is

$$\mathbf{H} = I (\omega_p \sin \theta) \hat{\mathbf{j}}' + I^s (\omega_s + \omega_p \cos \theta) \hat{\mathbf{k}}'. \quad (16.32)$$

To find the time rate of change of the angular momentum, we must recognize that the $\hat{\mathbf{j}}'$ and $\hat{\mathbf{k}}'$ are not constant; they rotate with the angular velocity of the $x'y'z'$ coordinate system. From Eqs. (b) and (c), this is

$$\begin{aligned} \boldsymbol{\omega}_3 &= \boldsymbol{\omega}_2 + \boldsymbol{\omega}_{3/2} = \boldsymbol{\omega}_p \\ &= (\omega_p \sin \theta) \hat{\mathbf{j}}' + (\omega_p \cos \theta) \hat{\mathbf{k}}'. \end{aligned} \quad (d)$$

Thus, the time rate of change of $\hat{\mathbf{j}}'$ and $\hat{\mathbf{k}}'$ are

$$\frac{d\hat{\mathbf{j}}'}{dt} = \boldsymbol{\omega}_p \times \hat{\mathbf{j}}' = -(\omega_p \cos \theta) \hat{\mathbf{i}}', \quad (e)$$

$$\frac{d\hat{\mathbf{k}}'}{dt} = \boldsymbol{\omega}_p \times \hat{\mathbf{k}}' = (\omega_p \sin \theta) \hat{\mathbf{i}}'. \quad (f)$$

Finally, using Eqs. (e) and (f) to differentiate Eq. (16.32), the net external moment that must be applied to the rotor to sustain the steady precession motion can be written as

$$\begin{aligned} \sum \mathbf{M} &= \frac{d\mathbf{H}}{dt} = \boldsymbol{\omega}_p \times \mathbf{H} \\ &= -I (\omega_p \sin \theta) (\omega_p \cos \theta) \hat{\mathbf{i}}' + I^s (\omega_s + \omega_p \cos \theta) (\omega_p \sin \theta) \hat{\mathbf{i}}', \end{aligned} \quad (16.33)$$

or

$$\sum \mathbf{M} = [I^s \omega_s + (I^s - I) \omega_p \cos \theta] \omega_p \sin \theta \hat{\mathbf{i}}', \quad (16.34)$$

* We should note here that the true instantaneous axis of rotation of the rotor is not the spin axis but accounts for the precession rotation also; this axis is shown in Fig. 16.14. It may be our tendency to picture the spin axis as the true axis of rotation of the rotor that leads us to the intuitive feeling that a gyroscope does not follow the laws of mechanics.

or as

$$\sum \mathbf{M} = \left[I^s + (I^s - I) \frac{\omega_p}{\omega_s} \cos \theta \right] (\boldsymbol{\omega}_p \times \boldsymbol{\omega}_s). \quad (16.35)$$

Any one of these equations can be referred to as a gyroscopic formula.

We note that when the angular velocity of spin, ω_s , is much larger than that of precession, ω_p , which is usually the case, then the second term in the square brackets of Eqs. (16.34) and (16.35) is negligible with respect to the first. Therefore, the two equations reduce to

$$\sum \mathbf{M} = I^s \omega_p \omega_s \sin \theta \hat{\mathbf{i}}' = I^s (\boldsymbol{\omega}_p \times \boldsymbol{\omega}_s). \quad (16.36)$$

Figure 16.15 shows the same rotor in the same orientation as Fig. 16.14, but this time the angular momentum vector $\mathbf{H}$ rather than the angular velocities is shown. We have already noted that the angular momentum vector includes the effects of both the spin and the precession angular velocities. We can also see that it continually precesses, sweeping out a cone with apex angle θ about the precession axis.

In Fig. 16.15, we see the angular momentum vector $\mathbf{H}$ at some instant t and also its changed orientation, $\mathbf{H}'$, after a short time interval, Δt , and we note that it has not changed magnitude, only orientation. Spanning the tips of these two vectors, we see the vector change in angular momentum, $\Delta \mathbf{H}$ over this short time interval. We note that, in the limit as Δt approaches zero, the direction of the $\Delta \mathbf{H}$ vector approaches the direction of the positive x and x' axes. This is totally consistent with Eq. (16.34), which indicates that $\Delta \mathbf{H}/\Delta t$ is a vector in the $\hat{\mathbf{i}}'$ direction.

To maintain a steady precession, Eq. (16.35) demonstrates that an external moment must continually be applied to the rotor; if this moment is not applied, regular precession does not continue. Note that the axis of the applied moment must be along the x' axis, which

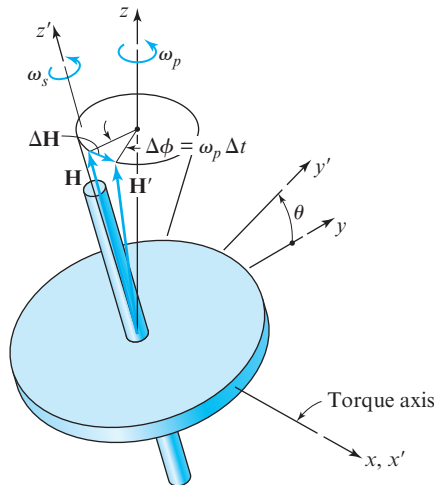


Figure 16.15 Angular momentum vector.

is continually changing during the precession. Note also that the sense of the moment is the same as that which would seem required to increase the angle θ . Thus, we see that there really is no paradox at all; the externally applied moment, $\sum \mathbf{M}$, about the x axis causes a change in angular momentum in exactly the same direction as the moment is applied.

16.14 FORCED PRECESSION

We have already observed that as speeds are increased in modern machinery, the engineer must be mindful of the increased importance of gyroscopic moments. Common mechanical equipment, which, in the past, were not thought of as exhibiting gyroscopic effects, do in fact often experience such moments, and these will become more significant as speeds increase. The purpose of this section is to present a few examples that, although they may not look like the standard gyroscope, do present gyroscopic moments that should be considered during the design of the equipment.

Much was presented in the earlier sections on the phenomenon of precession. This type of motion is almost certainly accompanied by gyroscopic moments. Yet, lest we think that precession is only an unintended wobbling motion of a toy top, we will look at examples where this precession is recognized and even designed into the operation of some mechanical devices.

Let us first consider a vehicle such as a train, automobile, or racing car, moving at high speed on a straight road. The gyroscopic effect of the spinning wheels is to keep the vehicle moving straight ahead and to resist changes in direction. But when an external moment is applied that forces the wheel to change its direction, gyroscopic reaction forces immediately come into play.

To study this gyroscopic reaction in the case of a vehicle, let us consider the following example.

EXAMPLE 16.2

Consider a pair of wheels connected by a straight axle, rounding a curve of constant radius, as shown in Fig. 16.16. For simplicity, the roadbed is assumed to be not banked. This wheel ensemble may be considered a gyroscope. Such rounding of the curve is a *forced* precession of the wheel-axle assembly around a vertical axis through the center of curvature of the track.

Find the gyroscopic torque exerted on the wheel-axle assembly.

SOLUTION

The notation used here is: r is the radius of the wheels, I^s is the combined mass moment of inertia about the axis of rotation, R is the radius of the curve, and $\mathbf{V}$ is the velocity of the center of the axle.

The angular velocity of precession of the assembly is

$$\omega_p = V/R. \quad (1)$$

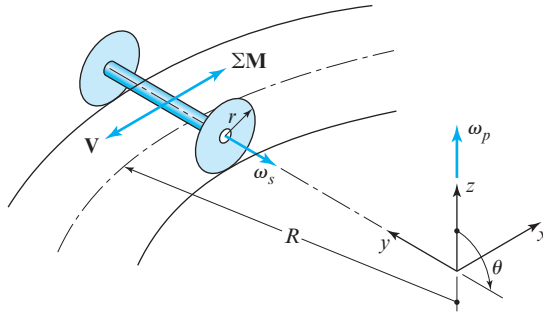


Figure 16.16 Axle and wheels following a curved path.

Similarly, the average angular velocity of spin of the two wheels is

$$\omega_s = V/r. \quad (2)$$

These are shown in Fig. 16.16.

The angle of the precession cone, that is, the angle between ω_p and ω_s , is $\theta = 90^\circ$. Substituting this value into Eq. (16.35) gives

$$\sum \mathbf{M} = I^s \omega_p \times \omega_s. \quad (3)$$

Then, substituting Eqs. (1) and (2) into Eq. (3) and performing the cross-product gives

$$\sum \mathbf{M} = \frac{I^s V^2}{Rr} \hat{\mathbf{i}}. \quad \text{Ans. (4)}$$

This is the additional external moment applied on the wheel-axle assembly caused by the gyroscopic action of the wheels while rounding the turn (forced precession) over and above the static and other steady-state dynamic loads. We note that the direction of this additional moment is such as to increase the tendency of the vehicle to roll over during the turn. This moment is applied by the tires by increasing the upward force on the outside tire and decreasing the upward force on the inside tire.

To make a quantitative comparison of the gyroscopic and centrifugal moments on the vehicle, let us consider the case of a racing car. The weight of the car including the driver is $W = 1\,800$ lb, and the height of the center of mass above the road is $h = 20$ in. We estimate that the radius of each wheel is $r = 18$ in, the radius of gyration is $k = 15$ in, and the weight of each wheel is $w = 100$ lb. From these data, we can determine the centrifugal moment $\mathbf{M}^*$ tending to cause rollover of the vehicle. Taking moments about the contact point of the outer tire, the result is

$$\mathbf{M}^* = \frac{hWV^2}{gR} \hat{\mathbf{i}} = \frac{(20 \text{ in})(1\,800 \text{ lb})V^2}{gR} \hat{\mathbf{i}} = 36\,000 \frac{V^2}{gR} \hat{\mathbf{i}} \text{ in} \cdot \text{lb}. \quad (5)$$

Since the vehicle has two axles and each has two wheels, the combined mass moment of inertia is

$$I^s = 2mk^2 = \frac{2(100 \text{ lb})(15 \text{ in})^2}{g} = \frac{45\,000 \text{ in}^2 \cdot \text{lb}}{g}. \quad (6)$$

Then, using Eq. (4) for two axles, the additional external moment applied on the wheel-axle assembly caused by the gyroscopic action of the wheels while rounding the turn is

$$\sum \mathbf{M} = 2 \frac{I^s V^2}{Rr} \hat{\mathbf{i}} = 2 \frac{(45\,000 \text{ in}^2 \cdot \text{lb}) V^2}{gR(18 \text{ in})} \hat{\mathbf{i}} = (5\,000 \text{ in} \cdot \text{lb}) \frac{V^2}{gR} \hat{\mathbf{i}} = 0.1389 \mathbf{M}^*. \quad (7)$$

Thus, we see that the gyroscopic effects of the tires add almost 14% to the tendency of the vehicle to roll over in a turn, and we see that this is independent of both the radius of the turn and the speed of the vehicle.

We note from this example that the problem itself looks nothing like a gyroscope; yet it has gyroscopic effects, and these may be missed in an oversimplified analysis. We note also that the precession motion is not just an unexpected result of the motion characteristics of the system. It is deliberately caused by the driver who steers the car; it is a forced precession.

Another gyroscopic effect in automobiles is that caused by the flywheel. Since the rotation of the flywheel of a rear-wheel-drive vehicle is along the longitudinal axis, and since the flywheel rotates counterclockwise as viewed from the rear, the spin vector $\boldsymbol{\omega}_s$ points toward the rear of the vehicle. When the vehicle makes a turn, the axis of the flywheel is forced to precess about a vertical axis as in the previous example. This forced precession brings into existence an applied gyroscopic moment about a horizontal axis through the center of the turn. The effect of this gyroscopic moment is to produce a bending moment on the driveshaft, tending to bend it in a vertical plane. The size of this gyroscopic moment causing bending in the driveshaft is usually of minor importance compared with the torsion loading because of the relatively small mass of the flywheel.

Lest we conclude that gyroscopic forces are always a disadvantage with which we must cope, consider the following example in which the gyroscopic effect is put to good use.

EXAMPLE 16.3

The edge mill shown schematically in Fig. 16.17 is a gyroscopic grinder used for crushing ore, seeds, grain, and such. It consists of a large steel pan in which one or more heavy conical rollers, called *mullers*, roll without slipping on the bottom of the pan, and, at the same time, revolve about a vertical shaft passing through the central axis of the pan. The mullers rotate about either horizontal or inclined axles that are attached to the vertical shaft, which is rotated under power. The radius of a muller is $r = 18 \text{ in}$, and the length of its axle is $l = 30 \text{ in}$; its weight is $W = 2\,200 \text{ lb}$, and its mass moment of inertia can be approximated by that of a cylindrical disk. Assume that the vertical shaft is driven at a constant angular velocity of $\omega_p = 40 \text{ rev/min}$ and that the muller rolls without slipping. Determine the optimum inclination angle θ for the muller axle, which maximizes the crushing force between the muller and the pan.

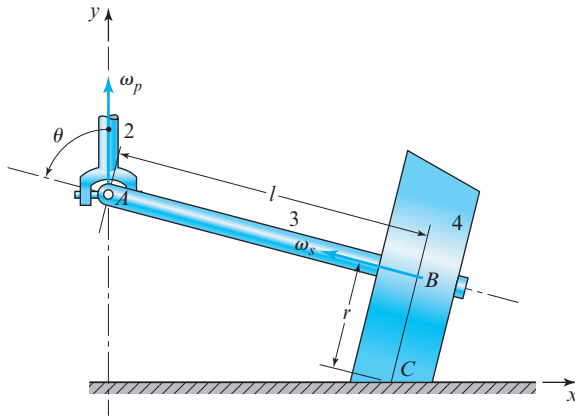


Figure 16.17 Schematic diagram of an edge mill.

SOLUTION

First, we use the assumption of rolling without slip to find the angular velocity of the muller. From Fig. 16.17, the angular velocity of the muller axis, link 3, is

$$\omega_p = \omega_3 = 40\hat{\mathbf{j}} \text{ rev/min} = 4.19\hat{\mathbf{j}} \text{ rad/s}, \quad (1)$$

whereas the angular velocity of spin of the muller about its axle is

$$\omega_s = \omega_{4/3} = \omega_s (-\sin\theta\hat{\mathbf{i}} + \cos\theta\hat{\mathbf{j}}). \quad (2)$$

Since the muller undergoes precession in addition to spin about its axle, the absolute angular velocity of the muller is

$$\omega_4 = \omega_3 + \omega_{4/3} = (-\omega_s \sin\theta)\hat{\mathbf{i}} + (4.19 \text{ rad/s} + \omega_s \cos\theta)\hat{\mathbf{j}}. \quad (3)$$

Since point A is fixed, the velocity of point B of link 3 is

$$\begin{aligned} \mathbf{V}_B &= \omega_3 \times \mathbf{R}_{BA} \\ &= (4.19\hat{\mathbf{j}} \text{ rad/s}) \times (30\sin\theta\hat{\mathbf{i}} - 30\cos\theta\hat{\mathbf{j}} \text{ in}) \\ &= -125.7\sin\theta\hat{\mathbf{k}} \text{ in/s}. \end{aligned} \quad (4)$$

Also, since there is no slip at point C, the velocity of point B of link 4 is

$$\begin{aligned} \mathbf{V}_B &= \omega_4 \times \mathbf{R}_{BC} \\ &= [(-\omega_s \sin\theta)\hat{\mathbf{i}} + (4.19 \text{ rad/s} + \omega_s \cos\theta)\hat{\mathbf{j}}] \times (18\cos\theta\hat{\mathbf{i}} + 18\sin\theta\hat{\mathbf{j}} \text{ in}) \\ &= -18(\omega_s + 4.19\cos\theta \text{ rad/s})\hat{\mathbf{k}}. \end{aligned} \quad (5)$$

Equating Eqs. (4) and (5), the angular velocity of spin of the muller about its axle is

$$\omega_s = 6.98\sin\theta - 4.19\cos\theta \text{ rad/s}. \quad (6)$$

From Eqs. (1) and (2), the cross-product is

$$\begin{aligned}(\boldsymbol{\omega}_p \times \boldsymbol{\omega}_s) &= (4.19 \hat{\mathbf{j}} \text{ rad/s}) \times \omega_s (-\sin \theta \hat{\mathbf{i}} + \cos \theta \hat{\mathbf{j}}) \\ &= 4.19 \omega_s \sin \theta \hat{\mathbf{k}} \text{ rad/s}.\end{aligned}$$

Then, substituting Eq. (6) into this result gives

$$(\boldsymbol{\omega}_p \times \boldsymbol{\omega}_s) = 4.19 (6.98 \sin \theta - 4.19 \cos \theta) \sin \theta \hat{\mathbf{k}} \text{ rad/s}^2. \quad (7)$$

Using the formula for a round disk from Table 5, Appendix A, the mass moment of inertia of the muller is

$$\begin{aligned}I^s &= \frac{mr^2}{2} = \frac{(2 \text{ 200 lb})(18 \text{ in})^2}{(386 \text{ in/s}^2) 2} = 923.3 \text{ in} \cdot \text{lb} \cdot \text{s}^2, \\ I &= \frac{mr^2}{4} + ml^2 = 5 \text{ 590 in} \cdot \text{lb} \cdot \text{s}^2.\end{aligned} \quad (8)$$

Then substituting Eqs. (7) and (8) into Eq. (16.35), the net externally applied moment on the muller is

$$\sum \mathbf{M} = \left[923.3 + \frac{-4 \text{ 667}(4.19) \cos \theta}{6.98 \sin \theta - 4.19 \cos \theta} \right] (6.98 \sin \theta - 4.19 \cos \theta) 4.19 \sin \theta \hat{\mathbf{k}} \text{ in} \cdot \text{lb}.$$

Rearranging this equation, the moment applied to the muller can be written as

$$\sum \mathbf{M} = (27 \text{ 001} \sin^2 \theta - 98 \text{ 082} \sin \theta \cos \theta) \hat{\mathbf{k}} \text{ in} \cdot \text{lb}. \quad (9)$$

This moment must be externally applied to the muller in order to sustain the forced precession. It must come from the net effect of the weight of the muller and the crushing force between the muller and the pan. Formulating these effects for the other side of the equation we obtain

$$\begin{aligned}\sum \mathbf{M} &= \mathbf{R}_{BA} \times \mathbf{W}_4 + \mathbf{R}_{CA} \times \mathbf{F}_c \\ &= (30 \sin \theta \hat{\mathbf{i}} - 30 \cos \theta \hat{\mathbf{j}} \text{ in}) \times (-2 \text{ 200} \hat{\mathbf{j}} \text{ lb}) \\ &\quad + [(30 \sin \theta - 18 \cos \theta) \hat{\mathbf{i}} + (18 \sin \theta - 30 \cos \theta) \hat{\mathbf{j}} \text{ in}] \times (F_c \hat{\mathbf{j}}) \\ &= (-66 \text{ 000} \sin \theta + 30 F_c \sin \theta - 18 F_c \cos \theta) \hat{\mathbf{k}} \text{ in} \cdot \text{lb}.\end{aligned} \quad (10)$$

Equating Eqs. (9) and (10), the crushing force becomes

$$F_c = \frac{27 \text{ 001} \sin^2 \theta - 98 \text{ 082} \sin \theta \cos \theta + 66 \text{ 000} \sin \theta \text{ in} \cdot \text{lb}}{30 \sin \theta - 18 \cos \theta \text{ in}}. \quad (11)$$

We note that the crushing force is a function of the angle θ of inclination of the muller axis; we now hope to choose this angle θ to maximize the crushing force. To do this we

differentiate Eq. (11) with respect to θ and set the result equal to zero; that is

$$\begin{aligned} & (54\,002 \sin \theta \cos \theta - 98\,082 \cos^2 \theta + 98\,082 \sin^2 \theta + 66\,000 \cos \theta) (30 \sin \theta - 18 \cos \theta) \\ & - (30 \cos \theta + 18 \sin \theta) (27\,001 \sin^2 \theta - 98\,082 \sin \theta \cos \theta + 66\,000 \sin \theta) = 0. \end{aligned}$$

This equation simplifies to

$$\begin{aligned} & 2\,456\,442 \sin^3 \theta + 810\,030 \sin^2 \theta \cos \theta \\ & - 972\,036 \sin \theta \cos^2 \theta + 1\,765\,476 \cos^3 \theta - 1\,188\,000 = 0, \end{aligned}$$

and can be solved numerically (the equation has multiple roots). When this is done, the root that maximizes the crushing force F_c of Eq. (11) is found to be

$$\theta = 115.9^\circ. \quad \text{Ans.}$$

This is the optimum angle of inclination of the muller axle. Substituting this value into Eq. (11), the crushing force of the mill is

$$F_c = 3\,437 \text{ lb.} \quad \text{Ans.}$$

We should note that this crushing force is more than 50% larger than the weight of the muller; the additional force is attributable to choosing the inclination angle θ for which both the gyroscopic and the centrifugal force effects contribute as much as possible to the crushing effect of the mill. There are other designs for crushing machines in which the pan rotates under a muller that has a fixed axis. In such a design, there is no gyroscopic action, and the crushing action is due solely to the weight of the muller.

An airplane propeller spinning at high speed is another example of a mechanical system exhibiting gyroscopic torque effects. In the case of a single propeller airplane, a turn in compass heading, for example, is a forced precession of the propeller's spin axis about a vertical axis. If the propeller is rotating clockwise as viewed from the rear, this forced precession induces a gyroscopic moment, causing the nose of the plane to move upward or downward as the heading is changed to the left or right, respectively. Turning the nose upward rather suddenly causes the plane to turn to the right, whereas a sudden turn downward causes a turn to the left.

Note that it is not the propeller forces that cause this effect, but the spinning mass and its gyroscopic effect during a change in direction of the plane (forced precession). The very substantial spinning mass of a radial engine (Fig. 1.22*b*) of early aircraft markedly exaggerated this gyroscopic effect and was, at least in part, responsible for the disappearance of use of rotary engines on airplanes. Since the effect comes from the precession of the spinning mass and not from the propeller, does the same danger not exist from the spinning mass of the rotor of a turbojet engine? When an airplane is equipped with two propellers rotating at equal speeds in opposite directions (or with counter-rotating turbines), however, the gyroscopic torques of the two can annul each other and leave no net effect on the plane as a whole.

16.15 REFERENCES

- [1] Bohnenburger, J. G. F., 1817. Beschreibung einer Maschine zur Erläuterung der Geseze der Umdrehung der Erde um ihre Axe, um der Veränderung der Lage der letzteren, *Tübinger Blätter für Naturwissenschaften und Arzneikunde*, 3:72–83.
- [2] Bollinger, J. G., and N. A. Duffie, 1988. *Computer Control of Machines and Processes*, Reading, MA: Addison-Wesley.
- [3] Euler, L., 1765. Theoria motus corporum solidorum seu rigidorum, *Commentarii Academiae Scientiarum Imperialis Petropolitanae* Translated by R. W. Willis, 1841. *Principles of Mechanisms*, John W. Parker, West Strand, London: Cambridge University Press.
- [4] Scarborough, J. B., 1958. *The Gyroscope: Theory and Applications*, New York: Interscience Publishers.

PROBLEMS

16.1 Table P16.1 lists the output torque for a one-cylinder engine running at 4 600 rev/min.

- (a) Find the mean output torque.
- (b) Determine the mass moment of inertia of an appropriate flywheel using $C_s = 0.025$.

Table P16.1 Torque data for Problem P16.1

θ_i deg	T_i N · m	θ_i deg	T_i N · m	θ_i deg	T_i N · m	θ_i deg	T_i N · m
0	0	180	0	360	0	540	0
10	17	190	−344	370	−145	550	−344
20	812	200	−540	380	−150	560	−540
30	963	210	−576	390	7	570	−577
40	1 016	220	−570	400	164	580	−572
50	937	230	−638	410	235	590	−643
60	774	240	−785	420	203	600	−793
70	641	250	−879	430	490	610	−893
80	697	260	−814	440	424	620	−836
90	849	270	−571	450	571	630	−605
100	1 031	280	−324	460	814	640	−379
110	1 027	290	−190	470	879	650	−264
120	902	300	−203	480	785	660	−300
130	712	310	−235	490	638	670	−368
140	607	320	−164	500	570	680	−334
150	594	330	−7	510	576	690	−198
160	544	340	150	520	540	700	−56
170	345	350	145	530	344	710	−2

16.2 Using the data of Table 16.2, determine the moment of inertia for a flywheel for a four-cylinder 90° V engine having a single crank. Use $C_s = 0.0125$ and a nominal speed of 4 600 rev/min. If a cylindric or

disk-type flywheel is to be used, what should be the thickness if it is made of steel and has an outside diameter of 400 mm? Use $\rho = 7.8 \text{ Mg/m}^3$ as the density of steel.

16.3 Using the data of Table 16.1, find the mean output torque and the flywheel inertia required for a three-cylinder in-line engine corresponding to a nominal speed of 2 400 rev/min. Use $C_s = 0.03$.

16.4 The load torque required by a 200-ton punch press is displayed in Table P16.4 for one revolution of the flywheel. The flywheel is to have a nominal angular velocity of 2 400 rev/min and to be designed for a coefficient of speed fluctuation of 0.075.

- (a) Determine the mean motor torque required at the flywheel shaft and the motor horsepower needed, assuming a constant torque-speed characteristic for the motor.
- (b) Find the moment of inertia needed for the flywheel.

Table P16.4 Torque data for Problem P16.4

θ_i deg	T_i in · lb	θ_i deg	T_i in · lb	θ_i deg	T_i in · lb	θ_i deg	T_i in · lb
0	857	90	7 888	180	1 801	270	857
10	857	100	8 317	190	1 629	280	857
20	857	110	8 488	200	1 458	290	857
30	857	120	8 574	210	1 372	300	857
40	857	130	8 403	220	1 115	310	857
50	1 287	140	7 717	230	1 029	320	857
60	2 572	150	3 515	240	943	330	857
70	5 144	160	2 144	250	857	340	857
80	6 859	170	1 972	260	857	350	857

16.5 Find T_m for the four-cylinder engine whose torque displacement is that of Fig. 16.4.

16.6 In a pendulum mill, shown schematically in Fig. P16.6, the grinding is by a conical muller that is free to spin about a pendulous axle that, in turn, is connected to a powered vertical shaft by a Hooke universal joint. The muller presses against the inner wall of a heavy steel pan, and it rolls around the inside of the pan without slipping. The weight of the muller is $W = 980$ lb; its principal mass moments of inertia are $I^s = 121$ in·lb·s² and $I = 88$ in·lb·s². The length of the muller axle is $l = R_{GA} = 40$ in, and the radius of the muller at its center of mass is $R_{GB} = 10$ in. Assuming that the vertical shaft is to be inclined at $\theta = 30^\circ$ and will be driven at a constant angular velocity of $\omega_p = 240$ rev/min, find the crushing force between the muller and the pan. Also determine the minimum angular velocity ω_p required to ensure contact between the muller and the pan.

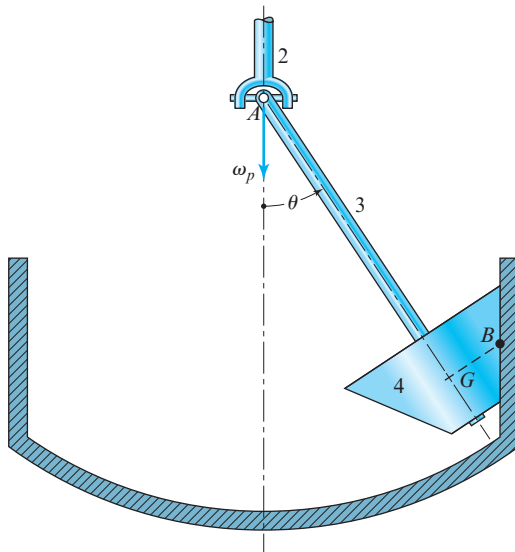


Figure P16.6

16.7 Using the gyroscopic formulae, Eqs. (16.33)–(16.35), solve the problem presented in Example 12.9 of Chap. 12.

16.8 The oscillating fan precesses sinusoidally according to the equation $\theta_p = \beta \sin 1.5t$, where $\beta = 30^\circ$; the fan blade spins at $\omega_s = 1\,800\hat{\mathbf{i}}$ rev/min. The weight of the fan and motor armature is 5.25 lb, and other masses can be assumed negligible; gravity acts in the $-\hat{\mathbf{j}}$ direction. The principal mass moments of inertia are $I^s = 0.065$ in·lb·s² and $I = 0.025$ in·lb·s²; the center of mass is located at $R_{GC} = 4$ in to the front of the precession axis. Determine the maximum moment, M^z , that must be accounted for in the clamped tilting pivot at C.

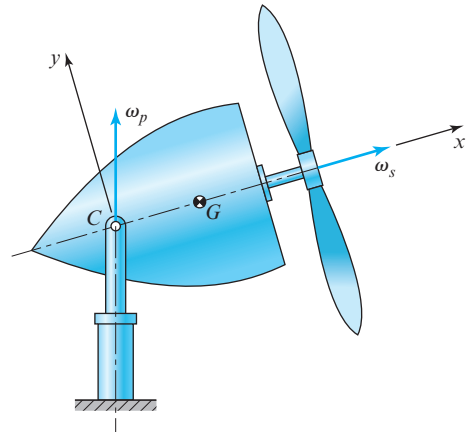


Figure P16.8

16.9 The propeller of an outboard motorboat is spinning at high speed and is caused to precess by steering to the right or left. Do the gyroscopic effects tend to raise or lower the rear of the boat? What is the effect, and is it of notable size?

16.10 A large and very high-speed turbine is to operate at an angular velocity of $\omega = 18\,000$ rev/min and will have a rotor with a principal mass moment of inertia of $I^s = 225$ in·lb·s². It has been suggested that since this turbine will be installed at the North Pole with its axis horizontal, perhaps the rotation of the earth will cause gyroscopic loads on its bearings. Estimate the size of these additional loads.

Appendix A: Tables

Table 1 Standard SI Prefixes^{*,†}

Name	Symbol	Factor
yotta	Y	1 000 000 000 000 000 000 000 000 = 10^{24}
zetta	Z	1 000 000 000 000 000 000 000 000 = 10^{21}
exa	E	1 000 000 000 000 000 000 000 = 10^{18}
peta	P	1 000 000 000 000 000 000 = 10^{15}
tera	T	1 000 000 000 000 000 = 10^{12}
giga	G	1 000 000 000 = 10^9
mega	M	1 000 000 = 10^6
kilo	k	1 000 = 10^3
hecto [‡]	h	100 = 10^2
deka [‡]	da	10 = 10^1
deci [‡]	d	0.1 = 10^{-1}
centi [‡]	c	0.01 = 10^{-2}
milli	m	0.001 = 10^{-3}
micro	μ	0.000 001 = 10^{-6}
nano	n	0.000 000 001 = 10^{-9}
pico	p	0.000 000 000 001 = 10^{-12}
femto	f	0.000 000 000 000 001 = 10^{-15}
atto	a	0.000 000 000 000 000 001 = 10^{-18}
zepto	z	0.000 000 000 000 000 000 000 = 10^{-21}
yocto	y	0.000 000 000 000 000 000 000 000 = 10^{-24}

^{*}If possible, multiple and submultiple prefixes are used in steps of 1 000. For example, lengths are specified in millimeters, meters, or kilometers. In a combination unit, prefixes are only used preceding the numerator; for example, meganewton per cubic meter (MN/m³) is used, but not Newton per cubic centimeter (N/cm³).

[†]Spaces are used in SI instead of commas to group numbers to avoid confusion with the practice in some countries of using commas for decimal points.

[‡]Not recommended but sometimes encountered.

Table 2 Conversion from US Customary Units to SI Units

To convert from	To	Multiply by	
		Accurate*	Common
Foot (ft)	Meter (m)	0.304 800*	0.305
Foot · pound (ft · lb)	Newton · meter (N · m)	1.355 818	1.36
	Joule (J)	1.355 818	1.36
Foot · pound/second (ft · lb/s)	Watt (W)	1.355 818	1.36
Horsepower (hp)	Watt (W)	745.699 9	746
Inch (in)	Meter (m)	0.025 400*	0.025 4
Inch · pound (in · lb)	Newton · meter (N · m)	0.112 984 8	0.113
	Joule (J)	0.112 984 8	0.113
Inch · pound/second (in · lb/s)	Watt (W)	0.112 984 8	0.113
Mile, U.S. statute (mi)	Meter (m)	1 609.344*	1 610
Pound force (lb)	Newton (N)	4.448 222	4.45
Pound mass (lb)	Kilogram (kg)	0.453 592 4	0.454
Pound/ft ² (lb/ft ²)	Pascal (Pa)	47.880 26	47.9
Pound/in ² (lb/in ²), (psi)	Pascal (Pa)	6 894.757	6 890
Revolutions/min (rev/min)	Radian/second (rad/s)	0.104 719 8	0.105
Ton, short (2 000 lb)	Kilogram (kg)	907.184 7	907

* In this table an asterisk indicates that the conversion is exact.

Table 3 Conversion from SI Units to US Customary Units

To convert from	To	Multiply by	
		Accurate	Common
Joule (J)	Foot · pound (ft · lb)	0.737 562 1	0.738
	Inch · pound (in · lb)	88.507 44	8.85
Kilogram (kg)	Pound mass (lb)	2.204 622	2.20
	Ton, short (2 000 lb)	0.001 102 311	0.001 10
Meter (m)	Foot (ft)	3.280 840	3.28
	Inch (in)	39.370 08	39.4
	Mile (mi)	0.000 621 371	0.000 621
Newton (N)	Pound (lb)	0.224 808 9	0.225
Newton · meter (N · m)	Foot · pound (ft · lb)	0.737 562 1	0.738
	Inch · pound (in · lb)	8.850 744	8.85
Newton · meter/second (N · m/s)	Horsepower (hp)	0.001 341 022	0.001 34
Pascal (Pa)	Pound/foot ² (lb/ft ²)	0.020 885 43	0.020 9
	Pound/inch ² (lb/in ²), (psi)	0.000 145 037 0	0.000 145
Radian/second (rad/s)	Revolutions/minute (rev/min)	9.549 297	9.55
Watt (W)	Horsepower (hp)	0.001 341 022	0.001 34
	Foot · pound /second (ft · lb/s)	0.737 562 1	0.738
	Inch · pound/second (in · lb/s)	8.850 744	8.85

Table 4 Properties of Areas

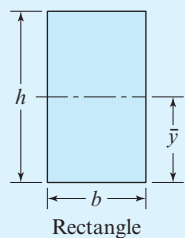
A = area

I = area moment of inertia

J = polar area moment of inertia

k = centroidal radius of gyration

$\bar{y}$ = centroidal distance from bottom



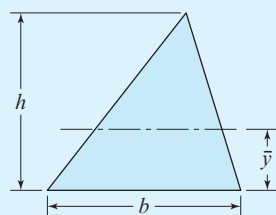
Rectangle

$$A = bh$$

$$I = bh^3/12$$

$$k = 0.289h$$

$$\bar{y} = h/2$$



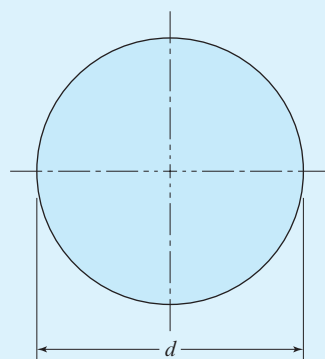
Triangle

$$A = bh/2$$

$$I = bh^3/36$$

$$k = 0.236h$$

$$\bar{y} = h/3$$



Circle

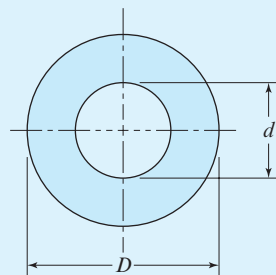
$$A = \pi d^2/4$$

$$I = \pi d^4/64$$

$$J = \pi d^4/32$$

$$k = d/4$$

$$\bar{y} = d/2$$



Annulus

$$A = \pi(D^2 - d^2)/4$$

$$I = \pi(D^4 - d^4)/64$$

$$J = \pi(D^4 - d^4)/32$$

$$k = \sqrt{D^2 + d^2}/4$$

$$\bar{y} = D/2$$

Table 5 Mass Moments of Inertia

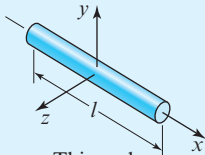
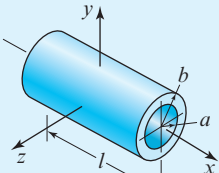
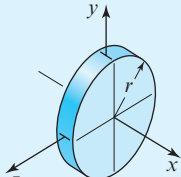
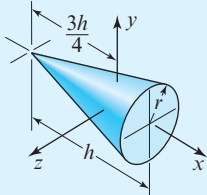
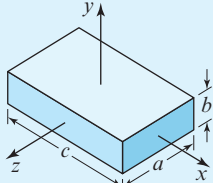
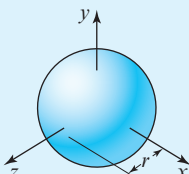
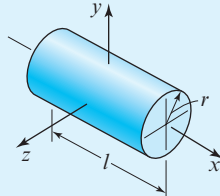
 Thin rod	$m = \rho \pi r^2 l$ $I^{yy} = I^{zz} = ml^2/12$	 Annular cylinder	$m = \rho \pi (b^2 - a^2) l$ $I^{xx} = m(a^2 + b^2)/2$ $I^{yy} = I^{zz} = m(3a^2 + 3b^2 + l^2)/12$
 Circular disk	$m = \rho \pi r^2 t$ $I^{xx} = mr^2/2$ $I^{yy} = I^{zz} = mr^2/4$	 Cone	$m = \rho \pi r^2 h/3$ $I^{xx} = 3mr^2/10$ $I^{yy} = I^{zz} = m(12r^2 + 3h^2)/80$
 Rectangular prism	$m = \rho abc$ $I^{xx} = m(a^2 + b^2)/12$ $I^{yy} = m(a^2 + c^2)/12$ $I^{zz} = m(b^2 + c^2)/12$	 Sphere	$m = \rho 4\pi r^3/3$ $I^{xx} = I^{yy} = I^{zz} = 2mr^2/5$
 Cylinder	$m = \rho \pi r^2 l$ $I^{xx} = mr^2/2$ $I^{yy} = I^{zz} = m(3r^2 + l^2)/12$		

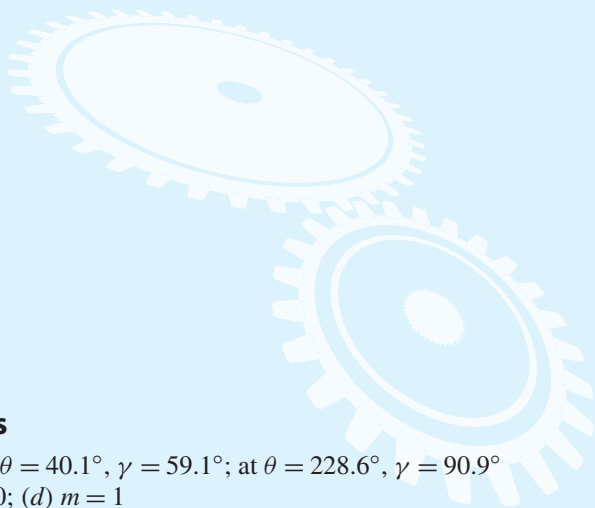
Table 6 Involute Function

φ (deg)	Inv (φ)	Inv ($\varphi + 0.1^\circ$)	Inv ($\varphi + 0.2^\circ$)	Inv ($\varphi + 0.3^\circ$)	Inv ($\varphi + 0.4^\circ$)
0.0	0.000 000	0.000 000	0.000 000	0.000 000	0.000 000
0.5	0.000 000	0.000 000	0.000 000	0.000 000	0.000 001
1.0	0.000 002	0.000 002	0.000 003	0.000 004	0.000 005
1.5	0.000 006	0.000 007	0.000 009	0.000 010	0.000 012
2.0	0.000 014	0.000 016	0.000 019	0.000 022	0.000 025
2.5	0.000 028	0.000 031	0.000 035	0.000 039	0.000 043
3.0	0.000 048	0.000 053	0.000 058	0.000 064	0.000 070
3.5	0.000 076	0.000 083	0.000 090	0.000 097	0.000 105
4.0	0.000 114	0.000 122	0.000 132	0.000 141	0.000 151
4.5	0.000 162	0.000 173	0.000 184	0.000 197	0.000 209
5.0	0.000 222	0.000 236	0.000 250	0.000 265	0.000 280
5.5	0.000 296	0.000 312	0.000 329	0.000 347	0.000 366
6.0	0.000 384	0.000 404	0.000 424	0.000 445	0.000 467
6.5	0.000 489	0.000 512	0.000 536	0.000 560	0.000 586
7.0	0.000 612	0.000 638	0.000 666	0.000 694	0.000 723
7.5	0.000 753	0.000 783	0.000 815	0.000 847	0.000 880
8.0	0.000 914	0.000 949	0.000 985	0.001 022	0.001 059
8.5	0.001 098	0.001 137	0.001 178	0.001 219	0.001 283
9.0	0.001 305	0.001 349	0.001 394	0.001 440	0.001 488
9.5	0.001 536	0.001 586	0.001 636	0.001 688	0.001 740
10.0	0.001 794	0.001 849	0.001 905	0.001 962	0.002 020
10.5	0.002 079	0.002 140	0.002 202	0.002 265	0.002 329
11.0	0.002 394	0.002 461	0.002 528	0.002 598	0.002 668
11.5	0.002 739	0.002 812	0.002 894	0.002 962	0.003 039
12.0	0.003 117	0.003 197	0.003 277	0.003 360	0.003 443
12.5	0.003 529	0.003 615	0.003 712	0.003 792	0.003 882
13.0	0.003 975	0.004 069	0.004 164	0.004 261	0.004 359
13.5	0.004 459	0.004 561	0.004 664	0.004 768	0.004 874
14.0	0.004 982	0.005 091	0.005 202	0.005 315	0.005 429
14.5	0.005 545	0.005 662	0.005 782	0.005 903	0.006 025
15.0	0.006 150	0.006 276	0.006 404	0.006 534	0.006 665
15.5	0.006 799	0.006 934	0.007 071	0.007 209	0.007 350
16.0	0.007 493	0.007 637	0.007 784	0.007 932	0.008 082
16.5	0.008 234	0.008 388	0.008 544	0.008 702	0.008 863
17.0	0.009 025	0.009 189	0.009 355	0.009 523	0.009 694
17.5	0.009 866	0.010 041	0.010 217	0.010 396	0.010 577
18.0	0.010 760	0.010 946	0.011 133	0.011 323	0.011 515
18.5	0.011 709	0.011 906	0.012 105	0.012 306	0.012 509
19.0	0.012 715	0.012 923	0.013 134	0.013 346	0.013 562
19.5	0.013 779	0.013 999	0.014 222	0.014 447	0.014 674
20.0	0.014 904	0.015 137	0.015 372	0.015 609	0.015 850
20.5	0.016 092	0.016 337	0.016 585	0.016 836	0.017 089
21.0	0.017 345	0.017 603	0.017 865	0.018 129	0.018 395
21.5	0.018 665	0.018 937	0.019 212	0.019 490	0.019 770
22.0	0.020 054	0.020 340	0.020 630	0.020 921	0.021 216
22.5	0.021 514	0.021 815	0.022 119	0.022 426	0.022 736

Table 6 (continued)

φ (deg)	Inv (φ)	Inv ($\varphi + 0.1^\circ$)	Inv ($\varphi + 0.2^\circ$)	Inv ($\varphi + 0.3^\circ$)	Inv ($\varphi + 0.4^\circ$)
23.0	0.023 049	0.023 365	0.023 684	0.024 006	0.024 332
23.5	0.024 660	0.024 992	0.025 326	0.025 664	0.026 005
24.0	0.026 350	0.026 697	0.027 048	0.027 402	0.027 760
24.5	0.028 121	0.028 485	0.028 852	0.029 223	0.029 598
25.0	0.029 975	0.030 357	0.030 741	0.031 130	0.031 521
25.5	0.031 917	0.032 315	0.032 718	0.033 124	0.033 534
26.0	0.033 947	0.034 364	0.034 785	0.035 209	0.035 637
26.5	0.036 069	0.036 505	0.036 945	0.037 388	0.037 835
27.0	0.038 287	0.038 696	0.039 201	0.039 664	0.040 131
27.5	0.040 602	0.041 076	0.041 556	0.042 039	0.042 526
28.0	0.043 017	0.043 513	0.044 012	0.044 516	0.045 024
28.5	0.045 537	0.046 054	0.046 575	0.047 100	0.047 630
29.0	0.048 164	0.048 702	0.049 245	0.049 792	0.050 344
29.5	0.050 901	0.051 462	0.052 027	0.052 597	0.053 172
30.0	0.053 751	0.054 336	0.054 924	0.055 519	0.056 116
30.5	0.056 720	0.057 267	0.057 940	0.058 558	0.059 181
31.0	0.059 809	0.060 441	0.061 779	0.061 721	0.062 369
31.5	0.063 022	0.063 680	0.064 343	0.065 012	0.065 685
32.0	0.066 364	0.067 048	0.067 738	0.068 432	0.069 133
32.5	0.069 838	0.070 549	0.071 266	0.071 988	0.072 716
33.0	0.073 449	0.074 188	0.074 932	0.075 683	0.076 439
33.5	0.077 200	0.077 968	0.078 741	0.079 520	0.080 305
34.0	0.081 097	0.081 974	0.082 697	0.083 506	0.084 321
34.5	0.085 142	0.085 970	0.086 804	0.087 644	0.088 490
35.0	0.089 342	0.090 201	0.091 066	0.091 938	0.092 816
35.5	0.093 701	0.094 592	0.095 490	0.096 395	0.097 306
36.0	0.098 224	0.099 149	0.100 080	0.101 019	0.101 964
36.5	0.102 916	0.103 875	0.104 841	0.105 814	0.106 795
37.0	0.107 782	0.108 777	0.109 779	0.110 788	0.111 805
37.5	0.112 828	0.113 860	0.114 899	0.115 945	0.116 999
38.0	0.118 060	0.119 130	0.120 207	0.121 291	0.122 384
38.5	0.123 484	0.124 592	0.125 709	0.126 833	0.127 965
39.0	0.129 106	0.130 254	0.131 411	0.132 576	0.133 749
39.5	0.134 931	0.136 122	0.137 320	0.138 528	0.139 743
40.0	0.140 968	0.142 201	0.143 443	0.144 694	0.145 954
40.5	0.147 222	0.148 500	0.149 787	0.151 082	0.152 387
41.0	0.153 702	0.155 025	0.156 358	0.157 700	0.159 052
41.5	0.160 414	0.161 785	0.163 165	0.164 556	0.165 956
42.0	0.167 366	0.168 786	0.170 216	0.171 656	0.173 106
42.5	0.174 566	0.176 037	0.177 518	0.179 009	0.180 511
43.0	0.182 023	0.183 546	0.185 080	0.186 625	0.188 180
43.5	0.189 746	0.191 324	0.192 912	0.194 511	0.196 122
44.0	0.197 744	0.199 377	0.201 022	0.202 678	0.204 346
44.5	0.206 026	0.207 717	0.209 420	0.211 135	0.212 863
45.0	0.214 602				

Appendix B



Answers to Selected Problems

- 1.3** $\gamma_{\min} = 53.1^\circ$; $\gamma_{\max} = 98.1^\circ$; at $\theta = 40.1^\circ$, $\gamma = 59.1^\circ$; at $\theta = 228.6^\circ$, $\gamma = 90.9^\circ$
1.5 (a) $m = 1$; (b) $m = 1$; (c) $m = 0$; (d) $m = 1$
1.7 7 variations
1.9 $m = 1$
1.11 $m = 1$
1.13 $j_1 = 5, j_2 = 1$; no slipping at A
1.15 $Q = 1.099$
1.17 0.104 17 in to left
1.19 One solution is $r_1 = 0.823$ m, $r_2 = 0.250$ m, $r_3 = 0.700$ m, $r_4 = 0.500$ m.
1.21 $m = 1$
1.23 $m = 1$
1.25 $m = 1$
1.27 $MA = 1.86$
1.29 $m = 1$
1.31 One solution is $r_1 = 102.5$ mm, $r_2 = O_2A = 222$ mm, $r_3 = AB = 340.5$ mm.
1.33 $m = 1$
1.35 Dead-centers: $\theta_2 = 114.05^\circ$, $\theta_4 = 162.82^\circ$ and $\theta_2''' = -114.05^\circ$, $\theta_4''' = -162.82^\circ$.
 Toggles: $\theta_2' = 56.50^\circ$, $\theta_4' = 133.14^\circ$, and $\theta_2'' = -56.50^\circ$, $\theta_4'' = -133.14^\circ$.
- 2.1** Spiral
2.3 $\mathbf{R}_{QP} = -7\hat{\mathbf{i}} - 14\hat{\mathbf{j}}$
2.5 $\mathbf{R}_A = -4.5a\hat{\mathbf{i}}$
2.7 Clockwise; $\mathbf{R}(0) = 4\angle 0^\circ$; $\mathbf{R}(20) = 404\angle 0^\circ$; $\Delta\mathbf{R} = 400\angle 0^\circ$
2.9 $\Delta\mathbf{R}_{P_3} = -2.121\hat{\mathbf{i}}_1 + 3.879\hat{\mathbf{j}}_1$; $\Delta\mathbf{R}_{P_3/2} = 3\hat{\mathbf{i}}_2$
2.11 $\Delta\mathbf{R}_Q = 1.902\hat{\mathbf{i}} + 1.098\hat{\mathbf{j}}$ in = 2.196 in $\angle 30^\circ$
2.13 $R_C = 2.5 \cos \theta_2 + \sqrt{48 - 5 \sin \theta_2 - 6.25 \sin^2 \theta_2}$ in
2.15 one loop; one constraint
2.17 one loop; one constraint
2.19 two loops; one constraint
2.21 two loops; one constraint
2.25 $\theta_2 = \pm(2k+1)\pi/2 = \pm 90^\circ, \pm 270^\circ, \dots$
2.27 three loops, three constraints
2.29 three loops, one constraint
2.31 $\theta_4 = 14.179^\circ$, $\theta_4 = -98.21^\circ$.

- 2.33 (a) $\theta_3 = 11^\circ$, $\theta_4 = 95^\circ$; $\theta'_3 = -71^\circ$, $\theta'_4 = -155^\circ$;
 (b) $\theta_3 = 11^\circ$, $\theta_4 = 94.90^\circ$; $\theta'_3 = -71^\circ$, $\theta'_4 = -154.90^\circ$;
 (c) $\theta_3 = 11.00^\circ$, $\theta_4 = 94.90^\circ$; $\theta'_3 = -71.00^\circ$, $\theta'_4 = -154.90^\circ$;
 (d) $\theta_3 = 11.0026^\circ$, $\theta_4 = 94.9037^\circ$.
- 3.1 $\dot{\mathbf{R}} = 314.159 \text{ in/s} \angle 162^\circ$.
- 3.3 $\mathbf{V}_{BA} = \mathbf{V}_{B_3/2} = 82.6 \text{ mi/h N } 24.8^\circ \text{ E}$
- 3.5 (a) $d = 1400 \text{ mm}$; (b) $\mathbf{V}_{AB} = 60\hat{\mathbf{j}} \text{ m/s}$; $\mathbf{V}_{BA} = -60\hat{\mathbf{j}} \text{ m/s}$; $\omega_2 = 200 \text{ rad/s cw}$
- 3.7 (a) Straight line at N 48° E ; (b) no change
- 3.9 $\omega_3 = 1.43 \text{ rad/s ccw}$; $\omega_4 = 15.40 \text{ rad/s ccw}$
- 3.11 $\mathbf{V}_C = 23.7 \text{ ft/s} \angle 284^\circ$; $\omega_3 = 0.335 \text{ rad/s ccw}$
- 3.13 $\mathbf{V}_C = 0.402 \text{ m/s} \angle 151^\circ$; $\mathbf{V}_D = 0.290 \text{ m/s} \angle 249^\circ$
- 3.15 $\mathbf{V}_B = 4.79 \text{ m/s} \angle 96^\circ$; $\omega_3 = \omega_4 = 22 \text{ rad/s ccw}$
- 3.17 $\omega_6 = 4.03 \text{ rad/s ccw}$; $\mathbf{V}_B = 0.964 \text{ ft/s} \angle 180^\circ$; $\mathbf{V}_C = 2.02 \text{ ft/s} \angle 208^\circ$;
 $\mathbf{V}_D = 2.02 \text{ ft/s} \angle 206^\circ$
- 3.19 $\omega_3 = 3.23 \text{ rad/s ccw}$; $\mathbf{V}_B = 16.9 \text{ ft/s} \angle -56^\circ$
- 3.21 $\mathbf{V}_C = 9.03 \text{ m/s} \angle 138^\circ$
- 3.23 $\mathbf{V}_B = 35.5 \text{ ft/s} \angle 240^\circ$; $\mathbf{V}_C = 40.6 \text{ ft/s} \angle 267^\circ$; $\mathbf{V}_D = 31.6 \text{ ft/s} \angle -60^\circ$
- 3.25 $\mathbf{V}_B = 1.04 \text{ ft/s} \angle -23^\circ$
- 3.27 $\mathbf{V}_C = 76.3 \text{ in/s} \angle 203.2^\circ$; $\omega_5 = \omega_6 = 9.67 \text{ rad/s cw}$; $\mathbf{V}_E = 77.4 \text{ in/s} \angle -99^\circ$
- 3.29 $\omega_4 = 4.36 \text{ rad/s ccw}$
- 3.31 $\mathbf{V}_A = 1.30 \text{ in/s} \angle 180^\circ$; $\mathbf{V}_B = 5.78 \text{ in/s} \angle 180^\circ$
- 3.32 $\omega_3 = 30.0 \text{ rad/s cw}$
- 3.41 $\theta'_4 = -0.0577 \text{ rad/in}$, $r'_4 = -1.1547 \text{ in/in}$; $\omega_3 = 0.520 \text{ rad/s ccw}$,
 $\omega_4 = 0.173 \text{ rad/s cw}$
- 3.43 $\theta'_3 = 0.0577 \text{ rad/in}$, $r'_4 = 1.1547 \text{ in/in}$; $\omega_3 = 0.289 \text{ rad/s cw}$, $V_{3/4} = -5.774 \text{ in/s}$
- 3.45 $r'_3 = -1.1547 \text{ m/m}$, $\theta'_3 = 14.43 \text{ rad/m}$, $\theta'_4 = -43.30 \text{ rad/m}$;
 $\omega_3 = 2.165 \text{ rad/s ccw}$, $\omega_4 = 6.495 \text{ rad/s cw}$, $\mathbf{V}_E = 0.150 \text{ m/s} \angle 150^\circ$,
 $\mathbf{V}_{B_4/3} = 0.173 \text{ m/s} \angle -60^\circ$
- 3.47 $\theta'_3 = -0.25 \text{ rad/in}$, $\theta'_4 = 0$; $\omega_3 = 3.75 \text{ rad/s cw}$, $\omega_4 = 0$; $\Delta = 0$ when
 $\theta_3 = \theta_4 = 68.91^\circ$
- 3.49 $\theta'_3 = -0.200 \text{ rad/rad}$, $\theta'_4 = 0$, $\theta'_5 = -0.500 \text{ rad/rad}$; $\omega_3 = 1.00 \text{ rad/s cw}$,
 $\omega_4 = 0$, $\omega_5 = 2.50 \text{ rad/s cw}$
- 3.51 $r'_2 = -100 \text{ mm/rad}$, $\theta'_3 = -8.33 \text{ rad/rad}$, $\omega_3 = 250 \text{ rad/s cw}$, $\mathbf{V}_{A_3/2} = -3.00 \text{ m/s}$.
- 3.53 $r'_3 = 0.259 \text{ m/m}$, $\theta'_3 = -1.288 \text{ rad/m}$; (a) $\omega_3 = 0.386 \text{ rad/s cw}$;
 (b) $\mathbf{V}_{A_2/3} = 0.0776 \text{ m/s} \angle 45^\circ$; (c) $\mathbf{V}_B = 0.483 \text{ m/s} \angle -45^\circ$
- 3.55 $\omega_3 = 30 \text{ rad/s cw}$, $\omega_4 = 20 \text{ rad/s cw}$, $\omega_5 = 12 \text{ rad/s cw}$.
- 3.57 $\theta'_3 = -0.577 \text{ rad/in}$, $R'_4 = 0.577 \text{ in/in}$, $\omega_3 = 2.771 \text{ rad/s cw}$, $\mathbf{V}_4 = 2.771 \text{ in/s}$.
- 3.59 $\omega_3 = \omega_4 = 6.667 \text{ rad/s cw}$; $\mathbf{V}_{D_4C_3} = 29.81 \text{ in/s} \angle 26.57^\circ$.
- 3.61 $\theta'_3 = \theta'_4 = 1 \text{ rad/rad}$; $\mathbf{V}_C = 252.34 \text{ in/s} \angle 162^\circ$.
- 4.1 $\ddot{\mathbf{R}} = -4\hat{\mathbf{i}} \text{ in/s}^2$
- 4.3 $\hat{\mathbf{u}}^t = 0.30071\hat{\mathbf{i}} - 0.95372\hat{\mathbf{j}}$; $A^n = -43.678 \text{ mm/s}^2$; $A^t = 12.581 \text{ mm/s}^2$;
 $\rho = -405.4 \text{ mm}$

- 4.5 $\mathbf{A}_A = -7\,200\hat{\mathbf{i}} + 2\,400\hat{\mathbf{j}} \text{ m/s}^2$
- 4.7 $\mathbf{V}_B = 12.0 \text{ ft/s} \angle 270^\circ$; $\mathbf{V}_C = 8.367 \text{ ft/s} \angle 12.1^\circ$; $\mathbf{A}_B = 395.1 \text{ ft/s}^2 \angle 165^\circ$;
 $\mathbf{A}_C = 210.2 \text{ ft/s}^2 \angle 240.3^\circ$
- 4.9 $\omega_2 = 38.64 \text{ rad/s}$ cw; $\alpha_2 = 5\,571.3 \text{ rad/s}^2$ cw
- 4.11 $\alpha_3 = 563.3 \text{ rad/s}^2$ ccw; $\alpha_4 = 123.7 \text{ rad/s}^2$ ccw.
- 4.13 $\mathbf{A}_C = 3\,104.4 \text{ ft/s}^2 \angle 114.4^\circ$; $\alpha_3 = 1\,741.6 \text{ rad/s}^2$ ccw; $\alpha_4 = 3\,055.8 \text{ rad/s}^2$ ccw
- 4.15 $\mathbf{A}_C = 2\,604 \text{ ft/s}^2 \angle -68.9^\circ$; $\alpha_4 = 1\,495 \text{ rad/s}^2$ ccw
- 4.17 $\mathbf{A}_B = 16.7 \text{ ft/s}^2 \angle 0^\circ$; $\alpha_3 = 17.49 \text{ rad/s}^2$ ccw; $\alpha_6 = 10.81 \text{ rad/s}^2$ cw
- 4.19 $\alpha_2 = 4\,181 \text{ rad/s}^2$ ccw
- 4.21 $\mathbf{A}_C = 450.6 \text{ m/s}^2 \angle 255.6^\circ$; $\alpha_3 = 74.08 \text{ rad/s}^2$ cw
- 4.23 $\mathbf{A}_B = 2\,440.6 \text{ ft/s}^2 \angle 240^\circ$; $\mathbf{A}_D = 4\,031 \text{ ft/s}^2 \angle 120^\circ$
- 4.25 $\theta_3 = 171^\circ$; $\theta_4 = 196^\circ$; $\omega_3 = 70.5 \text{ rad/s}$ ccw; $\omega_4 = 47.6 \text{ rad/s}$ ccw;
 $\alpha_3 = 3\,197 \text{ rad/s}^2$ ccw; $\alpha_4 = 3\,331 \text{ rad/s}^2$ ccw
- 4.27 $\theta_3 = 28.3^\circ$; $\theta_4 = 55.9^\circ$; $\omega_3 = 0.633 \text{ rad/s}$ cw; $\omega_4 = 2.16 \text{ rad/s}$ cw;
 $\alpha_3 = 7.82 \text{ rad/s}^2$ ccw; $\alpha_4 = 6.70 \text{ rad/s}^2$ ccw
- 4.29 $\theta_3 = 38.4^\circ$; $\theta_4 = 155.6^\circ$; $\omega_3 = 6.86 \text{ rad/s}$ cw; $\omega_4 = 1.23 \text{ rad/s}$ cw;
 $\alpha_3 = 62.5 \text{ rad/s}^2$ ccw; $\alpha_4 = 96.5 \text{ rad/s}^2$ cw
- 4.31 $\mathbf{V}_B = 184 \text{ in/s} \angle -19.1^\circ$; $\mathbf{A}_B = 2\,703 \text{ in/s}^2 \angle -172.5^\circ$; $\omega_4 = 6.57 \text{ rad/s}$ cw;
 $\alpha_4 = 86.4 \text{ rad/s}^2$ ccw
- 4.33 $\mathbf{A}_E = 602.7 \text{ ft/s}^2 \angle -107.2^\circ$
- 4.35 $\mathbf{A}_B = 1.69 \text{ ft/s}^2 \angle -102.1^\circ$; $\alpha_3 = 1.247 \text{ rad/s}^2$ cw.
- 4.37 $\mathbf{A}_{C_4} = 50.0 \text{ ft/s}^2 \angle 91.1^\circ$; $\alpha_3 = 6.0 \text{ rad/s}^2$ ccw
- 4.39 $\mathbf{A}_G = 29.3 \text{ ft/s}^2 \angle -64.5^\circ$; $\alpha_5 = 0$; $\alpha_6 = 110 \text{ rad/s}^2$ cw
- 4.41 $\theta_3'' = 0.240 \text{ rad/rad}^2$, $\theta_4'' = 0.150 \text{ rad/rad}^2$, $\theta_5'' = 0$; $\alpha_3 = 6.0 \text{ rad/s}^2$ ccw,
 $\alpha_4 = 3.75 \text{ rad/s}^2$ ccw; $\alpha_5 = 0$
- 4.47 $\rho_C = 43.78 \text{ m}$; $v = 56.55 \text{ m/s}$; $A_I = 2\,070 \text{ m/s}^2$; $V_A = 48.97 \text{ m/s}$; $V_C = 44.10 \text{ m/s}$;
 $A_A = 757.89 \text{ m/s}^2$; $A_C = 378.95 \text{ m/s}^2$.
- 4.49 $\rho_B = 3 \text{ in}$; $\rho_E = 4.24 \text{ in}$; $V_B = 10.0 \text{ in/s}$; $V_E = 14.1 \text{ in/s}$; $v = 13.3 \text{ in/s}$;
 $A_B = 56.00 \text{ in/s}^2$; $A_E = 165.58 \text{ in/s}^2$; $A_I = 133.33 \text{ in/s}^2$.
- 4.51 $\hat{\mathbf{u}}_D^t = -\hat{\mathbf{j}}$, $\hat{\mathbf{u}}_D^n = \hat{\mathbf{i}}$, $\rho_D = -90 \text{ mm}$, $x_{CC} = -120 \text{ mm}$, $y_{CC} = 60 \text{ mm}$,
 $\mathbf{V}_D = 7.500 \text{ m/s} \angle -90^\circ$, $\mathbf{A}_D = 667\,306 \text{ m/s}^2 \angle -159.53^\circ$.
- 4.53 $x_B' = -646.41 \text{ mm/rad}$, $y_B' = 0$, $x_B'' = 2\,785.6 \text{ mm/rad}^2$,
 $y_B'' = 900.0 \text{ mm/rad}^2$, $\hat{\mathbf{u}}_B^t = \hat{\mathbf{i}}$, $\hat{\mathbf{u}}_B^n = \hat{\mathbf{j}}$, $\rho_B = 464.27 \text{ mm}$, $x_{CC} = 200 \text{ mm}$,
 $y_{CC} = 364.27 \text{ mm}$, $\mathbf{A}_B = 421.543 \text{ m/s}^2 \angle 17.91^\circ$.
- 4.55 $\omega_3 = 100 \text{ rad/s}$ cw, $\omega_4 = 66.67 \text{ rad/s}$ cw, $\omega_5 = 40 \text{ rad/s}$ cw,
 $\alpha_3 = 30 \text{ rad/s}^2$ cw, $\alpha_3 = 20 \text{ rad/s}^2$ cw, $\alpha_3 = 12 \text{ rad/s}^2$ cw.
- 4.57 $\rho_C = -114.53 \text{ mm}$, $x_{CC} = 0$, $y_{CC} = 75 \text{ mm}$, $\mathbf{V}_C = 504.1 \text{ mm/s} \angle -130.9^\circ$,
 $\mathbf{A}_C = 617.44 \text{ m/s}^2 \angle -151.94^\circ$.
- 4.59 $\rho_B = -70.71 \text{ mm}$, $x_{CC} = -50 \text{ mm}$, $y_{CC} = -50 \text{ mm}$, $\mathbf{A}_C = 7.906 \text{ m/s}^2 \angle 161.57^\circ$.
- 4.61 $x_C' = -16.000 \text{ in/rad}$, $y_C' = 5.196 \text{ in/rad}$, $x_C'' = -35.218 \text{ in/rad}^2$,
 $y_C'' = -16.000 \text{ in/rad}^2$, $\hat{\mathbf{u}}_C^t = -0.951\,10\hat{\mathbf{i}} + 0.308\,88\hat{\mathbf{j}}$,
 $\hat{\mathbf{u}}_C^n = -0.308\,88\hat{\mathbf{i}} - 0.951\,10\hat{\mathbf{j}}$, $\rho_C = 10.844 \text{ in}$, $x_{CC} = 1.847 \text{ in}$,
 $y_{CC} = 5.687 \text{ in}$, $\mathbf{V}_C = 252.34 \text{ in/s} \angle 162.01^\circ$, $\mathbf{A}_C = 8\,705.12 \text{ in/s}^2 \angle -155.57^\circ$.

- 5.1 $V_{P_4} = 2.355 \text{ m/s} \angle 15.6^\circ$, $A_{P_4} = 125.73 \text{ m/s}^2 \angle -66.6^\circ$.
- 5.3 $\omega_3 = 41.64 \text{ rad/s}$ cw, $\omega_5 = 47.20 \text{ rad/s}$ ccw
- 5.5 $V_{P_3} = 3.93 \text{ ft/s} \angle -175.13^\circ$, $A_{P_3} = 37.27 \text{ ft/s}^2 \angle -79.70^\circ$.
- 5.7 $\theta_4 = 101.39^\circ$, $\theta_{5/4} = 61.45^\circ$.
- 5.9 $\omega_4 = 10.07 \text{ rad/s}$ cw, $\omega_{5/4} = 17.18 \text{ rad/s}$ ccw.
- 5.11 $\alpha_4 = 91.944 \text{ rad/s}^2$ ccw, $\alpha_4 = 37.818 \text{ rad/s}^2$ ccw.
- 5.13 $x'_{P_x} = 1.667 \text{ m/m}$, $y'_{P_x} = 0$, $x'_{P_y} = 0$, $y'_{P_y} = 1.667 \text{ m/m}$.
- 5.15 $\theta'_{62} = 2.734 \text{ rad/rad}$, $\theta'_{32} = -1.000 \text{ rad/rad}$, $\theta'_{42} = 8.660 \text{ rad/rad}$, $\theta'_{35} = 0$, $\theta'_{65} = 0$, $\theta'_{45} = -2.166 \text{ rad/rad}$, $\theta''_{322} = -9.474 \text{ rad/rad}^2$, $\theta''_{622} = 14.533 \text{ rad/rad}^2$, $\theta''_{422} = 46.021 \text{ rad/rad}^2$, $\theta''_{325} = 0$, $\theta''_{625} = 0$, $\theta''_{425} = 0$, $\theta''_{355} = 0$, $\theta''_{655} = 0$, $\theta''_{455} = 0$, $\omega_3 = 50 \text{ rad/s}$ cw, $\omega_4 = 487.18 \text{ rad/s}$ ccw, $\alpha_3 = 23 \text{ 685 rad/s}^2$ cw, $\alpha_4 = 115 \text{ 009 rad/s}^2$ ccw.
- 6.3 Face = 195 mm from pivot
- 6.5 $\dot{y}\left(\frac{\beta}{2}\right) = \frac{\pi L}{2\beta}\omega$; $\ddot{y}\left(\frac{\beta}{2}\right) = -\frac{\pi^3 L}{2\beta^3}\omega^3$; $\ddot{y}(0) = \frac{\pi^2 L}{2\beta^2}\omega^2$; $\ddot{y}(\beta) = -\frac{\pi^2 L}{2\beta^2}\omega^2$.
- 6.7 AB: dwell, $L_1 = 0$, $\beta_1 = 60^\circ$; BC: full-rise eighth-order polynomial motion, Eq. (6.14), $L_2 = 2.500 \text{ 0 in}$, $\beta_2 = 62.442^\circ$; CD: half-harmonic return motion, Eq. (6.20), $L_3 = 0.082 \text{ 4 in}$, $\beta_3 = 7.750^\circ$; DE: uniform motion, $L_4 = 1.000 \text{ 0 in}$, $\beta_4 = 60^\circ$; EA: half-cycloidal return motion, Eq. (6.25), $L_5 = 1.417 \text{ 6 in}$, $\beta_5 = 169.808^\circ$.
- 6.9 $t_{AB} = 0.025 \text{ s}$; $\dot{y}_{\max} = 506.2 \text{ in/s}$; $\dot{y}_{\min} = -40 \text{ in/s}$; $\ddot{y}_{\max} = 19 \text{ 457 in/s}^2$; $\ddot{y}_{\min} = -19 \text{ 457 in/s}^2$
- 6.11 $\dot{y}_{\max} = 41.888 \text{ rad/s}$; $\ddot{y}_{\max} = 7 \text{ 896 rad/s}^2$
- 6.13 Face width = 2.200 in; $\rho_{\min} = 3.000 \text{ in}$
- 6.15 $R_0 > 9.089 \text{ in}$; face width $> 13.342 \text{ in}$
- 6.17 $\phi_{\max} = 12^\circ$; $R_r < 14.5 \text{ in}$
- 6.19 $R_0 > 56 \text{ mm}$; $\ddot{y}_{\max} = 37.0 \text{ m/s}^2$
- 6.21 $R_0 > 71 \text{ mm}$; $\ddot{y}_{\max} = 39.5 \text{ m/s}^2$
- 6.23 $u = (R_0 + R_C + Y) \sin \theta + y' \cos \theta$; $v = (R_0 + R_C + Y) \cos \theta - y' \sin \theta$
- 6.27 (a) $y' = 21.65 \text{ mm/rad}$, $y'' = 25.0 \text{ mm/rad}^2$, $y''' = -86.60 \text{ mm/rad}^3$;
(b) $u_{\text{cam}} = 36.56 \text{ mm}$, $v_{\text{cam}} = 8.76 \text{ mm}$; (c) $\rho = -87.02 \text{ mm}$;
(d) $\hat{\mathbf{u}}^t = 0.746\hat{\mathbf{i}} - 0.666\hat{\mathbf{j}}$, $\hat{\mathbf{u}}^n = 0.666\hat{\mathbf{i}} + 0.746\hat{\mathbf{j}}$; (e) $\phi = 8.2^\circ$.
- 6.31 (a) $y' = -3.450 \text{ 65 in/rad}$, $y'' = -6.736 \text{ 18 in/rad}^2$; (b) $u_{\text{cam}} = -8.543 \text{ in}$, $v_{\text{cam}} = -4.601 \text{ in}$; (c) $\rho_{\text{cam}} = -5.945 \text{ in}$; (d) $\phi = 32.19^\circ$.
- 6.33 (a) $y' = 1.527 \text{ 88 in/rad}$, $y'' = 0$, $y''' = -27.501 \text{ 78 in/rad}^3$; (b) $\rho_{\text{cam}} = -2.723 \text{ in}$;
(c) $\hat{\mathbf{u}}^t = -0.674 \text{ 55}\hat{\mathbf{i}} - 0.738 \text{ 23}\hat{\mathbf{j}}$, $\hat{\mathbf{u}}^n = 0.738 \text{ 23}\hat{\mathbf{i}} - 0.674 \text{ 55}\hat{\mathbf{j}}$;
(d) $u_{\text{cam}} = 0.990 \text{ in}$, $v_{\text{cam}} = -2.878 \text{ in}$; (e) $\phi = 17.58^\circ$.
- 6.35 (a) $y' = -1.145 \text{ 91 in/rad}$, $y'' = -3.969 \text{ 55 in/rad}^2$, $y''' = 13.750 \text{ 89 in/rad}^3$;
(b) $\rho_{\text{cam}} = -1.452 \text{ in}$; (c) $\hat{\mathbf{u}}^t = -0.307 \text{ 01}\hat{\mathbf{i}} + 0.951 \text{ 70}\hat{\mathbf{j}}$, $\hat{\mathbf{u}}^n = -0.951 \text{ 70}\hat{\mathbf{i}} - 0.307 \text{ 01}\hat{\mathbf{j}}$; (d) $u_{\text{cam}} = -2.599 \text{ 51 in}$, $v_{\text{cam}} = -1.954 \text{ in}$;
(e) $\phi = 22.12^\circ$.
- 6.37 (a) $F = m\omega^2 e \cos \omega t + mg$; (b) $\omega = 19.65 \text{ rad/s}$.
- 6.39 (a) $F_{c,\max} = 51.1 \text{ N}$, $F_{c,\min} = 0$; (b) $\Theta = 121.39^\circ$.
- 6.41 $\omega = 138.2 \text{ rev/min}$.

- 7.1** $P = 16$ teeth/in.
7.3 $m = 2$ mm/tooth
7.5 $P = 0.897$ 6 teeth/in; $D = 44.563$ in
7.7 $m = 12.732$ mm/tooth; $D = 458.4$ mm
7.9 $D = 9.191$ in
7.11 $N_2 = 17$ teeth; $N_3 = 51$ teeth
7.13 $a = 0.250$ in; $d = 0.312$ 5 in; $c = 0.062$ 5 in; $p_c = 0.785$ 4 in/tooth; $p_b = 0.738$ 0 in/tooth; $t = 0.392$ 7 in; $R_2 = 3.000$ in, $R_3 = 4.500$ in; $r_2 = 2.819$ in; $r_3 = 4.229$ in; $CP = 0.625$ in; $PD = 0.591$ in; $m_c = 1.647$ teeth.
7.15 $\alpha_2 = 18.6^\circ$; $\alpha_3 = 6.3^\circ$; $\beta_2 = 16.0^\circ$; $\beta_3 = 5.5^\circ$; $m_c = 1.63$ teeth
7.17 $CP = 18.930$ mm; $PD = 16.243$ mm; $m_c = 1.54$ teeth
7.19 (a) $CP = 1.462$ in; $PD = 1.196$ in; $m_c = 1.80$ teeth (b) $CP = 1.096$ in; $PD = 1.196$ in; $m_c = 1.55$ teeth; no change in pressure angle
7.25 $t_b = 17.142$ mm; $t_a = 6.737$ mm; $\varphi_a = 32.78^\circ$
7.27 $t_b = 1.146$ in
7.29 $t_b = 0.159$ 4 in; $\varphi_a = 35.35^\circ$; $t_a = 0.041$ 5 in
7.31 (a) 0.168 2 in; (b) 9.827 0 in
7.33 $m_c = 1.664$ teeth
7.35 $m_c = 1.771$ teeth
7.37 $\phi = 26.24^\circ$
7.39 $a_3 = 1.344$ in
7.41 $a'_3 = 0.712$ 3 in, $d'_3 = 1.637$ 7 in, $a'_2 = 1.287$ 7 in, $d'_2 = 1.062$ 3 in, $e_2 = -0.062$ 3 in, $e_3 = -0.637$ 7 in, $m'_c = 1.49$ teeth
7.45 $\omega_5 = 16.80$ rev/min cw, $\omega_7 = 31.50$ rev/min cw
7.47 One solution: $N_6 = 20$, $N_5 = 30$, $N_8 = 35$, $N_3 = 30$, $N_4 = N_7 = 25$, $N_{10} = 40$ teeth.
7.49 $\omega_7 = 222.1$ rev/min ccw
7.51 $\omega_A = 644.8$ rev/min cw
7.53 77.3%, opposite to input sense.
7.55 $N_5 = 84$ teeth, $R_5 = 156$ mm, $\omega_3 = 8.08$ rev/min ccw
7.57 $\omega_A = (9/34) \omega_2$; Levai type F
7.59 $\Delta t_D = 60$ min/rev, $\Delta t_H = 12$ hr/rev, $\omega_F = 2$ rev/day.
- 8.1** $p_t = 0.523$ 6 in/tooth, $p_n = 0.370$ 2 in/tooth, $P_n = 8.485$ teeth/in, $R_2 = 1.250$ in, $R_3 = 2.000$ in, $N_{e2} = 42.43$ teeth, $N_{e3} = 67.88$ teeth
8.3 $\psi_b = 33.34^\circ$, $m_x = 1.67$ teeth
8.5 $N_3 = 72$ teeth, $R_2 = 2.0$ in, $R_3 = 9.0$ in, $p_n = 0.723$ in/tooth, $P_n = 4.345$ teeth/in, $F > 3.7$ in
8.7 $m_n = 1.79$ teeth, $m_{\text{total}} = 2.87$ teeth
8.9 $\psi_3 = 15^\circ$ RH, $N_3 = 39$ teeth, $R_2 = 0.966$ in, $R_3 = 1.682$ in
8.11 $\gamma_2 = 18.43^\circ$, $\gamma_3 = 71.57^\circ$
8.13 $\gamma_2 = 27.0^\circ$, $\gamma_3 = 93.0^\circ$
8.15 $R_2 = 1.063$ in, $R_3 = 1.750$ in, $\gamma_2 = 34.83^\circ$, $\gamma_3 = 70.17^\circ$, $a_2 = 0.168$ 1 in, $a_3 = 0.081$ 9 in, $d_2 = 0.105$ 4 in, $d_3 = 0.191$ 6 in, $F \approx 0.558$ in, $N_{e2} = 20.71$, $N_{e3} = 82.54$
8.17 $N_2 = 1$ tooth, $R_2 = 1.725$ in, $N_3 = 60$ teeth, $R_3 = 4.775$ in

- 8.19** $N_3 = 123.6$ teeth, $R_3 = 7.871$ in, $\psi = 20.0^\circ$, $R_2 = 0.525$ in, $R_2 + R_3 = 8.396$ in
8.21 59.1%
8.23 $\omega_6 = 651.3$ rev/min, $\omega_5 = 693.3$ rev/min, $\omega_3 = 672.3$ rev/min
- 9.1** For six points, 0.170, 1.464, 3.706, 6.294, 8.536, and 9.830.
9.3 Typical solution: $r_2 = 7.06$ in, $r_3 = 22.75$ in.
9.5 Typical solution: $r_1 = 7.63$ ft, $r_2 = 3.22$ ft, $r_3 = 8.48$ ft.
9.7 Typical solution: O_2 at $x = -1\,979$ mm, $y = 1\,094$ mm, $r_2 = 346$ mm, $r_3 = 2\,288$ mm.
9.9 Typical solution: $O_2A = AB = O_4B = O_2O_4$ and spring AO_4 with chosen free length.
9.13 and **9.23** $r_2/r_1 = 1.834$, $r_3/r_1 = 2.238$, $r_4/r_1 = -0.693$.
9.15 and **9.25** $r_2/r_1 = -3.499$, $r_3/r_1 = 0.878$, $r_4/r_1 = 3.399$.
9.17 and **9.27** $r_2/r_1 = 0.625$, $r_3/r_1 = 1.309$, $r_4/r_1 = -0.401$.
9.19 and **9.29** $r_2/r_1 = -1.801$, $r_3/r_1 = 0.908$, $r_4/r_1 = 1.274$.
9.21 and **9.31** $r_2/r_1 = -0.610$, $r_3/r_1 = 0.565$, $r_4/r_1 = 0.380$.
- 10.1** $m = 2$ including one idle freedom. Path of B is the intersection of cylinder of radius BA about y axis and sphere of radius BO_3 about O_3 .
10.3 $\omega_2 = -2.570\hat{\mathbf{j}}$ rad/s; $\omega_3 = 1.158\hat{\mathbf{i}} - 0.086\hat{\mathbf{j}} + 0.643\hat{\mathbf{k}}$ rad/s; $\mathbf{V}_B = -96.361\hat{\mathbf{i}} - 50.000\hat{\mathbf{j}} + 166.903\hat{\mathbf{k}}$ m/s.
10.5 and **10.7** $\omega_3 = \omega_4 = -25.7\hat{\mathbf{i}}$ rad/s; $\mathbf{V}_A = 180\hat{\mathbf{j}}$ in/s; $\mathbf{V}_B = -231.4\hat{\mathbf{k}}$ in/s; $\alpha_3 = 1\,543\hat{\mathbf{j}}$ rad/s²; $\alpha_4 = -343\hat{\mathbf{i}}$ rad/s²; $\mathbf{A}_A = 10\,800\hat{\mathbf{i}}$ in/s²; $\mathbf{A}_B = -5\,951\hat{\mathbf{j}} - 3\,087\hat{\mathbf{k}}$ in/s².
10.9 $\Delta\theta_4 = 46.5^\circ$; $Q = 1.01$.
10.11 and **10.13** $\omega_3 = 4.083\hat{\mathbf{i}} - 7.071\hat{\mathbf{j}} + 3.334\hat{\mathbf{k}}$ rad/s, $\omega_4 = 10.797\hat{\mathbf{i}}$ rad/s, $\mathbf{V}_A = -1.169\hat{\mathbf{i}} - 0.675\hat{\mathbf{j}}$ m/s, $\mathbf{V}_B = 1.207\hat{\mathbf{j}} + 2.559\hat{\mathbf{k}}$ m/s, $\alpha_3 = 273\hat{\mathbf{i}} + 115\hat{\mathbf{j}} - 148\hat{\mathbf{k}}$ rad/s², $\alpha_4 = 130\hat{\mathbf{i}}$ rad/s², $\mathbf{A}_A = 24.30\hat{\mathbf{i}} - 42.09\hat{\mathbf{j}}$ m/s², $\mathbf{A}_B = -13.04\hat{\mathbf{j}} + 43.91\hat{\mathbf{k}}$ m/s².
10.15 $\omega_2 = 20.78\hat{\mathbf{i}} - 12.00\hat{\mathbf{j}}$ rad/s, $\omega_3 = 2.31\hat{\mathbf{i}} + 6.66\hat{\mathbf{j}} - 3.23\hat{\mathbf{k}}$ rad/s, $\mathbf{V}_A = 12.00\hat{\mathbf{i}} + 20.78\hat{\mathbf{j}} + 41.57\hat{\mathbf{k}}$ m/s, $\mathbf{V}_B = 13.82\hat{\mathbf{i}}$ m/s.
10.17 and **10.19** $\omega_2 = 12.00\hat{\mathbf{i}} - 20.78\hat{\mathbf{j}}$ rad/s, $\omega_3 = 1.33\hat{\mathbf{i}} + 6.91\hat{\mathbf{j}} - 1.82\hat{\mathbf{k}}$ rad/s, $\mathbf{V}_A = 20.78\hat{\mathbf{i}} + 12.00\hat{\mathbf{j}} + 41.57\hat{\mathbf{k}}$ m/s, $\mathbf{V}_B = 26.11\hat{\mathbf{i}}$ m/s.
10.21 and **10.23** $\mathbf{V}_B = 112$ in/s; $\omega_3 = -4.19\hat{\mathbf{i}} + 19.44\hat{\mathbf{j}} - 5.47\hat{\mathbf{k}}$ rad/s, $\omega_4 = 19.44\hat{\mathbf{j}}$ rad/s.
10.24 $T_{15} = \begin{bmatrix} 0.866 & -0.500 & 0 & 17.32 \\ -0.500 & -0.866 & 0 & 0 \\ 0 & 0 & -1 & 8.00 \\ 0 & 0 & 0 & 1 \end{bmatrix}$; $\mathbf{R} = 17.32\hat{\mathbf{i}}_1 + 6.50\hat{\mathbf{k}}_1$ in.
10.26 $T_{15} = \begin{bmatrix} 0 & -1 & 0 & 180 \text{ mm} \\ 0 & 0 & -1 & -100 \text{ mm} \\ 1 & 0 & 0 & 450 \text{ mm} \\ 0 & 0 & 0 & 1 \end{bmatrix}$; $\mathbf{R} = 180\hat{\mathbf{i}}_1 - 145\hat{\mathbf{j}}_1 + 450\hat{\mathbf{k}}_1$ mm.
10.28 $\dot{\mathbf{R}}_5 = -1.750\hat{\mathbf{i}}_1 + 0.433\hat{\mathbf{j}}_1$ in/s; $\ddot{\mathbf{R}}_5 = -0.541\hat{\mathbf{i}}_1 - 0.088\hat{\mathbf{j}}_1$ in/s²

- 10.30** $\phi_2 = \cos^{-1}(0.020t^2 + 0.050t - 0.875)$ with $-180^\circ \leq \phi_2 \leq 0$
 $\phi_1 = \tan^{-1} \left[\frac{3(1 + \cos \phi_2) - 4 \sin \phi_2}{4(1 + \cos \phi_2) + 3 \sin \phi_2} \right]; \phi_3 = 8 \text{ in}; \phi_4 = \phi_1 + \phi_2 - 36.87^\circ$
- 10.32** $\tau_1 = 25 \text{ in} \cdot \text{lb}; \tau_2 = 100 \sin \phi_1 + 25 \text{ in} \cdot \text{lb}; \tau_3 = -5 \text{ lb};$
 $\tau_4 = -100 \sin \phi_1 - 100 \sin(\phi_1 + \phi_2) - 25 \text{ in} \cdot \text{lb}$
- 11.3** $P = 1\,463 \text{ N}.$
- 11.5** $\mathbf{T}_{12} = 90.6\hat{\mathbf{k}} \text{ in} \cdot \text{lb}.$
- 11.7** $\mathbf{T}_{12} = 383\hat{\mathbf{k}} \text{ in} \cdot \text{lb}.$
- 11.9** $\mathbf{T}_{12} = 252\hat{\mathbf{k}} \text{ in} \cdot \text{lb}.$
- 11.11** $\mathbf{T}_{12} = -756\hat{\mathbf{k}} \text{ in} \cdot \text{lb}; \mathbf{F}_{12} = \mathbf{F}_{23} = 226 \text{ lb} \angle 56.7^\circ; \mathbf{F}_{14} = 318 \text{ lb} \angle -61.4^\circ,$
 $\mathbf{F}_{34} = 189 \text{ lb} \angle 88.8^\circ.$
- 11.13** $\mathbf{F}_{12} = \mathbf{F}_{23} = 307 \text{ kN} \angle 230.4^\circ; \mathbf{F}_{14} = \mathbf{F}_{43} = 387 \text{ kN} \angle 59.7^\circ; \mathbf{M}_{12} = -113\hat{\mathbf{k}} \text{ kN} \cdot \text{m}$
- 11.15** $\mathbf{T}_{12} = 426\hat{\mathbf{k}} \text{ in} \cdot \text{lb}$
- 11.17** (a) $\mathbf{F}_{13} = 2\,520 \text{ lb} \angle 0^\circ$; (b) $\mathbf{F}_{13} = 1\,050 \text{ lb} \angle 225^\circ$; (c) $\mathbf{F}_{13} = 2\,250 \text{ lb} \angle 135^\circ$
- 11.19** $\mathbf{F}_C = 216 \text{ lb} \angle 189^\circ; \mathbf{F}_D = 350 \text{ lb} \angle 163.4^\circ$
- 11.21** $\mathbf{F}_C = 118\hat{\mathbf{i}} + 140\hat{\mathbf{j}} - 251\hat{\mathbf{k}} \text{ lb}; \mathbf{F}_D = -71\hat{\mathbf{i}} - 155\hat{\mathbf{k}} \text{ lb}$
- 11.23** $\mathbf{F}_E = 162\hat{\mathbf{i}} - 191\hat{\mathbf{j}} + 354\hat{\mathbf{k}} \text{ lb}; \mathbf{F}_F = 110\hat{\mathbf{j}} + 144\hat{\mathbf{k}} \text{ lb}$
- 11.25** $\mathbf{T}_{12} = -61.5\hat{\mathbf{k}} \text{ N} \cdot \text{m}$
- 11.29** $\mathbf{F}_F = 378\hat{\mathbf{i}} + 1\,106\hat{\mathbf{j}} \text{ lb}; \mathbf{F}_R = 1\,239\hat{\mathbf{j}} \text{ lb}; \mathbf{F}_T = -378\hat{\mathbf{i}} + 655\hat{\mathbf{j}} \text{ lb}; \mu \geq 0.34$
- 11.31** $\mathbf{T}_{12} = 0.170\hat{\mathbf{k}} \text{ N} \cdot \text{m}$
- 11.33** $T = F_B d (R/x)^2; x \geq R\sqrt{1 + 1/\mu^2}$
- 11.35** (a) $T_{12} = 0.908 \text{ N} \cdot \text{m}$ cw; (b) $\mathbf{F}_{14} = 18.23 \text{ N} \angle -106.7^\circ,$
 $\mathbf{F}_{34} = \mathbf{F}_{23} = \mathbf{F}_{12} = 34.92 \text{ N} \angle 30^\circ$; (c) slipping
- 11.37** $F_{12} = F_{32} = F_{23} = F_{43} = F_{34} = 8 \text{ N}; \mathbf{P} = 11.31 \text{ N} \angle 45^\circ; \mathbf{F}_{14}^B = 5.33 \text{ N} \angle 90^\circ,$
 $\mathbf{F}_{14}^C = 13.33 \text{ N} \angle -90^\circ$, tipping
- 11.40** $d = 66.4 \text{ mm}$
- 11.42** (a) $P_B = 60\,000 \text{ lb};$ (b) $P_{\text{cr}} = 89\,717 \text{ lb}, P_{\text{cr}}/A = 28\,558 \text{ lb/in}^2, N = 1.15;$
(c) $D = 2.77 \text{ in}$
- 11.44** (a) $(S_r)_D = 141.18;$ (b) $P_{\text{cr}} = 239.14 \text{ kN}, N = 4.78;$ (c) $b = 25.5 \text{ mm}$
- 11.46** (a) $(S_r)_D = 140.50;$ (b) $P_{\text{cr}} = 63\,213 \text{ lb};$ (c) $F = 42\,142 \text{ lb};$ (d) $L = 16.5 \text{ ft}$
- 11.48** (a) $N = 6.46;$ (b) $(S_r)_{D3} = 99\,346;$ (c) $t = 0.459 \text{ in};$ (d) $N = 0.28.$
- 11.50** (a) $S_r = 400, (S_r)_D = 147.90;$ (b) $P_{\text{cr}}/A = 19.97 \text{ MPa};$ (c) Johnson
- 12.1** $I_O = 0.030\,8 \text{ in} \cdot \text{lb} \cdot \text{s}^2$
- 12.3** $\mathbf{T}_{12} = -192\hat{\mathbf{k}} \text{ in} \cdot \text{lb}$
- 12.5** $\mathbf{F}_{23} = \mathbf{F}_{12} = 1\,525 \text{ lb} \angle -177^\circ; \mathbf{F}_{14} = 444 \text{ lb} \angle -90^\circ; \mathbf{F}_{34} = 826 \text{ lb} \angle 147.5^\circ,$
 $\mathbf{T}_{12} = 3\,080\hat{\mathbf{k}} \text{ in} \cdot \text{lb}.$
- 12.7** $\mathbf{F}_{23} = \mathbf{F}_{12} = 9.827 \text{ kN} \angle -20.5^\circ; \mathbf{F}_{14} = 11.521 \text{ kN} \angle -154.7^\circ,$
 $\mathbf{F}_{34} = 11.171 \text{ kN} \angle 14.5^\circ; \mathbf{T}_{12} = -2.907\hat{\mathbf{k}} \text{ N} \cdot \text{m}$
- 12.9** $\omega_3 = 3.02\hat{\mathbf{k}} \text{ rad/s}, \omega_4 = 3.61\hat{\mathbf{k}} \text{ rad/s}, \alpha_3 = -0.39\hat{\mathbf{k}} \text{ rad/s}^2, \alpha_4 = -40.82\hat{\mathbf{k}} \text{ rad/s}^2,$
 $\mathbf{A}_{G_3} = 40.37 \text{ m/s}^2 \angle -17.16^\circ, \mathbf{A}_{G_4} = 19.27 \text{ m/s}^2 \angle -8.48^\circ, \mathbf{F}_{23} = \mathbf{F}_{12} =$
 $2.56 \text{ kN} \angle -129.7^\circ, \mathbf{F}_{14} = 6.97 \text{ kN} \angle -84.1^\circ, \mathbf{F}_{34} = 4.34 \text{ kN} \angle 164.1^\circ,$
 $\mathbf{T}_{12} = 668\hat{\mathbf{k}} \text{ N} \cdot \text{m}$

- 12.11** $\mathbf{F}_{23} = \mathbf{F}_{12} = 9.43 \text{ kN} \angle 10.7^\circ$, $\mathbf{F}_{14} = 9.26 \text{ kN} \angle -98.8^\circ$, $\mathbf{F}_{34} = 2.70 \text{ kN} \angle 156^\circ$, $\mathbf{T}_{12} = 2\,780 \hat{\mathbf{k}} \text{ N} \cdot \text{m}$
- 12.13** $\omega_3 = 10.6 \hat{\mathbf{k}} \text{ rad/s}$, $\omega_4 = -4.3 \hat{\mathbf{k}} \text{ rad/s}$, $\alpha_3 = -200 \hat{\mathbf{k}} \text{ rad/s}^2$, $\alpha_4 = -352 \hat{\mathbf{k}} \text{ rad/s}^2$, $\mathbf{A}_{G_3} = 102.4 \text{ m/s}^2 \angle 59.00^\circ$, $\mathbf{A}_{G_4} = 43.94 \text{ m/s}^2 \angle 43.92^\circ$, $\mathbf{F}_{23} = \mathbf{F}_{12} = 169 \text{ N} \angle 47.0^\circ$, $\mathbf{F}_{34} = 659 \text{ N} \angle -39.9^\circ$, $\mathbf{F}_{14} = 655 \text{ N} \angle 134.4^\circ$, $\mathbf{T}_{12} = 11.0 \hat{\mathbf{k}} \text{ N} \cdot \text{m}$
- 12.15** $\omega_3 = -2.57 \hat{\mathbf{k}} \text{ rad/s}$, $\mathbf{V}_B = 1.18 \hat{\mathbf{i}} \text{ m/s}$, $\alpha_3 = 77.19 \hat{\mathbf{k}} \text{ rad/s}^2$, $\mathbf{A}_B = 25.16 \hat{\mathbf{i}} \text{ m/s}^2$, $\mathbf{A}_{G_3} = 16.52 \text{ m/s}^2 \angle -28.85^\circ$, $\mathbf{F}_{23} = \mathbf{F}_{12} = 3.25 \text{ kN} \angle -12.42^\circ$, $\mathbf{F}_{34} = 2.16 \text{ kN} \angle -17.21^\circ$, $\mathbf{F}_{14} = 0.639 \hat{\mathbf{j}} \text{ kN}$, $\mathbf{T}_{12} = -241 \hat{\mathbf{k}} \text{ N} \cdot \text{m}$
- 12.17** $\mathbf{F}_{23} = 191 \hat{\mathbf{i}} - 78 \hat{\mathbf{j}} \text{ N}$, $\mathbf{F}_{14} = -1 \hat{\mathbf{j}} \text{ N}$
- 12.19** $\omega_3 = 0.611 \hat{\mathbf{k}} \text{ rad/s}$, $\mathbf{V}_B = -1.16 \hat{\mathbf{i}} \text{ m/s}$, $\alpha_3 = 6.28 \hat{\mathbf{k}} \text{ rad/s}^2$, $\mathbf{A}_B = 5.46 \hat{\mathbf{i}} \text{ m/s}^2$, $\mathbf{F}_{12} = \mathbf{F}_{23} = 55.6 \text{ kN} \angle 78.41^\circ$, $\mathbf{F}_{14} = 14.3 \hat{\mathbf{j}} \text{ kN}$, $\mathbf{T}_{12} = -9.240 \hat{\mathbf{k}} \text{ kN} \cdot \text{m}$
- 12.21** $\mathbf{F}_{12} = \mathbf{F}_{23} = 8.83 \text{ lb} \angle -120^\circ$, $\mathbf{F}_{14} = \mathbf{F}_{43} = 41.41 \text{ lb} \angle 79.4^\circ$, $\mathbf{T}_{14} = 15.77 \hat{\mathbf{k}} \text{ in} \cdot \text{lb}$
- 12.23** At $\theta_2 = 0^\circ$, $\theta_3 = 120^\circ$, $\theta_4 = 141.8^\circ$, $\omega_3 = 6.67 \text{ rad/s cw}$, $\omega_4 = 6.67 \text{ rad/s cw}$, $\alpha_3 = 141 \text{ rad/s}^2 \text{ cw}$, $\alpha_4 = 64.1 \text{ rad/s}^2 \text{ cw}$, $\mathbf{F}_{21} = 6\,734 \text{ lb} \angle -56^\circ$, $\mathbf{F}_{21} = 7\,883 \text{ lb} \angle 142.8^\circ$, $\mathbf{T}_{12} = 7\,468 \hat{\mathbf{k}} \text{ ft} \cdot \text{lb}$
- 12.25** $\mathbf{T}_{12} = 0.168 \hat{\mathbf{k}} \text{ N} \cdot \text{m}$
- 12.27** $\alpha_2 = -261 \hat{\mathbf{k}} \text{ rad/s}^2$, $\mathbf{F}_{12} = 600 \hat{\mathbf{i}} + 180 \hat{\mathbf{j}} \text{ lb}$
- 12.29** (a) $R'_S = 1 \text{ m/m}$, $R'_C = 1.732 \text{ m/m}$, (b) $m_{\text{EQ}} = 11.25 \text{ kg}$, (c) $P = -733.93 \text{ N}$
- 12.31** (c) $\mathbf{F}_{12} = -81 \hat{\mathbf{i}} + 628 \hat{\mathbf{j}} \text{ N}$, $\mathbf{F}_{23} = -54 \hat{\mathbf{i}} + 523 \hat{\mathbf{j}} \text{ N}$, $\mathbf{F}_{34} = 361 \hat{\mathbf{j}} \text{ N}$, $\mathbf{F}_{14} = \pm 361 \hat{\mathbf{i}} \text{ N}$ (ccw couple). (d) $\mathbf{T}_2 = 230 \hat{\mathbf{k}} \text{ N} \cdot \text{m}$. (e) Contact at top right and bottom left.
- 12.33** (c) $\mathbf{F}_{12} = 119.2 \hat{\mathbf{j}} \text{ N}$, $\mathbf{F}_{23} = 67.7 \hat{\mathbf{i}} + 89.8 \hat{\mathbf{j}} \text{ N}$, $\mathbf{F}_{13} = 35.5 \text{ N} \angle -120^\circ$. (d) $\mathbf{P} = 97.7 \hat{\mathbf{i}} \text{ N}$. (e) Contact at 0.125 m to right of G_3 .
- 12.35** (a) $r'_S = 1 \text{ m/m}$, $r'_C = -1 \text{ m/m}$, $x'_{G_3} = 1 \text{ m/m}$, $y'_{G_3} = 0$, $x''_{G_3} = -1.5 \text{ m/m}^2$, $y''_{G_3} = -2.598 \text{ m/m}^2$. (b) $m_{\text{EQ}} = 2 \text{ kg}$. (d) $\mathbf{P} = 97.1 \hat{\mathbf{i}} \text{ N}$.
- 12.37** $w_B = w_F = 1.33 \text{ lb}$.
- 12.39** $w_B = w_E = 2.67 \text{ lb}$, $\theta_B = \theta_E = 180^\circ$.
- 12.41** $I_G = 2.89 \text{ in} \cdot \text{lb} \cdot \text{s}^2$, $k_G = 5.22 \text{ in}$.
- 12.45** $\omega_3 = (1 - 3R/4r)\omega_2$; $\omega_3 = 0$ for $r = 3R/4$.
- 12.47** $\alpha_{in} = 512 \text{ rad/s}^2$, $F_{2A} = 338.1 \text{ lb}$, $F_{11} = 221.4 \text{ lb}$; unbalanced moment on housing of 4 190 in · lb must be absorbed by mounting bolts during startup.
- 12.49** $\mathbf{F}_{12} = -87.4 \hat{\mathbf{j}} \text{ lb}$, $\mathbf{T}_{12} = -1\,200 \hat{\mathbf{j}} \text{ in} \cdot \text{lb}$, $\mathbf{F}_{35} = -238 \hat{\mathbf{i}} - 300 \hat{\mathbf{k}} \text{ lb}$, $\mathbf{M}_{35} = 258 \hat{\mathbf{k}} \text{ in} \cdot \text{lb}$, $\mathbf{F}_{45} = 238 \hat{\mathbf{i}} + 300 \hat{\mathbf{k}} \text{ lb}$, $\mathbf{M}_{45} = -258 \hat{\mathbf{k}} \text{ in} \cdot \text{lb}$, $\mathbf{F}_{15} = \mathbf{0}$, $\mathbf{M}_{15} = 2\,400 \hat{\mathbf{j}} \text{ in} \cdot \text{lb}$.
- 12.51** $T_{12} = 26 \text{ N} \cdot \text{m}$, $\mu \geq 0.26$
- 12.53** $\mathbf{F}_{35} = 443.8 \text{ N} \angle 131^\circ$, bottom of slot, $\mathbf{F}_{25} = 619.2 \text{ N} \angle -56.5^\circ$
- 12.55** $R'_S = 0$, $\theta'_S = 0.80 \text{ rad/m}$, $R'_C = 1.73 \text{ m/m}$, $T = 171.6 \text{ N} \cdot \text{m}$, $P = 424.5 \text{ N}$
- 12.57** (a) $x'_{G_2} = 0.866 \text{ m/m}$, $y'_{G_2} = 0.500 \text{ m/m}$, $x''_{G_2} = y''_{G_2} = 0$, (b) $R'_S = 1.000 \text{ m/m}$, $R'_C = -1.000 \text{ m/m}$, (c) $\ddot{R}_2 = -29.8 \text{ m/s}^2$
- 12.59** (a) $R'_S = 50 \text{ mm/rad}$, $\theta'_S = 0$, $R'_C = -37.5 \text{ mm/rad}$, $\theta'_C = -0.25 \text{ rad/rad}$, (b) $T = 69.7 \text{ N} \cdot \text{m}$, (c) $\mathbf{F} = -1\,465 \hat{\mathbf{i}} \text{ N}$
- 13.3** (a) 632 N/m; (b) 15.5 mm; (c) 0.25 s/rev; (d) 4 Hz; (e) $\dot{x} = 0.448 \text{ m/s}$; $\ddot{x} = 5.74 \text{ m/s}^2$; (f) -5.74 N .

- 13.5 3.010 Hz
- 13.7 (a) $\omega_n = \sqrt{\frac{g}{\ell} \left[\frac{(k_1 + k_2)a^2}{W\ell} - 1 \right]}$, (b) $\frac{\ell}{a} \geq (k_1 + k_2) \frac{a}{W}$
- 13.9 (a) 1st era: $0 \leq t \leq 0.040$ s, $x = (1.429 \text{ mm})(1 - \cos 171t)$;
2nd era: $t > 0.040$ s, $x = (0.773 \text{ mm}) \cos(171t - 74.3^\circ)$
(b) 1st era: $X = 1.429$ mm; 2nd era: $X = 0.773$ mm
- 13.11 (a) 47.7 rev/min; (b) 16.7 rad/s; (c) 1.71° , 3.18°
- 13.13 0.177 in
- 13.15 (a) $c_c = 2\,993 \text{ N} \cdot \text{s/m}$; (b) $\omega_n = 37.4 \text{ rad/s}$; (c) $\omega_d = 36.7 \text{ rad/s}$; (d) $\delta = 1.28$
- 13.17 (a) $\omega_d = 35.15 \text{ rad/s}$, (b) $\tau = 0.179 \text{ s/cycle}$, $\delta = 1.28$
- 13.21 $X = 1.280 \text{ mm}$, $\phi = 55.8^\circ$
- 13.23 $k = 2\,637 \text{ kN/m}$, $X = 0.151 \text{ mm}$
- 13.25 (a) Upper limit ($\omega_1 = 112 \text{ rad/s}$) is Rayleigh-Ritz, lower limit ($\omega_1 = 100 \text{ rad/s}$) is Dunkerley. Exact is $\omega_1 = 110 \text{ rad/s}$. (b) $m = 0.5 \text{ kg}$ (c) $a_{12} = 5.94 \times 10^{-5} \text{ m/N}$
- 13.27 (a) $\omega_1 = 665.6 \text{ rad/s}$ (b) $\omega_1 = 667.2 \text{ rad/s}$
- 13.29 $\omega_1 = 512.98 \text{ rad/s}$, $\omega_2 = 2\,835.07 \text{ rad/s}$; (a) $\omega_1 = 513.45 \text{ rad/s}$;
(b) $\omega_1 = 504.78 \text{ rad/s}$
- 13.31 $a_{22} = 7.17 \times 10^{-4} \text{ mm/N}$
- 13.33 (a) $m = 274.35 \text{ lb}$; (b) $\omega_2 = 286.0 \text{ rad/s}$
- 13.35 $\omega_1 = 145.4 \text{ rad/s}$
- 13.37 (a) $\omega_1 = 144.25 \text{ rad/s}$; (b) $\omega_1 = 149.20 \text{ rad/s}$
- 13.39 $\omega_1 = 471.95 \text{ rad/s}$, $\omega_2 = 809.42 \text{ rad/s}$; (a) $\omega_1 = 494.08 \text{ rad/s}$;
(b) $\omega_1 = 407.75 \text{ rad/s}$
- 13.41 (a) $\omega_1 = 152 \text{ rad/s}$; (b) $m = 0.074 \text{ kg}$; (c) $a_{12} = 1.27 \times 10^{-4} \text{ m/N}$
- 14.1 At $X = 30\%$, $p_e = 1\,512 \text{ kPa}$, $p_c = 338 \text{ kPa}$.
- 14.3 See figure.

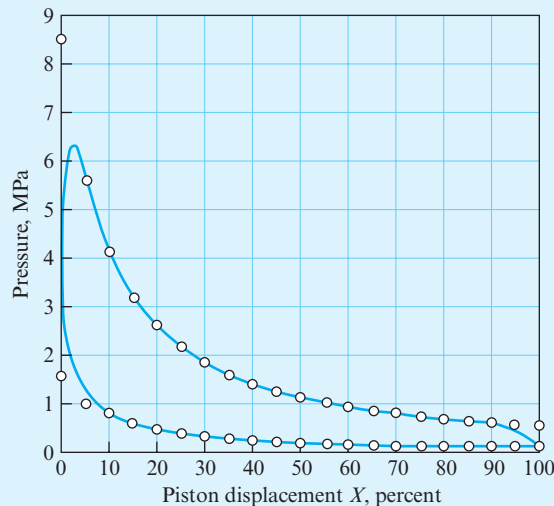


Figure AP14.3

- 14.5 $\mathbf{F}_{41} = -1\,010\hat{\mathbf{j}}\text{ N}$, $\mathbf{F}_{34} = 4\,254\text{ N}\angle -13.7^\circ$, $\mathbf{F}_{32} = 4\,196\text{ N}\angle 163.7^\circ$,
 $\mathbf{T}_{21} = 89.59\hat{\mathbf{k}}\text{ N}\cdot\text{m}$.
- 14.9 $\mathbf{F}_{41} = -574\hat{\mathbf{j}}\text{ N}$, $\mathbf{F}_{34} = 2\,999\text{ N}\angle 191^\circ$, $\mathbf{F}_{32} = 9\,918\text{ N}\angle -39.0^\circ$, $\mathbf{T}_{21} = 183\hat{\mathbf{k}}\text{ N}\cdot\text{m}$.
- 14.11 At $\omega t = 60^\circ$, $X = 40.9\%$, $P = 4\,445\text{ N}$, $\ddot{x} = -1\,335\text{ m/s}^2$, $F_{14} = 561\text{ N}$,
 $T_{21} = 110.6\text{ N}\cdot\text{m}$.
- 15.1 $\mathbf{F}_A = 64.7\angle 76.1^\circ\text{ N}$, $\mathbf{F}_B = 16.2\angle 76.1^\circ\text{ N}$, $m_c = 1.64\text{ kg}$, $\theta_C = 76.1^\circ$.
- 15.3 $\mathbf{F}_A = 8.06\angle -14.4^\circ\text{ lb}$, $\mathbf{F}_B = 2.69\angle 165.6^\circ\text{ lb}$, $w_C = 2.63\text{ lb}$ at $\theta_C = -14.4^\circ$.
- 15.5 $\mathbf{F}_A = 13.16\angle 90^\circ\text{ N}$, $\mathbf{F}_B = 0$.
- 15.7 $m_L\mathbf{R}_L = 5.98\text{ oz}\cdot\text{in}\angle -16.5^\circ$, $m_R\mathbf{R}_R = 7.33\text{ oz}\cdot\text{in}\angle 136.8^\circ$.
- 15.9 Remove $m_L\mathbf{R}_L = 782.1\text{ g}\cdot\text{mm}\angle 180.4^\circ$ and $m_R\mathbf{R}_R = 236.8\text{ g}\cdot\text{mm}\angle 301.2^\circ$.
- 15.11 See Sec. 15.8 for answers.
- 15.13 $(m_1)_{\text{new}} = 1.5\text{ kg}\angle \theta_1 = 27.3^\circ$; $(m_3)_{\text{new}} = 1.113\text{ kg}\angle \theta_3 = 300^\circ$.
- 15.15 $m_{C1} = -10.286\text{ kg}\angle 75^\circ$; $m_{C2} = -7.500\text{ kg}\angle -120^\circ$
- 15.17 $\mathbf{F}_B = 3.766\text{ N}\angle 108.49^\circ$; $\mathbf{F}_A = 6.220\text{ N}\angle -141.14^\circ$; $m_{C1} = 1.016\text{ kg}$,
 $m_{C2} = 0.561\text{ kg}$
- 15.19 $F_A = 105.20\text{ lb}\angle -30^\circ$, $F_B = 26.90\text{ lb}\angle -141^\circ$
- 15.21 $\mathbf{F}_A = 60.51\text{ N}\angle -153^\circ$, $\mathbf{F}_B = 33.32\text{ N}\angle -85^\circ$
- 15.23 (a) $F_P^x = 36\,000\cos\theta + 44\,785\sin\theta\text{ N}$, $F_P^y = 3\,215.4\cos\theta - 12\,000\sin\theta\text{ N}$;
 (b) $m_{C1} = 15.0\text{ kg}$, $\gamma_1 = 120^\circ$, $m_{C2} = 21.21\text{ kg}$, $\gamma_2 = 135^\circ$;
 (c) $F_P^x = 53\,569\text{ N}$, $F_P^y = -3\,215.4\text{ N}$
- 15.25 (a) $F_1 = 0$, $F_2 = 1\,012.5\text{ N}$; (b) $m_{C1} = 0$, $m_{C2} = 2.5\text{ kg}$; $\gamma_1 = \text{undefined}$, $\gamma_2 = 90^\circ$
- 15.27 (a) $S_P^x = -0.116P\cos\theta - 0.790P\sin\theta$, $S_P^y = -1.299P\cos\theta + 2.250P\sin\theta$
 (b) $F_1 = 1.097P$, $F_2 = 1.578P$,
- 15.29 (a) $S_P^x = 184\,312.5\cos\theta - 117\,937.5\sin\theta\text{ N}$, $S_P^y = 63\,187.5\cos\theta + 51\,750\sin\theta\text{ N}$;
 (b) $m_{C1} = 47.61\text{ kg}$, $m_{C2} = 22.95\text{ kg}$; $\gamma_1 = +217.5^\circ$, $\gamma_2 = +202.4^\circ$;
 (c) $S_P = 128.88\text{ kN}$, $\tau = 38.65^\circ$
- 16.1 $T_m = 70.9\text{ N}\cdot\text{m}$, $I = 0.154\text{ kg}\cdot\text{m}^2$.
- 16.3 $T_m = 278\text{ in}\cdot\text{lb}$, $I = 1.842\text{ in}\cdot\text{lb}\cdot\text{s}^2$.
- 16.5 $T_m = 1\,111\text{ in}\cdot\text{lb}$.
- 16.6 $F = 3\,319\,000\text{ lb}$; $\omega_p \geq 33.9\text{ rev/min}$
- 16.8 $M^z = 2.2\text{ in}\cdot\text{lb}$
- 16.10 $\sum M = -4.9\hat{\mathbf{k}}\text{ in}\cdot\text{lb}$ (negligible)

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